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TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 3 – Software Requirements Specification

Automatic Attendance System Based on Real-Time Location and Face Recognition

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Definitions & Acronyms

No	Acronyms & Abbreviations	Definitions
1	**AAS**	**The Automatic Attendance System Based on Real-Time Location and Face Recognition**
2	**PMP**	Project Management Plan
3	**SRS**	Software Requirements Specification
4	**SDD**	Software Design Description
5	**STD**	Software Testing Documentation
6	**SUG**	Software User Guides
7	**WBS**	Work Breakdown Structure
8	**BR**	Business Rule
9	**ERD**	Entity Relationship Diagram
10	**GUI/UI**	Graphical User Interface
11	**UX**	User Experience
12	**JWT**	JSON Web Token
13	**HTTPS**	Hypertext Transfer Protocol Secure
14	**UC**	Use Case
15	**API**	Application Program Interface
16	**E**	Exception
17	**RP**	Release Package
18	**MKT**	Marketing Module
19	**AUT**	Authentication Module
20	**USR**	User Profile Management Module
21	**PRJ**	Projects Management Module
22	**SCE**	Scenarios Management Module
23	**BOA**	Blocks of Action Module
24	**AST**	Assets Management Module
25	**LOG**	Logs Management Module
26	**SUB**	Subscription Management Module
27	**SYS**	System Management Module
28	**APS**	App Settings Module
29	**CI/CD**	Continuous Integration/Continuous Deployment
30	**CRUD**	Create Read Update Delete
31	**URL**	Uniform Resource Locator
32	**VPS**	Virtual Private Server
33	**JSON**	JavaScript Object Notation
34	**DOM**	Document Oriented Model
35	**CMS**	Content Management System
36	**ADB**	Android Debug Bridge

Table 1: Definitions & Acronyms

1. Overview

1.1. Product Perspective

The Automatic Attendance System (AAS) is a new, independent software solution developed to enhance and automate the attendance-taking process in educational environments. It operates as a system comprising a mobile application for students and lecturers, and a web-based portal for lecturers and administrators. The system integrates seamlessly with institutional IT infrastructure but does not rely on existing third-party attendance solutions.

AAS leverages Bluetooth Low Energy (BLE) and on-device face recognition to detect student presence and authenticate identity during class sessions. The mobile application plays a dual role: broadcasting BLE signals and performing liveness-verified face recognition. The backend system, hosted on cloud infrastructure, synchronizes attendance data, supports real-time analytics, and enforces business rules such as session schedules, thresholds, and complaint resolution.

This product follows an offline-first architecture, allowing attendance data to be captured and cached locally when internet access is unavailable, with automatic synchronization when the connection is restored. Unlike traditional attendance systems that require fixed hardware (e.g., biometric kiosks, RFID scanners), AAS minimizes deployment and maintenance costs by utilizing students' personal devices.

The system is modular, extensible, and aligned with "Smart Campus" initiatives, supporting future enhancements such as Android compatibility, advanced biometric spoofing prevention, and integration with learning management systems (LMS).

1.2. System Interfaces

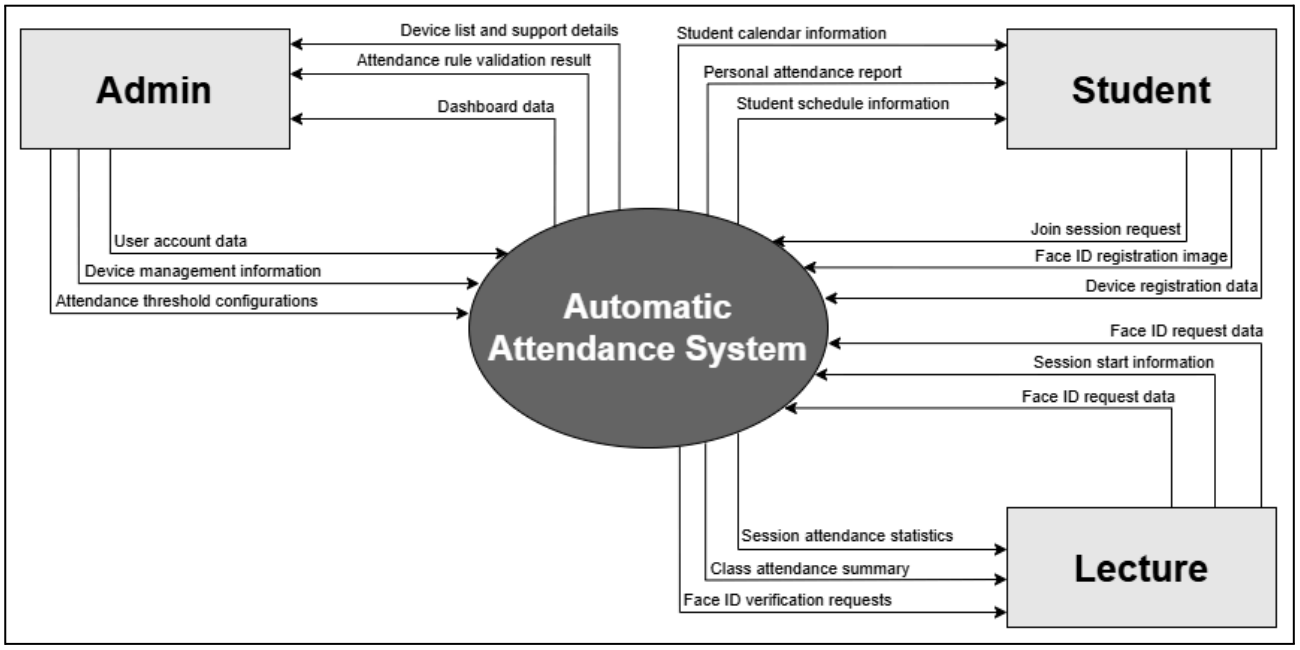


Figure 1: Context Diagram

1.3. User Interfaces

1.3.1. Web Interface

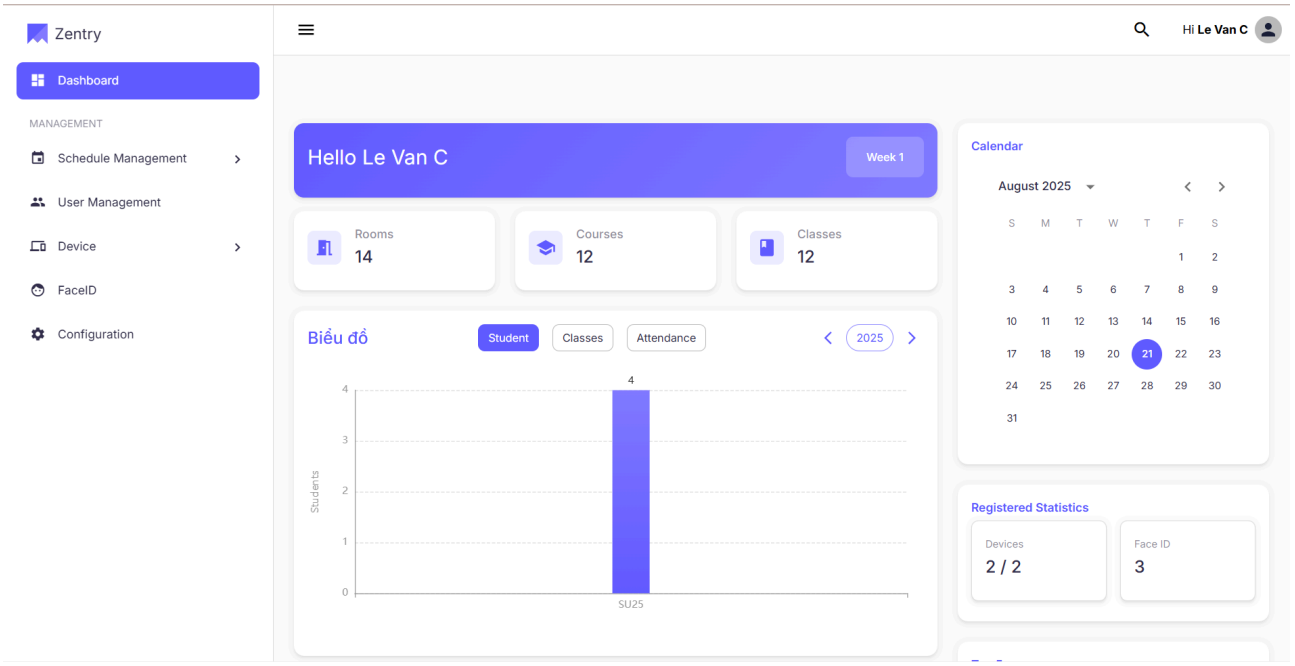


Figure 2: Web Interface

The web interface of **the AAS system** is designed exclusively for **Admin users**, allowing them to manage the entire attendance infrastructure.

Key Features:

Dashboard: View attendance trends, system status, and performance metrics.

User Management: Create, update, or remove student and lecturer accounts.

Schedule Management: Import and manage class schedules.

Device Management: Register, update, or delete attendance devices.

Attendance Rules: Configure thresholds and attendance detection rules to ensure accurate attendance tracking.

All admin functions are accessible via the web with secure login and role-based access control.

1.3.2. Mobile Interface

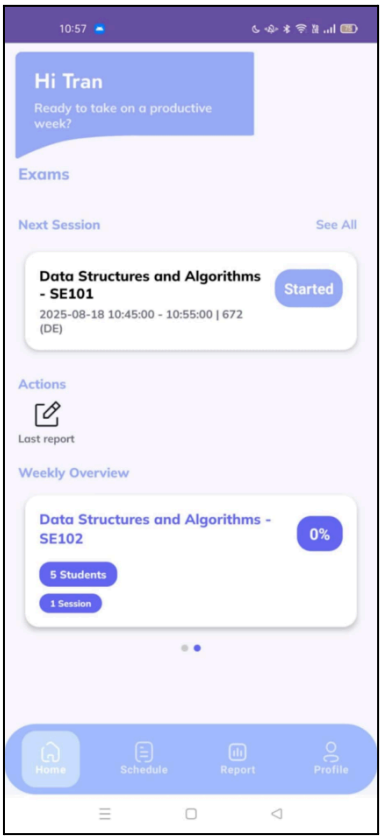


Figure 3: Mobile Interface

The mobile interface of **the AAS system** is designed to be role-aware and dynamically adjusts its features based on the logged-in user's role (Student, Lecturer, or Admin). The interface is optimized for Android devices with a clean, grid-based layout and icon-driven navigation.

Key Features (Displayed based on role):

- **Attendance Sessions**
  - Student: Join sessions using Bluetooth or face recognition.
  - Lecturer: Start and manage attendance sessions.
- **Attendance Reports**
  - Student: View personal attendance history.
  - Lecturer/Admin: View class/session statistics.
- **Manage Users**
  - Admin only: Manage student and lecturer accounts.
- **Schedule**
  - Student: View timetable.
  - Lecturer/Admin: Manage or import class schedules.
- **Face ID**

  Register, update, or verify face ID for attendance purposes.
- **Settings**

  Configure preferences, notifications, and attendance options.

1.4. Hardware Interfaces

The system interacts with various hardware to ensure compatibility and functionality across devices and environments.

Mobile Devices:

- Android smartphones and tablets (Android 6.0 or later) with standard hardware configurations, including touchscreen and sensors.

- Internet connectivity for specific external device integration.

Web Browsers:

- Optimized for modern browsers like Chrome, Edge, and Firefox for web-based administration.

API Servers:

Requires servers with Docker capabilities for deployment of the API and backend systems.

2. User Requirements

2.1. Actors

1	**Admin**	A system administrator responsible for managing users, devices, schedules, and configuring attendance rules.
2	**Authorized User**	A verified user who has signed in and can manage settings, register devices, and receive notifications.
3	**Student**	A user who inherits all functions of an Authorized User and can join attendance sessions, view personal attendance.
4	**Lecturer**	A user who inherits all functions of an Authorized User and can manage attendance sessions, view teaching schedules.

Table 2: Actors Description

2.2. Use Cases

2.2.1. Diagrams

2.2.1.1. Admin

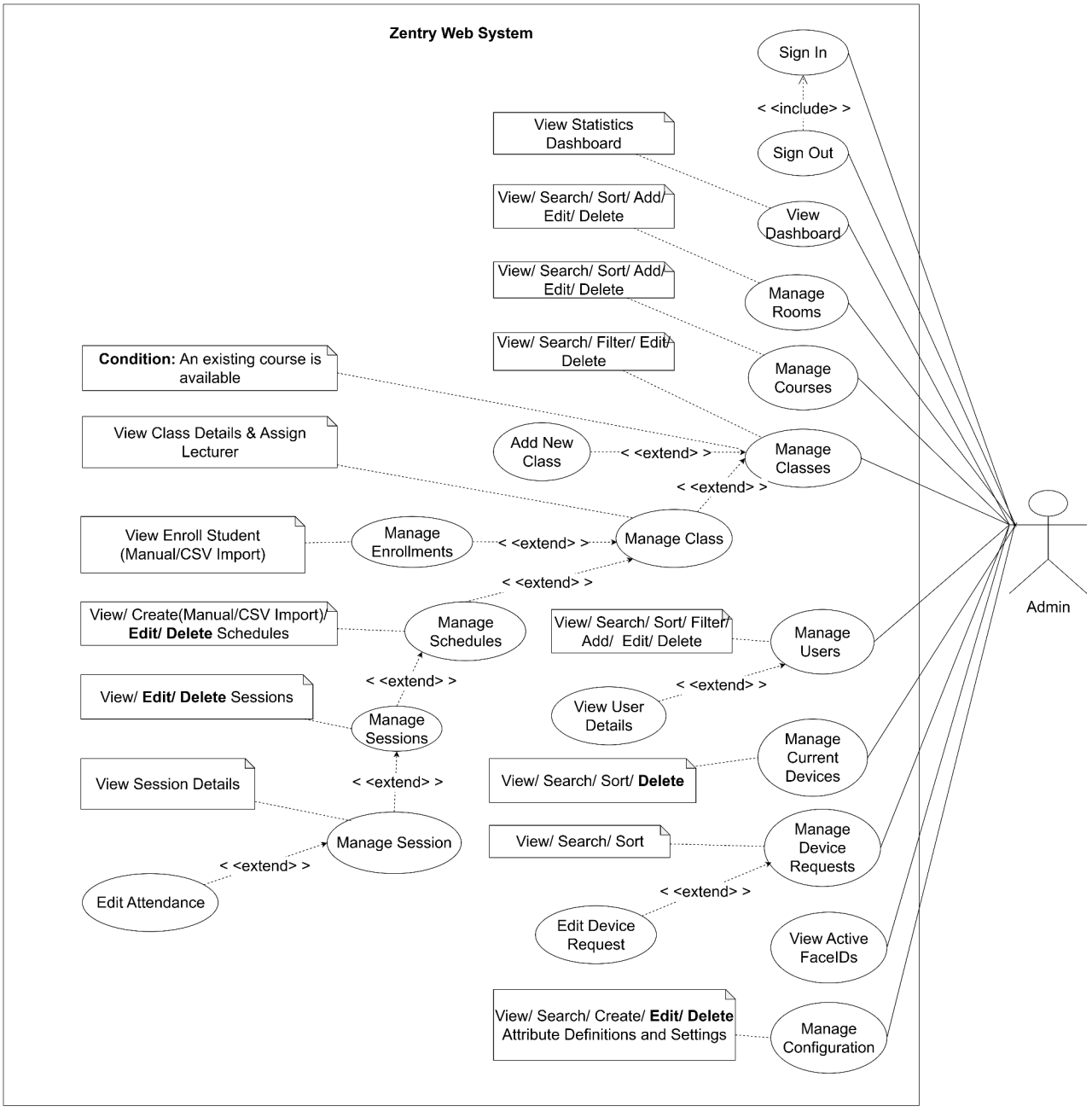


Figure 4: Use Cases Diagram - Admin

2.2.1.2. Authorized User

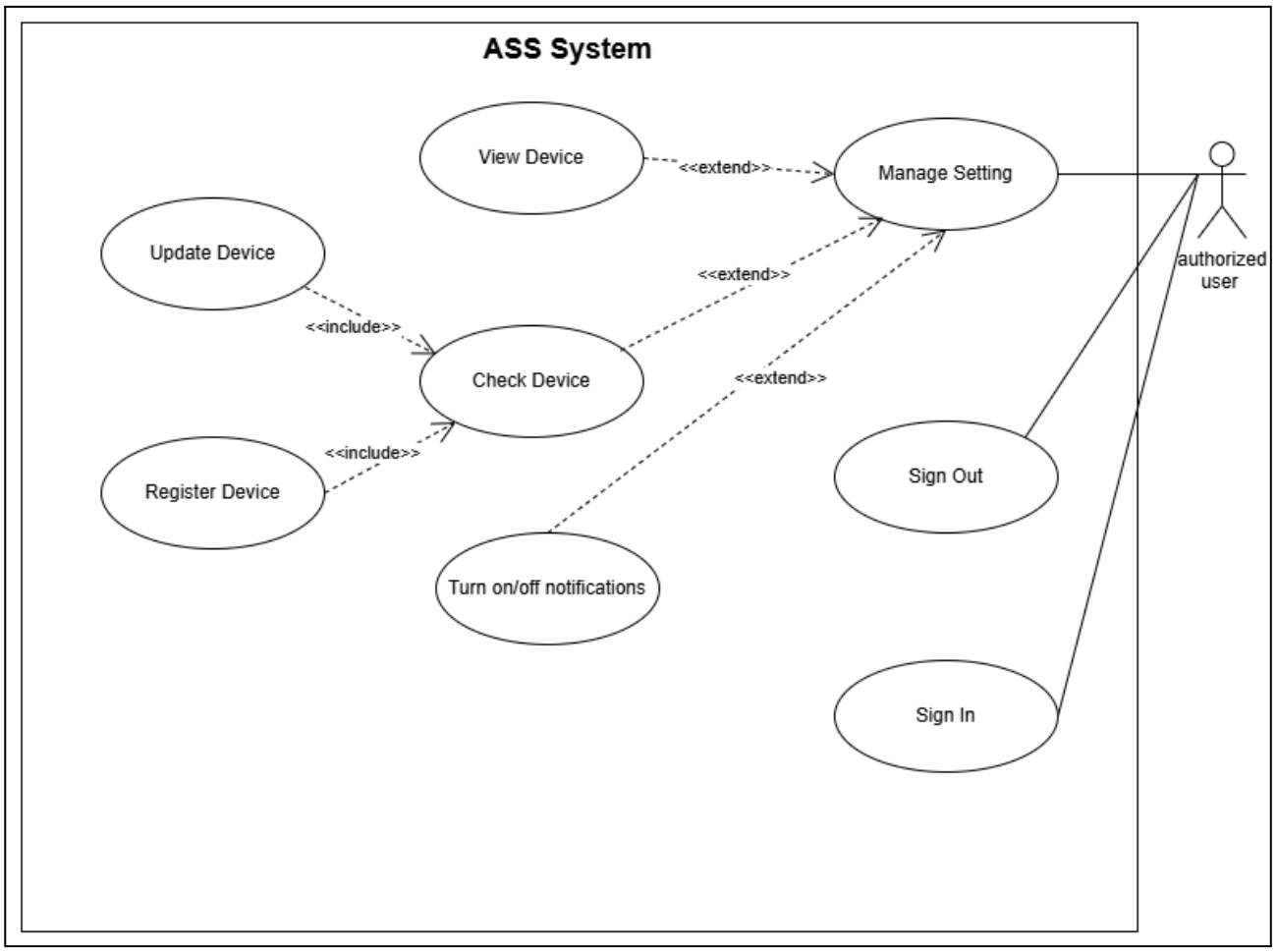


Figure 5: Use Cases Diagram - Authorized User

2.2.1.3. Student

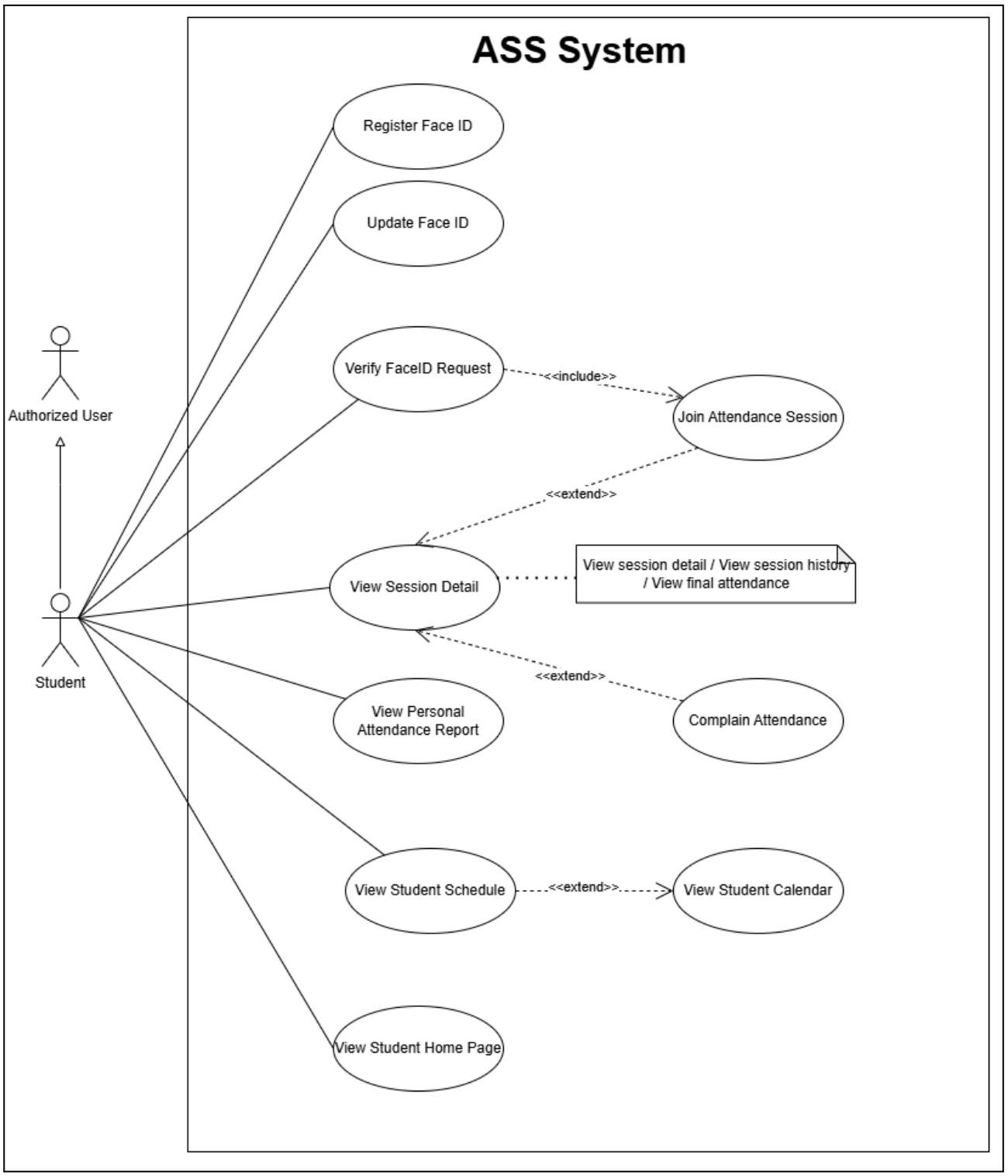


Figure 6: Use Cases Diagram – Student

2.2.1.4. Lecturer

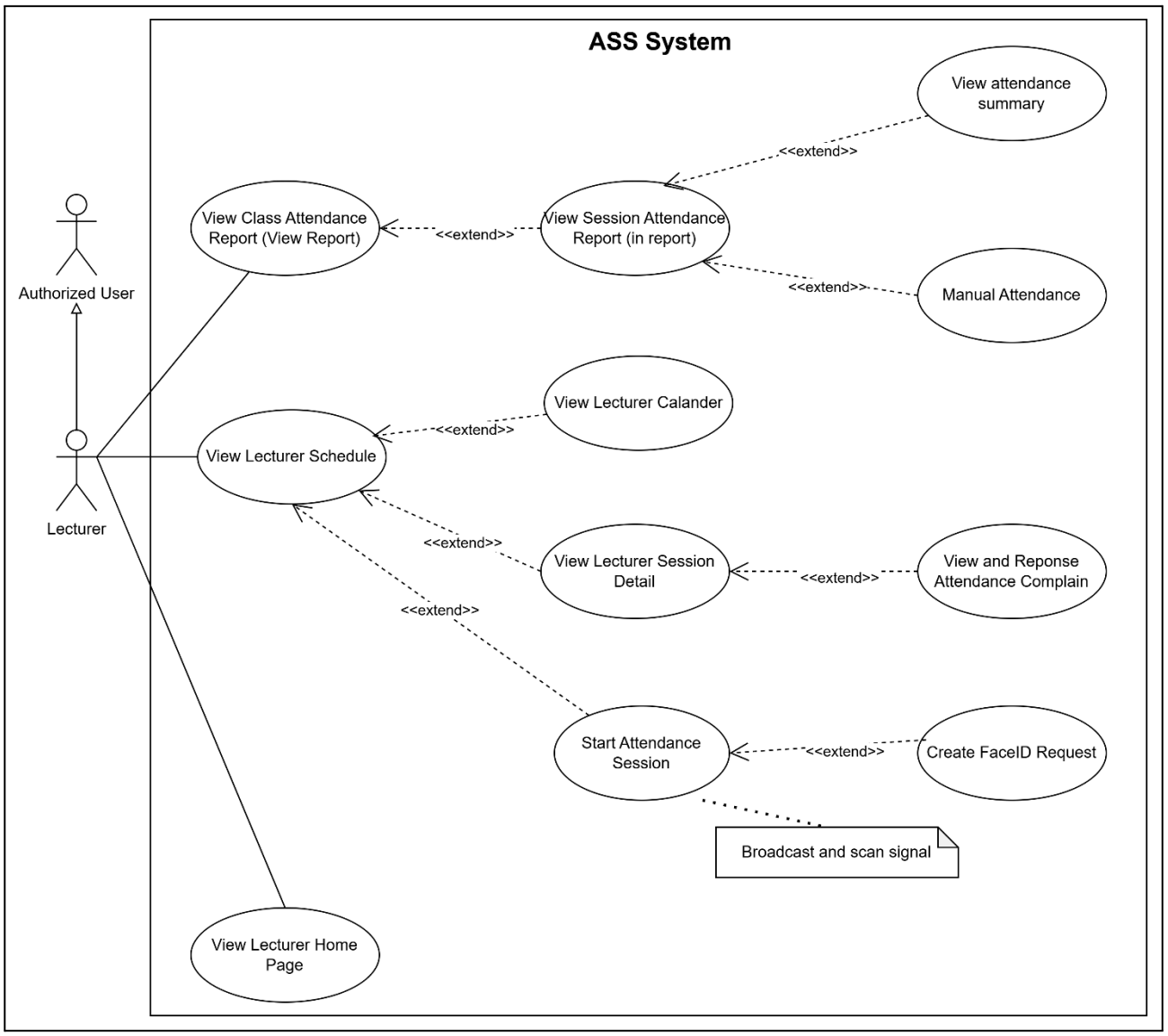


Figure 7: Use Cases Diagram – Lecturer

2.2.2. Use case Descriptions

No	Use Case	Actor	Definitions
UC-01	Sign In	Student, Lecturer, Admin	Login using registered credentials or linked OAuth account.
UC-02	Sign Out	Student, Lecturer, Admin	Securely sign out of the application.
UC-03	View Student Setting	Student	View/manage settings: profile, notifications, device, FaceID, logout.
UC-04	View Lecture Setting	Lecturer	View/manage settings: profile, notifications, device, logout.
UC-05	View Profile	Student, Lecturer	Display user profile details.
UC-06	View Device	Student, Lecturer	View registered devices and update info.
UC-07	Register Device	Student, Lecturer	Register Bluetooth-capable devices for attendance.
UC-08	View FaceID	Student, Lecturer	View and manage registered Face ID.
UC-09	Register Face ID	Student	Register facial data for secure login/attendance.
UC-10	Update Face ID	Student	Update Face ID if already registered.
UC-11	View Student Home Page	Student	Display upcoming schedule and announcements.
UC-12	View Student Schedule	Student	Show student timetable and sessions.
UC-13	View Student Calendar	Student, Lecturer	View calendar format with classes and session details.
UC-14	View Personal Attendance Report	Student	View attendance percentage per subject/semester.
UC-15	Backfill Data	System	Backfill offline data when reconnected.
UC-16	Join Attendance Session	Student	Join an active roll call session.
UC-17	View Session Detail	Student	View session info, history, and final attendance.
UC-18	Verify FaceID Request	Student	Verify identity via Face ID during session.
UC-19	View Lecturer Home Page	Lecturer	Teaching overview: subjects, student stats, progress.
UC-20	View Lecturer Schedule	Lecturer	View teaching schedule and access sessions.
UC-21	View Lecturer Calendar	Lecturer	Calendar of scheduled classes and activities.
UC-22	View Lecturer Session Detail	Lecturer	Detailed session info: history, total duration, stats.
UC-23	View Class Attendance Report	Lecturer	View class-level attendance summaries.
UC-24	View Session Attendance Report	Lecturer	View detailed report of each session.
UC-25	Broadcast and Scan Signal	Student, Lecturer	Start Bluetooth scan to trigger attendance.

UC-26	Create FaceID Request	Lecturer	Request Face ID verification during the session.
UC-27	Automotive Attendance	Lecturer	Auto-take attendance via Bluetooth.
UC-28	Manual Attendance	Lecturer	Mark attendance manually (within 24h).
UC-29	Start Attendance Session	Lecturer	Start a new attendance session (manual/auto).
UC-30	View Dashboard	Admin	Provide an overview of key statistics, schedules, and summary cards, charts, and highlights.
UC-31	View Rooms	Admin	Admin views all rooms with search, filter, and pagination.
UC-32	Add Room	Admin	Admin adds a new room (name, building, capacity, status).
UC-33	Edit Room	Admin	Admin edits an existing room's details.
UC-34	Delete Room	Admin	Admin deletes a room (if not referenced or with safe checks).
UC-35	View Courses	Admin	Admin views all courses with search/filter.
UC-36	Add Course	Admin	Admin creates a new course (code, name, description, semester).
UC-37	Edit Course	Admin	Admin updates course information.
UC-38	Delete Course	Admin	Admin deletes a course (with dependency validation).
UC-39	View Classes	Admin	Admin views all classes (by course/semester/lecturer).
UC-40	Edit Class	Admin	Admin edits class info (lecturer, section, schedule links).
UC-41	Delete Class	Admin	Admin deletes a class (respecting constraints/history).
UC-42	Add Class	Admin	Admin adds a new class and links it to a course/semester.
UC-43	View Class Details	Admin	Admin views detailed class info (students, sessions, stats).
UC-44	Assign Lecturer	Admin	Admin assigns a lecturer to a class (or reassigns).
UC-45	View Enrolled Student	Admin	Admin views enrolled students of a class.

UC-46	Import Student	Admin	Admin imports students in bulk (e.g., CSV/Excel).
UC-47	Enroll Student	Admin	Admin enrolls a student into a class.
UC-48	View Schedules	Admin	Admin views all schedules (filters by room/class/date).
UC-49	Add Schedule	Admin	Admin creates a schedule entry (time, room, class).
UC-50	Edit Schedule	Admin	Admin edits a schedule entry.
UC-51	Delete Schedule	Admin	Admin deletes a schedule entry.
UC-52	View Sessions	Admin	Admin views sessions (planned and completed).
UC-53	Edit Session	Admin	Admin edits a session (time/room/status/notes).
UC-54	Delete Session	Admin	Admin deletes a session (with audit constraints).
UC-55	View Session Details	Admin	Admin views session details (attendance list, metrics).
UC-56	Edit Attendance	Admin	Admin edits attendance records for a session (audited).
UC-57	View Users	Admin	Admin views users (students/lecturers/admins) with filters.
UC-58	Add User	Admin	Admin adds a new user account.
UC-59	Import User	Admin	Admin imports users in bulk (CSV/Excel).
UC-60	Edit User	Admin	Admin edits user information (name, email, role, etc.).
UC-61	Change User Status	Admin	Admin changes a user’s status (activate/deactivate/lock).
UC-62	View User Details	Admin	Admin views detailed user profile and activity.
UC-63	View Active Devices	Admin	Admin views active devices linked to users.
UC-64	Inactive Device	Admin	Admin marks a device as inactive/disabled.

UC-65	View Device Requests	Admin	Admin views incoming device registration requests.
UC-66	Edit Device Request	Admin	Admin reviews and updates a device request (approve/reject, notes).
UC-67	View Active FaceIDs	Admin	Admin views active FaceIDs (approved/usable records).
UC-68	View Attribute Definitions	Admin	Admin views attribute definitions used by the system.
UC-69	Create Attribute Definition	Admin	Admin creates a new attribute definition.
UC-70	Update Attribute Definition	Admin	Admin updates an existing attribute definition.
UC-71	Delete Attribute Definition	Admin	Admin deletes an attribute definition (with validation).
UC-72	View Settings	Admin	Admin views global/system settings.
UC-73	Create Setting	Admin	Admin creates a new setting (key/value or structured).
UC-74	Update Setting	Admin	Admin updates an existing setting.
UC-75	Delete Setting	Admin	Admin deletes a setting (with safety checks).
UC-76	Process Attendance Data	System	Validate and store submitted attendance records.
UC-77	Store Local Attendance Data	System (Mobile)	Temporarily store attendance offline when the device is disconnected.

Table 3: Use Cases Descriptions

2.3. Business Rules

BR-01	User Authentication Required	All users must sign in using valid credentials or OAuth before accessing the system.
BR-02	Role-Based Access Control	Only authorized roles (Student, Lecturer, Admin) can access specific functions.
BR-03	Device Registration Required	Only approved devices can be used for attendance and FaceID verification.
BR-04	FaceID Registration Required	Students must register FaceID before joining FaceID-based attendance.
BR-05	Attendance Session Timeframe	Joining or broadcasting an attendance session is only allowed within the configured time slots.
BR-06	Only One Active Attendance Session	Each class can only have one active attendance session at a time.

BR-07	Manual Attendance Adjustment Deadline	Manual attendance must be completed within 24 hours after the session ends.
BR-08	Admin Approval for Device Updates	Device update/registration must be approved by Admin before use.
BR-09	Only Lecturer Can Modify Attendance	Only Lecturers are allowed to modify attendance records. Students and Admins cannot make changes.
BR-10	Automatic Data Synchronization	Offline attendance data must be automatically synchronized once the connection is restored.
BR-11	Attendance Threshold Configurable	Admin can define attendance thresholds (e.g., 10%, 20%) for monitoring purposes.
BR-12	Attendance History Must Be Stored	All attendance session data must be stored for reporting and auditing purposes.
BR-13	Admin Manages Schedules and Users	Only Admin has CRUD rights over schedules and user accounts.
BR-14	Attendance Data Backup Required	The system must ensure automatic backup and recovery of attendance data.
BR-15	Hidden Device Scanning Time	The system automatically scans nearby devices for attendance purposes, but Lecturers and Students are not informed of when the scan occurs to ensure security and prevent cheating.

Table 4: Business Rules

No	Actor	Description
1	**Admin**	A system administrator responsible for managing users, devices, schedules, and configuring attendance rules.
2	**Authorized User**	A verified user who has signed in and can manage settings, register devices, and receive notifications.
3	**Student**	A user who inherits all functions of an Authorized User and can join attendance sessions, view personal attendance.
4	**Lecturer**	A user who inherits all functions of an Authorized User and can manage attendance sessions, view teaching schedules.

Table 5: Actors Description

3. Functional Requirements

3.1. System Functional Overview

3.1.1. Screens Flow

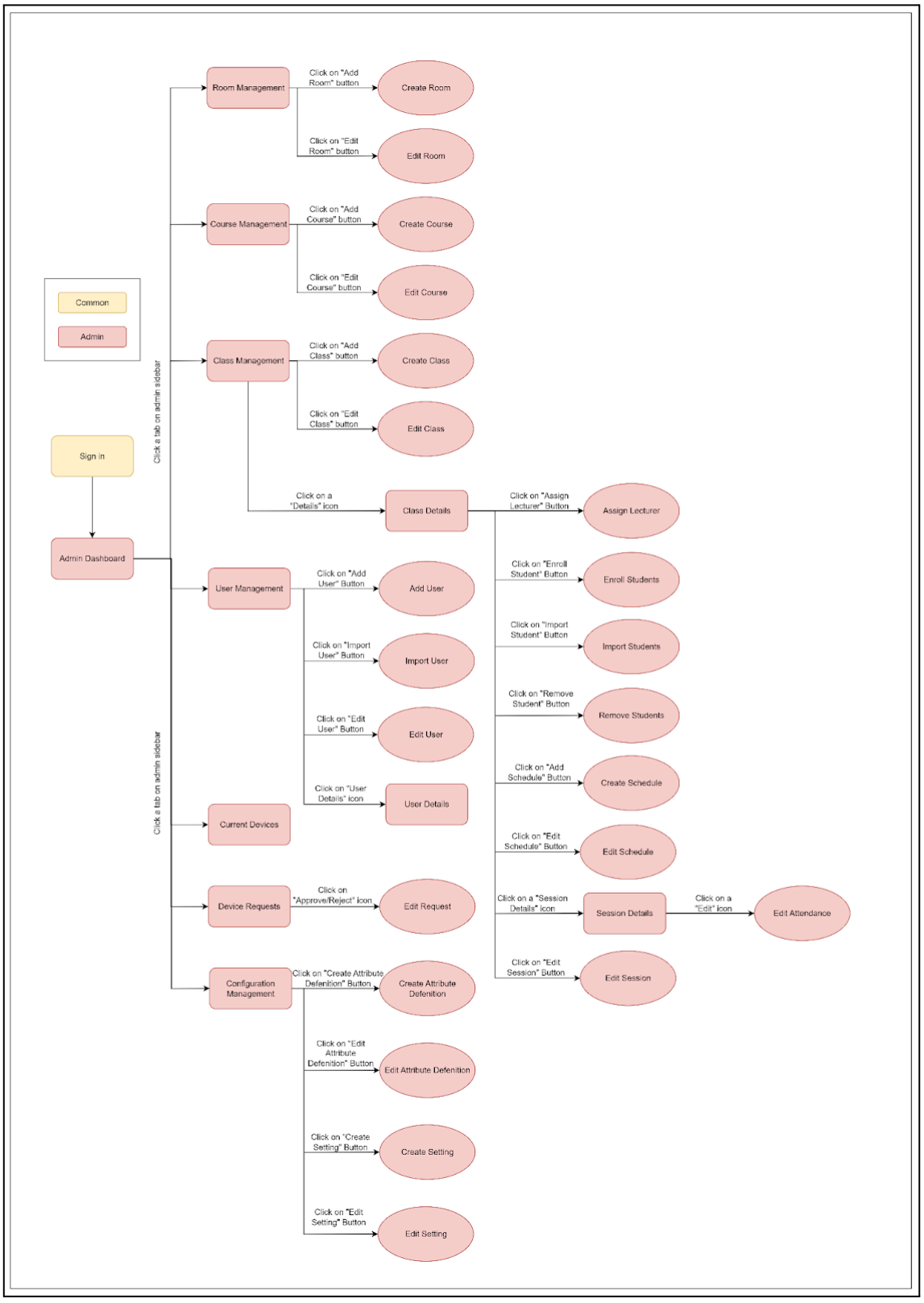


Figure 8: AAS Web System Screens Flow Diagram

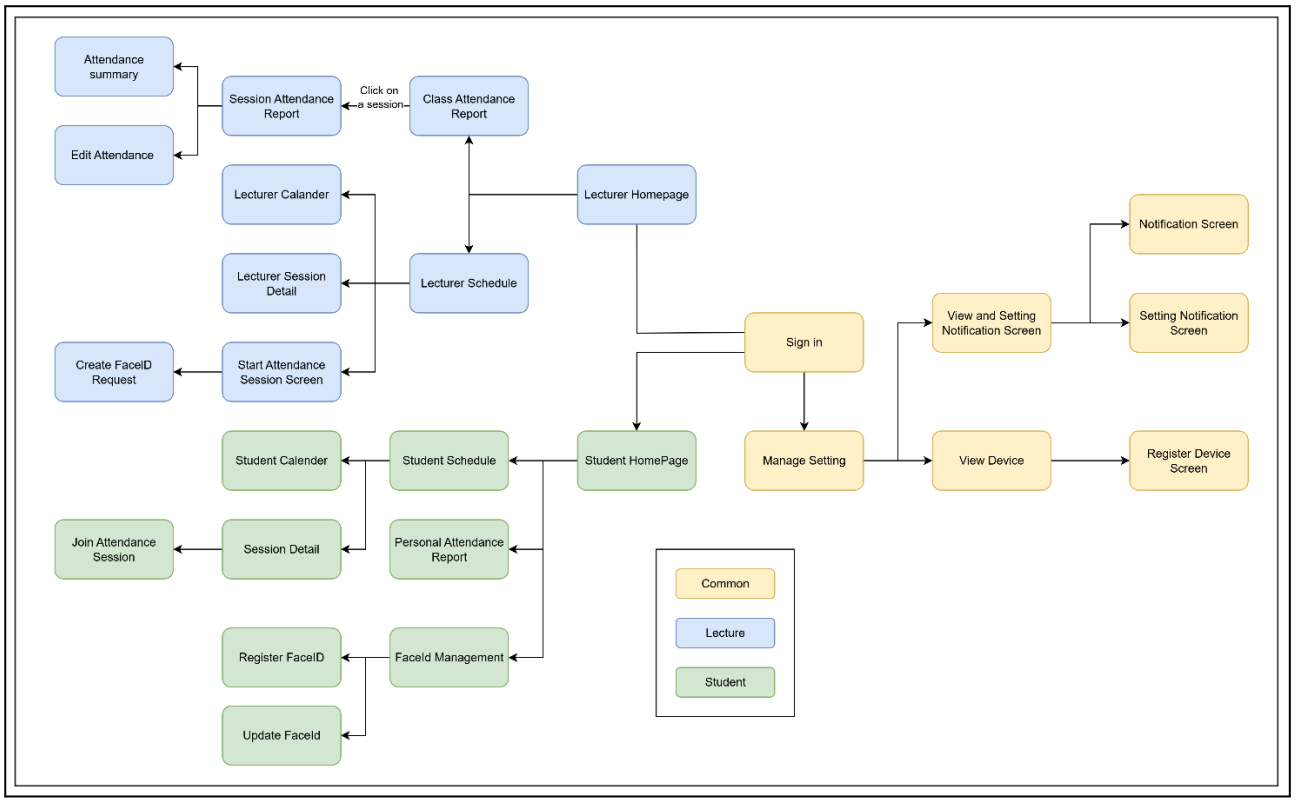


Figure 9: AAS Mobile System Screens Flow Diagram

3.1.2. Screen Descriptions

No	Name	Role	Description
Website Screen			
			Description
1	Sign In	Admin	Allows admin to securely log into the system using credentials.
2	Admin Dashboard	Admin	Displays system overview including statistics of rooms, courses, classes, users, devices, and schedules.
3	Room Management	Admin	Provides a list of all available rooms with options to view, filter, and manage.
4	Create Room	Admin	Enables admin to add a new room with details such as name, building, and capacity.
5	Edit Room	Admin	Allows modification of existing room information.
6	Course Management	Admin	Provides a list of courses with search and filtering functions.

8	Create Course	Admin	Enables admin to create a new course by defining code, name, description, and semester.
9	Edit Course	Admin	Allows editing course information already stored in the system.
10	Class Management	Admin	Displays list of classes grouped by semester, with options for detail view.
11	Create Class	Admin	Supports creating new classes linked to specific courses and lecturers.
12	Edit Class	Admin	Allows updating class details such as section code, semester, or assigned lecturer.
13	Class Details	Admin	Provides detailed information of a selected class including enrolled students and sessions.
14	Assign Lecturer	Admin	Enables assigning lecturers to a specific class.
15	Enroll Students	Admin	**Enroll Students** – Allows admin to manually enroll students into a class.
16	Import Students	Admin	**Import Students** – Supports bulk student enrollment via file import.
17	Remove Students	Admin	**Remove Students** – Allows removal of selected students from a class.
18	Create Schedule	Admin	**Create Schedule** – Enables creation of session schedules for a class.
19	Edit Schedule	Admin	**Edit Schedule** – Provides options to update existing session schedules.
20	Session Details	Admin	**Session Details** – Displays details of a specific session including attendance records.
21	Edit Session	Admin	**Edit Session** – Allows admin to adjust session details such as date, time, or room.

22	Edit Attendance	Admin	**Edit Attendance** – Enables correction or update of student attendance records.
23	User Management	Admin	**User Management** – Displays list of system users including Admin, Lecturer, and Student roles.
24	Add User	Admin	**Add User** – Allows manual addition of a new user with account details.
25	Import User	Admin	**Import User** – Supports bulk creation of users via file upload.
26	Edit User	Admin	**Edit User** – Provides functionality to update existing user information.
27	User Details	Admin	**User Details** – Displays detailed profile information of a selected user.
28	Current Devices	Admin	**Current Devices** – Shows list of devices currently approved and linked to users.
29	Device Requests	Admin	**Device Requests** – Lists pending device registration requests awaiting admin approval.
30	Edit Request	Admin	**Edit Request** – Allows admin to approve or reject device requests.
31	Configuration Management	Admin	**Configuration Management** – Provides system-wide configuration options for attributes and definitions.
32	Create Attribute	Admin	**Create Attribute** – Enables creation of new custom attributes in the system.
33	Edit Attribute	Admin	**Edit Attribute** – Allows modification of existing attributes.
34	Create Definition	Admin	**Create Definition** – Supports creation of definitions for use in system configurations.
35	Edit Definition	Admin	**Edit Definition** – Allows editing of definitions already created.

Mobile Screen			
01	Sign in	Common	Login screen for both students and lecturers
02	Manage Setting	Common	Manage personal preferences and application settings
03	View and Setting Notification Screen	Common	View the list of notifications and configure preferences
04	Setting Notification Screen	Common	Configure individual notification types and behaviors
05	View Device	Common	Display the registered device information
06	Register Device Screen	Common	Register a new device to the system
07	Update Device Screen	Common	Edit or update information of an existing device
08	Student HomePage	Student	Main landing screen after login, showing summary information
09	Student Calendar	Student	Calendar view of scheduled sessions
10	Student Schedule	Student	Detailed list view of class schedule
11	Session Detail	Student	View full details about a selected session
12	Join Attendance Session	Student	Join and check in to an active session
13	Personal Attendance Report	Student	View personal attendance history and statistics
14	FaceID Management	Student	Manage registered FaceID data for attendance
15	Register FaceID	Student	Register new facial data for the attendance system
16	Update FaceID	Student	Update existing facial recognition data
17	Lecturer Homepage	Lecturer	Main dashboard for lecturers after logging in
18	Lecturer Schedule	Lecturer	View the list of upcoming teaching sessions
19	Lecturer Calendar	Lecturer	Calendar view of lecturer’s schedule
20	Lecturer Session Detail	Lecturer	Detailed information about a specific teaching session
21	Start Attendance Session Screen	Lecturer	Start a new attendance-taking session
22	Create FaceID Request	Lecturer	Submit request to register FaceID for a student
23	Manual Attendance	Lecturer	Manually mark attendance for students
24	Session Attendance Report	Lecturer	View attendance report for a single session
25	Class Attendance Report	Lecturer	Aggregate attendance report for an entire class

26	Attendance Summary	Lecturer	Summary of attendance across all sessions

Table 6: Screens Description

3.1.3. Screen Authorization

No	Name	Platform	Admin	Student	Lecturer
1	Sign in	Web/Mobile	✔	✔	✔
2	Admin Dashboard	Web	✔	✖	✖
3	Room Management	Web	✔	✖	✖
4	Create Room	Web	✔	✖	✖
5	Edit Room	Web	✔	✖	✖
6	Course Management	Web	✔	✖	✖
7	Create Course	Web	✔	✖	✖
8	Edit Course	Web	✔	✖	✖
9	Class Management	Web	✔	✖	✖
10	Create Class	Web	✔	✖	✖
11	Edit Class	Web	✔	✖	✖
12	Class Details	Web	✔	✖	✖
13	Assign Lecturer	Web	✔	✖	✖
14	Enroll Students	Web	✔	✖	✖
15	Import Students	Web	✔	✖	✖
16	Remove Students	Web	✔	✖	✖
17	Create Schedule	Web	✔	✖	✖
18	Edit Schedule	Web	✔	✖	✖

19	Session Details	Web	✔	✖	✖
20	Edit Session	Web	✔	✖	✖
21	Edit Attendance	Web	✔	✖	✖
22	User Management	Web	✔	✖	✖
23	Add User	Web	✔	✖	✖
24	Import User	Web	✔	✖	✖
25	Edit User	Web	✔	✖	✖
26	User Details	Web	✔	✖	✖
27	Current Devices	Web	✔	✖	✖
28	Device Requests	Web	✔	✖	✖
29	Edit Request	Web	✔	✖	✖
30	Configuration Management	Web	✔	✖	✖
31	Create Attribute Definition	Web	✔	✖	✖
32	Edit Attribute Definition	Web	✔	✖	✖
33	Create Setting	Web	✔	✖	✖
34	Edit Setting	Web	✔	✖	✖
35	Manage Setting	Mobile	✖	✔	✔
36	View and Setting Notification	Mobile	✖	✔	✔
38	Setting Notification	Mobile	✖	✔	✔
39	View Device	Mobile	✖	✔	✔
40	Register Device	Mobile	✖	✔	✔
41	Update Device	Mobile	✖	✔	✔
42	Student Homepage	Mobile	✖	✔	✖
43	Student Calendar	Mobile	✖	✔	✖
44	Student Schedule	Mobile	✖	✔	✖

45	Session Detail	Mobile	✖	✔	✔
46	Join Attendance Session	Mobile	✖	✔	✖
48	Personal Attendance Report	Mobile	✖	✔	✖
49	FaceID Management	Mobile	✖	✔	✖
50	Register FaceID	Mobile	✖	✔	✖
51	Update FaceID	Mobile	✖	✔	✖
52	Lecturer Homepage	Mobile	✖	✖	✔
53	Lecturer Schedule	Mobile	✖	✖	✔
54	Lecturer Calendar	Mobile	✖	✖	✔
55	Lecturer Session Detail	Mobile	✖	✖	✔
56	Start Attendance Session	Mobile	✖	✖	✔
57	Create FaceID Request	Mobile	✖	✖	✔
59	Manual Attendance	Mobile	✖	✖	✔
60	Session Attendance Report	Mobile	✖	✖	✔
61	Class Attendance Report	Mobile	✖	✖	✔
62	Attendance Summary	Mobile	✖	✖	✔

Table 7: Screen Authorization

3.1.5. Entity Relationship

3.1.5.1. Diagram

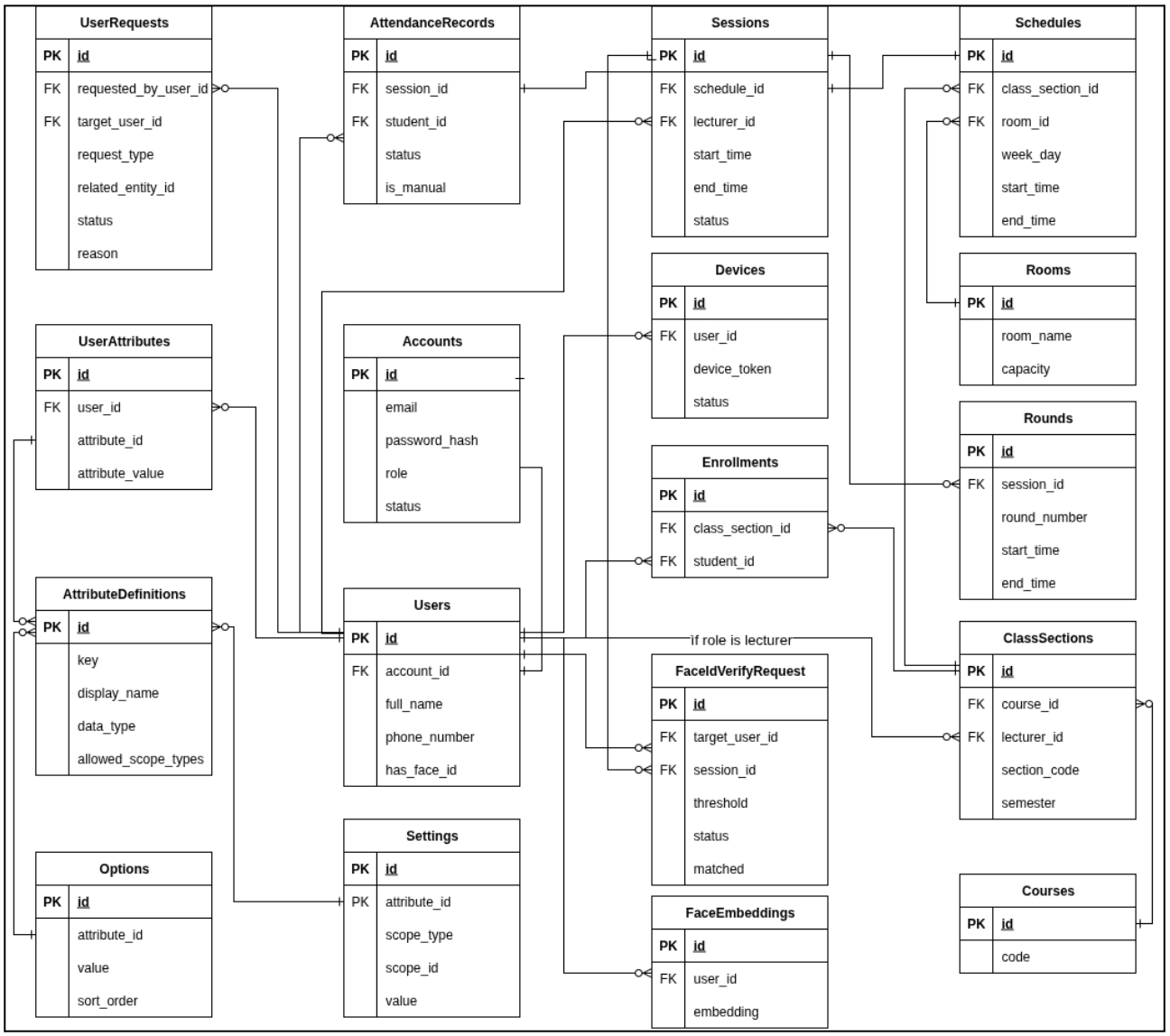


Figure 10.1: SQL Diagram

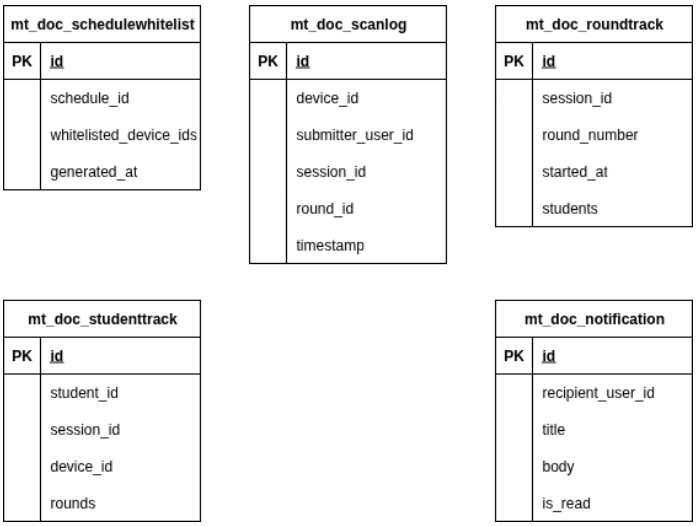


Figure 10.2: NoSQL Diagram

3.1.5.2. Description

No	Name	Description
1	Users	List of users in the system
2	Accounts	Login account information for users
3	UserRequests	Requests created by users
4	UserAttributes	Custom attributes associated with users
5	AttributeDefinitions	Definitions of user attribute types
6	AttendanceRecords	Attendance logs of users in sessions/events
7	Devices	Devices used for attendance or system management
8	Sessions	Specific session of a course
9	Rounds	Attendance rounds within a session
10	Enrollments	Records of users enrolled in courses
11	Courses	Information about courses
12	Schedules	Course schedules
13	Rooms	Classrooms where sessions are held
14	Settings	System settings or configuration data
15	Options	Shared selectable values or configuration options
16	ClassSections	Course sections with lecturer assignments
17	FaceIdVerifyRequests	Face recognition verification requests
18	FaceEmbeddings	Encrypted facial recognition data for user identification
19	mt_doc_schedulewhitelist	Whitelisted devices for schedule access
20	mt_doc_scanlog	Device scanning activity logs during sessions
21	mt_doc_roundtrack	Attendance round progress tracking
22	mt_doc_studenttrack	Individual student attendance patterns
23	mt_doc_notification	User notifications and system messages

Table 8: Entity description

3.2. System Functional Details

3.2.1. Website

3.2.1.1 Sign in

a. Screen Layouts

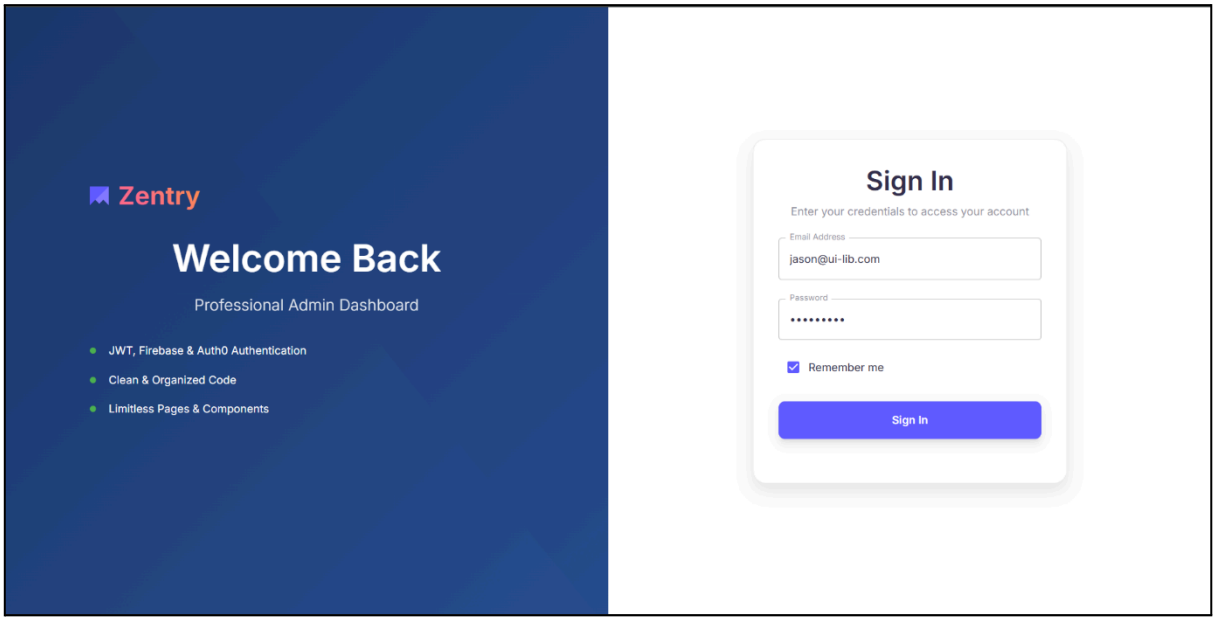


Figure 11: Sign in - Sign in form

No	Field Name	Field Type	Description
1	**Welcome Back**	Label	Greeting text on the left panel.
2	**Subtitle Text**	Label	Informational bullets (e.g., “Professional Admin Dashboard”).
3	**Sign In**	Label	Header text for the form.
4	**Instruction**	Label	“Enter your credentials to access your account.”
5	**Email Address**	Textbox	User email. Validation: must be a valid email format.
6	**Password**	Password	User password. Validation: not empty; min length per BR.
7	**Remember me**	Checkbox	Persist session on this device.
8	**Sign In**	Button	Submit credentials for verification.
9	**Error message area**	Label	Displays errors (invalid credentials, system issues, timeout).

Table 9: Sign In - Sign in form

b. Use Case Specifications

ID and Name	UC01 - Sign in		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user opens the website and attempts to log in using username/email and password.		
Description	As a user, I want to log in securely so that I can access system features according to my role.		
Preconditions	**PRE-1.** The user has a registered account. **PRE-2.** The system is online and operational.		
Postconditions	**POST- 1.** The user is redirected to the corresponding dashboard based on their role.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Open a website
	2	User	Enter email and password (optionally check “Remember me”).
	3	System	Validate credentials against the user store.
	4	System	If valid, identify the user’s role.
	5	System	Redirect the user to the corresponding dashboard.
Alternative Flows	**Alt Step**	**Condition**	**Description**
Exceptions	• **E1:** Invalid credentials → show error message and allow retry. • **E2:** System/database error → show “temporarily unavailable”. • **E3:** Login request timeout/session expired → prompt user to try again.		
Frequency of Use	High		
Business Rules	BR-01, BR-02		
Other Information	All login attempts are logged for security auditing and monitoring.		
Assumptions	Users are pre-registered by Admin or via batch import. Stable internet connection during login.		

Table 10: Sign In Specification

3.2.1.2. Sign out

a. Screen Layouts

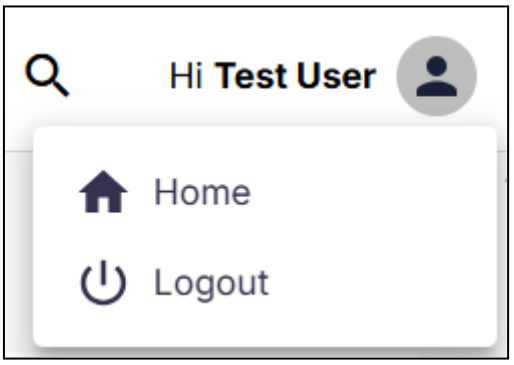


Figure 12: Sign Out - Avatar Dropdown

No	Field Name	Field Type	Description
1	**Avatar Icon**	Icon/Button	Displays the user avatar; when clicked, it opens the dropdown menu.
2	**Home Option**	Menu Item	Navigation option that redirects the user to the Home page.
3	**Logout Option**	Menu Item	Allows the user to sign out from the system.
4	**Dropdown Menu**	Container	The component that contains and displays the list of options (Home, Logout).

Table 11: Sign out

b. Use Case Specifications

ID and Name	UC02 - Sign out		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user opens the header user menu and selects **Logout**.		
Description	As a user, I want to securely sign out so that my session is terminated and my account cannot be accessed from this device without signing in again.		
Preconditions	**PRE-1.** The user is authenticated. **PRE-2.** The system is online and operational.		
Postconditions	**POST- 1.** Current session and any refresh token are invalidated/revoked server-side. **POST- 2.** Client-side auth artifacts (cookies, local storage) are cleared.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC02 - Sign out</td></tr>
<tr><td></td><td colspan="3">POST- 3. User is redirected to the Sign In page.</td></tr>
<tr><td rowspan="6">Normal Flows</td><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>Clicks User avatar → menu opens.</td></tr>
<tr><td>2</td><td>User</td><td>Click Logout.</td></tr>
<tr><td>3</td><td>User</td><td>Clicks Sign out in the dialog.</td></tr>
<tr><td>4</td><td>System</td><td>Invalidates server session/refresh token; clears client cookies/storage. (Logs the event.)</td></tr>
<tr><td>5</td><td>System</td><td>Redirects to Sign In and shows optional toast “Signed out.”</td></tr>
<tr><td rowspan="2">Alternative Flows</td><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>Session already expired</td><td>System clears local artifacts and redirects to Sign In with notice “Session expired.”</td></tr>
<tr><td>Exceptions</td><td colspan="3">• E1: Server error while invalidating session → show error “Sign-out failed, please try again”; user stays signed in.
• E2: Network error → show warning; allow retry. If the user chooses Force sign out, clear local artifacts and redirect (best-effort log).
• E3: Client storage cannot be cleared → show instruction to refresh/close browser and try again.</td></tr>
<tr><td>Frequency of Use</td><td colspan="3">High</td></tr>
<tr><td>Business Rules</td><td colspan="3">BR-01, BR-02</td></tr>
<tr><td>Other Information</td><td colspan="3">• Use secure, HttpOnly cookies for session/refresh tokens where applicable.
• Set cache headers to prevent back-button access to protected pages after sign-out.</td></tr>
<tr><td>Assumptions</td><td colspan="3">• The application uses cookies/session or JWT with refresh tokens; both are revoked on sign-out.
• “Remember me” also clears any persistent refresh token on sign-out.</td></tr>
</table>

Table 12: Sign Out Specification

3.2.1.3 View Dashboard

a. Screen Layouts



Figure 13: Dashboard

No	Field Name	Field Type	Description
1	**Page Title (“Hello <User>”)**	Label	Personalized greeting header.
2	**Week Selector**	Panel	Shows current week (e.g., “Week 11”); optional quick jump.
3	**Summary Cards Area**	Panel	Hosts *Rooms*, *Courses*, *Classes* cards.
4	**Performance Chart**	Panel	Hosts Chart widget + tabs + year navigation.
5	**Calendar**	Panel	Monthly calendar widget.
6	**Registered Statistics**	Panel	Devices / Face ID counters.
7	**Top 5 Courses**	Panel	Ranked course activity list.

Table 13: View Dashboard Layout



Figure 14: Dashboard

No	Field Name	Field Type	Description
1	**Rooms**	Card	Displays total rooms in the system.
2	**Courses**	Card	Displays total available courses.
3	**Classes**	Card	Displays total scheduled classes.

Table 14: View Dashboard 2

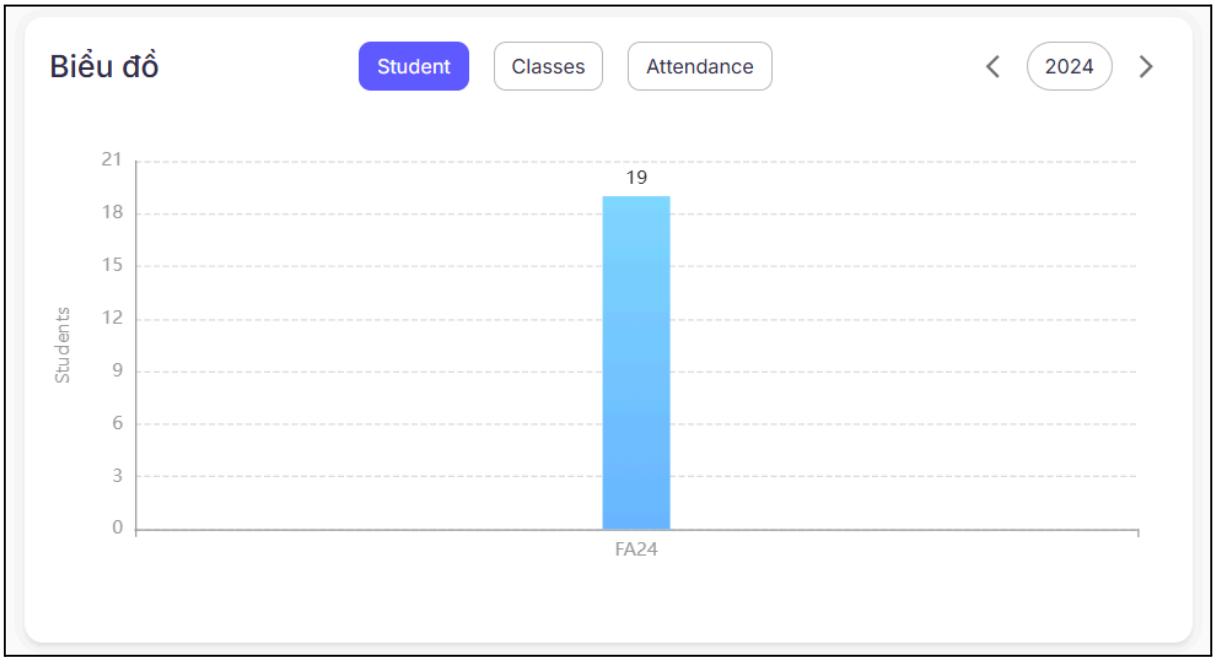


Figure 15: Dashboard

No	Field Name	Field Type	Description
1	**Title (“Chart”)**	Label	Section title.
2	**Tab: Student**	Tab	Shows student statistics over time.
3	**Tab: Classes**	Tab	Shows class statistics over time.
4	**Tab: Attendance**	Tab	Shows attendance trend/volume.
5	**Year Selector**	Button/Label	Displays selected year (e.g., “2025”).
6	**Prev Year**	Icon Button	Navigate to the previous year.
7	**Next Year**	Icon Button	Navigate to next year.
8	**Chart Canvas**	Chart	Renders series for the selected tab & year (with value labels).
9	**Empty/No-data State**	Label	Message shown when there is no data for selection.

Table 15: View Dashboard 3

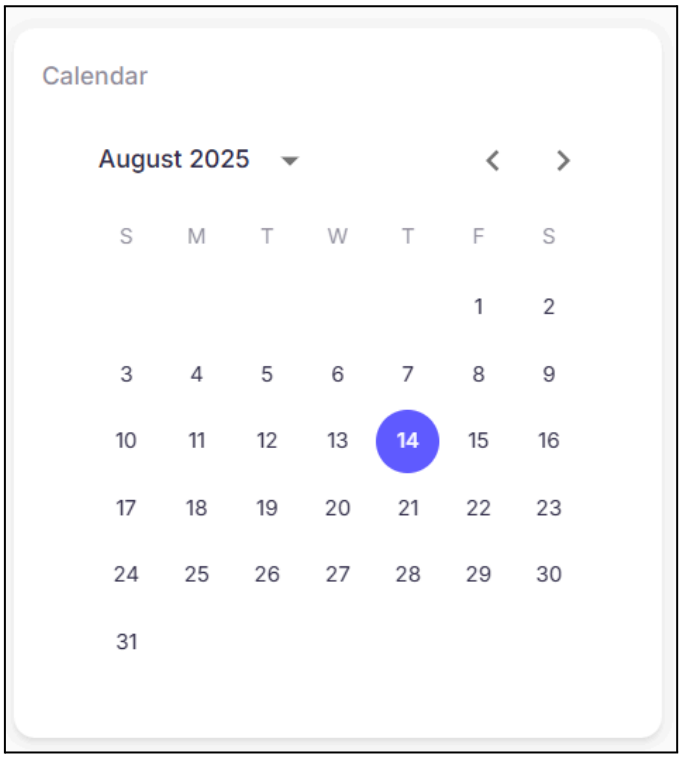


Figure 16: Dashboard

No	Field Name	Field Type	Description
1	**Month Caption**	Label	Current month & year (e.g., “August 2025”).
2	**Month Dropdown**	Dropdown	Select month/year.
3	**Prev Month**	Icon Button	Navigate to the previous month.
4	**Next Month**	Icon Button	Navigate to next month.
5	**Day Grid**	Calendar	Shows month days; **today** highlighted.
6	**Day Badge/Events (optional)**	Badge	Indicates scheduled classes/events for that day.

Table 16: View Dashboard 4

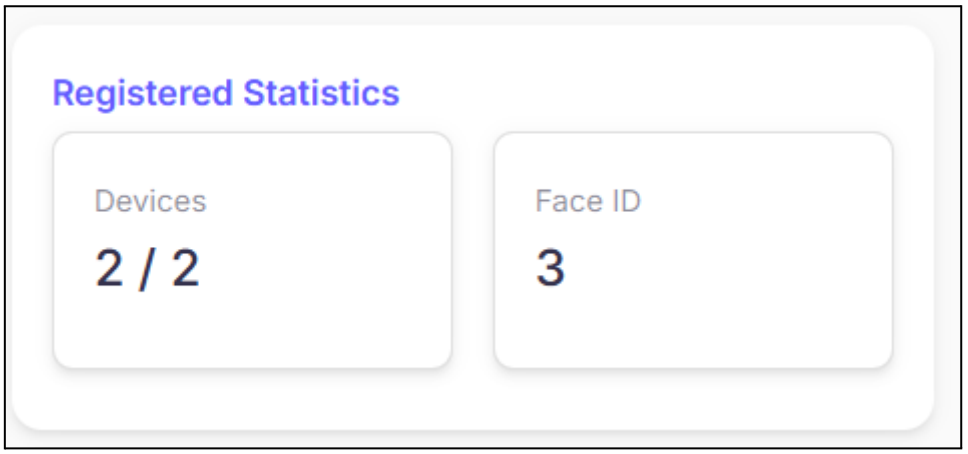


Figure 17: Dashboard

No	Field Name	Field Type	Description
1	**Devices**	Stat Card	Total number of registered devices.
2	**Face ID**	Stat Card	Total number of registered Face IDs.

Table 17: View Dashboard 5

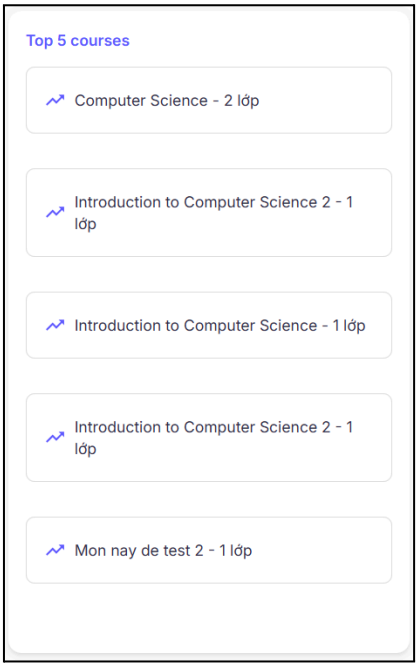


Figure 18: Dashboard

No	Field Name	Field Type	Description
1	**Section Title**	Label	“Top 5 courses”.
2	**Course Item**	List Item	“<Course name> – <n> classes”; ordered by activity desc.
3	**Empty/No-data State**	Label	Message if ranking unavailable.

Table 18: View Dashboard 6

b. Use Case Specifications

ID and Name	**UC30 – View Dashboard**		
Created By	DungND	**Date Created**	28/06/2025
Primary Actor	Admin	**Secondary Actors**	System
Trigger	The user lands on Dashboard after signing in or navigates to Dashboard from the side menu.		
Description	The Dashboard provides an overview of key statistics, upcoming schedules, and quick navigation to important modules. It displays summary cards, a calendar, performance charts, registered statistics, and the top 5 most active courses.		
Preconditions	**PRE-1.** The user is authenticated. **PRE-2.** User role is authorized to view dashboard data **PRE-3.** Relevant data exists (rooms/courses/classes, devices, Face IDs) or graceful empty states are supported.		
Postconditions	**POST- 1.** No data is modified; users may navigate to detailed modules from dashboard widgets.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	**View Summary Cards** (Rooms, Courses, Classes). Each card shows a total and links to its module.
	2	User	**View Chart** section.
	2.1	User	Switch chart **tabs** (Student / Classes / Attendance).
	2.2	System	Render data trends for selected **category & year**.
	2.3	User	Navigate **years** with left/right arrows.
	3	User	**Check Calendar** for the current month; **today** is highlighted. Use arrows to change month.
	4	User	**Review Registered Statistics** (Devices, Face ID totals).
	5	User	**View Top 5 Courses** ranked by activity (#classes). Click an item to open its course page (if enabled).
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Partial API failure	Affected widget shows error + **Retry** action; other widgets still load.
Exceptions	• **E1:** Backend unavailable → display toast/banner and per-widget error states; allow manual **Retry**.		

ID and Name	UC30 – View Dashboard
	• **E2:** Data inconsistency (e.g., counts mismatch) → log and show safe fallback (0 or "—"). • **E3:** Network timeout → show retry option; preserve user selections (tab/year).
Frequency of Use	High
Business Rules	BR-01, BR-02
Other Information	• Metrics are read-only aggregates; the refresh strategy may be configurable (auto refresh interval / manual Refresh). • Accessibility: keyboard navigation for tabs and calendar; ARIA roles on chart/controls.
Assumptions	• Data services expose efficient aggregate endpoints for counts and chart series. • Timezone and locale are derived from the user profile or the browser. • If caching is enabled, metrics are eventually consistent within an acceptable delay.

Table 19: View Dashboard Specification

3.2.1.4 View Rooms

a. Screen Layouts

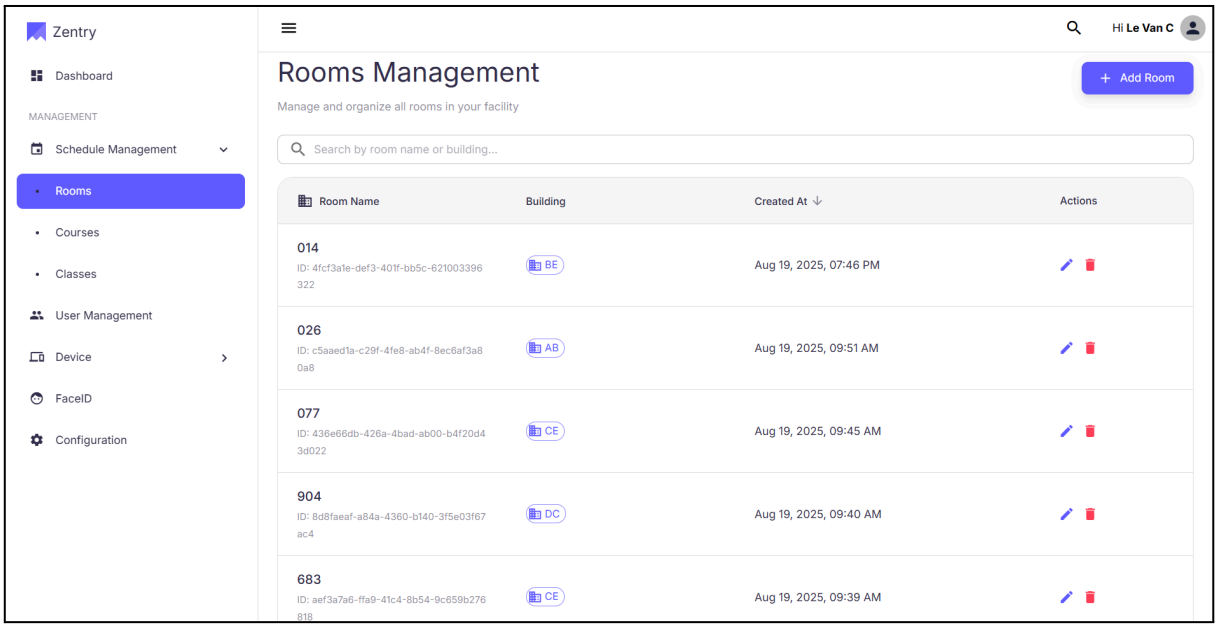


Figure 19: View room layout

No	Field Name	Field Type	Description
1	Search Box	Textbox	Allows users to search rooms by name or building.

No	Field Name	Field Type	Description
2	Add Room Btn	Button	Opens a form to add a new room.
3	Room Name	Text Label	Displays the room name and unique ID.
4	Building	Badge/Label	Shows the building code of the room.
5	Created At	Label	Shows the date and time the room was created.
6	Edit Room Btn	Icon Button	Allows users to edit the details of a specific room.
7	Delete Room Btn	Icon Button	Allows users to delete a room from the system.
8	Pagination	Navigation	Navigates through pages if the number of rooms exceeds one page.

Table 20: View Rooms Layout

b. Use Case Specifications

ID and Name	UC33 – Manage Rooms (View Rooms)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user opens the Rooms Management Page (either by entering a URL or navigating Schedule Management → Rooms)		
Description	As a user, I want to view all rooms within the facility so that I can easily identify available spaces, check their details, and manage attendance or scheduling effectively.		
Preconditions	**PRE-1.** The user is authenticated and authorized (Admin or Staff role). **PRE-2.** The system is online and operational.		
Postconditions	**POST- 1.** The user can successfully view all rooms. **POST- 2.** The user can add, edit, or delete room records as needed.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Navigate to the Rooms Management page
		System	Display all existing rooms with its name, code, building code, capacity and creation time
	2	User	Search for a room by its name or building code using the search bar
	3	System	Display all satisfying rooms according to the search
	4	User	Customize the pagination view by changing page size, navigate among pages
	5	System	Display the rooms according to the specified pagination
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No rooms available	Show empty state text (e.g., “No rooms available”) and keep other widgets functional

ID and Name	UC33 – Manage Rooms (View Rooms)
Exceptions	• **E1:** Backend unavailable → display toast/banner and error states; allow manual **Retry**. • **E2:** Data inconsistency (e.g., counts mismatch) → log and show safe fallback (0 or "—"). • **E3:** Network timeout → show retry option; preserve user selections (search/pagination).
Frequency of Use	High
Business Rules	BR-01, BR-02
Other Information	• Supports server-side pagination and sorting by Created At/Name. • Times are displayed in the user's locale/time zone. • Table remembers last search/sort on return navigation.
Assumptions	• Building codes exist and are preconfigured. • Room IDs are immutable. • Network/database are stable during listing.

Table 21: View Rooms Specification

3.2.1.5 Add Room

a. Screen Layouts

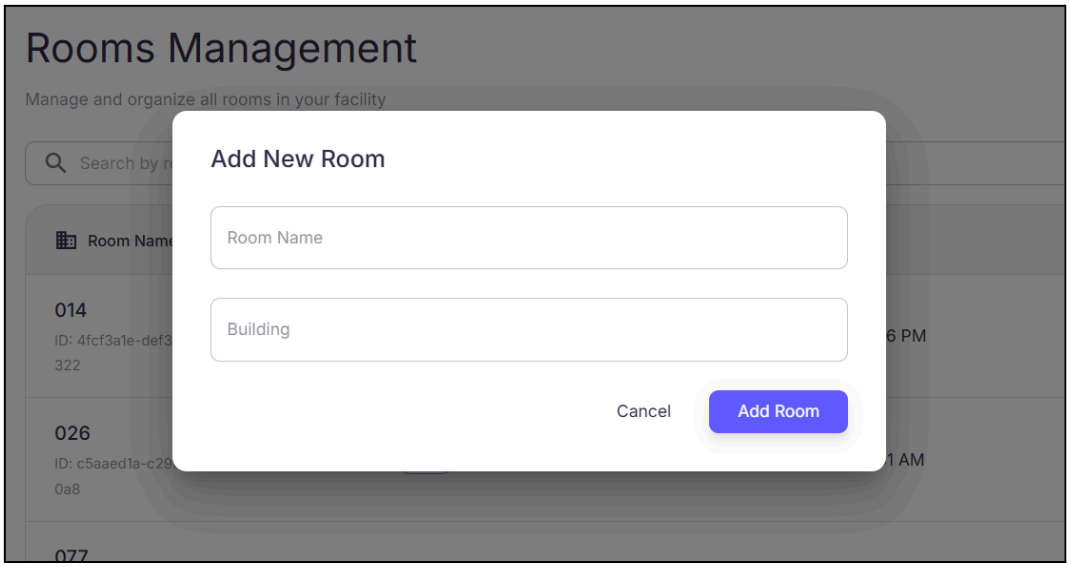


Figure 20: Add room

No	Field Name	Field Type	Description
1	Title	Label	Displays: "Add New Room"
2	Room Name	Text Input	User enters the name of the room (e.g., "Meeting Room A", "Lab 201")
3	Building	Text Input	User specifies the building where the room is located
4	Cancel	Button	Closes the popup without saving changes

No	Field Name	Field Type	Description
5	Add Room	Button	Saves the room details and closes the popup (if room added successfully)

Table 22: Add room layout

b. Use Case Specifications

ID and Name	UC32 – Manage Rooms (Add Rooms)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user opens the Rooms Management Page (either by entering a URL or navigating Schedule Management → Rooms)		
Description	As a user, I want to add a new room with its name, building, and capacity so that the room can be properly managed and scheduled in the system.		
Preconditions	**PRE-1.** The user is authenticated and authorized (Admin or Staff role). **PRE-2.** The system is online and operational.		
Postconditions	**POST- 1.** The new room is added and visible in the room list.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Click on button “Add Room”
	2	System	Display the “Add New Room” popup.
	3	User	Enter room name, building.
	4	User	Click “Add Room”
	5	System	Validates the inputs.
	6	System	Saves the new room and updates the room list.
	7	System	Displays a success message.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Invalid Input	Show an error message and prompt the user to correct.
	A2	Cancel	Popup closes without saving
Exceptions	• **E1:** Backend unavailable → display toast/banner and error states; allow manual **Retry** • **E2:** Validation errors → highlight fields with error messages and prevent saving. • **E3:** Network timeout → show retry option; preserve user selections (search/pagination).		
Frequency of Use	Medium		
Business Rules	BR-01, BR-02		
Other Information	• All “add room” activities are logged for auditing. • Supports both client-side and server-side validation.		

ID and Name	UC32 – Manage Rooms (Add Rooms)
Assumptions	• Building lists are already available in the system. • Room IDs are immutable once created. • Network/database are stable during the creation process.

Table 23: Add room specification

3.2.1.6 Edit Room

a. Screen Layouts

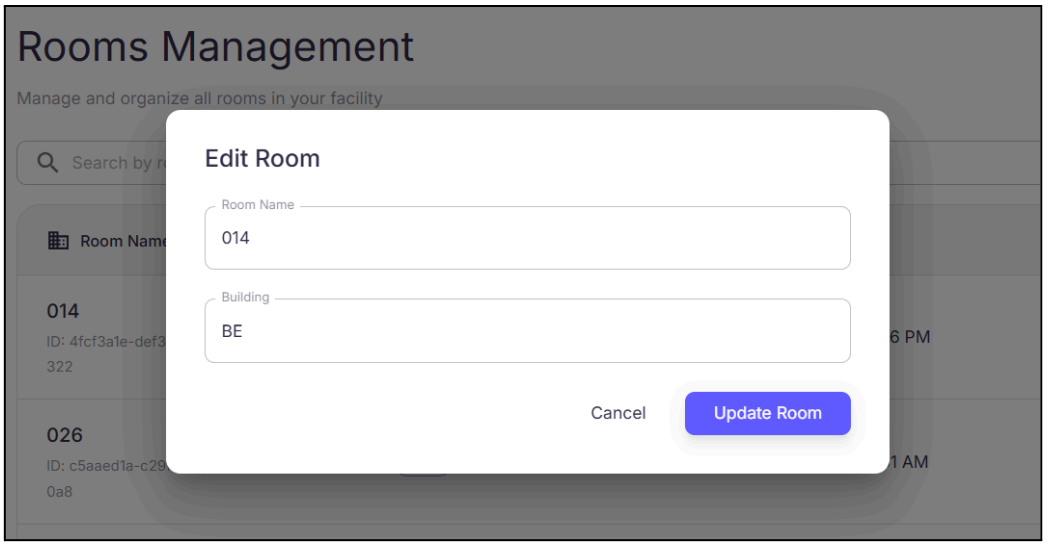


Figure 21: Edit Room

No	Field Name	Field Type	Description
1	Room Name	Text Field	The name of the room (e.g., “AL101”). Required.
2	Building	Dropdown	Select the building where the room is located.
3	Save Button	Button	Saves the changes and updates the room details.
4	Cancel Button	Button	Cancels the edit and returns to the Rooms Management Page without saving.

Table 24: Edit room layout

b. Use Case Specifications

ID and Name	UC33 – Manage Rooms (Edit Rooms)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user selects a room from the “Rooms Management” page and clicks on the edit button.		
Description	As a user, I want to edit the details of an existing room so that I can keep room information up to date and accurate for scheduling and attendance management.		
Preconditions	**PRE-1.** The user is authenticated and authorized (Admin). **PRE-2.** The system is online and operational.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC33 – Manage Rooms (Edit Rooms)</td></tr>
<tr><td></td><td colspan="3">PRE-3. The room to be edited already exists in the system.</td></tr>
<tr><th>Postconditions</th><td colspan="3">POST-1. Room details are updated in the database.
POST-2. Updated details are immediately reflected in the room listing.</td></tr>
<tr><th rowspan="7">Normal Flows</th><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>Clicks on the Edit button for a specific room.</td></tr>
<tr><td>2</td><td>System</td><td>Displays the Edit Room Form pre-filled with current room details.</td></tr>
<tr><td>3</td><td>User</td><td>Updates one or more fields and clicks Save.</td></tr>
<tr><td>4</td><td>System</td><td>Validates inputs (required fields, unique constraints, numeric fields).</td></tr>
<tr><td>5</td><td>System</td><td>Saves updated details into the database.</td></tr>
<tr><td>6</td><td>System</td><td>Displays confirmation message and returns to the Rooms Management Page with updated details.</td></tr>
<tr><th rowspan="2">Alternative Flows</th><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>Cancel</td><td>Discard changes and returns to Rooms Management Page.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Backend unavailable → display toast/banner and error states; allow manual Retry
• E2: Validation errors → highlight fields with error messages and prevent saving.
• E3: Network timeout → show retry option; preserve user selections (search/pagination).</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Medium</td></tr>
<tr><th>Business Rules</th><td colspan="3">BR-01, BR-02</td></tr>
<tr><th>Other Information</th><td colspan="3">• All edits are logged with timestamp and user ID for auditing.
• Supports inline editing in future versions.
• Admin can edit multiple rooms in succession.</td></tr>
<tr><th>Assumptions</th><td colspan="3">• Admin has correct privileges to edit room details.
• The system always loads the latest version of the room before editing.
• Network/database stability is assumed during save operation.</td></tr>
</table>

Table 25: Edit room specification

3.2.1.7 Delete Room

a. Screen Layouts

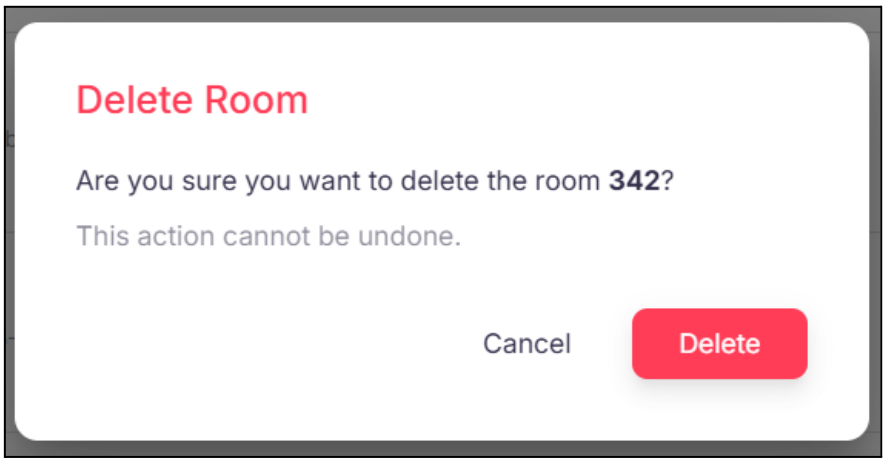


Figure 22: Delete Room

No	Field Name	Field Type	Description
1	Delete Room	Heading	The title of the confirmation dialog.
3	Confirmation Message	Text Label	Asks the user to confirm the deletion of the room
4	Warning Message	Text Label	A warning that the action is irreversible
5	Delete Button	Button	Confirms the action and deletes the room.
6	Cancel Button	Button	Allows the user to cancel the action and close the dialog.

Table 26: Delete Room Layout

b. Use Case Specifications

ID and Name	UC34 – Manage Rooms (Delete Rooms)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user selects a room from the “Rooms Management” page and clicks on the delete button.		
Description	As a user, I want to delete a room from the system to remove outdated or incorrect room information.		
Preconditions	**PRE-1.** The user is authenticated and authorized (Admin). **PRE-2.** The system is online and operational. **PRE-3.** The room to be deleted already exists in the system.		
Postconditions	**POST-1**: The room is removed from the database. **POST-2**: The room is no longer visible in the room listing.		
Normal Flows	**Step**	**By**	**Step Details**

ID and Name	UC34 – Manage Rooms (Delete Rooms)		
	1	User	Click on the **Delete** button for a specific room.
	2	System	Display a confirmation dialog with the room ID.
	3	User	Click the Delete button within the dialog.
	4	System	Remove the room from the database.
	5	System	Display a confirmation message and return to the Rooms Management page.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Cancel	The system discards the action and closes the dialog, returning to the Rooms Management page without making any changes.
Exceptions	• **E1:** Backend unavailable → display toast/banner and error states; allow manual **Retry** • **E2**: Room cannot be deleted (e.g., currently has upcoming bookings) → System shows a warning message and prevents the deletion. • **E3:** Network timeout → show retry option; preserve user selections (search/pagination).		
Frequency of Use	Low		
Business Rules	BR-01, BR-02		
Other Information	• All deletions are logged with a timestamp and user ID for auditing. • Only Admin can delete rooms.		
Assumptions	• The system always verifies if the room can be deleted before proceeding. • Network/database stability is assumed during the delete operation.		

Table 27: Delete Rooms Specification

3.2.1.8 View Courses

a. Screen Layouts

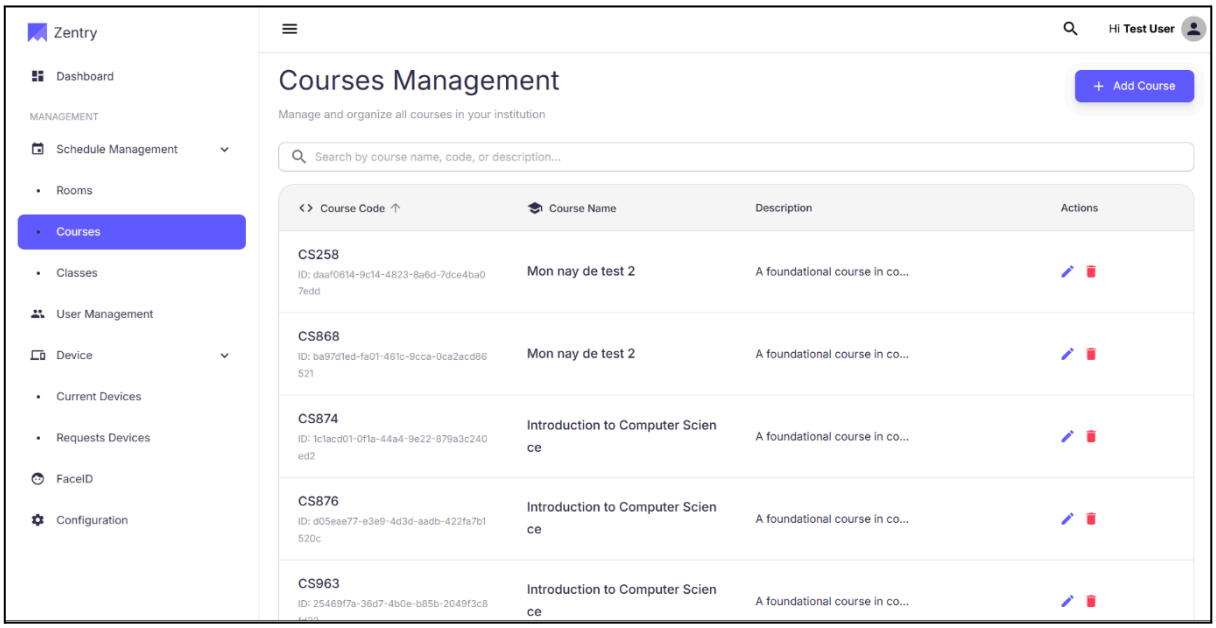


Figure 23: View Courses

No	Field Name	Field Type	Description
1	Search Box	Textbox	Allows users to search courses by course code, name, or description.
2	Add Course	Button	Opens a form to add a new course.
3	Course Code	Text	The unique code for each course.
4	Course Name	Text	The name of each course.
5	Description	Text	A brief description of the course.
6	Actions	Button	Buttons for editing or deleting a course.
7	Edit Course	Icon Button	Allows the user to edit a specific course.
8	Delete Course	Icon Button	Allows the user to delete a specific course.

Table 28: Manage Course (View Course) layout

b. Use Case Specifications

ID and Name	UC35 – Manage Course (View Course)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user navigate to the Course Management page		
Description	As a user, I want to view a list of all courses in the system so that I can manage and organize courses in the institution.		
Preconditions	**PRE-1.** The user has a registered account. **PRE-2.** The system is online and operational.		
Postconditions	**POST-1:** The system displays the list of courses.		

ID and Name	UC35 – Manage Course (View Course)		
	POST-2: The user can interact with the displayed list (e.g., search, sort, add, edit, or delete).		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Navigate to the Course Management page
		System	Display all existing courses with its name, code, and description
	2	User	Search for a course by its name, code or description using the search bar
	3	System	Display all satisfying courses according to the search
	4	User	Customize the pagination view by changing page size, navigate among pages
	5	System	Display the courses according to the specified pagination
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No courses available	Show empty state text (e.g., “No courses available”) and keep other widgets functional
Exceptions	• **E1:** Backend unavailable → display toast/banner and error states; allow manual **Retry**. • **E2:** Data inconsistency (e.g., counts mismatch) → log and show safe fallback (0 or “—”). • **E3:** Network timeout → show retry option; preserve user selections (search/pagination).		
Frequency of Use	High		
Business Rules	BR-01, BR-02		
Other Information	• The list should be paginated if the number of courses is large. • The system should support search functionality across multiple fields (course code, name, description).		
Assumptions	• The system's API for retrieving courses is fast and reliable. • The user is logged in and has the necessary permissions to view the course list.		

Table 29: Manage Course (View Course) Specification

3.2.1.9 Add Course

a. Screen Layouts

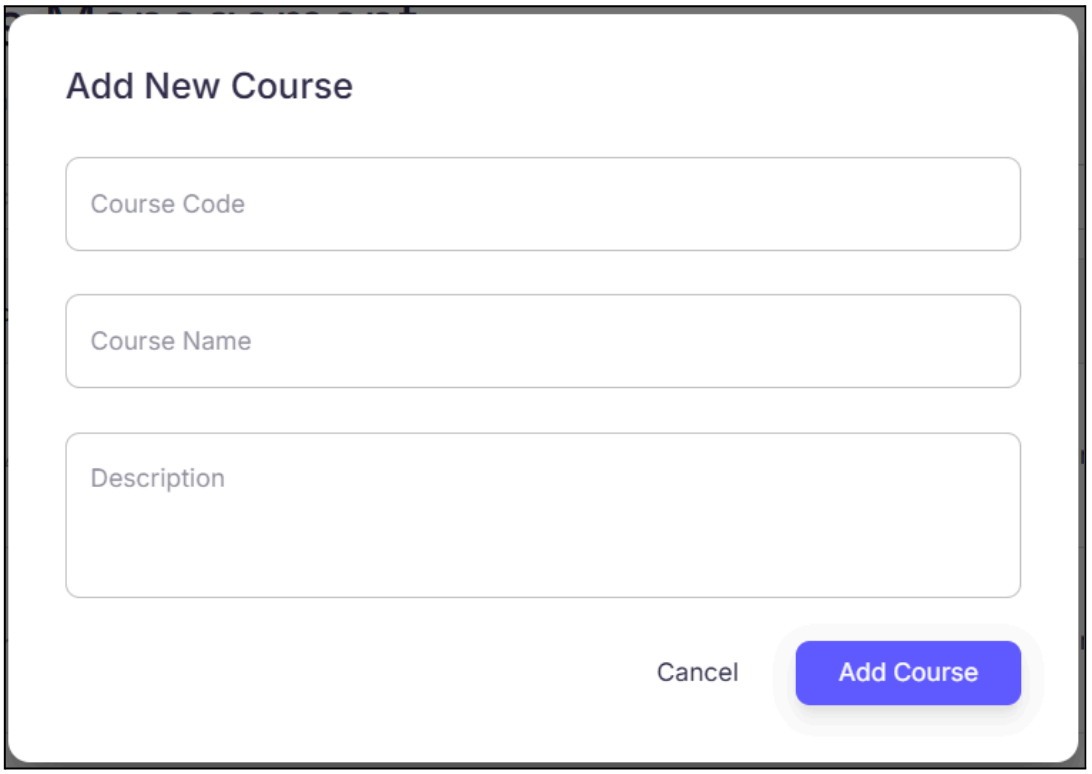


Figure 24 -Add new course

No	Field Name	Field Type	Description
1	Add New Course	Heading	The title of the form for adding a new course.
2	Course Code	Textbox	The unique code of the course
3	Course Name	Textbox	The name of the course
4	Description	Textbox	Detailed description for the new course.
5	Cancel	Button	Closes the form and discards any changes.
6	Add Course	Button	Confirms and submits the new course to the system.

Table 30: Manage Course (Add Course)

b. Use Case Specifications

ID and Name	UC36 – Manage Course (Add Course)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user clicks the "Add Course" button on the "Courses Management" page.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC36 – Manage Course (Add Course)</td></tr>
<tr><td></td><td colspan="3"></td></tr>
<tr><th>Description</th><td colspan="3">As a user, I want to add a new course to the system so that new classes can be created and managed under it.</td></tr>
<tr><th>Preconditions</th><td colspan="3">PRE-1. The user is authenticated and authorized (Admin or Staff role).
PRE-2. The system is online and operational.
PRE-3: The user has opened the "Add New Course" form.</td></tr>
<tr><th>Postconditions</th><td colspan="3">POST-1: A new course record is created in the database.
POST-2: A confirmation message is displayed, and the user is redirected to the "Courses Management" page.
POST-3: The newly added course is visible in the course listing.</td></tr>
<tr><th rowspan="6">Normal Flows</th><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>Fill in required fields</td></tr>
<tr><td></td><td>User</td><td>Click the “Add Course” button</td></tr>
<tr><td>2</td><td>System</td><td>Validate the inputs</td></tr>
<tr><td>3</td><td>System</td><td>Create the new course record in the database.</td></tr>
<tr><td>4</td><td>System</td><td>Displays a confirmation message ("Course added successfully") and returns to the Courses Management page with the new course visible in the list.</td></tr>
<tr><th rowspan="2">Alternative Flows</th><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>Cancel</td><td>Discards all entered data and returns the user to the "Courses Management" page.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Backend unavailable → display toast/banner and error states; allow manual Retry.
• E2: Data inconsistency (e.g., counts mismatch) → log and show safe fallback (0 or “—”).
• E3: Network timeout → show retry option; preserve user selections (search/pagination).</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Medium</td></tr>
<tr><th>Business Rules</th><td colspan="3">BR-01, BR-02</td></tr>
<tr><th>Other Information</th><td colspan="3">• All new course creations are logged with a timestamp and user ID for auditing.</td></tr>
<tr><th>Assumptions</th><td colspan="3">• All required fields will be clearly marked for the user.
• The system can automatically handle the unique ID generation (if not manually entered).</td></tr>
</table>

Table 31: Manage Course (Add Course)

3.2.1.10 Edit Course

a. Screen Layouts

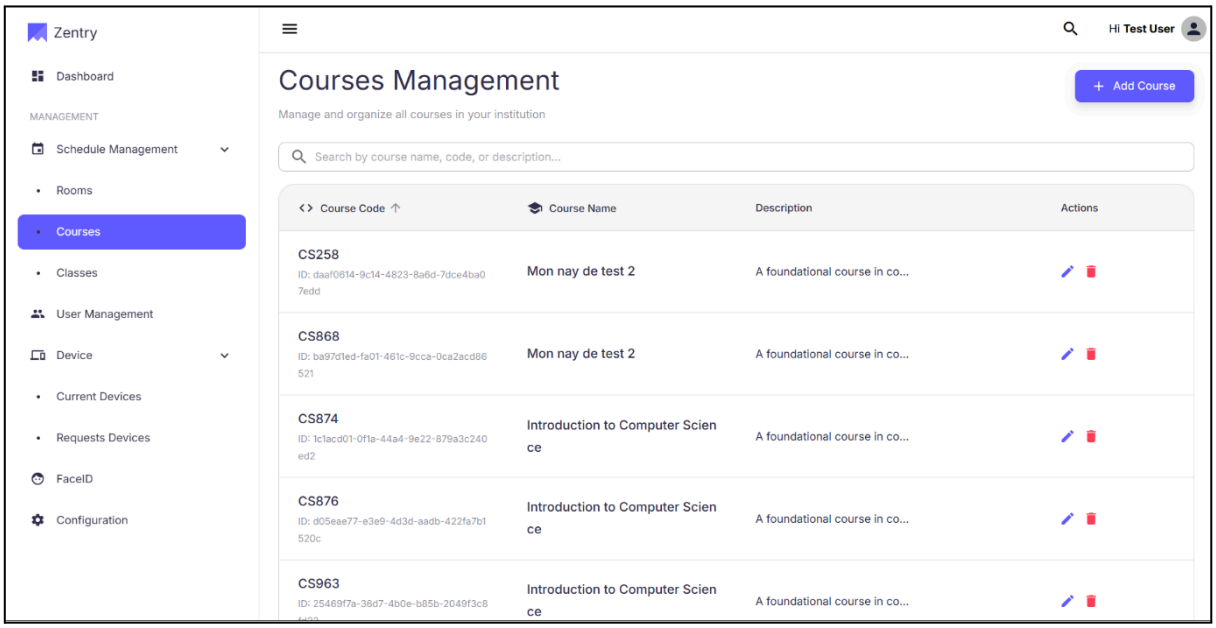


Figure 25 - Edit course

No	Field Name	Field Type	Description
1	Courses Table	Table	Displays **Course Code, Course Name, Description, Actions**.
2	Actions (per row)	Icon buttons	**Edit** (pencil) opens the Edit Course dialog.

Table 32: Manage Course (Edit Course) layout

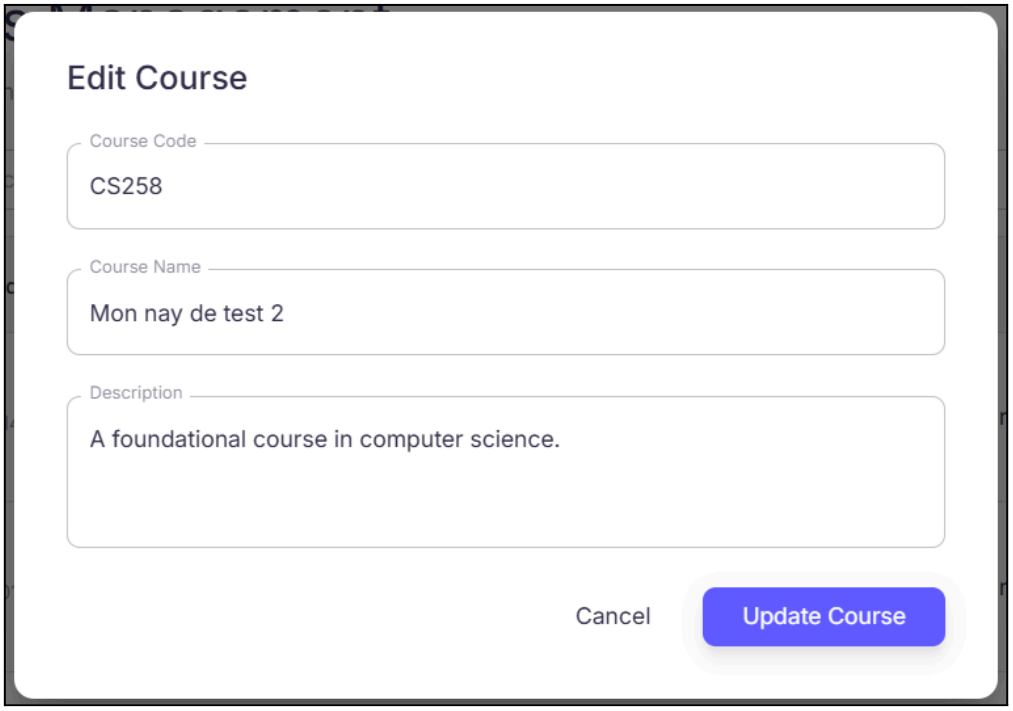


Figure 26 - Edit course dialog

No	Field Name	Field Type	Description
1	Title (“Edit Course”)	Label	Dialog header.
2	Course Code	Text field	Editable course code (must be unique).
3	Course Name	Text field	Editable course name.
4	Description	Text area	Editable description of the course.
5	Cancel	Button (secondary)	Close dialog without saving.
6	Update Course	Button (primary)	Save changes and update course. Disabled until changes are valid.
7	Error Area (optional)	Label	Displays error messages if update fails.

Table 33: Manage Course (Edit Course 2)

b. Use Case Specifications

ID and Name	UC37 – Manage Course (Edit Course)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	User clicks the **Edit** (pencil) icon for a course in the **Course List**.		
Description	As an authorized user, I want to edit course details so that I can correct mistakes or update course information.		
Preconditions	**PRE-1.** User is authenticated (BR-01). **PRE-2.** User has permission to manage courses (BR-02, BR-16). **PRE-3.** The target course exists in the system.		
Postconditions	**POST-1.** The course record is updated in the system. **POST-2.** A success notification is displayed and changes are visible immediately in the course table.		
Normal Flows	**Step**	**By**	**Step Details**

ID and Name	UC37 – Manage Course (Edit Course)		
	1	User	Locate the course in the **Course List** and click **Edit** (pencil).
	2	System	Open **Edit Course dialog** with current values prefilled.
	3	User	Modify one or more fields (**Course Code, Course Name, Description**).
	4	User	Click **Update Course**.
	5	System	Validate input and update course in database.
	6	System	If successful: close dialog, refresh table, show green toast **"Course edited successfully."** If unsuccessful: keep dialog open, display error (e.g., **"Course code already exists."**).
	Step	**By**	**Step Details**
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	User clicks **Cancel**	Dialog closes; no changes saved.
	A2	Validation fails	System highlights invalid fields (e.g., empty name, duplicate code).
Exceptions	• **E1:** Permission error → "You don't have permission to edit courses." • **E2:** Course not found (concurrent deletion) → Refresh list, show error. • **E3:** Network/Server error → Show error toast; retry possible.		
Frequency of Use	High		
Business Rules	BR-01, BR-02		
Other Information	• Inline validation improves usability (e.g., prevent saving empty Course Code). • Keep audit logs of edits (extend BR-15 Session History).		
Assumptions	• Course IDs are immutable. • Network/database are stable during listing.		

Table 34: Manage Course (Edit Course) Specification

3.2.1.11 Delete Course

a. Screen Layouts



Figure 27 - Delete course icon

No	Field Name	Field Type	Description
1	Courses Table	Table	Displays **Course Code, Course Name, Description, Actions**.
2	Actions (per row)	Icon buttons	**Edit** (pencil) opens Edit Course dialog.

Table 35: Manage Course (Delete Course) layout

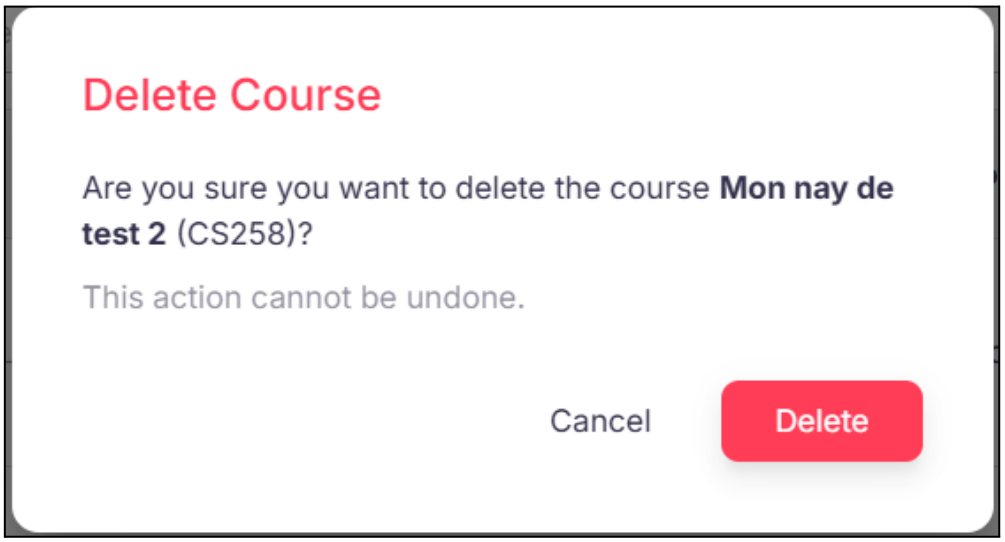


Figure 28 - Delete course confirm pop up

No	Field Name	Field Type	Description
1	Delete Icon	Button/Icon	Trash bin icon in the **Actions** column of the course table.
2	Confirmation Dialog	Modal	Dialog displayed to confirm course deletion.
3	Warning Message	Text	Message informing user of the consequences of deletion.
4	Delete Button	Button	Confirms deletion of the course.
5	Cancel Button	Button	Cancels the deletion action and closes the dialog.
6	Notification	Toast/Alert	Displays success or error message after action.

Table 36: Manage Course (Delete Course) layout

b. Use Case Specifications

ID and Name	UC38 – Manage Course (Delete Course)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user opens the Rooms Management Page (either by entering a URL or navigating **Schedule Management** → **Rooms**)		
Description	As a user, I want to delete irrelevant course from the system		
Preconditions	**PRE-1.** User is authenticated and authorized (Admin or Staff role). **PRE-2.** The system is online and operational. **PRE-3.** The course is not currently linked to any active classes or other dependencies.		
Postconditions	**POST- 1.** The course is successfully deleted from the system. **POST-2**. A success notification is displayed.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Click on the Delete button next to the course.
		System	Shows a confirmation popup
	2	User	Confirms deletion
	3	System	Deletes the course from the database.
	4	System	Displays a success message (e.g., “Course deleted successfully”).
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Cancel	Returns to the Course Management Page without deleting the course.
Exceptions	• **E1:** Backend unavailable → display toast/banner and error states; allow manual **Retry**. • **E2:** Data inconsistency (e.g., counts mismatch) → log and show safe fallback (0 or “—”). • **E3:** Network timeout → show retry option; preserve user selections (search/pagination).		
Frequency of Use	High		
Business Rules	BR-01, BR-02		
Other Information	• Audit logs should record which Admin deleted the course and when.		

ID and Name	UC38 – Manage Course (Delete Course)
Assumptions	• Course IDs are immutable. • Network/database are stable during listing.

Table 37: Manage Course

3.2.1.12 View Classes

a. Screen Layouts

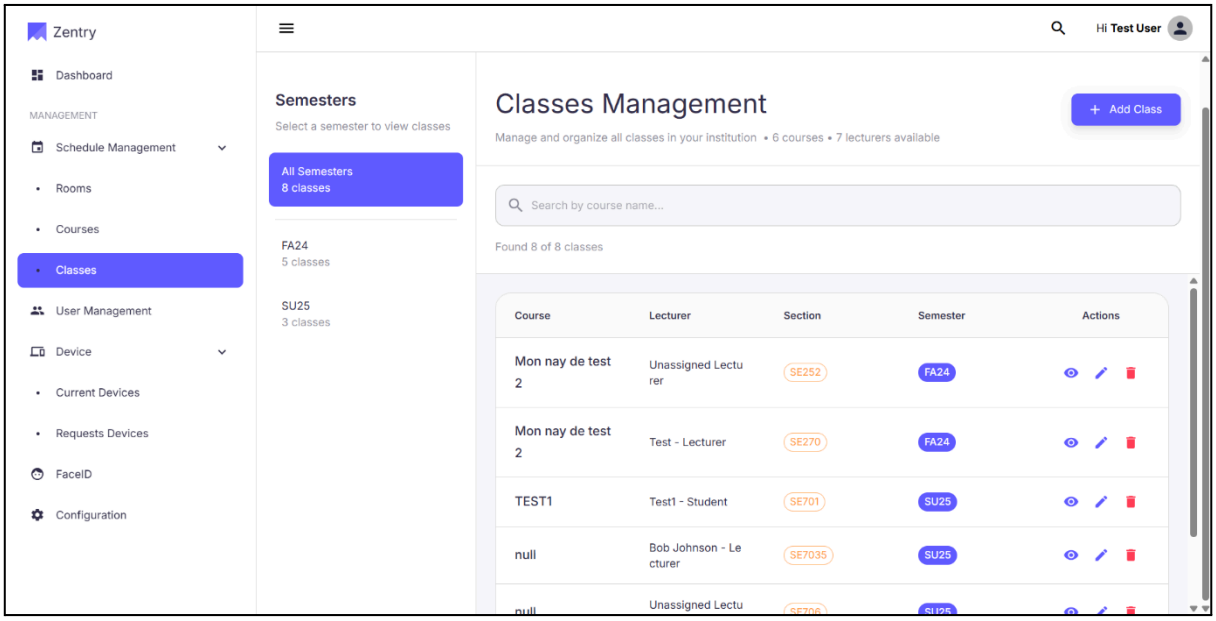


Figure 29 - Class management layout

No	Field Name	Field Type	Description
1	**Page Title ("Classes Management")**	Label	Section heading with context summary (e.g., "6 courses • 7 lecturers available").
2	**Add Class**	Button	Opens Add Class flow (handled in a separate UC). Visible to Admin (BR-02, BR-16).
3	**Search bar**	Searchbox	Search classes by **course name** (debounced).
4	**Result Counter**	Label	"Found x of y classes".
5	**Semesters Panel**	Panel	Left filter panel listing **All Semesters** and individual terms (FA24, SU25…).
6	**Classes Table**	Table	Main list of classes with columns below.
7	**Pagination**	Control	Rows per page selector, Previous/Next arrows.

Table 38: Class Management layout

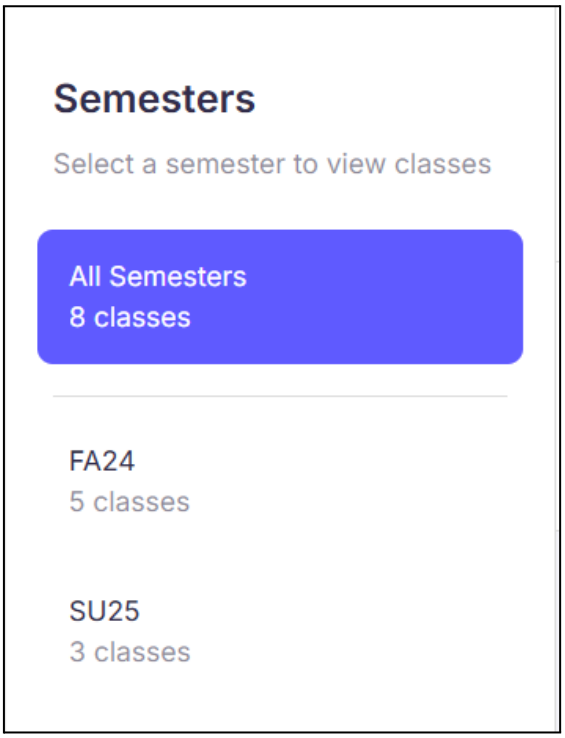


Figure 30 - Class management screen - Semester

No	Field Name	Field Type	Description
1	**All Semesters**	List item	Shows all classes in the system. Selected by default.
2	**<Term> (e.g., FA24)**	List item	Filter the table to the selected term. Shows class count badge.
3	**Active Indicator**	Indicator	Highlights the currently selected semester.

Table 39: Semester layout

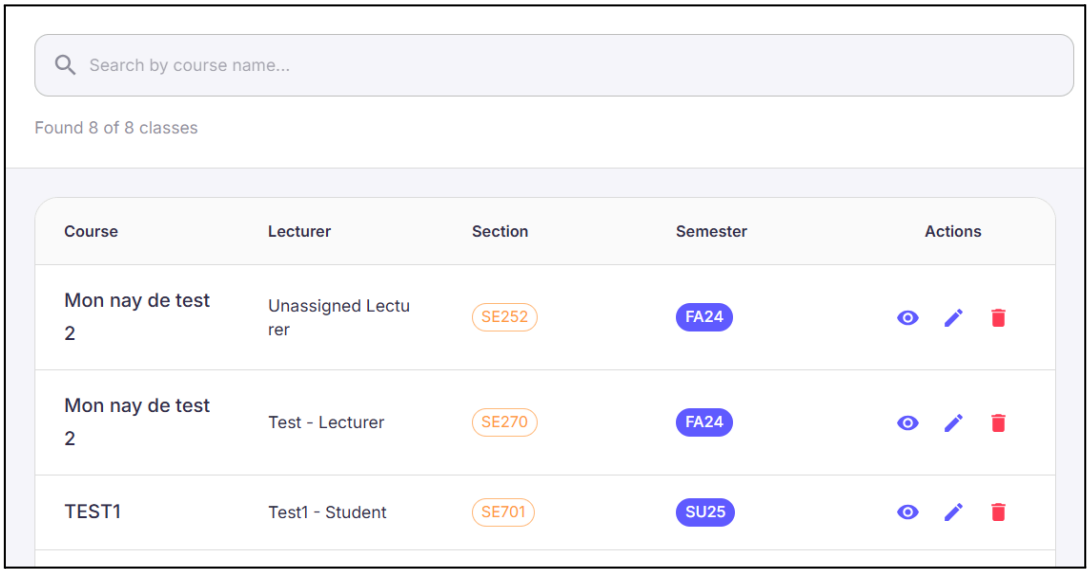


Figure 31 -Class management screen - Class Table

No	Field Name	Field Type	Description
1	**Course**	Text (link)	Course name of the class; may link to course page.
2	**Lecturer**	Text	Assigned lecturer's name; shows **"Unassigned Lecturer"** if none.
3	**Section**	Badge	Section code (e.g., SE252).
4	**Semester**	Badge	Term label (e.g., FA24, SU25).
5	**Actions**	Icon buttons	**View** (eye), **Edit** (pencil), **Delete** (trash). View navigates to Class Details; Edit/Delete invokes their own UCs and permissions (BR-02, BR-16).

Table 40: Class table layout

b. Use Case Specifications

ID and Name	UC39 - Manage Class (View Classes)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user navigates to **Schedule Management** → **Classes** from the sidebar.		
Description	The Classes Management screen allows users to view all classes in the institution, organized by semester. Each row shows the course, assigned lecturer, section, semester, and available actions. The left panel lists semesters to filter the table.		
Preconditions	**PRE-1.** User is authenticated **PRE-2.** User is authorized to view class data **PRE-3.** Class, course, and lecturer data exist—or empty states are handled gracefully.		
Postconditions	**POST- 1.** No data is modified by viewing alone. **POST- 2.** If a row action is selected, the user is routed to the corresponding UC (View/Edit/Delete Class).		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	In the sidebar, click **Schedule Management** to expand.
	2	User	Click **Classes** to open Classes Management.

ID and Name	UC39 - Manage Class (View Classes)		
	3	User	In the **Semesters** panel: choose **All Semesters** to view all classes, or select a specific term (e.g., **FA24**, **SU25**) to filter.
	4	User	Review the **Classes Table** columns: **Course**, **Lecturer**, **Section**, **Semester**, **Actions** (View / Edit / Delete).
	5	User	Use the **Search** bar to find classes by **course name**.
	6	User	Use **Pagination**: adjust **Rows per page** and navigate with **Previous/Next** arrows.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Click **View** action	Navigate to **UC—View Class Details**
	A2	Click **Edit** action	Navigate to **UC—Edit Class** if permitted
	A3	Click **Delete** action	Invoke **UC—Delete Class** (confirmation dialog, permissions required).
Exceptions	• **E1:** Backend/API error while loading list → show inline error with **Retry**; keep filters and pagination state. • **E2:** Network timeout → show toast/banner; allow retry; do not reset current selection. • **E3:** Data inconsistency (e.g., missing course/lecturer reference) → display placeholder (“Unknown”) and log for admin review.		
Frequency of Use	High		
Business Rules	BR-01, BR-02		
Other Information	• The table may support future enhancements: sort by columns, multi-filter (semester + lecturer), and bulk actions (admin only). • Accessibility: keyboard navigation, visible focus on row actions, ARIA labels on icons.		
Assumptions	• Backend provides paginated, filterable endpoints by semester and search query. • Terms (FA24, SU25…) follow a consistent naming convention.		

Table 41: View Class

3.2.1.13 Edit Class

a. Screen Layouts

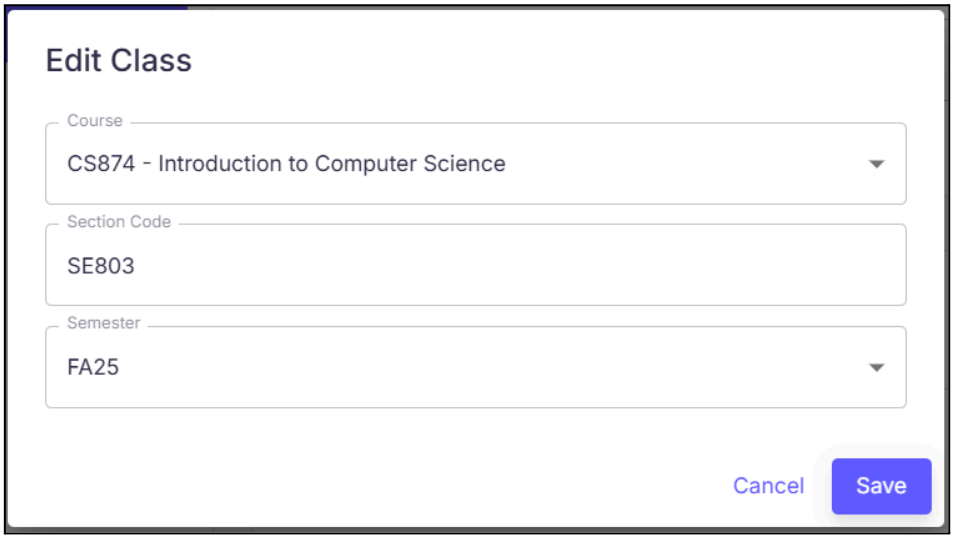


Figure 32 - Edit Class

No	Field Name	Field Type	Description
1	**Title ("Edit Class")**	Label	Dialog header.
2	**Course**	Dropdown	Select a course to associate with the class. Pre-filled with the current value. Required.
3	**Section Code**	Textbox	Edit the section code (e.g., SE803). Required: validation rules per system config.
4	**Semester**	Dropdown	Select the term (e.g., FA25). Pre-filled with the current value. Required.
5	**Cancel**	Button	Close the dialog without saving.
6	**Save**	Button	Validate and persist updates. Disabled until form is valid or changes are detected.
7	**Error/Validation Area**	Label	Inline messages for invalid fields or server errors.

Table 42: Edit class layout

b. Use Case Specifications

ID and Name	UC-40: Manage Classes (Edit class)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	From **Schedule Management** → **Classes**, the user clicks the **Edit** (pencil) icon in the row of a class.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC-40: Manage Classes (Edit class)</td></tr>
<tr><th>Description</th><td colspan="3">As an authorized user (Admin), I want to modify an existing class’s course, section code, and semester so that the class information remains accurate and up to date.</td></tr>
<tr><th>Preconditions</th><td colspan="3">PRE-1. User is authenticated
PRE-2. User is authorized to view class data
PRE-3. Target class exists and is retrievable; reference lists for Course and Semester are available.</td></tr>
<tr><th>Postconditions</th><td colspan="3">POST- 1. Class record is updated with the new values.
POST- 2. The Classes table reflects the changes and shows a success notification.
POST- 3. Edit action is logged for auditing</td></tr>
<tr><th rowspan="8">Normal Flows</th><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>Locate the class to edit in the list.</td></tr>
<tr><td>2</td><td>User</td><td>Click the Edit icon (pencil) in Actions.</td></tr>
<tr><td>3</td><td>System</td><td>Open Edit Class dialog pre-filled with current values.</td></tr>
<tr><td>4</td><td>User</td><td>Update one or more fields: Course (dropdown), Section Code, Semester (dropdown).</td></tr>
<tr><td>5</td><td>User</td><td>Click Save to apply changes.</td></tr>
<tr><td>6</td><td>System</td><td>Validate inputs and persist updates.</td></tr>
<tr><td>7</td><td>System</td><td>If successful, close dialog, refresh table, and show green toast: “Class edited successfully.” If unsuccessful, display an error message and keep dialog open.</td></tr>
<tr><th rowspan="3">Alternative Flows</th><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>User clicks Cancel</td><td>Dialog closes; no changes saved.</td></tr>
<tr><td>A2</td><td>No field changed</td><td>Save is disabled or results in no-op; dialog stays open.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Validation error (missing required field, invalid format) → inline field errors; prevent save.
• E2: Data conflict (e.g., duplicate section code within same course/semester if enforced) → show conflict message; prevent save.
• E3: Network/Server error → show error toast/banner; allow retry without losing user input.
• E4: Missing reference (course/semester no longer available) → show message and force re-selection.</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Medium</td></tr>
</table>

ID and Name	UC-40: Manage Classes (Edit class)
Business Rules	BR-01, BR-02
Other Information	• Form validation rules (length/pattern for Section Code) follow system configuration; errors appear inline. • Keyboard accessibility: Tab/Shift+Tab navigation, Enter to save, Esc to cancel.
Assumptions	• Backend exposes an update endpoint with optimistic locking or equivalent conflict handling. • The list view refreshes automatically after a successful edit.

Table 43 - Edit class specification

3.2.1.14 Delete Class

a. Screen Layouts

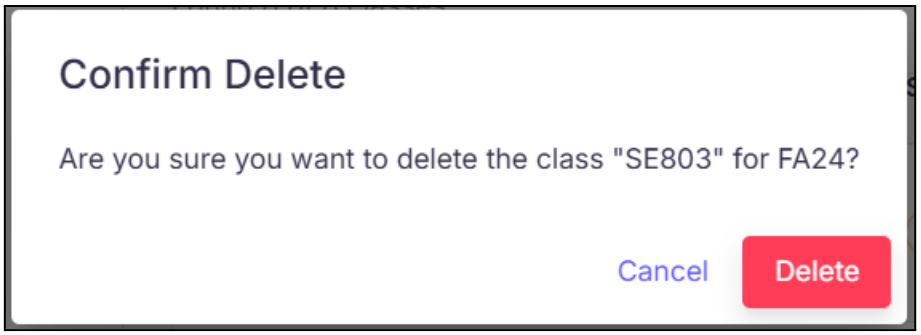


Figure 33 - Delete class confirm dialog

No	Field Name	Field Type	Description
1	**Title (“Confirm Delete”)**	Label	Dialog header.
2	**Message**	Label	Shows the target **Section Code** and **Semester**, e.g., *“Are you sure you want to delete the class ‘SE803’ for FA24?”*
3	**Cancel**	Button	Closes dialog, **no** changes applied.
4	**Delete**	Button (danger)	Confirms deletion; proceeds to remove the class. Disabled during request.
5	**Error area (optional)**	Label	Displays server/validation errors if deletion fails.

Table 44 - Delete class confirm layout

b. Use Case Specifications

<table>
<tr><th>ID and Name</th><td colspan="3">UC-41: Manage Classes (Delete class)</td></tr>
<tr><th>Created By</th><td>DungND</td><th>Date Created</th><td>28/06/2025</td></tr>
<tr><th>Primary Actor</th><td>Admin</td><th>Secondary Actors</th><td>System</td></tr>
<tr><th>Trigger</th><td colspan="3">From Schedule Management → Classes, the user clicks the Delete (trash) icon for a class.</td></tr>
<tr><th>Description</th><td colspan="3">As an authorized user (Admin), I want to remove a class from the system so that outdated or incorrect class records are not shown to users.</td></tr>
<tr><th>Preconditions</th><td colspan="3">PRE-1. User is authenticated
PRE-2. User is authorized to view class data
PRE-3. Target class exists and is retrievable.</td></tr>
<tr><th>Postconditions</th><td colspan="3">POST- 1. The class record is removed
POST- 2. The list view refreshes and a success notification is shown.
POST- 3. Deletion event is logged for audit</td></tr>
<tr><th rowspan="6">Normal Flows</th><th>Step</th><th>By</th><th>Step Details</th></tr>
<tr><td>1</td><td>User</td><td>Locate the class in the list.</td></tr>
<tr><td>2</td><td>User</td><td>Click the Delete icon in the Actions column.</td></tr>
<tr><td>3</td><td>System</td><td>Open Confirm Delete dialog showing Section Code and Semester of the selected class.</td></tr>
<tr><td>4</td><td>User</td><td>Click Delete to confirm (or Cancel to abort).</td></tr>
<tr><td>5</td><td>System</td><td>If successful, remove the class, close dialog, refresh table, and show green toast: “Class deleted successfully.” If unsuccessful, keep dialog open and display the error reason.</td></tr>
<tr><th rowspan="2">Alternative Flows</th><th>Alt Step</th><th>Condition</th><th>Description</th></tr>
<tr><td>A1</td><td>User clicks Cancel</td><td>Dialog closes; no changes saved.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Permission error → “You don’t have permission to delete this class.”
• E2: Concurrency conflict (class already deleted/modified) → show conflict message and refresh list.
• E3: Network/Server error → show error toast, allow retry.</td></tr>
</table>

ID and Name	UC-41: Manage Classes (Delete class)
Frequency of Use	Medium
Business Rules	BR-01, BR-02
Other Information	• Consider double-confirmation or reason capture if policy requires. • Ensure protected pages are not cached so the deleted class cannot appear via back navigation.
Assumptions	• Backend exposes a delete endpoint with proper referential checks and returns actionable error messages on constraint violations.

Table 45: Delete class specification

3.2.1.15 Add Class

a. Screen Layouts

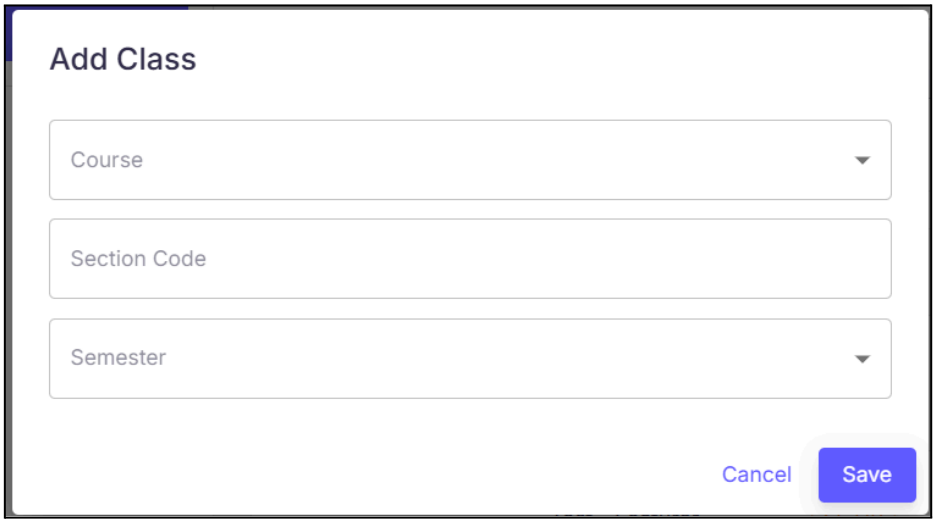


Figure 34: Add new class layout

No	Field Name	Field Type	Description
1	**Title (“Add Class”)**	Label	Dialog header.
2	**Course**	Dropdown	Required. Select a course from the available list.
3	**Section Code**	Textbox	Required. Enter section code (e.g., **SE101**). Validation per system policy.
4	**Semester**	Dropdown	Required. Select a term (e.g., **FA25**, **SU25**).
5	**Cancel**	Button	Close dialog without saving.
6	**Save**	Button	Validate and create the class. Disabled until form is valid.

No	Field Name	Field Type	Description
7	**Error/Validation Area**	Label	Inline messages for invalid fields or server errors.

Table 46: Add a new class

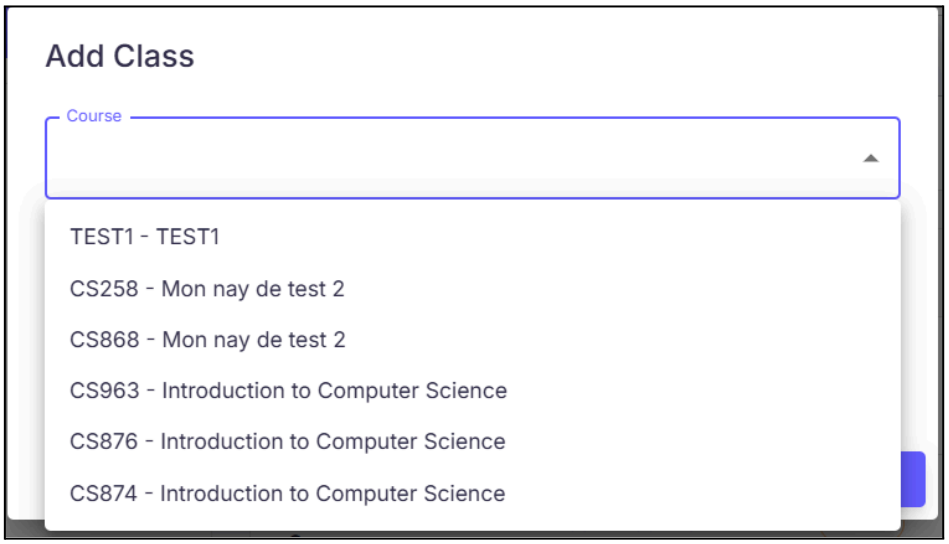


Figure 35 - Add new class selection

No	Field Name	Field Type	Description
1	**Course Options**	List	e.g., TEST1 – TEST1, CS258 – …, CS963 – Introduction to Computer Science, …

Table 47: Add new class selection layout



Figure 36 - Add semester Class Layout

No	Field Name	Field Type	Description
1	**Semester**	List	Predefined term values selectable for the class.

Table 48: Add semester Class

b. Use Case Specifications

ID and Name	UC42 – Add new class section (Add new class)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	On **Schedule Management** → **Classes**, the user clicks **Add Class** (top-right).		
Description	As an authorized user (Admin), I want to create a new class by selecting a course, entering a section code, and choosing a semester so that the class appears in the list and can be scheduled and managed.		
Preconditions	**PRE-1.** The user has a registered account. **PRE-2.** User has permission to create classes **PRE-3.** Reference data is available: list of **Courses** and **Semesters**.		
Postconditions	**POST- 1.** A new class record is created with the provided **Course**, **Section Code**, and **Semester**. **POST- 2.** The Classes table refreshes to include the new class and a success notification is shown. **POST- 3.** Creation event is logged for auditing		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	On the Classes Management page, click **Add Class**.
	2	User	In the dialog, provide details: **Course** (dropdown), **Section Code**, **Semester** (dropdown)
	3	User	Click **Save** to add the class.
	4	System	Validate input and create the record.
	5	System	If successful, close dialog, refresh table, and show green toast **"Class created successfully."** If not, keep the dialog open and display an error message.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	User clicks **Cancel**	Dialog closes; no data is saved.
	A2	Duplicate or conflicting class (policy)	System blocks creation (e.g., duplicate **Section Code** within the same **Course/Semester**) and shows a clear error; users can adjust inputs.

ID and Name	UC42 – Add new class section (Add new class)
Exceptions	• **E1:** Validation errors (missing required fields, invalid format) → inline field errors; prevent save. • **E2:** Network/Server error → error toast; allow retry without losing entered data. • **E3:** Missing reference (selected course/semester removed) → show message; force re-selection.
Frequency of Use	Medium
Business Rules	BR-01, BR-02
Other Information	• Keyboard shortcuts: **Enter** to Save (when valid), **Esc** to Cancel. • Consider optimistic UI: pre-append the new row then reconcile on server response.
Assumptions	• Backend provides a create endpoint and enforces uniqueness/consistency constraints for section per course/semester.

Table 49: Add semester Class Specification

3.2.1.16 View Class Details

a. Screen Layouts

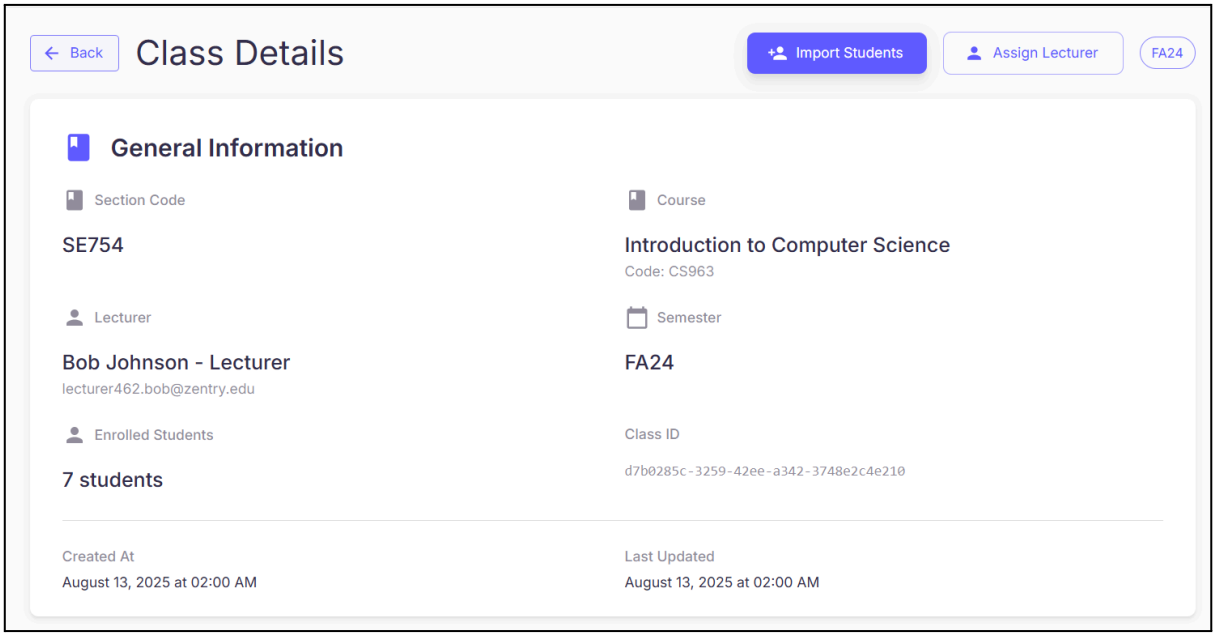


Figure 37 - View Class detail Layout

No	Field Name	Field Type	Description
1	**Back**	Button	Returns to **Classes Management**.
2	**Page Title (“Class Details”)**	Label	Header of the page.

No	Field Name	Field Type	Description
3	**Import Students**	Button	Opens bulk import & enroll flow (separate UC). Visible per permissions (BR-02).
4	**Assign Lecturer**	Button	Opens lecturer assignment flow (separate UC). Visible per permissions (BR-02).
5	**Term Chip**	Badge	Displays the class semester (e.g., **FA24**).
6	**General Information Card**	Panel	Container for class attributes below.
7	**Section Code**	Text	e.g., **SE754**.
8	**Course**	Text	Course name and code (e.g., **Introduction to Computer Science – CS963**).
9	**Lecturer**	Text	Lecturer's display name and email. Shows **"Unassigned Lecturer"** if none.
10	**Semester**	Text	e.g., **FA24**.
11	**Enrolled Students**	Text	Total number of enrolled students (e.g., **7 students**).
12	**Class ID**	Text	Unique identifier (UUID).
13	**Created At**	Text	Timestamp when the class was created.
14	**Last Updated**	Text	Timestamp of last modification.

Table 50 - View Class detail layout

b. Use Case Specifications

ID and Name	UC43 – Manage Class (View class detail)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	From Schedule Management → Classes, the user clicks the View (eye) icon for a class.		
Description	As an authorized user, I want to view all details of a specific class (section, course, lecturer, semester, enrolled students, identifiers, and timestamps) so that I can verify information and take follow-up actions such as importing students or assigning a lecturer.		
Preconditions	**PRE-1.** The user has a registered account. **PRE-2.** User has permission to view class details. **PRE-3.** The target class exists.		
Postconditions	**POST- 1.** No data is modified by viewing. **POST- 2.**User may choose optional actions (Import Students / Assign Lecturer) that lead to their respective UCs.		
Normal Flows	**Step**	**By**	**Step Details**

<table>
<tr><th>ID and Name</th><td colspan="3">UC43 – Manage Class (View class detail)</td></tr>
<tr><td rowspan="5"></td><td>1</td><td>User</td><td>On the Classes Management page, click the View icon in the row of the desired class.</td></tr>
<tr><td>2</td><td>System</td><td>Navigate to Class Details. The page shows: Section Code, Course (name & code), Lecturer (name & email), Semester, Enrolled Students (count), Class ID, Created At, Last Updated.</td></tr>
<tr><td>3</td><td>User</td><td>(Optional) Click Import Students to upload and enroll multiple students.</td></tr>
<tr><td>4</td><td>User</td><td>(Optional) Click Assign Lecturer to change or assign a lecturer to the class.</td></tr>
<tr><td>5</td><td>User</td><td>Click Back to return to Classes Management.</td></tr>
<tr><th rowspan="2">Alternative Flows</th><th>Alt Step</th><th>Condition</th><th>Description</th></tr>
<tr><td>A1</td><td>No lecturer assigned</td><td>The page shows “Unassigned Lecturer” and disables the “email” link.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Class not found or deleted → show “Class not found” and provide a link back to Classes.
• E2: Backend/API error → show error state with Retry; do not navigate away automatically.
• E3: Permission error → show “You don’t have permission to view this class” and offer navigation back.</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Medium–High</td></tr>
<tr><th>Business Rules</th><td colspan="3">BR-01, BR-02</td></tr>
<tr><th>Other Information</th><td colspan="3">• Dates/times follow user locale/timezone.
• Copy-to-clipboard can be enabled for Class ID.
• Accessibility: buttons and links have ARIA labels; keyboard focus order follows the layout.</td></tr>
<tr><th>Assumptions</th><td colspan="3">• Service provides a single “get class by ID” endpoint returning all shown fields and counts.
• “Import Students” and “Assign Lecturer” are specified in separate use cases.</td></tr>
</table>

Table 51: Add new class section (View class detail) Specification

3.2.1.17 Assign Lecturer

a. Screen Layouts

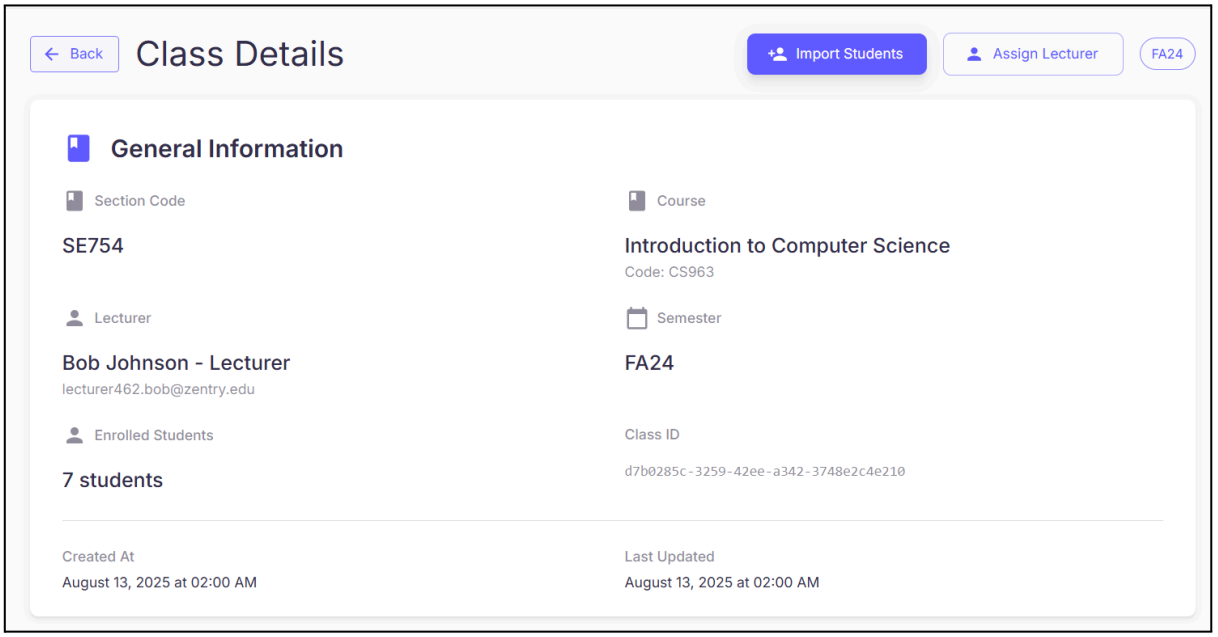


Figure 38 - Assign lecturer

No	Field Name	Field Type	Description
1	**Title ("Assign Lecturer for <Section Code>")**	Label	Dialog header, e.g., **Assign Lecturer for SE754**.
2	**Lecturer**	Searchable Dropdown	Required. Lists available **Lecturer** accounts (name + email). Supports type-ahead and scroll.
3	**Cancel**	Button	Close dialog without saving.
4	**Assign Lecturer**	Button (primary)	Confirms the assignment; disabled until a lecturer is selected.
5	**Error/Validation Area**	Label	Inline error for missing/invalid selection or server errors.

Table 52: Add new class section (Assign Lecturer)

b. Use Case Specifications

ID and Name	UC44 – Add new class section (Assign Lecturer)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	On Class Details, the user clicks Assign Lecturer (top-right).		
Description	As an authorized user (Admin), I want to assign a lecturer to a specific class by selecting from a list of available lecturers so that the class has a responsible instructor.		
Preconditions	**PRE-1.** The user has a registered account.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC44 – Add new class section (Assign Lecturer)</td></tr>
<tr><td></td><td colspan="3">PRE-2. User has permission to manage classes/lecturers.
PRE-3. The class exists and is loaded in Class Details.
PRE-4. Lecturer directory is available (accounts with role Lecturer).</td></tr>
<tr><td>Postconditions</td><td colspan="3">POST- 1. The selected lecturer is set as the class lecturer.
POST- 2. Class Details updates to show the new lecturer (name & email).
POST- 3. Success notification displayed and action logged for audit.</td></tr>
<tr><td rowspan="8">Normal Flows</td><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>On Class Details, click Assign Lecturer.</td></tr>
<tr><td>2</td><td>System</td><td>Open the dialog.</td></tr>
<tr><td>3</td><td>User</td><td>Click the Lecturer dropdown to view available lecturers.</td></tr>
<tr><td>4</td><td>User</td><td>Scroll or type to search, then select the desired lecturer.</td></tr>
<tr><td>5</td><td>User</td><td>Click Assign Lecturer to confirm.</td></tr>
<tr><td>6</td><td>System</td><td>Validate selection and persist the assignment.</td></tr>
<tr><td>7</td><td>System</td><td>If successful, close the dialog, refresh Class Details, and show green toast “Lecturer assigned successfully.” If unsuccessful, keep the dialog open and display the error.</td></tr>
<tr><td rowspan="3">Alternative Flows</td><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>Lecturer already assigned</td><td>Dialog preselects current lecturer; choosing a different one replaces the assignment.</td></tr>
<tr><td>A2</td><td>User clicks Cancel</td><td>Dialog closes; no changes saved.</td></tr>
<tr><td>Exceptions</td><td colspan="3">• E1: Network/Server error → show error toast and allow retry.
• E2: Concurrency conflict (class updated by another user) → show conflict message and offer Reload.
• E3: Directory lookup failure → show inline error and disable dropdown until recovered.</td></tr>
<tr><td>Frequency of Use</td><td colspan="3">Medium</td></tr>
<tr><td>Business Rules</td><td colspan="3">BR-01, BR-02</td></tr>
<tr><td>Other Information</td><td colspan="3">• Search should match on name and email; display Full Name – Lecturer and secondary line as email.</td></tr>
</table>

ID and Name	UC44 – Add new class section (Assign Lecturer)
	• Consider throttled server-side search for large directories. • Accessibility: dropdown supports keyboard navigation; **Enter** to confirm, **Esc** to cancel.
Assumptions	• Backend exposes an endpoint to assign/update lecturer for a class and returns updated class metadata.

Table 53: Add new class section (Assign Lecturer) Specification

3.2.1.18 View Enrolled Student

a. Screen Layouts

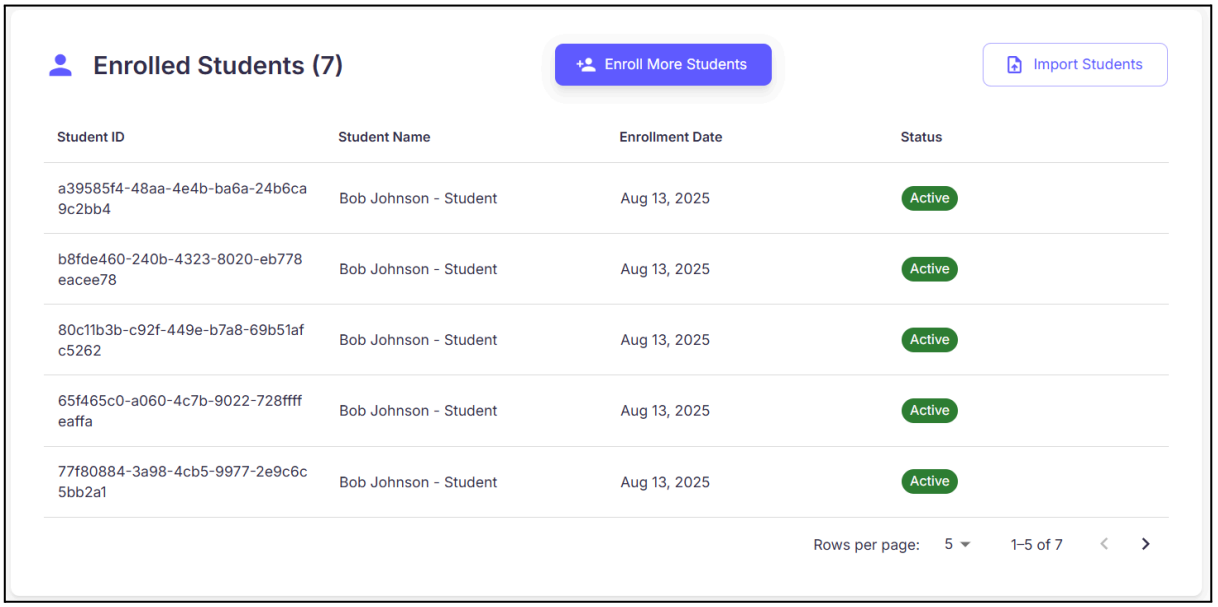


Figure 38 - Enroll Management (View Enrolled Student)

No	Field Name	Field Type	Description
1	**Section Title ("Enrolled Students (n)")**	Label	Header with total count.
2	**Enroll More Students**	Button	Opens manual enrollment flow (separate UC). Visible per permissions (BR-02, BR-16).
3	**Import Students**	Button	Opens CSV bulk import flow (separate UC). Visible per permissions (BR-02, BR-16).
4	**Students Table**	Table	Read-only list of current enrollments with columns below.
5	**Pagination**	Control	Rows per page selector and Previous/Next navigation.

Table 54: Enroll Management (View Enrolled Student) layout

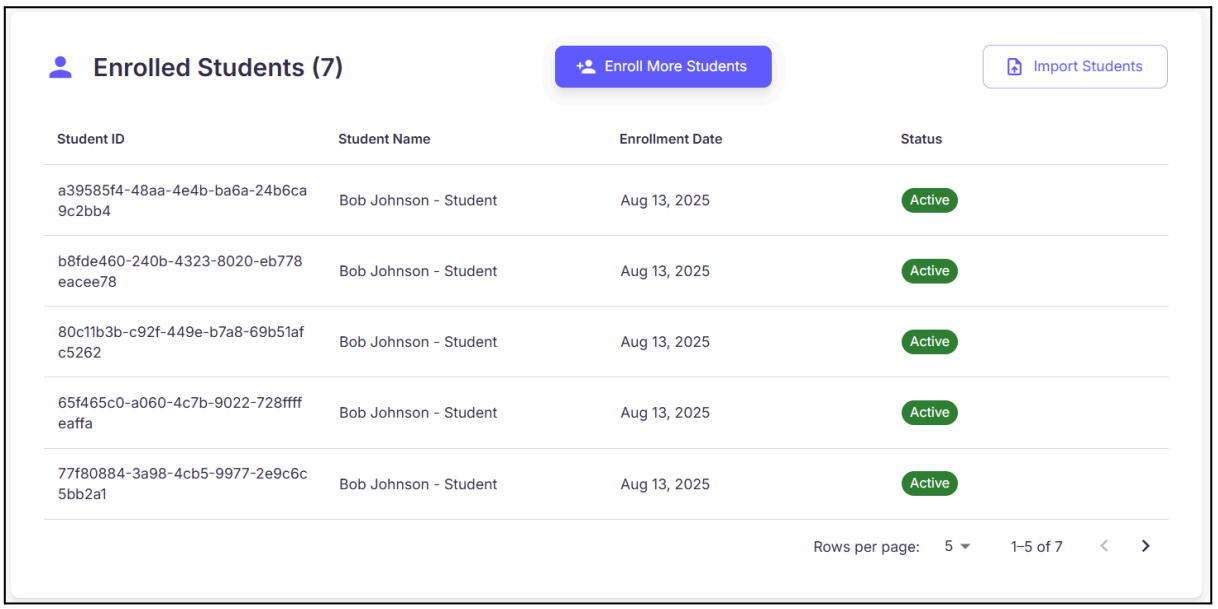


Figure 39 - Enroll Management (Enrolled Student Table) layout

No	Field Name	Field Type	Description
1	**Student ID**	Text	Unique identifier for each student.
2	**Student Name**	Text	Full name and role (e.g., "Bob Johnson – Student").
3	**Enrollment Date**	Date	Date the student was added to the class.
4	**Status**	Badge	Current enrollment status (e.g., **Active**).

Table 55: Enroll Management (Enrolled Student) layout

b. Use Case Specifications

ID and Name	UC45 - Manage Enrollments (View Enrolled Student)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	User opens Class Details and scrolls to the Enrolled Students section.		
Description	As an authorized user, I want to view the list of students enrolled in a specific class—along with their IDs, names, enrollment dates, and statuses—so that I can verify enrollment and take follow-up actions (enroll more or import).		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to view class enrollments. **PRE-3.** The target class exists and has enrollment data (or supports empty state).		
Postconditions	**POST- 1.** No data is modified by viewing alone. **POST- 2.** After enrollment/import actions complete, the table auto-refreshes and the count updates.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	On **Class Details**, scroll to **Enrolled Students**.

ID and Name	UC45 - Manage Enrollments (View Enrolled Student)		
	2	User	Review table columns: **Student ID**, **Student Name**, **Enrollment Date**, **Status**.
	3	User	Click **Enroll More Students** to add individual students (separate UC).
	4	User	Click **Import Students** to bulk add from a CSV file (separate UC).
	5	User	If many records, use **Rows per page** and **Previous/Next** to navigate.
	6	System	Data updates automatically after students are added/removed; show green toast on success (e.g., **"Student added successfully."**).
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No students enrolled	Show empty state "No students enrolled yet" with quick actions to **Enroll** or **Import** (if permitted).
	A2	Role-limited actions	Lecturer may **view** but cannot **Import**/**Enroll** unless granted permission.
	A3	Pagination edge	Disable **Previous** on first page and **Next** on last page; remember user's page size selection in session.
Exceptions	• **E1:** Backend/API error → show inline error with **Retry**; keep table state. • **E2:** Data changed concurrently (another user added/removed) → show non-blocking "Data updated" banner and refresh. • **E3:** Import/Enroll flow fails → show error reason (e.g., invalid CSV), remain on section.		
Frequency of Use	Medium–High		
Business Rules	BR-01, BR-02		
Other Information	• The table may later support sorting/filtering (by status/date). • Accessibility: table headers announced to screen readers; pagination controls are keyboard accessible.		
Assumptions	• The service provides a paginated endpoint for enrolled students of a class. • After successful enroll/import/delete operations, the section refreshes automatically and updates the count.		

Table 56: View Enrolled Student

3.2.1.19 Import Student

a. Screen Layouts

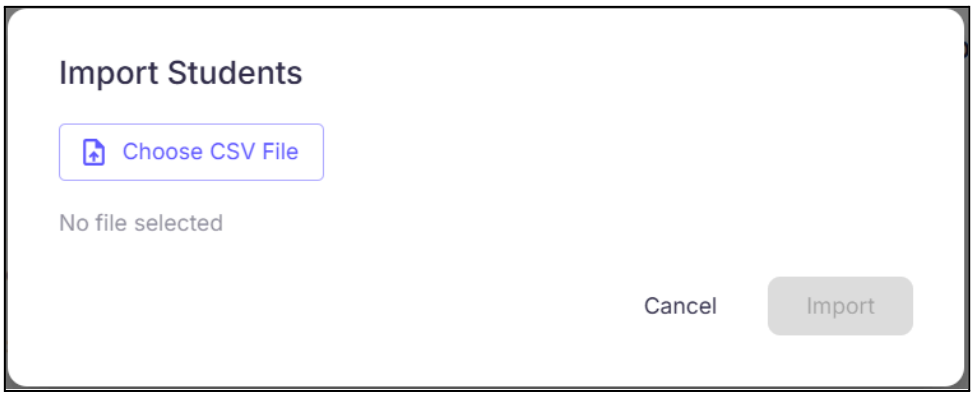


Figure 40 - Import student

No	Field Name	Field Type	Description
1	**Title ("Import Students")**	Label	Dialog header.
2	**Choose CSV File**	Button	Open OS file picker to select a .csv. Accepts only CSV MIME/type.
3	**Selected File Name**	Label	Shows the chosen file or "No file selected".
4	**File Requirements Hint**	Helper text	Required headers: **Student ID**. Optional: **Full Name**, **Email**, **Enrollment Date** (YYYY-MM-DD).
5	**Cancel**	Button	Close dialog without importing.
6	**Import**	Button (primary)	Uploads and processes the file. Disabled until a valid file is chosen.
7	**Progress / Result Area**	Label	Shows upload progress, success count, and/or row-level errors after processing.

Table 57: Import Student Layout

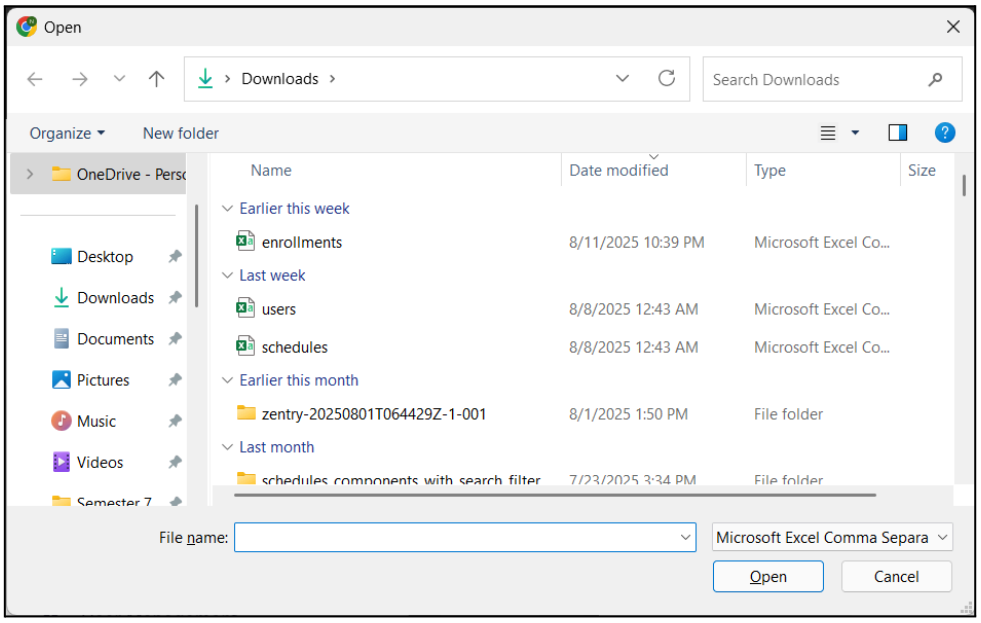


Figure 41 - Import student (2)

No	Field Name	Field Type	Description
1	**File list**	Native control	User selects a .csv file from local machine.
2	**Open**	Button	Confirms file selection.
3	**Cancel**	Button	Closes picker without selection.

Table 58: Import Student Layout 2

b. Use Case Specifications

ID and Name	UC46 - Manage Enrollments (Import Student)		
Created By	DungND	**Date Created**	28/06/2025
Primary Actor	Admin	**Secondary Actors**	System
Trigger	On Class Details, the user clicks Import Students (top-right of the page or Enrolled Students section).		
Description	As an authorized user, I want to upload a CSV file to bulk enroll students into a class so that I can add many students efficiently and consistently.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to import/enroll students. **PRE-3.** The target class exists and is loaded in **Class Details**. **PRE-4.** CSV uses the required schema: **Student ID** (required), **Full Name** (optional), **Email** (optional), **Enrollment Date**.		
Postconditions	**POST- 1.** Valid rows are enrolled into the class; the **Enrolled Students** table is refreshed. **POST- 2.** A success toast is shown (e.g., “Students imported successfully”). **POST- 3.** Import activity (counts, actor, class) is logged for audit.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC46 - Manage Enrollments (Import Student)</td></tr>
<tr><th rowspan="6">Normal Flows</th><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>On Class Details, click Import Students.</td></tr>
<tr><td>2</td><td>User</td><td>Click Choose CSV File, pick the CSV that follows the required format.</td></tr>
<tr><td>3</td><td>User</td><td>Click Import to upload and process the file.</td></tr>
<tr><td>4</td><td>System</td><td>Parse CSV, validate headers & rows, then enroll valid students into the class.</td></tr>
<tr><td>5</td><td>System</td><td>On success, show green notification “Students imported successfully” and update the Enrolled Students list automatically. On failure, show an error explaining what went wrong (e.g., missing columns, invalid format).</td></tr>
<tr><th rowspan="5">Alternative Flows</th><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>No file selected</td><td>Import remains disabled; users may Cancel.</td></tr>
<tr><td>A2</td><td>Partial success</td><td>System enrolls valid rows, lists failed rows with reasons (e.g., “Student not found”, “Invalid date”). Optionally provide a downloadable error report.</td></tr>
<tr><td>A3</td><td>Duplicate / already enrolled</td><td>System skips duplicates and reports them as “Already enrolled”.</td></tr>
<tr><td>A4</td><td>Policy requires existing students only</td><td>Rows whose Student ID does not exist are rejected with a clear message; none of those rows are imported. (If policy allows auto-create, prompt or document separately.)</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Missing/incorrect headers → show “Required column ‘Student ID’ is missing.”
• E2: File type/size invalid → show constraint message and block import.
• E3: Network/server error during upload → show error toast; allow retry without reselecting file if possible.
• E4: Concurrency/conflict (class deleted/closed) → abort and show error; return to Classes page.</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Medium</td></tr>
</table>

ID and Name	UC46 - Manage Enrollments (Import Student)
Business Rules	BR-01, BR-02
Other Information	• Recommended CSV encoding: UTF-8, comma-separated, first row headers (case-insensitive). • **Enrollment Date** defaults to the current date if omitted. • Timezone/locale applied when displaying dates. • Consider idempotency: re-importing the same file should not create duplicates.
Assumptions	• Student identities are keyed by **Student ID**; name/email in CSV is used for verification/display only unless account creation is enabled by policy. • Backend exposes a bulk import endpoint that validates and returns per-row results.

Table 59: Manage Enrollments (Import Student) Specification

3.2.1.20 Enroll Student

a. Screen Layouts

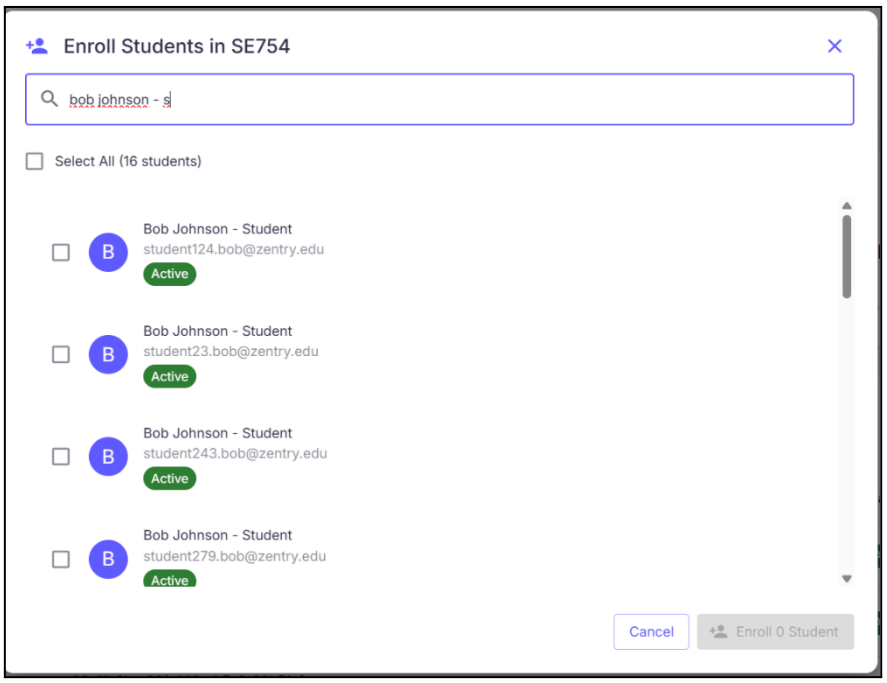


Figure 42 - Enroll Student

No	Field Name	Field Type	Description
1	**Title (“Enroll Students in <Section Code>”)**	Label	Dialog header, e.g., **Enroll Students in SE754**.
2	**Search Bar**	Searchbox	Filter available students by **name**, **email**, or **ID** (debounced).

No	Field Name	Field Type	Description
3	**Select All (n students)**	Checkbox	Select/deselect **all students on the current page**. Shows total count in view.
4	**Student List**	Table / List	Scrollable list of students (see columns below).
5	**Import Students**	Button (secondary)	Opens **Import Students** flow (CSV) — optional shortcut.
6	**Cancel**	Button	Close dialog without changes.
7	**Enroll X Student(s)**	Button (primary)	Disabled when 0 selected; label updates with selected count.

Table 60: Enroll Student Layout

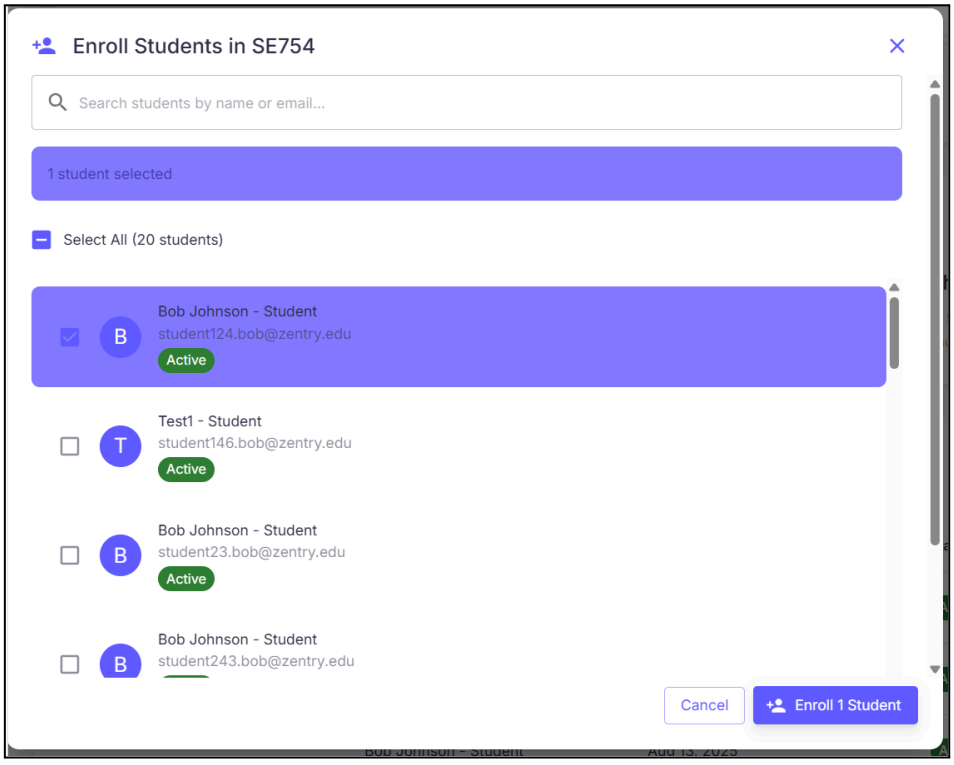


Figure 43 - Enroll Student (2)

No	Field Name	Field Type	Description
1	**Row Select**	Checkbox	Select an individual student.
2	**Student ID**	Text	Unique identifier.
3	**Student Name**	Text	Full name and role, e.g., “Bob Johnson – Student”.
4	**Email**	Text (secondary)	Contact email (if available).
5	**Status**	Badge	Current status (e.g., **Active**). Non-active entries may be disabled per policy.

Table 61: Enroll Student Layout 2

b. Use Case Specifications

ID and Name	UC47 - Manage Enrollments (Enroll Student)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	On Class Details → Enrolled Students, the user clicks Enroll More Students.		
Description	As an authorized user, I want to add one or more students to a class by selecting them from an available list (or by importing a CSV) so that the class roster is updated efficiently.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to manage enrollments. **PRE-3.** Target class exists and is loaded in **Class Details**. **PRE-4.** Directory of eligible students is available.		
Postconditions	**POST- 1.** Selected students are enrolled into the class; the **Enrolled Students** table/count updates. **POST- 2.** A success notification is shown and the action is logged.		

Normal Flows	Step	By	Step Details
	1	User	On **Class Details**, click **Enroll More Students** to open the dialog.
	2	User	In the dialog: use the **Search Bar** to filter by name or ID; optionally use **Select All (n students)** to select all on the current page.
	3	User	Review list columns: **Student ID**, **Student Name**, **Status**.
	4	User	Select one or more students. When none are selected, the **Enroll** button is disabled; when selected, its label shows the count (e.g., **Enroll 3 Students**).
	5	User	(Optional) Click **Import Students** to bulk upload via CSV (see UC11).
	6	User	Click **Enroll X Students** to confirm.
	7	System	Validate selections and enroll them into the class.
	8	System	If successful, close dialog, refresh the **Enrolled Students** table, and show green toast **“Student(s) enrolled successfully.”** If unsuccessful, show error and keep dialog open.

Alternative Flows	Alt Step	Condition	Description
	A1	No results after search	Show empty state “No students found”; allow clearing the query.

ID and Name	UC47 - Manage Enrollments (Enroll Student)		
	A2	Some selected students already enrolled	System skips duplicates and reports them as **Already enrolled**; successful rows proceed.
	A3	User cancels	Close dialog; no changes saved.
Exceptions	• **E1:** Network/Server error → show error toast; allow retry without losing selection. • **E2:** Directory fetch failure → show inline error and a **Retry** control. • **E3:** Concurrency conflict (class modified/closed) → abort with message and return to **Class Details**.		
Frequency of Use	Medium–High		
Business Rules	BR-01, BR-02		
Other Information	• Multi-select supports Shift/keyboard navigation; list is virtualized for large directories. • The dialog may show a small banner with "X student(s) selected". • Pagination or infinite scroll may apply; **Select All** refers to the **current page** only (unless "select across pages" is implemented).		
Assumptions	• Backend provides a bulk-enroll endpoint accepting a list of Student IDs and the Class ID, returning per-ID results.		

Table 62: Manage Enrollments (Enroll Student) Specification

3.2.1.21 View Schedules

a. Screen Layouts

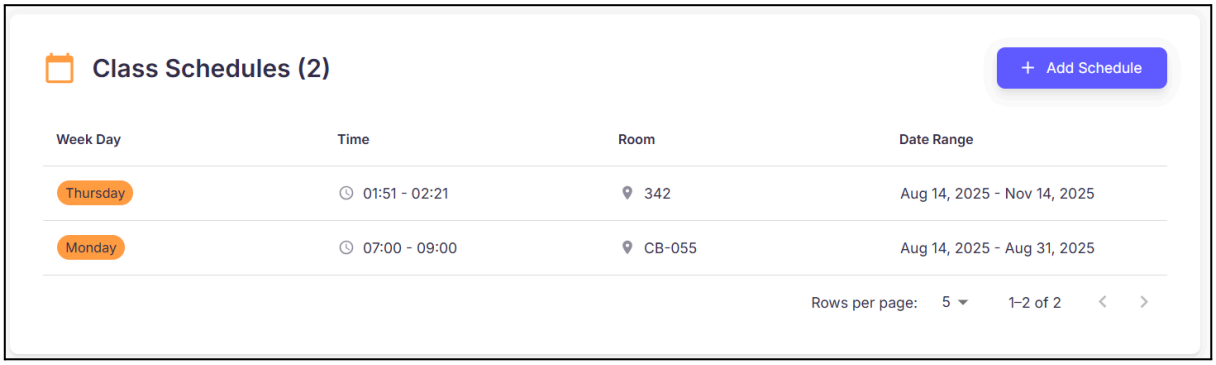


Figure 44 - View schedule layout

No	Field Name	Field Type	Description
1	**Section Title ("Class Schedules (n)")**	Label	Header with total count.
2	**Add Schedule**	Button	Opens **Add Schedule** flow (separate UC). Visible per permissions.

No	Field Name	Field Type	Description
3	**Schedules Table**	Table	Read-only list of schedules with the columns below.
4	**Pagination**	Control	Rows per page selector, Previous/Next arrows.
5	**Empty State**	Label	“No schedules have been created for this class yet.” (when the list is empty).

Table 63: Manage Schedules (View schedule) layout

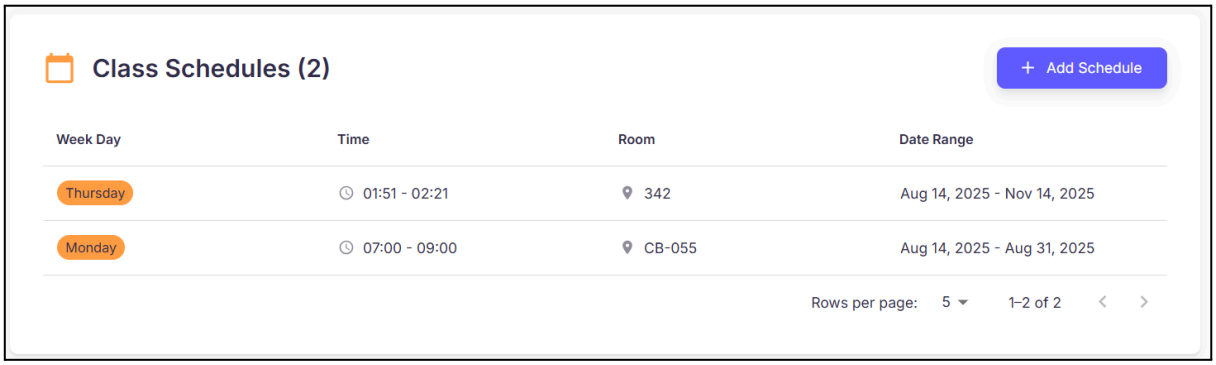


Figure 45 - View schedule table

No	Field Name	Field Type	Description
1	**Week Day**	Badge/Text	Day of week when the class occurs (e.g., Monday, Thursday).
2	**Time**	Text	Start and end time of the session (HH:MM – HH:MM).
3	**Room**	Text	Room/location (e.g., CB-055).
4	**Date Range**	Text	Start date ⟶ end date for the repeating schedule.

Table 64: Manage Schedules (View schedule detail) layout

b. Use Case Specifications

ID and Name	UC – 48: Manage Schedules (View scheduke)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	Manage Schedules (View schedule) layout		
Description	As a user, I want to view all schedules for a class—showing weekday, time, room, and date range—so that I can understand when and where sessions occur and create new schedules if needed.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to view schedules for the class. **PRE-3.** The target class exists.		
Postconditions	**POST- 1.** No data is modified by viewing. **POST- 2.** If a new schedule is added (or one is edited/deleted), the table refreshes automatically.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC – 48: Manage Schedules (View scheduke)</td></tr>
<tr><th rowspan="6">Normal Flows</th><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>On Class Details, scroll to Class Schedules.</td></tr>
<tr><td>2</td><td>User</td><td>Review table columns: Week Day, Time, Room, Date Range.</td></tr>
<tr><td>3</td><td>User</td><td>Click Add Schedule to create a new schedule (opens separate UC).</td></tr>
<tr><td>4</td><td>User</td><td>If there are many records, use Rows per page and Previous/Next pagination controls.</td></tr>
<tr><td>5</td><td>System</td><td>The table updates automatically after a schedule is added/edited/deleted and shows a green toast on success (e.g., “Schedule added successfully.”).</td></tr>
<tr><th rowspan="4">Alternative Flows</th><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>No schedules exist</td><td>Show empty state with a quick action Add Schedule (if permitted).</td></tr>
<tr><td>A2</td><td>Role-limited view</td><td>Students can view only; Add Schedule is hidden.</td></tr>
<tr><td>A3</td><td>Timezone/locale differences</td><td>Times render in the user’s timezone/locale; a tooltip may show server time.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Backend/API error → show inline error with Retry; keep pagination and scroll position.
• E2: Data inconsistency (e.g., room not found) → show placeholder (“Unknown room”) and log for admin.
• E3: Network timeout → show toast; allow retry without reloading the page.</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Medium–High</td></tr>
<tr><th>Business Rules</th><td colspan="3">BR-01, BR-02</td></tr>
<tr><th>Other Information</th><td colspan="3">• The table may later include actions (Edit/Delete) per row for authorized roles.
• Accessibility: table headers announced to screen readers; pagination and the Add button are keyboard accessible.</td></tr>
</table>

ID and Name	UC – 48: Manage Schedules (View scheduke)
Assumptions	• Backend provides a paginated list of schedules for a class, including weekday, start/end time, room, and date range.

Table 65: Manage Schedules (View schedule) layout

3.2.1.22 Add Schedule

a. Screen Layouts

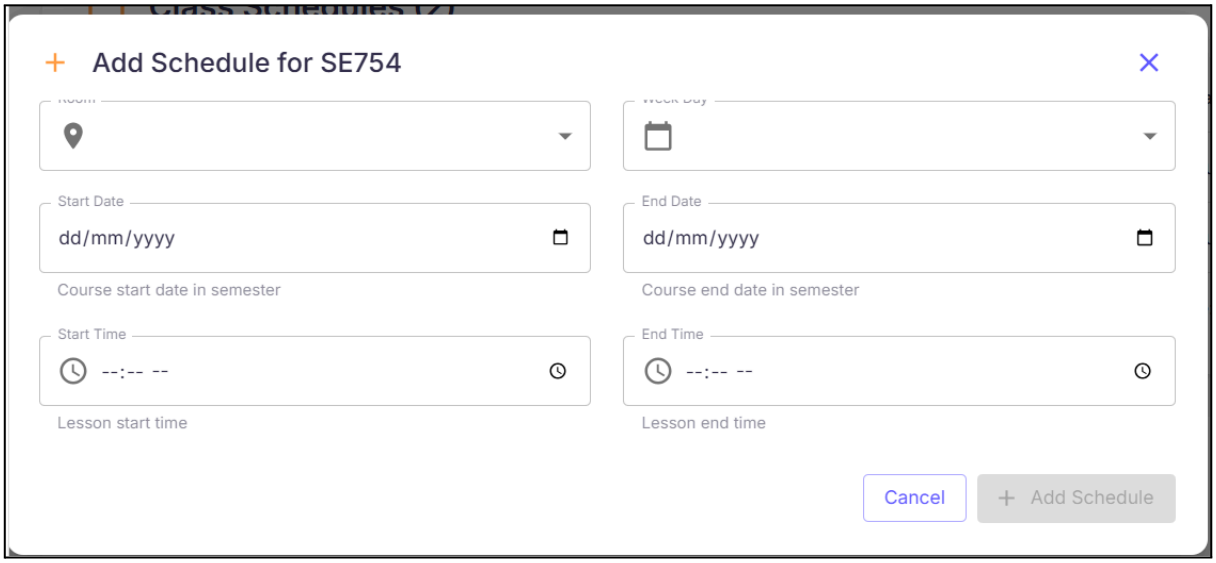


Figure 47 - Add schedule

No	Field Name	Field Type	Description
1	**Title ("Add Schedule for <Section Code>")**	Label	Dialog header, e.g., **Add Schedule for SE754**.
2	**Room**	Dropdown	**Required.** Select from available rooms; each option shows **name • building • capacity**.
3	**Week Day**	Dropdown	**Required.** Choose the day of week (Monday…Sunday).
4	**Start Date**	DatePicker	**Required.** Course start date within the semester.
5	**End Date**	DatePicker	**Required.** Course end date within the semester. Must be ≥ Start Date.
6	**Start Time**	TimePicker	**Required.** Lesson start time (HH:MM).
7	**End Time**	TimePicker	**Required.** Lesson end time (HH:MM). Must be > Start Time.
8	**Schedule Summary**	Panel	Read-only preview of chosen **Week Day, Time, Room, Date Range**. Updates live.

No	Field Name	Field Type	Description
9	**Cancel**	Button	Close dialog; do not save.
10	**Add Schedule**	Button (primary)	Validate & create schedule; disabled until form is valid.
11	**Error/Validation Area**	Label	Inline field errors or server messages.

Table 66: Add schedule Layout (add schedule) layout

b. Use Case Specifications

ID and Name	UC49 - Manage Schedules (Add Schedule)		
Created By	DungND	**Date Created**	28/06/2025
Primary Actor	Admin	**Secondary Actors**	System
Trigger	In Class Details ⟶ Class Schedules, the user clicks Add Schedule (top-right).		
Description	As an authorized user, I want to add a new schedule to a class by specifying the room, weekday, date range, and lesson times so that the class timetable is accurate and visible in the schedules list.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to manage schedules. **PRE-3.** Target class exists; room catalog and semester dates are available.		
Postconditions	**POST- 1.** A weekly recurring schedule (on the chosen weekday between the selected dates) is created for the class. **POST- 2. Class Schedules** table refreshes and shows the new row; a success toast is displayed. **POST- 3.** Action is logged for auditing.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	In **Class Schedules**, click **Add Schedule**.
	2	User	Fill the form: **Room**, **Week Day**, **Start Date / End Date**, **Start Time / End Time**.
	3	User	Review **Schedule Summary** to confirm selections.
	4	User	Click **Add Schedule** to save.
	5	System	Validate inputs, create the schedule, and update the table. On success, show green toast **"Schedule added successfully."** On failure, display an error and keep the dialog open.

<table>
<tr><th>ID and Name</th><td colspan="3">UC49 - Manage Schedules (Add Schedule)</td></tr>
<tr><th rowspan="3">Alternative Flows</th><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>User clicks Cancel</td><td>Close dialog; no data saved.</td></tr>
<tr><td>A2</td><td>Dates outside semester / End < Start</td><td>System blocks save; show inline validation and highlight offending fields.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Network/Server error → show error toast; allow retry without losing input.
• E2: Missing reference data (rooms/semester) → show blocking message and disable Add Schedule.
• E3: Permission error → show “You don’t have permission to add schedules.”</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Medium</td></tr>
<tr><th>Business Rules</th><td colspan="3">BR-01, BR-02</td></tr>
<tr><th>Other Information</th><td colspan="3">• Timezone/locale respected when rendering and validating times.
• Consider preventing past-date creation if policy requires.
• Form may prefill Room or Week Day from previous selection within the session.</td></tr>
<tr><th>Assumptions</th><td colspan="3">• Backend accepts: {classId, roomId, weekday, startDate, endDate, startTime, endTime} and returns the created schedule row.
• Schedules repeat weekly on the specified weekday within the given date range.</td></tr>
</table>

Table 67: Add Schedule

3.2.1.23 Edit Schedule

a. Screen Layouts

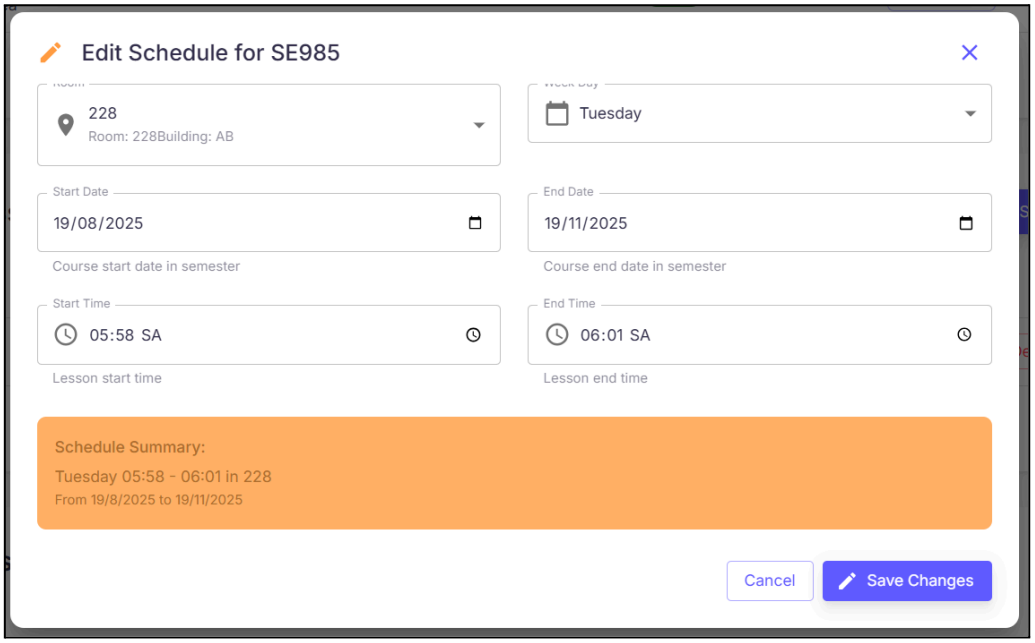


Figure 48 - Edit Schedule

No	Field Name	Field Type	Description
1	**Title ("Edit Schedule for <Section Code>")**	Label	Dialog header, e.g., **Edit Schedule for SE754**.
2	**Room**	Dropdown	**Required.** Select a room (shows **name • building • capacity**).
3	**Week Day**	Dropdown	**Required.** Choose the new weekday (Mon…Sun).
4	**Start Date**	DatePicker	**Required.** New start date within the semester.
5	**End Date**	DatePicker	**Required.** New end date (must be ≥ Start Date).
6	**Start Time**	TimePicker	**Required.** Lesson start time (HH:MM).
7	**End Time**	TimePicker	**Required.** Lesson end time (must be > Start Time).
8	**Schedule Summary**	Panel	Live preview of **Week Day, Time, Room, Date Range** after edits.
9	**Cancel**	Button	Close dialog; no changes saved.
10	**Save Changes**	Button (primary)	Validate and persist updates; disabled until form is valid or any field changed.
11	**Error/Validation Area**	Label	Inline field errors or server messages.

Table 68: Edit schedule layout

b. Use Case Specifications

ID and Name	UC49 - Manage Schedules (Add Schedule)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	User clicks the Edit (pencil) icon for a schedule in Class Details → Class Schedules.		
Description	As an authorized user, I want to update a class schedule's room, weekday, date range, and times so that the timetable remains accurate and conflicts are avoided.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to manage schedules. **PRE-3.** The target schedule exists and is loaded with its current values. **PRE-4.** Reference data (rooms, semester dates) is available.		
Postconditions	**POST- 1.** The schedule is updated with the new values. **POST- 2. Class Schedules** list refreshes and shows the updated row; a success toast is displayed. **POST- 3.** Update action is logged for audit.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	In **Class Schedules**, locate the schedule and click **Edit** (pencil).
	2	System	Open **Edit Schedule** dialog pre-filled with current data.
	3	User	Update fields as needed: **Room**, **Week Day**, **Start/End Date**, **Start/End Time**.
	4	User	Review **Schedule Summary** to verify the changes.
	5	User	Click **Save Changes** to apply updates.
	6	System	Validate input and persist the changes.
	7	System	If successful, close dialog, refresh the table, and show green toast **"Schedule updated successfully."** If unsuccessful, show an error (e.g., invalid time range, missing information) and keep the dialog open.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No fields changed	**Save Changes** remains disabled or results in no-op; dialog stays open.

ID and Name	UC49 - Manage Schedules (Add Schedule)		
	A2	Date outside semester / End < Start	Block save; highlight invalid fields with inline messages.
	A3	Time conflict (same class or room overlap)	System blocks save with clear message (e.g., “Overlaps with another schedule in CB-055”).
	A4	Room becomes unavailable during edit	Prompt user to reselect a room and refresh options.
	A5	User clicks **Cancel**	Close dialog; discard edits.
Exceptions	• **E1:** Permission error → “You don’t have permission to edit schedules.” • **E2:** Concurrency conflict (schedule modified/deleted by someone else) → show conflict notice and refresh list. • **E3:** Network/Server error → show error toast; allow retry without losing edits. • **E4:** Missing reference data (rooms/semester) → block save and instruct user to retry after data refresh.		
Frequency of Use	Medium		
Business Rules	BR-01, BR-02		
Other Information	• Times respect user timezone/locale; consider preventing edits to past occurrences if policy requires. • If the system supports recurring exceptions, this UC edits the *base weekly schedule* (not individual dates).		
Assumptions	• Backend exposes an update endpoint {scheduleId, roomId, weekday, startDate, endDate, startTime, endTime} with conflict detection and returns the updated schedule row.		

Table 69: Manage Schedule (Edit Schedule)

3.2.1.24 Delete Schedule

a. Screen Layouts

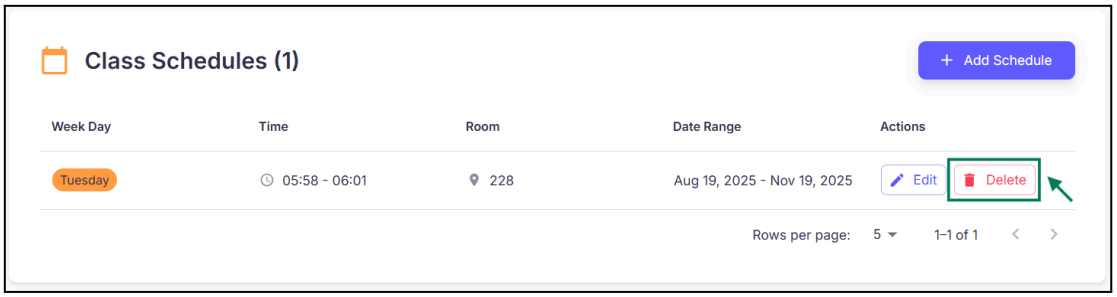


Figure 49 - Delete Schedule

No	Field Name	Field Type	Description
1	**Schedules Table**	Table	Lists **Week Day**, **Time**, **Room**, **Date Range**, **Actions**.
2	**Actions (per row)**	Icon buttons	**Edit** (pencil), **Delete** (trash bin). Delete opens confirmation dialog.
3	**Empty State**	Label	"No schedules have been created for this class yet." (when list is empty).

Table 70: Manage Schedule (Delete Schedule) layout

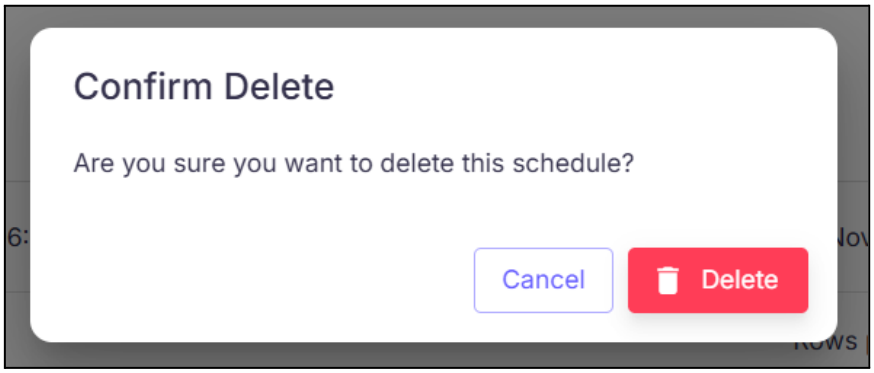


Figure 50 - Delete Schedule (2)

No	Field Name	Field Type	Description
1	**Title ("Confirm Delete")**	Label	Dialog header.
2	**Schedule Summary**	Label	Shows **Week Day, Time, Room, Date Range** for verification.
3	**Cancel**	Button (secondary)	Close dialog; no changes.
4	**Confirm**	Button (primary, red)	Permanently delete the schedule. Disabled while processing.
5	**Error Area (optional)**	Label	Displays server/validation errors if deletion fails.

Table 71: Manage Schedule (Delete Schedule) layout 2

b. Use Case Specifications

ID and Name	UC51 - Manage Schedule (Delete Schedule)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	User clicks the Delete (trash bin) icon for a schedule in Class Details → Class Schedules.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC51 - Manage Schedule (Delete Schedule)</td></tr>
<tr><th>Description</th><td colspan="3">As an authorized user, I want to permanently remove a schedule from a class so that outdated or incorrect timetable entries are no longer shown.</td></tr>
<tr><th>Preconditions</th><td colspan="3">PRE-1. User is authenticated.
PRE-2. User has permission to manage schedules.
PRE-3. Target schedule exists for the selected class.</td></tr>
<tr><th>Postconditions</th><td colspan="3">POST- 1. The schedule is removed (or marked deleted per policy) and no longer appears in the list.
POST- 2. A success notification is displayed; deletion is logged for audit.</td></tr>
<tr><th rowspan="6">Normal Flows</th><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>In Class Schedules, locate the schedule and click Delete (trash bin).</td></tr>
<tr><td>2</td><td>System</td><td>Show Delete Confirmation dialog with Week Day, Time, Room, Date Range.</td></tr>
<tr><td>3</td><td>User</td><td>Click Confirm to proceed (or Cancel to abort).</td></tr>
<tr><td>4</td><td>System</td><td>If successful, remove the schedule, refresh the table, and show green toast “Schedule deleted successfully.” If unsuccessful, keep dialog open and show error (e.g., “Unable to delete schedule. Please try again.”).</td></tr>
<tr><td>5</td><td>User</td><td>To delete more schedules, repeat Steps 1–4 for each row.</td></tr>
<tr><th rowspan="2">Alternative Flows</th><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>User clicks Cancel</td><td>Dialog closes; no changes saved.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Permission error → “You don’t have permission to delete schedules.”
• E2: Concurrency conflict (already deleted/edited) → show info and refresh list.
• E3: Network/Server error → show error toast; allow retry.</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Medium</td></tr>
<tr><th>Business Rules</th><td colspan="3">BR-01, BR-02</td></tr>
<tr><th>Other Information</th><td colspan="3">• Consider double-confirmation for destructive actions in production environments.
• Ensure protected pages aren’t cached so deleted schedules don’t reappear via back navigation.</td></tr>
<tr><th>Assumptions</th><td colspan="3">• Backend exposes a delete endpoint with proper referential checks and returns actionable error messages when constraints prevent deletion.</td></tr>
</table>

Table 72: Manage Schedule (Delete Schedule)

3.2.1.25 View Sessions

a. Screen Layouts



Figure 51 - View session

No	Field Name	Field Type	Description
1	**Section Title ("Sessions (n)")**	Label	Header with total count.
2	**Sessions Table**	Table	Read-only list of sessions with columns below.
3	**Pagination**	Control	Rows per page selector, Previous/Next arrows.
4	**Empty State**	Label	"No sessions available for this class." (shown when list is empty).

Table 73: Manage Sessions (View session) layout

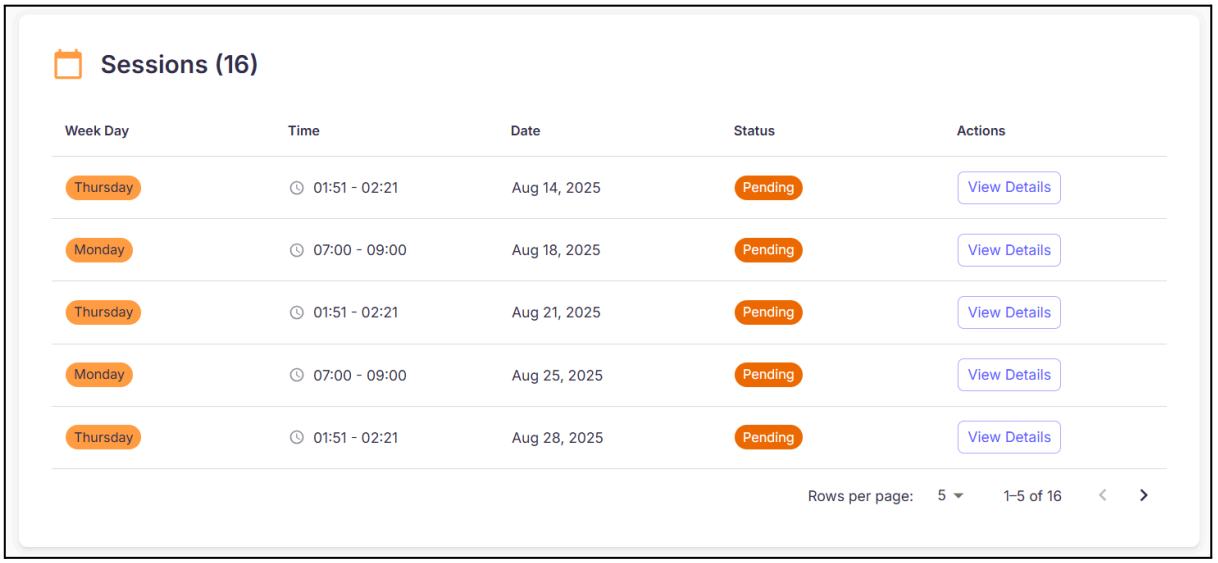


Figure 51 - View session (2)

No	Field Name	Field Type	Description
1	**Week Day**	Badge/Text	Day of the week the session occurs (e.g., Monday, Thursday).
2	**Time**	Text	Start–end time of the session (HH:MM – HH:MM).
3	**Date**	Date	Specific date of the session (localized).
4	**Status**	Badge	Session state (e.g., **Pending**, **Completed**, **Cancelled**).
5	**Actions**	Button	**View Details** → opens Session Details page. Edit and Delete Session.

Table 74: Manage Sessions (View session) layout 2

b. Use Case Specifications

ID and Name	UC40 – Manage Sessions		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	From Class Details, the user scrolls to the Sessions section.		
Description	As a user, I want to see all sessions for a class—showing weekday, time, date, status, and a link to details—so that I can review upcoming and past sessions and open a specific session for more information.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to view sessions for the class. **PRE-3.** The target class exists and sessions have been generated from its schedules (or an empty state is shown).		
Postconditions	**POST- 1.** No data is modified by viewing. **POST- 2.** When the user opens a row's **View Details**, the **Session Details** page is displayed.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	On **Class Details**, scroll to **Sessions**.
	2	User	Review table columns: **Week Day**, **Time**, **Date**, **Status**, **Actions**.
	3	User	Click **View Details** in a row to open the **Session Details** page for that session.
	4	User	If many records exist, use **Rows per page** and **Previous/Next** to navigate.
	5	System	The table reflects updates automatically when sessions are added/updated; opening **View Details** navigates to the

ID and Name	UC40 – Manage Sessions		
			detailed page with full information and related actions.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No sessions exist	Show empty state with guidance to create schedules which will generate sessions.
Exceptions	• **E1:** Backend/API error → show inline error with **Retry**; keep pagination state. • **E2:** Concurrency conflict (session removed/changed) → show notice and refresh list. • **E3:** Network timeout → show toast; allow retry.		
Frequency of Use	Medium–High		
Business Rules	BR-01, BR-02		
Other Information	• Dates/times render in user's timezone/locale. • Accessibility: table headers are announced to screen readers; **View Details** buttons are keyboard accessible.		
Assumptions	• Sessions are derived from schedules and stored with attributes: weekday, start/end time, date, and status. • The "Session Details" page is handled in a separate use case.		

Table 75: Manage Sessions (View session)

3.2.1.26 Edit Session

a. Screen Layouts

Updating

Figure 52 - Edit session

No	Field Name	Field Type	Description
1	**Title ("Edit Session")**	Label	Modal or page header.
2	**Week Day**	Dropdown	**Required.** Select Monday…Sunday.
3	**Date**	DatePicker	**Required.** Specific session date (localized). Must be valid for the class/semester window.
4	**Start Time**	TimePicker	**Required.** Session start (HH:MM).
5	**End Time**	TimePicker	**Required.** Session end (HH:MM). Must be after Start Time.

No	Field Name	Field Type	Description
6	**Room**	Dropdown	**Required.** Choose from available rooms (name • building • capacity).
7	**Status**	Dropdown	**Required.** Pending / Completed / Cancelled (and any other configured states).
8	**Validation/Conflict Area**	Inline message	Shows conflicts (e.g., overlapping room/time) or missing fields.
9	**Cancel**	Button	Close without saving.
10	**Save Changes**	Button (primary)	Validate and persist edits; disabled until form is valid or fields changed.

Table 76: Manage Sessions (edit session) layout

b. Use Case Specifications

ID and Name	UC53 - Manage Sessions (Edit Session)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	On the Sessions table, the user clicks Edit for a specific session.		
Description	As an authorized user, I want to update a session's weekday, time, date, room, and status so that the schedule reflects real-world changes and remains conflict-free.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to manage sessions. **PRE-3.** Target session exists and its current data is loaded. **PRE-4.** Room catalog and class/semester date constraints are available.		
Postconditions	**POST- 1.** The session is updated with new values. **POST- 2.** The **Sessions** table refreshes to show the changes and a success notification appears. **POST- 3.** Update is logged for auditing.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	In **Sessions**, locate the target row and click **Edit**.
	2	System	Open **Edit Session** form (modal/page) pre-filled with current values.
	3	User	Update fields: **Week Day**, **Time** (Start/End), **Date**, **Room**, **Status**.
	4	User	Click **Save Changes** to confirm (or **Cancel** to discard).
	5	System	Validate inputs (date in range, end > start, room availability, conflict checks).

ID and Name	UC53 - Manage Sessions (Edit Session)		
	6	System	If successful, persist updates, refresh the **Sessions** table, and show green toast **"Session updated successfully."** If not, show a red error (e.g., **"Selected time conflicts with another session."**) and keep the form open.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No fields changed	**Save Changes** disabled or no-op; remain on form.
	A2	Date outside class/semester	Block save; show inline message to choose a valid date.
	A3	Room/time conflict	Block save with clear details of the conflicting session/room.
	A4	Room becomes unavailable during edit	Prompt to reselect a room and reload availability.
	A5	User chooses **Cancel**	Close the dialog; no changes saved.
Exceptions	• **E1:** Permission error → "You don't have permission to edit sessions." • **E2:** Concurrency conflict (session modified/deleted by another user) → show notice and refresh the table. • **E3:** Network/Server error → show error toast; allow retry without losing edits. • **E4:** Missing references (rooms not loaded) → disable save and instruct user to retry after data refresh.		
Frequency of Use	Medium		
Business Rules	BR-01, BR-02		
Other Information	• Times and date pickers respect user timezone/locale. • If status is **Completed** or **Cancelled**, policy may restrict time/room edits—surface any limits in the UI.		
Assumptions	• Backend exposes an update endpoint {sessionId, weekday, date, startTime, endTime, roomId, status} with conflict detection and returns the updated session row.		

Table 77: Manage Sessions (edit session) specification

3.2.1.27 Delete Session

a. Screen Layouts

Updating

Figure 53 - Delete Session

No	Field Name	Field Type	Description
1	**Sessions Table**	Table	Columns: **Week Day**, **Time**, **Date**, **Room**, **Status**, **Actions**.
2	**Actions (per row)**	More menu (···)	Contains **Edit**, **View Details**, **Delete**. Choosing **Delete** opens confirmation dialog.
3	**Empty State**	Label	"No sessions available for this class." (when list is empty).

Table 78: Manage Session (delete session) layout

b. Use Case Specifications

ID and Name	UC54 – Manage Sessions (Delete Sessions)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	In Class Details → Sessions, the user opens the More (···) menu for a row and chooses Delete.		
Description	As an authorized user, I want to permanently remove a scheduled session from a class so that cancelled or obsolete sessions no longer appear in the timetable, while respecting policy checks for past or attended sessions.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to delete sessions. **PRE-3.** Target session exists for the class. **PRE-4.** Policy and validation rules are available (e.g., timeframe limits per BR-05, attendance constraints).		
Postconditions	**POST- 1.** The session is deleted (or soft-deleted per policy) and removed from the **Sessions** table. **POST- 2.** A success toast is shown; action is logged for audit (BR-15). **POST- 3.** If deletion is blocked by policy, no data changes occur and the user is informed.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Locate the target session and click **More (···) → Delete**.
	2	System	Show **Delete Session** confirmation dialog with **Week Day, Time, Date, Room, Status**.
	3	System	(Optional validations) • If attendance exists/locked, show warning and require **I understand the consequences** checkbox. • If session date is in the past and policy forbids deletion, disable **Delete** and show note.
	4	User	Click **Delete** to proceed, or **Cancel** to abort.

ID and Name	UC54 – Manage Sessions (Delete Sessions)		
	5	System	If successful, close dialog, remove the session from the table, and show green toast **“Session deleted successfully.”** If unsuccessful, show red error banner (e.g., **“You don’t have permission to delete completed sessions.”**) and make no changes.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	User cancels	Dialog closes; no changes saved.
Exceptions	• **E1:** Permission error → “You don’t have permission to delete sessions.” • **E2:** Concurrency conflict (session already deleted/changed) → show info and refresh table. • **E3:** Server/Network error → show error toast; allow retry. • **E4:** Business constraint violation (locked attendance, policy block) → show explanatory message; prevent deletion.		
Frequency of Use	Medium		
Business Rules	BR-01, BR-02		
Other Information	• Consider soft-delete with recovery window in production. • Ensure back-button/page cache doesn’t show deleted rows. • If associated attendance is removed, downstream reports should recalculate.		
Assumptions	• Backend provides a delete endpoint with checks for attendance linkage, status, and date, returning actionable error messages when blocked.		

Table 79: Manage Sessions (Delete Sessions)

3.2.1.28 View Session Details

a. Screen Layouts

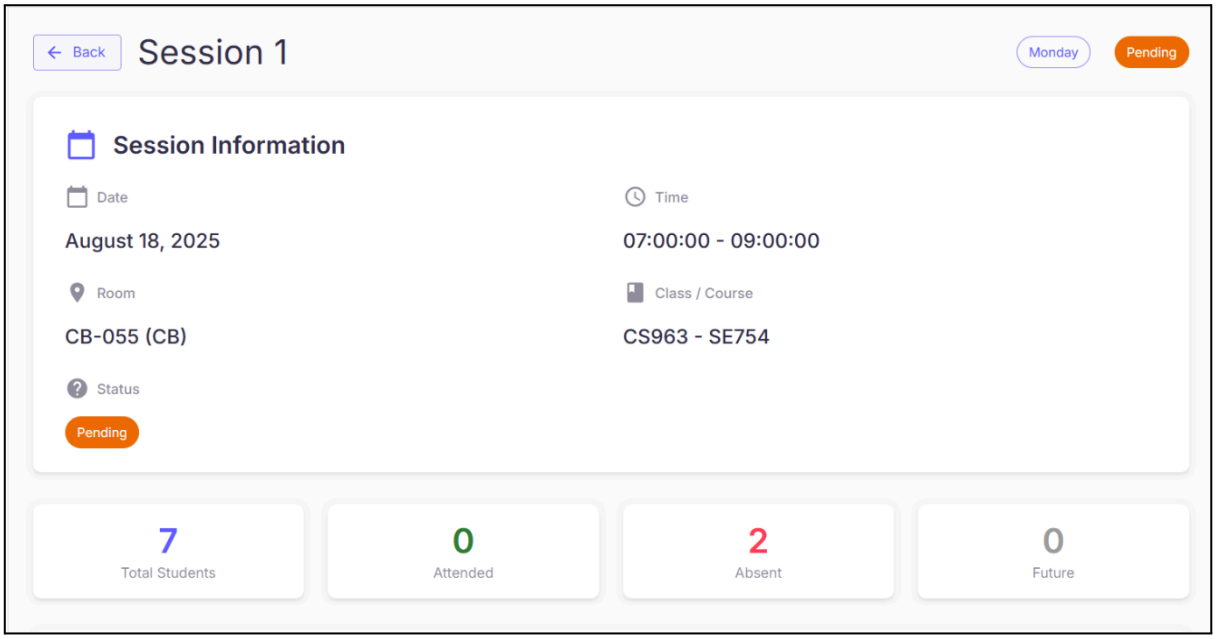


Figure 54 - View session details

No	Field Name	Field Type	Description
1	**Back**	Button	Returns to Sessions list.
2	**Page Title (“Session <n>”)**	Label	Header of the page.
3	**Weekday Tag**	Badge	e.g., **Monday**, **Thursday** (top-right).
4	**Status Tag**	Badge	**Pending / Ongoing / Completed / Cancelled** (top-right).
5	**Session Information Card**	Panel	Container for key fields below.
6	**Date**	Text	Scheduled date (localized).
7	**Time**	Text	Start–end time (HH:MM:SS – HH:MM:SS).
8	**Room**	Text	Room code and name (e.g., **CB-055 (CB)**).
9	**Class / Course**	Text	Associated identifiers (e.g., **CS963 – SE754**).
10	**Status (inside card)**	Badge	Current status; duplicates top-right for clarity.
11	**Attendance Summary**	Panel	Four stat tiles: **Total Students**, **Attended**, **Absent**, **Future**.

Table 80: Manage Session (View Session Detail)

b. Use Case Specifications

ID and Name	UC55 - Manage Session (View Session Detail)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	From Class Details → Sessions, the user clicks View Details on a session row.		
Description	As a user, I want to view complete information for a specific session—its date, time, room, related class/course, status, and attendance summary—so that I can verify the setup and quickly assess attendance.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User is authorized to view sessions for the class. **PRE-3.** Target session exists.		
Postconditions	**POST- 1.** No data is modified by viewing. **POST- 2.** User may navigate back to the Sessions list.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	In the Sessions list, click **View Details** for the target session.
	2	System	Open **Session Details** showing two sections: **Session Information** → **Date**, **Time**, **Room**, **Class/Course**, **Status**; **Attendance Summary** → **Total Students**, **Attended**, **Absent**, **Future**.
	3	User	Click **Back** to return to the Sessions list. Weekday and Status tags are shown at the top-right for quick identification.
	4	System	Load data in real time; reflect any updates to status or counts immediately.
Alternative Flows	**Alt Step**	**Condition**	**Description**
Exceptions	• **E1:** Session not found or deleted → redirect to Sessions list and show error **"Session not found or has been removed."** • **E2:** Permission error → show **"You don't have permission to view this session."** and return to previous page. • **E3:** Backend/Network error → show error state with **Retry**; do not navigate away automatically.		
Frequency of Use	Medium–High		
Business Rules	BR-01, BR-02		
Other Information	• Times/dates are localized to the user's timezone/locale. • Consider copy-to-clipboard for identifiers (class/course). • Accessibility: all tiles and tags are screen-reader friendly and keyboard navigable.		
Assumptions	None		

Table 81: Manage Session (View Session Detail) Specification

3.2.1.29 Edit Attendance

a. Screen Layouts

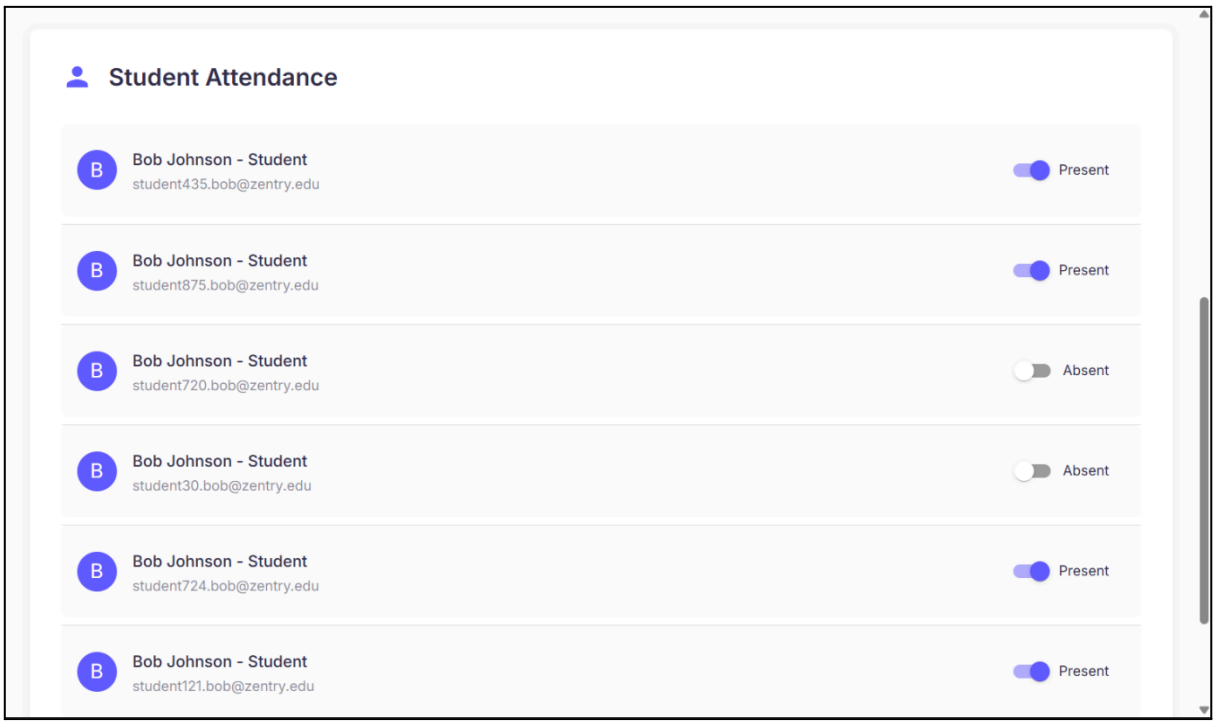


Figure 55 - Edit attendance

No	Field Name	Field Type	Description
1	**Section Title ("Student Attendance")**	Label	Header of the attendance list.
2	**Student Row**	List item	Shows **Avatar**, **Student Name – role**, secondary line **email/ID**.
3	**Status Chip**	Badge	Displays current state: **Present** / **Absent**. Auto-updates after save.
4	**Toggle**	Switch	Change a student's status (on → Present, off → Absent). Triggers confirmation dialog.
5	**Counters (top tiles)**	Stat tiles	**Attended**, **Absent**, **Future** – refresh after successful update.
6	**Empty State**	Label	"No students in this session." (if list is empty).

Table 82: Edit Attendance layout

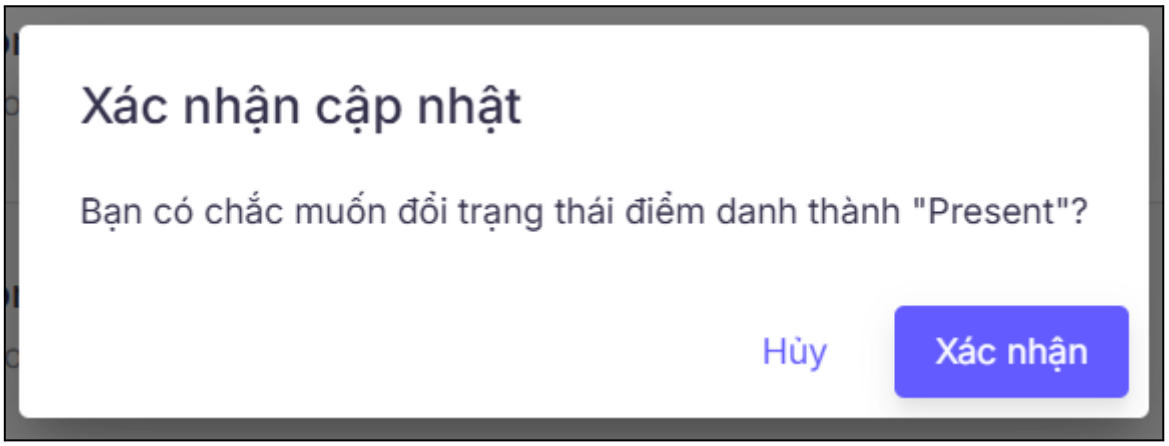


Figure 56 - Edit attendance (2)

No	Field Name	Field Type	Description
1	**Title (“Confirm update”)**	Label	Dialog header.
2	**Message**	Label	“Are you sure you want to change attendance to **Present/Absent**?”
3	**Cancel**	Button (secondary)	Close dialog; keep current status.
4	**Confirm**	Button (primary)	Apply the change. Disabled while processing.
5	**Error area (optional)**	Label	Shows validation/server errors; toggle reverts on failure.

Table 83: Manage Session (Edit Attendance) layout 2

b. Use Case Specifications

ID and Name	UC56 - Edit Attendance		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	On Session Details, the user toggles a student’s attendance in Student Attendance.		
Description	As an authorized user, I want to update a student’s attendance for a specific session by switching between Present and Absent, with a confirmation step, so that the attendance record is accurate and the session statistics stay up to date.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to modify attendance. **PRE-3.** Target session exists, and the student is enrolled in the class. **PRE-4.** The session is within the editable window (e.g., during session or within 24 hours after end, per policy). **PRE-5.** If FaceID is used, verification may be required by the lecturer.		
Postconditions	**POST- 1.** The student’s attendance status is updated and persisted.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC56 - Edit Attendance</td></tr>
<tr><td></td><td colspan="3">POST- 2. The Status chip and counters (Attended/Absent/Future) refresh immediately.
POST- 3. Action is logged for audit (BR-15).
POST- 4. (Optional) Notification may be sent if enabled (BR-12/BR-17).</td></tr>
<tr><td rowspan="7">Normal Flows</td><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>On Session Details, scroll to Student Attendance.</td></tr>
<tr><td>2</td><td>User</td><td>For the target student, toggle the switch: on = Present, off = Absent.</td></tr>
<tr><td>3</td><td>System</td><td>Show Confirm update dialog displaying the new status.</td></tr>
<tr><td>4</td><td>User</td><td>Click Confirm to proceed (or Cancel to keep the current status).</td></tr>
<tr><td>5</td><td>System</td><td>Validate permissions and time window; persist the new status.</td></tr>
<tr><td>6</td><td>System</td><td>If successful: update the row’s Status chip, refresh top counters, and show green toast “Attendance updated successfully.” If unsuccessful: show error, revert the toggle to the previous state, and keep counts unchanged.</td></tr>
<tr><td rowspan="2">Alternative Flows</td><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>User cancels in dialog</td><td>No changes saved; toggle remains in its original state.</td></tr>
<tr><td>Exceptions</td><td colspan="3">• E1: Network/Server error → show error toast; toggle reverts.
• E2: Concurrency conflict (record updated elsewhere) → inform user and refresh list.
• E3: Student not enrolled / record missing → show error and block change.</td></tr>
<tr><td>Frequency of Use</td><td colspan="3">High</td></tr>
<tr><td>Business Rules</td><td colspan="3">BR-01, BR-02</td></tr>
<tr><td>Other Information</td><td colspan="3">• All updates are timestamped with editor identity for audit.
• UI should debounce rapid toggles to avoid duplicate writes.
• Accessibility: toggle is keyboard operable; dialog has focus trapping and ARIA labels.</td></tr>
<tr><td>Assumptions</td><td colspan="3">• Backend provides an endpoint like {sessionId, studentId, status} and returns updated aggregates for counters.</td></tr>
</table>

Table 84: Manage Session (Edit Attendance) Specification

3.2.1.30 View Users

a. Screen Layouts

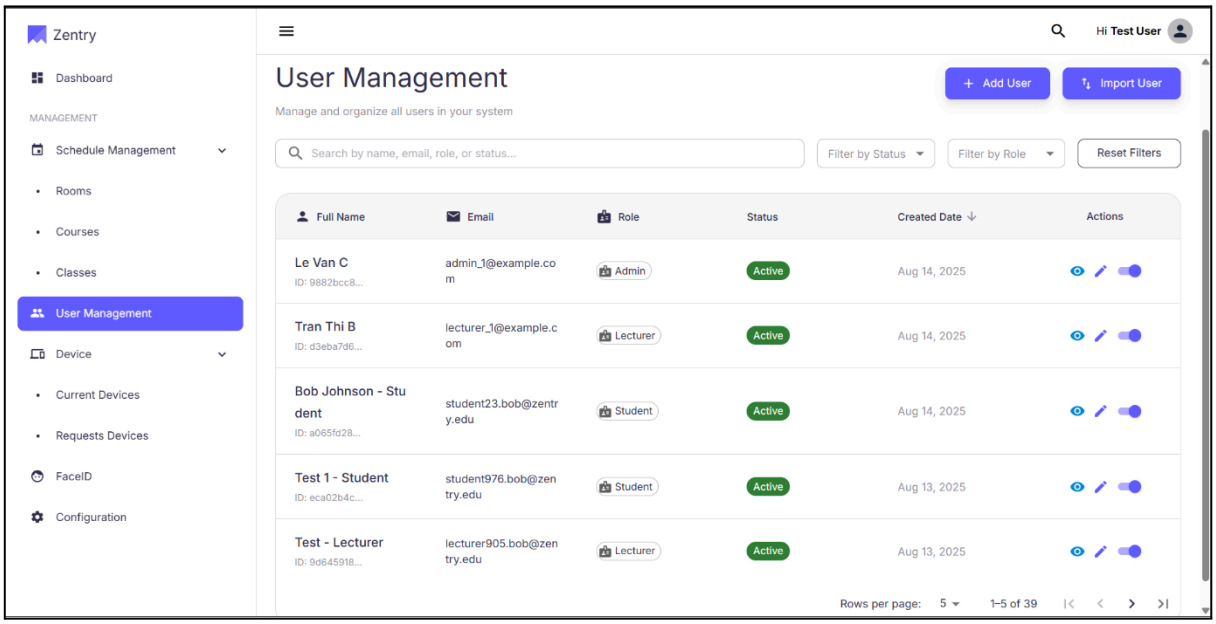


Figure 57 - View users

No	Field Name	Field Type	Description
1	**Sidebar Menu – User Management**	Menu Item	Navigation item in the left sidebar that redirects to the User Management page.
2	**Page Title**	Text	Displays "User Management" at the top of the page.
3	**Search Bar**	Input Field	Allows searching users by name, email, role, or status.
4	**Filter by Status**	Dropdown	Allows filtering users by status: Active / Inactive.
5	**Filter by Role**	Dropdown	Allows filtering users by role: Admin, Lecturer, Student.
6	**User Table**	Table	Displays the list of all users with details.
7	**Full Name & User ID**	Table Column	Shows the full name of the user along with their unique ID.
8	**Email**	Table Column	Displays the user's email address.
9	**Role**	Table Column	Displays the role of the user (Admin, Lecturer, Student).
10	**Status**	Table Column	Displays whether the user is Active or Inactive.

No	Field Name	Field Type	Description
11	**Created Date**	Table Column	Displays the date when the user account was created.
12	**Actions – View**	Button/Icon	Opens detailed information about the selected user.
13	**Actions – Edit**	Button/Icon	Allows editing the selected user's details.
14	**Actions – Change Status Toggle**	Toggle Button	Enables changing user status between Active and Inactive.

Table 85: Manage User (View User) layout

b. Use Case Specifications

ID and Name	UC57 – Manage Users (View Users)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user navigates to the User Management page from the dashboard.		
Description	As an authorized user, I want to view a list of all users and search for specific users so that I can manage their accounts.		
Preconditions	**PRE-1.** The user is authenticated and authorized **PRE-2.** The system is online and operational.		
Postconditions	**POST-1.** The user can successfully view a list of all user records. **POST-2.** The user can search and filter user records as needed.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	From the Dashboard, navigate to **User Management** in the left sidebar menu.
	2	System	The **User Management page** is displayed, containing a table of all users. Each row shows: Full Name & User ID, Email, Role, Status, Created Date, and Actions (View, Edit, Change Status).
	3	User	Use the **search bar** to find a user by name, email, role, or status.
	4	User	Apply **filters** to narrow results: by Status (Active/Inactive) or by Role (Admin, Lecturer, Student).
	5	**System**	**The table updates to reflect the applied search or filter criteria.**
Alternative Flows	**Alt Step**	**Condition**	**Description**

ID and Name	UC57 – Manage Users (View Users)		
	A1	No users found	If no matching users are found, the system displays a message: *"No users match the search/filter criteria."*
Exceptions	• **E1**: Database connection error → System displays error message and suggests retry later. **E2**: Unexpected system failure → Error logged for admin review, user notified. • **E3:** If the login session time goes out, the system prompts the user to re-enter their credentials.		
Frequency of Use	High		
Business Rules	BR-01, BR-02		
Other Information	• UI should support pagination for large user lists. Sorting should be available for columns (e.g., Created Date, Name).		
Assumptions	• The system already stores user information including name, email, role, status, and creation date. • Only authorized users (e.g., Admin, or specific Lecturer roles) can view the User Management page. • The search and filter functionalities are case-insensitive. • The table is paginated for performance when displaying large datasets.		

Table 86: Manage Users (View Users) Specification

3.2.1.31 Add User

a. Screen Layouts

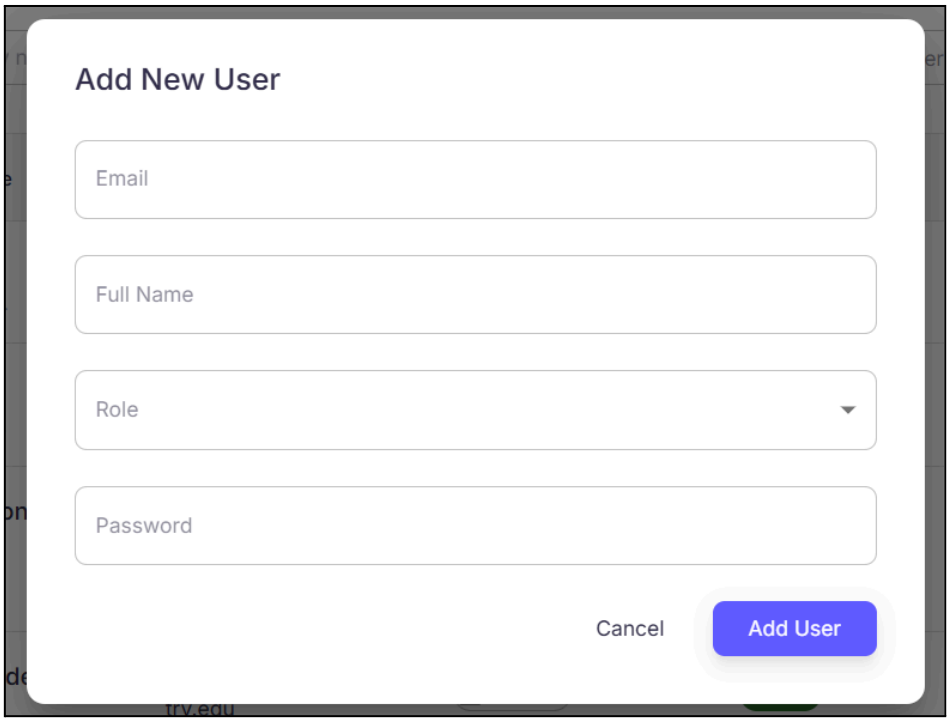


Figure 58 - Add user

No	Field Name	Field Type	Description
1	**Add User Button**	Button	Add a new user
2	**Email**	Text Field	Input field for user's email address. Must be unique and valid format.
3	**Full Name**	Text Field	Input field for the user's full name.
4	**Role**	Dropdown	Dropdown menu with options: Admin, Lecturer, Student.
5	**Password**	Password Box	Input field for temporary password. Must meet security policy.
6	**Cancel Button**	Button	Cancels the action and closes the dialog.

Table 87: Manage Users (Add Users)

b. Use Case Specifications

ID and Name	UC58 – Manage Users (Add Users)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user opens the website and attempts to log in using credentials or a linked OAuth account.		
Description	As a user, I want to log in securely so that I can access system features according to my role		
Preconditions	**PRE-1.** The user has a registered account. **PRE-2.** The system is online and operational.		
Postconditions	**POST- 1.** The user is redirected to the corresponding dashboard based on their role.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Admin	On the User Management page, click the **Add User** button in the top-right corner.
	2	Admin	In the **Add New User** dialog, fill in the fields: • **Email**: Enter the user's email address • **Full Name**: Enter the user's full name • **Role**: Open the dropdown and select Admin, Lecturer, or Student • **Password**: Enter a temporary password that meets the security policy.
	3	Admin	Click **Add User** to create the account (or **Cancel** to close without changes).

ID and Name	UC58 – Manage Users (Add Users)		
	4	System	Validates input data and processes the request. • If successful: A green notification appears — “User added successfully”. The dialog closes, and the new user appears in the list (status: Active by default). • If unsuccessful: An error message appears. Invalid fields are highlighted, and the dialog remains open for correction.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Email already exists in the system	System shows error: “Email already exists”. Admin must enter a different email.
	A2	Email format is invalid	System highlights email field and shows error: “Invalid email format”.
	A3	Password is weak or does not meet security policy	System highlights password field and shows error: “Password does not meet requirements”.
	A4	Admin clicks **Cancel** instead of **Add User**	Dialog closes without creating a new user.
Exceptions	• **E1**: Database connection error → System displays error message and suggests retry later. **E2**: Unexpected system failure → Error logged for admin review, user notified. • **E3:** If the login session time goes out, the system prompts the user to re-enter their credentials.		
Frequency of Use	High		
Business Rules	BR-01, BR-02		
Other Information	• Audit logs should capture the creation of new users (who created, when, and details).		
Assumptions	• Admin has the correct permissions to create users. • The system is online and connected to the database. • Role list (Admin, Lecturer, Student) is predefined and up-to-date. • The notification system is functional to display success/error messages.		

Table 88: Manage Users (Add Users) Specification

3.2.1.32 Import User

a. Screen Layouts

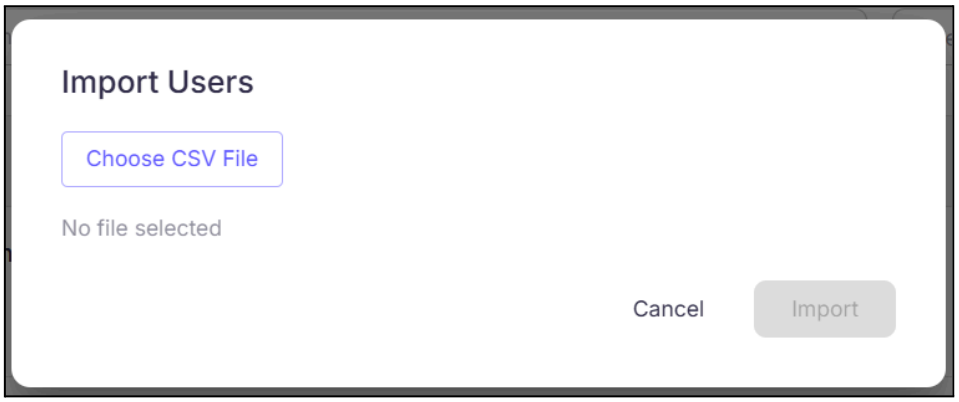


Figure 59 - Import Users layout

No	Field Name	Field Type	Description
1	**Import User Button**	Button	Located on the User Management page (top-right). Opens Import Users dialog.
2	**Import Users Dialog**	Modal/Dialog	Popup window displayed when clicking Import User button.
3	**Choose CSV File Button**	Button	Triggers file picker window to select CSV file from computer.
4	**Import Button**	Button	Confirms upload of selected CSV file.
5	**Cancel Button**	Button	Closes the Import Users dialog without making changes.

Table 89: Import User Field Description Table

b. Use Case Specifications

ID and Name	UC59 – Manage User (Import User)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user clicks on “Import Users” in the “User Management”		
Description	As a user, I want to import multiple users from a CSV file, so that I can quickly add many users at once without manual entry.		
Preconditions	**PRE-1.** Admin must be logged into the system with appropriate permissions. **PRE-2.** The CSV file must exist and follow the system’s required format (e.g., correct headers, valid data types).		
Postconditions	**POST- 1.** New users are added to the system with status = Active by default.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC59 – Manage User (Import User)</td></tr>
<tr><th rowspan="8">Normal Flows</th><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>Admin</td><td>On the User Management page, click the Import User button in the top-right corner.</td></tr>
<tr><td>2</td><td>System</td><td>Display the Import Users dialog.</td></tr>
<tr><td>3</td><td>Admin</td><td>Click Choose CSV File to open the file picker.</td></tr>
<tr><td>4</td><td>Admin</td><td>In the file picker, navigate to and select the desired CSV file from the computer. The selected file name appears in the dialog.</td></tr>
<tr><td>5</td><td>Admin</td><td>Click Import to upload the file (or Cancel to close without changes).</td></tr>
<tr><td>6</td><td>System</td><td>Process the file and validate its contents.</td></tr>
<tr><td>7</td><td>System</td><td>If successful: - Display green notification: “Users imported successfully”. - Close dialog. - Newly added users appear in the list with status: Active.</td></tr>
<tr><th rowspan="3">Alternative Flows</th><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>File is invalid (wrong format, missing required columns)</td><td>System rejects the file and displays an error message prompting re-upload.</td></tr>
<tr><td>A3</td><td>Admin clicks Cancel</td><td>System closes the dialog without making changes.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: If the uploaded CSV exceeds the maximum size, system displays an error and prevents upload.
E2: Unexpected system failure → Error logged for admin review, user notified.
• E3: If the login session time goes out, the system prompts the user to re-enter their credentials.</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Medium</td></tr>
<tr><th>Business Rules</th><td colspan="3">BR-01, BR-02</td></tr>
<tr><th>Other Information</th><td colspan="3">• All login activities are recorded in system logs for security auditing and system health monitoring.</td></tr>
</table>

ID and Name	UC59 – Manage User (Import User)
	• The system supports both internal (username/password) and external (OAuth) authentication methods.
Assumptions	• CSV file is UTF-8 encoded and follows the predefined template format. • Each user record must contain at least required fields (e.g., Email, Name, Role). • Admin has sufficient permissions to perform bulk import.

Table 90: Import User Specification

3.2.1.33 Edit User

a. Screen Layouts

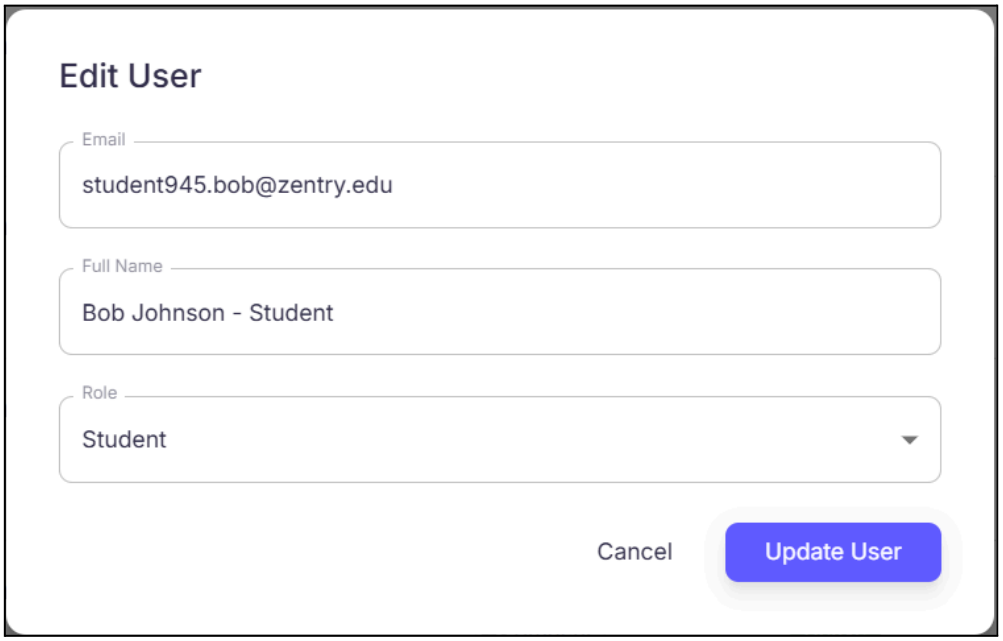


Figure 60: Edit User Layout

No	Field Name	Field Type	Description
1	**Edit User Icon**	Button/Icon	Pencil icon in the Actions column of the user table, used to open the Edit User dialog.
2	**Edit User Dialog**	Modal/Dialog	Dialog window that appears with the user's current information pre-filled for editing.
3	**Email**	Textbox/Input	Field for entering or modifying the user's email. Validates email format and uniqueness.
4	**Full Name**	Textbox/Input	Field for modifying the user's full name.
5	**Role**	Dropdown/List	Dropdown to select the user's role (Admin, Lecturer, Student).

No	Field Name	Field Type	Description
6	**Update User Button**	Button	Saves the changes and updates the user information.
7	**Cancel Button**	Button	Discards changes and closes the Edit User dialog.
8	**Success Notification**	Toast/Alert	Green notification displayed after a successful update, closes dialog and refreshes list.
9	**Error Message**	Text/Alert	Displays error message when update fails (e.g., invalid email, duplicate email). Highlights problematic fields for correction.

Table 91: Edit User Field Description Table

b. Use Case Specifications

ID and Name	UC- 60: Manage Users (Edit User)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user clicks the Edit User icon for a specific user in the User Management page.		
Description	As a user, I want to edit an existing user's information (email, full name, and role) so that the system reflects the most up-to-date and correct user data.		
Preconditions	**PRE-1.** The user has a registered account. **PRE-2.** The system is online and operational.		
Postconditions	**POST- 1.** The selected user's details are updated and displayed in the user list (if update is successful). If update fails, the user remains unchanged, and the admin is informed of the error		
Normal Flows	**Step**	**By**	**Step Details**
	1	Admin	On the **User Management** page, locate the target user and click the **Edit User icon (pencil)**.
	2	System	Displays the **Edit User dialog** with current details pre-filled (Email, Full Name, Role).
	3	Admin	Updates one or more fields: - Email - Full Name - Role (Admin, Lecturer, Student).
	4	Admin	Clicks **Update User** to confirm changes (or **Cancel** to discard).
	5	System	Validates input data (format, uniqueness of email, required fields).
	6	System	If validation passes, updates the user record in the database.

ID and Name	UC- 60: Manage Users (Edit User)		
	7	System	Displays a **green success notification**: "User updated successfully", closes the dialog, and refreshes the user list.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Admin clicks **Cancel** instead of Update	The system closes the dialog without saving changes.
Exceptions	• **E1**: Database connection error → System displays error message and suggests retry later. **E2**: Unexpected system failure → Error logged for admin review, user notified. • **E3:** If the login session time goes out, the system prompts the user to re-enter their credentials.		
Frequency of Use	High		
Business Rules	BR-01, BR-02		
Other Information	• Audit log should record the details of user updates (who edited, when, what fields changed). • UI design reference: Figure xx - Edit User Dialog (with updated fields). • Accessibility: Error messages must be screen-reader compatible.		
Assumptions	• Email is the unique identifier for users.		

Table 92: Manage Users (Edit User) Specification

3.2.1.34 Change User Status

a. Screen Layouts

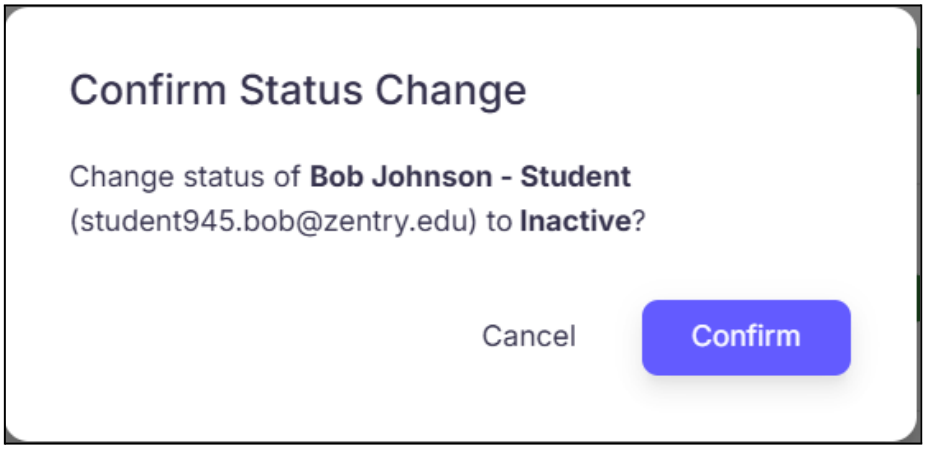


Figure 61 - Confirm status change layout

No	Field Name	Field Type	Description
1	**User List Table**	Table	Displays users with their details including current status.

No	Field Name	Field Type	Description
2	**Toggle Status Button**	Button/Icon	Switch icon located in the user's row for toggling status.
3	**Confirmation Dialog**	Modal	Dialog box confirming the status change action.
4	**Current Status Label**	Text/Label	Shows the user's current status (Active/Inactive).
5	**New Status Preview**	Text/Label	Shows what the status will change to (Active ↔ Inactive).
6	**Confirm Button**	Button	Executes the status change action.
7	**Cancel Button**	Button	Cancels the action and closes the dialog.
8	**Notification**	Toast/Alert	Displays success or error message after attempting status change.

Table 93: Confirm status change Field Description Table

b. Use Case Specifications

ID and Name	UC- 61: Manage User (Change User Status)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user clicks the Toggle Status button on a specific user in the User Management page		
Description	As a user, I want to change a user's status between Active and Inactive so that I can control whether a user can access the system.		
Preconditions	**PRE-1.** The user has a registered account. **PRE-2.** The system is online and operational.		
Postconditions	**POST- 1.** The user's status is updated in the system and displayed in the User Management table.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Admin	Locate the user in the User Management table and click the Toggle Status button.
	2	System	Display confirmation dialog showing current status and new status.
	3	Admin	Click Confirm in the dialog. System updates the user status, updates the status label in the table, and shows a green notification: "Status updated successfully".
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Cancel	Close the dialog without changing the user status.

ID and Name	UC- 61: Manage User (Change User Status)
Exceptions	• **E1**: Database connection error → System displays error message and suggests retry later. **E2**: Unexpected system failure → Error logged for admin review, user notified. • **E3:** If the login session time goes out, the system prompts the user to re-enter their credentials.
Frequency of Use	High
Business Rules	BR-01, BR-02
Other Information	• The toggle status action should be reflected immediately in the table without requiring a page refresh. • The system must log each status change for audit purposes, including the admin who performed the change and the timestamp. • Notifications should disappear automatically after a few seconds.
Assumptions	• The system's network connection is stable; otherwise, an error message will be shown. • Only admins have permission to change user status. • Users in the system already have either "Active" or "Inactive" status. • Status change cannot be performed on a user who is currently in the process of being deleted.

Table 94 - Manage User (Confirm status change) Specification

3.2.1.35 View User Details

a. Screen Layouts

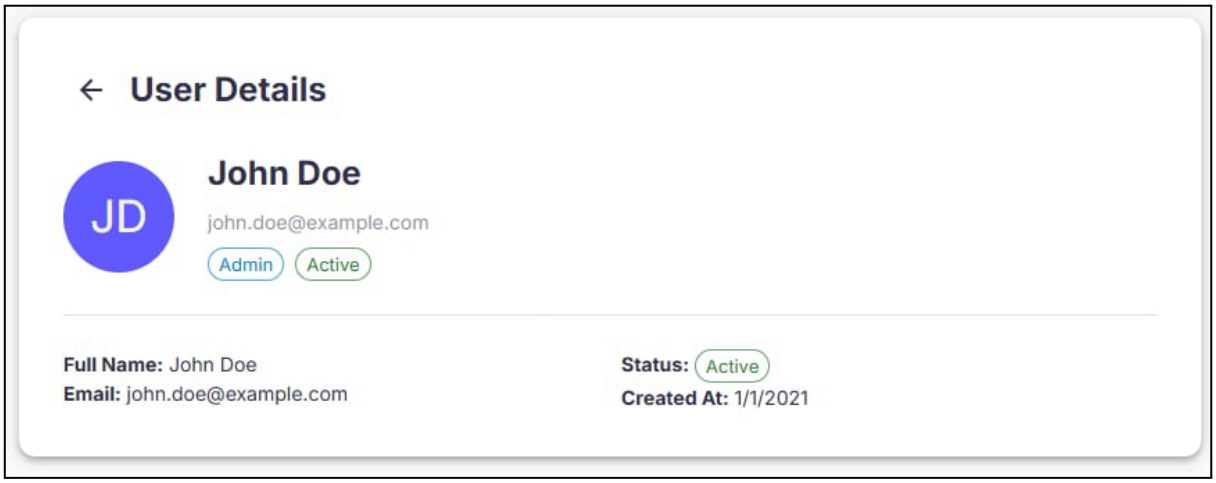


Figure 62 - User detail layout

No	Field Name	Field Type	Description
1	**View User Icon**	Button/Icon	Eye icon in the Actions column of User Management table. Opens User Details page.

No	Field Name	Field Type	Description
2	**Back Arrow**	Button/Icon	Located at the top-left of the User Details page. Navigates back to User Management list.
3	**Full Name**	Text Label	Displays the full name of the selected user.
4	**Email**	Text Label	Displays the email address of the user.
5	**Role**	Badge	Displays the user's role (Admin, Lecturer, or Student).
6	**Status**	Badge	Displays the current account status (Active/Inactive).
7	**Created At**	Text Label	Displays the date when the user account was created.
8	**Error Message**	Text	Displays "User not found." if the selected user does not exist.

Table 95: User detail Field Table

b. Use Case Specifications

ID and Name	**UC62 – View User Detail**		
Created By	DungND	**Date Created**	28/06/2025
Primary Actor	Admin	**Secondary Actors**	System
Trigger	The user clicks the View User icon (eye symbol) on a specific user in the User Management page.		
Description	As a user, I want to view detailed information about a user so that I can understand their account details, role, and status.		
Preconditions	**PRE-1.** The user has a registered account. **PRE-2.** The system is online and operational.		
Postconditions	**POST- 1.** User details are displayed on the User Details page.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Admin	Locate the user in the User Management table and click the View User icon.
	2	System	Load and display the User Details page with full name, email, role, status, and creation date.
	3	Admin	Review the information.
	4	Admin	Click the Back button to return to the User Management list.
Alternative Flows	**Alt Step**	**Condition**	**Description**

ID and Name	UC62 – View User Detail		
	A1	User does not exist (e.g., deleted)	System displays an error message: “User not found” and remains on the User Management page.
Exceptions	• **E1**: Database connection error → System displays error message and suggests retry later. **E2**: Unexpected system failure → Error logged for admin review, user notified. • **E3:** If the login session time goes out, the system prompts the user to re-enter their credentials.		
Frequency of Use	High		
Business Rules	BR-01, BR-02, BR-16		
Other Information	• The User Details page should always display the most up-to-date information from the database. • Role is displayed as a badge with color coding (optional for UI clarity). • Back button should preserve the previous table state (e.g., pagination, filters).		
Assumptions	• Only admins have permission to view detailed user information. • All users have valid data for full name, email, role, status, and created date. • Network connection is stable for loading details. • Deleted or non-existent users trigger the error flow.		

Table 96 - User detail specification

3.2.1.36 View Active Devices

a. Screen Layouts

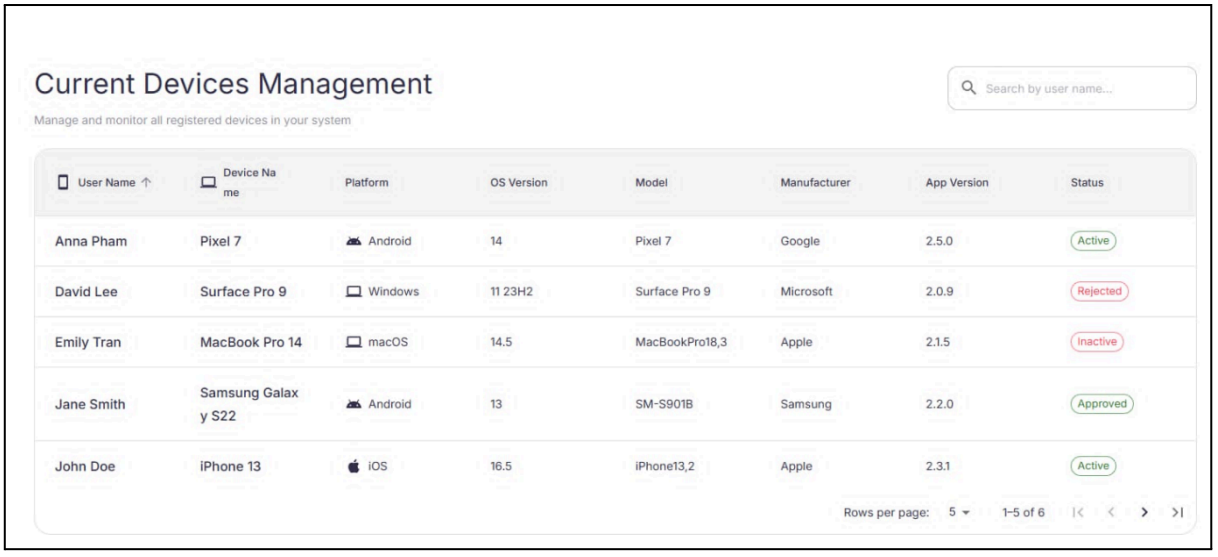


Figure 63: View Active Devices Layout

No	Field Name	Field Type	Description
1	**Page Title (“Current Devices Management”)**	Label	Header and short helper text (“Manage and monitor all registered devices”).
2	**Search by user name…**	Searchbox	Filter rows by user name (debounced).
3	**Status Filter**	Dropdown/Chips	Optional filter with values **Active, Approved, Inactive, Rejected**; defaults to **All** (or **Active** if configured).
4	**Devices Table**	Table	Lists device records (see columns below).
5	**Pagination**	Control	Rows per page selector; Previous/Next arrows.
6	**Empty State**	Label	“No active devices found.” shown when the filtered result set is empty.

Table 97: View Active Devices Field table

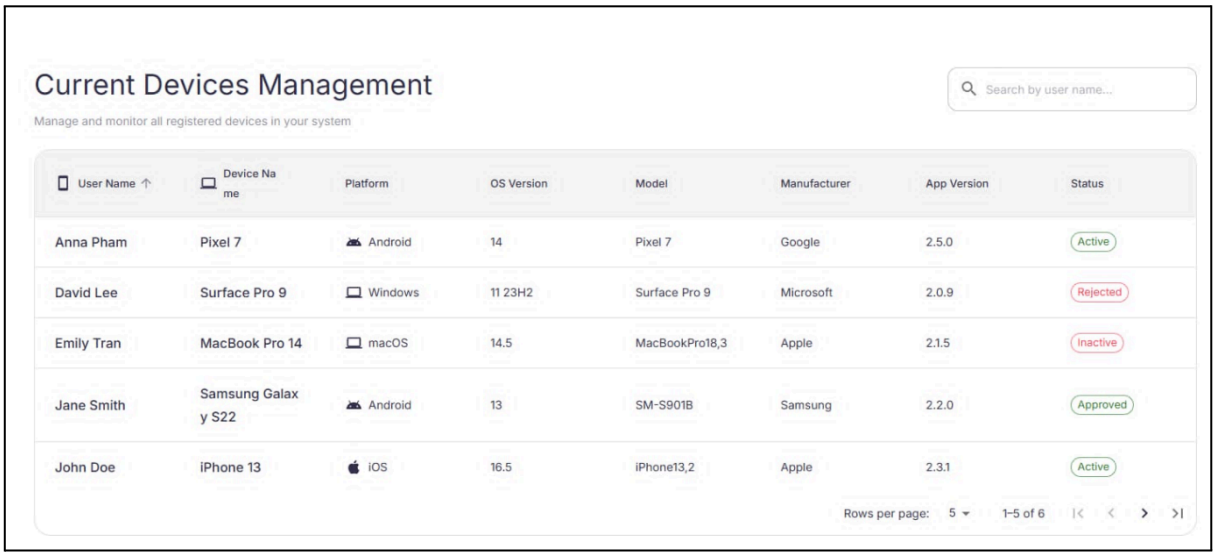


Figure 64: View Active Devices Table

No	Column	Field Type	Description
1	**User Name**	Text	Owner’s display name.
2	**Device Name**	Text	Device label/model name set by user or system.
3	**Platform**	Icon + Text	**Android / iOS / Windows / macOS**, etc.
4	**OS Version**	Text	Major/minor OS version (e.g., 16.5).
5	**Model**	Text	Manufacturer model identifier.
6	**Manufacturer**	Text	e.g., Apple, Samsung, Microsoft, Google.

No	Column	Field Type	Description
7	**App Version**	Text	Installed app version on the device.
8	**Status**	Badge	**Active** (green), plus other statuses for context. Only **Active** rows are in-scope for this UC when filter is applied.

Table 98: Current Devices Management Field table

b. Use Case Specifications

ID and Name	UC-63 Manage Active Devices (View Active Devices)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	The user navigates to Device → Current Devices from the sidebar.		
Description	As an admin, I want to view and monitor all devices currently marked as Active so that I can confirm which registered/approved devices are allowed to use the system.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to view device inventory. **PRE-3.** Devices are registered in the system (BR-03) and their approval states are recorded (BR-10).		
Postconditions	**POST- 1.** No data is modified by viewing; the admin may refine the list via search/filters. **POST- 2.** (Optional) The view remembers the last filter/pagination settings for the session.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	In the sidebar, under **Device**, click **Current Devices**.
	2	User	In the **Current Devices Management** table, locate the **Status** column.
	3	User	Use the **Search** bar or scroll the list; identify rows with the **Active** (green) status badge.
	4	User	*(Optional)* Use **Status Filter** and **Pagination** to refine the displayed devices.
	5	System	Display all devices matching **Active** (when filtered), including **User Name, Device Name, Platform, OS Version, Model,**

ID and Name	UC-63 Manage Active Devices (View Active Devices)		
			Manufacturer, App Version, Status. If none, show **"No active devices found."**
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Default view shows all statuses	Admin switches **Status Filter** to **Active** to focus only on active devices.
Exceptions	• **E1:** Backend/API error → inline error state with **Retry**; keep search/filter context. • **E2:** Network timeout → toast banner; allow retry without resetting pagination. • **E3:** Data inconsistency (e.g., missing platform) → display placeholder ("—") and log for review.		
Frequency of Use	Medium–High		
Business Rules	BR-01, BR-02		
Other Information	• Common table affordances: column sort (User Name, Platform, App Version), export CSV (admin only), and row click to open device details (future UC). • Consider periodic auto-refresh so statuses reflect recent approvals/revocations. • Accessibility: search and table controls are keyboard-friendly with ARIA labels.		
Assumptions	None		

Table 99: Current Devices Specification

3.2.1.37 Inactive Device

a. Screen Layouts

Figure 65: Inactive Devices

No	Field Name	Field Type	Description
1	**Page Title ("Current Devices Management")**	Label	Header and helper text ("Manage and monitor all registered devices").
2	**Search by user name…**	Searchbox	Filter rows by **user name** (debounced).
3	**Status Filter**	Dropdown/Chips	Choose **Inactive** (also supports Active/Approved/Rejected/All). For this UC, filter = **Inactive**.
4	**Devices Table**	Table	Lists device records (see columns below).

No	Field Name	Field Type	Description
5	**Pagination**	Control	Rows-per-page selector, Previous/Next arrows.
6	**Empty State**	Label	“No inactive devices found.” when result set is empty.

Table 100: Inactive Device Field Table

b. Use Case Specifications

ID and Name	UC-64 Manage Active Devices (Inactive Devices)		
Created By	DungND	**Date Created**	28/06/2025
Primary Actor	Admin	**Secondary Actors**	System
Trigger	User navigates to **Device** → **Current Devices** from the sidebar.		
Description	As an admin, I want to view and manage all devices currently marked as *Inactive* so that I can either re-activate eligible devices or remove those that should no longer be in the system.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to view/manage device inventory. **PRE-3.** Devices are registered and have status tracking.		
Postconditions	**POST- 1.** Viewing alone does not modify data. **POST- 2.** If the admin performs an action (reactivate/remove), the table reflects the change and the action is logged.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	In the sidebar under **Device**, click **Current Devices**.
	2	User	In the **Current Devices Management** table, focus on the **Status** column.
	3	User	Use the **Search** bar or **Status Filter** to locate devices; identify rows with the **Inactive** status badge (gray/red outline).
	4	User	*(Optional)* Click a specific inactive device to open **Device Details** (shows **Last Active**, **Registered User**, **Platform**, **OS Version**, etc.).

ID and Name	UC-64 Manage Active Devices (Inactive Devices)		
	5	System	The table lists all devices with status **Inactive**. If none, display **"No inactive devices found."**
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Reactivate device	Admin clicks **Reactivate** → confirmation dialog → on success, device status becomes **Active/Approved** and row moves out of the Inactive view; show success toast **"Device re-activated successfully."**
Exceptions	• **E1:** Permission error → "You don't have permission to manage devices." • **E2:** Network/Server error → inline error with **Retry**; preserve filters/pagination. • **E3:** Concurrency conflict (device already reactivated/removed) → show info and refresh list. • **E4:** Data inconsistency (missing attributes) → show placeholders ("—") and log for review.		
Frequency of Use	Medium		
Business Rules	BR-01, BR-02		
Other Information	• Auto-refresh interval can be enabled to reflect recent changes. • Accessibility: search/filter controls and table rows are keyboard/reader-friendly.		
Assumptions	None		

Table 101 - Inactive Device Specification

3.2.1.38 View Device Requests

a. Screen Layouts



Figure 66: Requests Devices Layout

No	Field Name	Field Type	Description
1	**Page Title (“Requests Devices”)**	Label	Header with helper text: “Review and manage device registration requests.”
2	**Search…**	Searchbox	Filter by **user name** or **device name** (debounced).
3	**Status Filter**	Dropdown/Chips	Defaults to **Pending**; may include **Approved**, **Rejected** (history view).
4	**Requests Table**	Table	List of pending requests (see columns below).
5	**Pagination**	Control	Rows-per-page selector; Previous/Next arrows.
6	**Empty State**	Label	“No pending device requests.” when list is empty.

Table 102 - Requests Devices Layout



Figure 67: Requests Devices Table

No	Column	Field Type	Description
1	**User Name**	Text	Requester's display name.
2	**Device Name**	Text	Submitted device label/model.
3	**Platform**	Icon + Text	Android / iOS / Windows / macOS, etc.
4	**OS Version**	Text	e.g., 16.5, 14, 11 23H2.
5	**Model**	Text	Manufacturer model identifier.
6	**Manufacturer**	Text	Apple, Samsung, Microsoft, Google, …
7	**App Version**	Text	Installed app version reported by client.
8	**Current Status**	Badge	**Active / Inactive** if the device already exists; otherwise **Pending**.
9	**Actions**	Icon buttons	✓ **Approve** and ✕ **Reject**. Optional "View details"/"History" in a row menu.

Table 103 - Requests Devices Table

b. Use Case Specifications

ID and Name	UC65 - Manage Device Requests (View Device Requests)		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	From the sidebar, the admin opens **Device** → **Requests Devices**.		
Description	As an admin, I want to review pending device registration requests and approve or reject them so that only eligible devices become active in the system.		
Preconditions	**PRE-1.** Admin is authenticated. **PRE-2.** Admin has permission to manage device requests. **PRE-3.** Requests are submitted by users and stored with device metadata.		
Postconditions	**POST- 1.** Approved requests move to **Current Devices** with appropriate status; rejected requests disappear from the pending list. **POST- 2.** Actions are logged for audit (BR-15) and (optionally) notifications are sent to the requester.		
Normal Flows	**Step**	**By**	**Step Details**

ID and Name	UC65 - Manage Device Requests (View Device Requests)		
	1	User	In the sidebar, go to **Device** → **Requests Devices**.
	2	System	Display **Pending Device Requests** with columns: **User Name, Device Name, Platform, OS Version, Model, Manufacturer, App Version, Current Status**.
	3	User	To decide on a request, click ✓ **(Approve)** or ✕ **(Reject)** in the **Actions** column.
	4	System	**If approved:** move device to **Current Devices**, set status (e.g., **Active/Approved**), and remove from Requests list. **If rejected:** remove from pending list and notify the user of rejection.
	5	System	Show a green notification confirming the action (e.g., **"Device approved successfully."** or **"Device request rejected."**).
Alternative Flows	**Alt Step**	**Condition**	**Description**
Exceptions	• **E1:** Permission error → "You don't have permission to approve/reject device requests." • **E2:** Network/Server error → inline error; keep table state and allow retry. • **E3:** Concurrency conflict (request already processed) → show info and refresh list. • **E4:** Invalid metadata → show warning; allow admin to reject with reason.		
Frequency of Use	Medium–High		
Business Rules	BR-01, BR-02		
Other Information	• Provide optional **Reject reason** field (template list + free text) to improve user guidance. • Consider SLA indicators (e.g., "Waiting 3 days") and sorting by **Submitted At**. • Accessibility: action icons have ARIA labels; table fully keyboard-navigable.		

ID and Name	UC65 - Manage Device Requests (View Device Requests)
Assumptions	None

Table 104 - View Device Request Specification

3.2.1.39 Edit Device Request

a. Screen Layouts

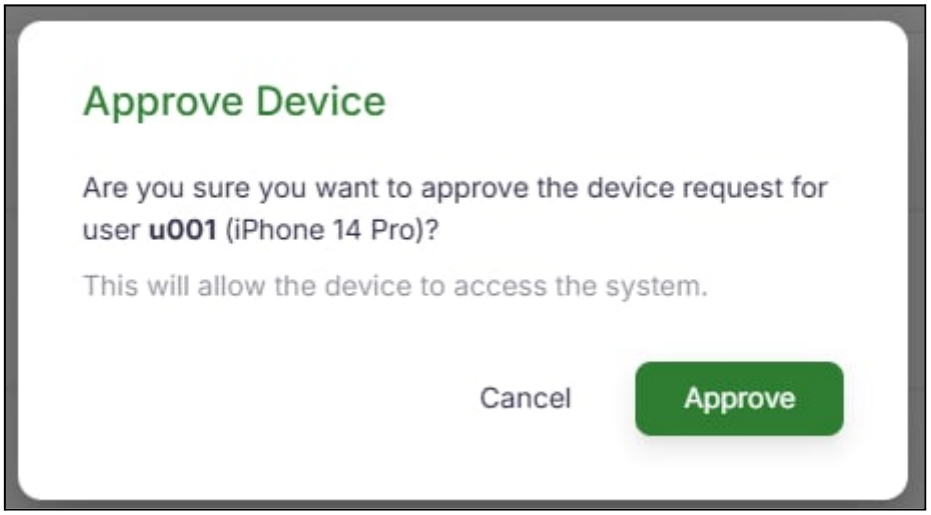


Figure 68: Approve Device Layout

No	Field Name	Field Type	Description
1	**Title ("Approve Device")**	Label	Dialog header.
2	**Request Summary**	Read-only block	Shows **User ID / User Name**, **Device Name**, **Platform**, **OS Version**, **Model**, **Manufacturer**, **App Version**.
3	**Note (optional)**	Textarea	Internal note stored with the decision.
4	**Cancel**	Button (secondary)	Close without changes.
5	**Approve**	Button (primary)	Confirms approval and moves device to **Current Devices**.
6	**Error Area**	Inline message	Displays server/policy errors if approval fails.

Table 105 - Approve Device Layout

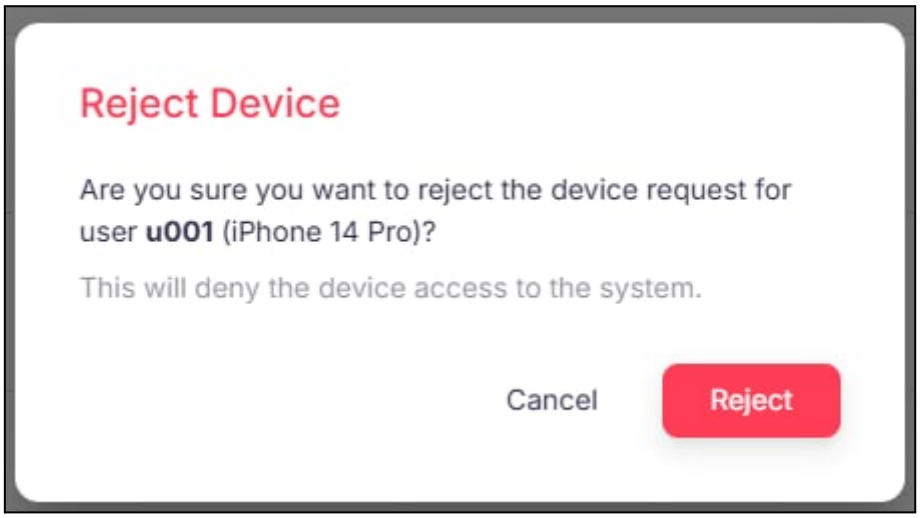


Figure 69: Reject Device Layout

No	Field Name	Field Type	Description
1	**Title ("Reject Device")**	Label	Dialog header.
2	**Request Summary**	Read-only block	Same fields as above to verify the target request.
3	**Reject Reason**	Textarea / Select+Textarea	**Required.** Reason to notify the requester (e.g., "OS version below minimum").
4	**Cancel**	Button (secondary)	Close without changes.
5	**Reject**	Button (danger)	Confirms rejection and removes request from the pending list.
6	**Error Area**	Inline message	Displays server/policy errors if rejection fails.

Table 106 - Reject Device Layout

b. Use Case Specifications

ID and Name	UC66 - Manage Device Requests (Edit Device Requests)		
Created By	DungND	**Date Created**	28/06/2025
Primary Actor	Admin	**Secondary Actors**	System
Trigger	On **Device** → **Requests Devices**, the admin clicks ✔ **(Approve)** or ✖ **(Reject)** in the **Actions** column for a pending request.		

<table>
<tr><td>ID and Name</td><td colspan="3">UC66 - Manage Device Requests (Edit Device Requests)</td></tr>
<tr><td></td><td colspan="3"></td></tr>
<tr><td>Description</td><td colspan="3">As an admin, I want to review a pending device request and confirm either approval or rejection—optionally adding a note/reason—so that the decision is finalized and reflected in device inventory.</td></tr>
<tr><td>Preconditions</td><td colspan="3">PRE-1. Admin is authenticated.
PRE-2. Admin is authorized to manage device requests.
PRE-3. Request exists in Pending state with complete device metadata.</td></tr>
<tr><td>Postconditions</td><td colspan="3">POST- 1. If approved, the device appears in Current Devices with the correct status (e.g., Active/Approved).
POST- 2. If rejected: the request is removed from the pending list and a rejection notice (with reason, if provided) is sent to the requester.
POST- 3. Decision (who/when/what) is recorded for audit.</td></tr>
<tr><td rowspan="6">Normal Flows</td><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>Click ✔ (Approve) or ✖ (Reject) in the row’s Actions.</td></tr>
<tr><td>2</td><td>System</td><td>Open the corresponding confirmation dialog showing User ID + device info.</td></tr>
<tr><td>3</td><td>User</td><td>In the dialog, click Approve or Reject to continue, or Cancel to abort. (For Reject, provide a Reject Reason.)</td></tr>
<tr><td>4</td><td>System</td><td>Persist decision and update lists.</td></tr>
<tr><td>5</td><td>System</td><td>Show notification: “Device approved successfully.” (green) or “Device rejected successfully.” (red). On error, display an inline error and keep the dialog open.</td></tr>
<tr><td rowspan="2">Alternative Flows</td><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>Exceptions</td><td colspan="3">• E1: Permission error → “You don’t have permission to approve/reject device requests.”
• E2: Concurrency conflict (already processed) → inform user and refresh list.
• E3: Network/Server error → show error and allow retry without losing entered notes/reason.
• E4: Validation error (missing reject reason when required) → highlight the field and prevent submission.</td></tr>
<tr><td>Frequency of Use</td><td colspan="3">Medium–High</td></tr>
<tr><td>Business Rules</td><td colspan="3">BR-01, BR-02</td></tr>
</table>

ID and Name	UC66 - Manage Device Requests (Edit Device Requests)
Other Information	• Capture **Approved By / Rejected By**, timestamp, and optional **note/reason** for reports. • Keyboard accessibility: **Enter** submits, **Esc** cancels; focus is trapped inside the dialog.
Assumptions	• Backend provides endpoints to **approve** and **reject** requests, returning updated device/request states and enforcing policy checks. • Rejection requires a reason (stored and included in the user notification). • On success, the requests table refreshes in place without a full page reload.

Table 107 - Sign In Specification

3.2.1.40 View Active FaceIDs

a. Screen Layouts

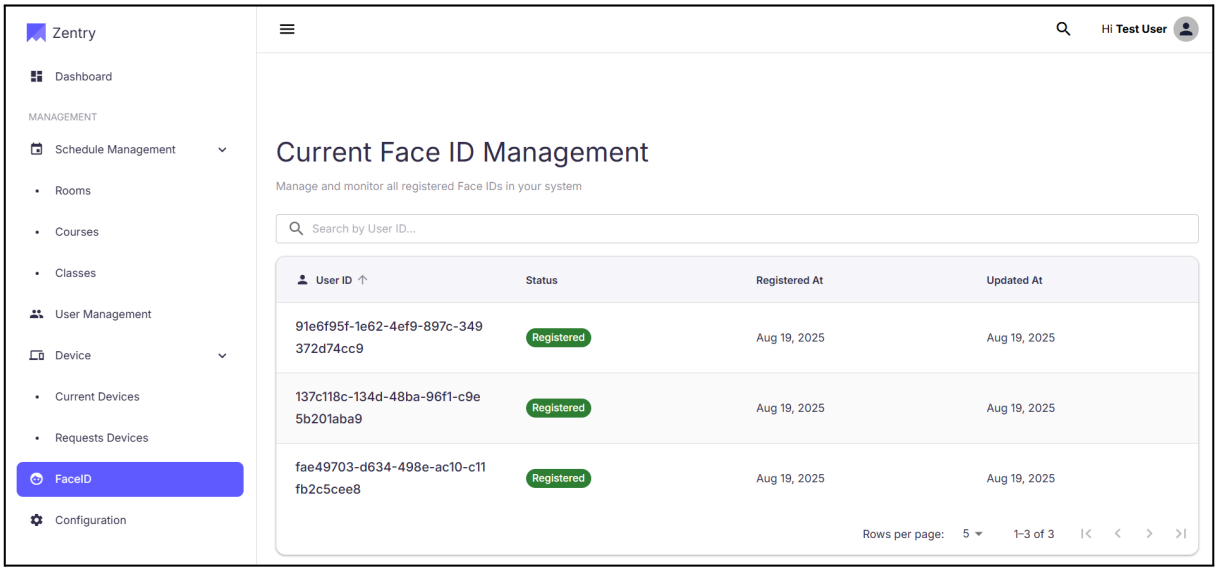


Figure 70: Current Face ID Management

No	Field Name	Field Type	Description
1	**Page Title (“Current Face ID Management”)**	Label	Header with helper text: *Manage and monitor registered Face IDs in your system*.
2	**Search by User ID…**	Searchbox	Filters rows by **User ID** (debounced).
3	**FaceID Table**	Table	Lists users and FaceID info (see columns below).
4	**Pagination**	Control	Rows-per-page selector; Previous/Next arrows at bottom-right.

No	Field Name	Field Type	Description
5	**Empty State**	Label	"No active FaceIDs found." when no **Registered** rows match.

Table 108 - Current Face ID Management Layout

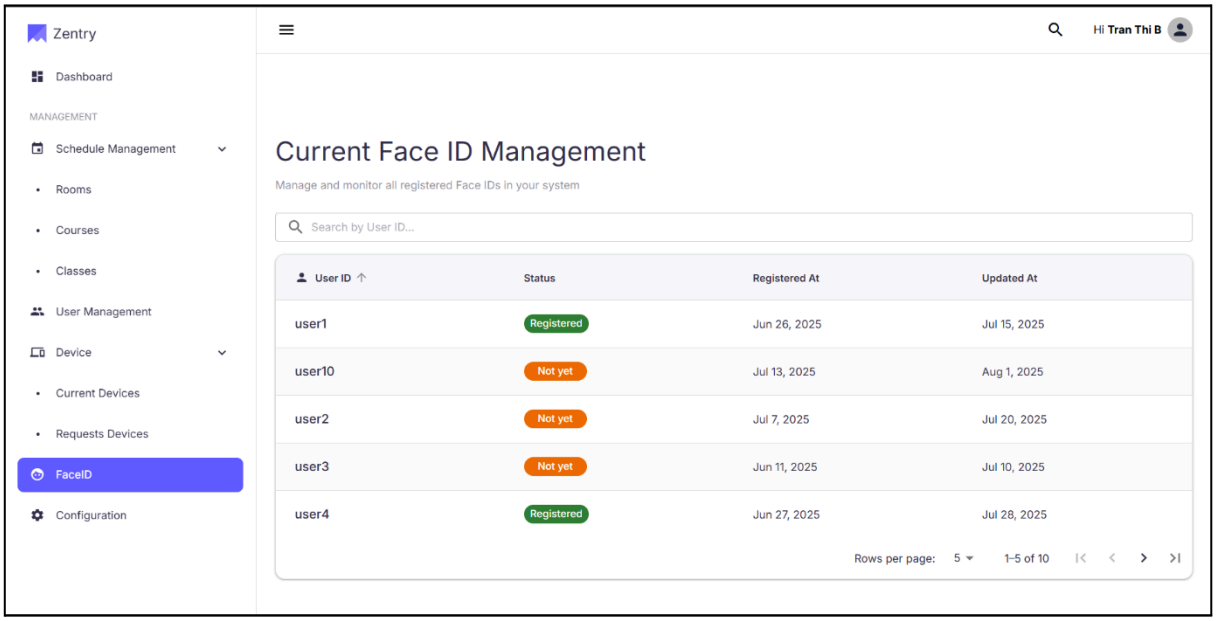


Figure 71: Current Face ID Management Table

No	Column	Field Type	Description
1	**User ID**	Text	Unique identifier of the user (e.g., user1).
2	**Status**	Badge	**Registered** (green) or **Not yet** (orange/gray).
3	**Registered At**	Date	Initial FaceID registration date.
4	**Updated At**	Date	Most recent update to FaceID data.

Table 109 - Current Face ID Management Table

b. Use Case Specifications

ID and Name	UC67 - View Active FaceIDs		
Created By	DungND	**Date Created**	28/06/2025
Primary Actor	Admin	**Secondary Actors**	System
Trigger	From the sidebar, the admin clicks **FaceID**.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC67 - View Active FaceIDs</td></tr>
<tr><th>Description</th><td colspan="3">As an admin, I want to view all users with active (registered) FaceIDs—along with registration and last-updated timestamps—so that I can verify who is enabled for FaceID and quickly find specific users.</td></tr>
<tr><th>Preconditions</th><td colspan="3">PRE-1. Admin is authenticated.
PRE-2. Admin has permission to view FaceID registry.
PRE-3. FaceID records exist for some users.</td></tr>
<tr><th>Postconditions</th><td colspan="3">POST- 1. No data is modified by viewing.
POST- 2. Filters/search and pagination reflect the current dataset.</td></tr>
<tr><th rowspan="6">Normal Flows</th><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>On the sidebar, click FaceID.</td></tr>
<tr><td>2</td><td>System</td><td>Display a table of all users with columns User ID, Status, Registered At, Updated At.</td></tr>
<tr><td>3</td><td>User</td><td>To focus on active FaceIDs, check the Status column and identify rows marked Registered (green).</td></tr>
<tr><td>4</td><td>User</td><td>Use the Search box to find a user by User ID quickly.</td></tr>
<tr><td>5</td><td>User</td><td>If records span multiple pages, use Pagination at the bottom-right to navigate.</td></tr>
<tr><th rowspan="4">Alternative Flows</th><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>Status filter is available</td><td>Set Status = Registered to display only active FaceIDs.</td></tr>
<tr><td>A2</td><td>Sorting is enabled</td><td>Sort by Registered At or Updated At to review newest/oldest registrations.</td></tr>
<tr><td>A3</td><td>Large dataset / server-side paging</td><td>Server-side search and pagination are used; results update without a full page reload and current filters are preserved.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Permission error → “You don’t have permission to view FaceID data.”
• E2: Network/Server error → inline error with Retry; keep search/pagination state.
• E3: Data inconsistency (missing timestamps) → show placeholder “—” and log for follow-up.</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Medium</td></tr>
<tr><th>Business Rules</th><td colspan="3">BR-01, BR-02</td></tr>
</table>

ID and Name	UC67 - View Active FaceIDs
Other Information	• All login activities are recorded in system logs for security auditing and system health monitoring. • The system supports both internal (username/password) and external (OAuth) authentication methods.
Assumptions	• Privacy: show metadata only; never expose biometric templates/images. • Sorting/Filtering: default **Updated At (desc)**; quick status chips (**All/Registered/Not yet**); search by **User ID**. • Time/Locale: dates shown in admin's timezone/locale.

Table 110 - View Active FaceIDs Specification

3.2.1.41 View Attribute Definitions

a. Screen Layouts

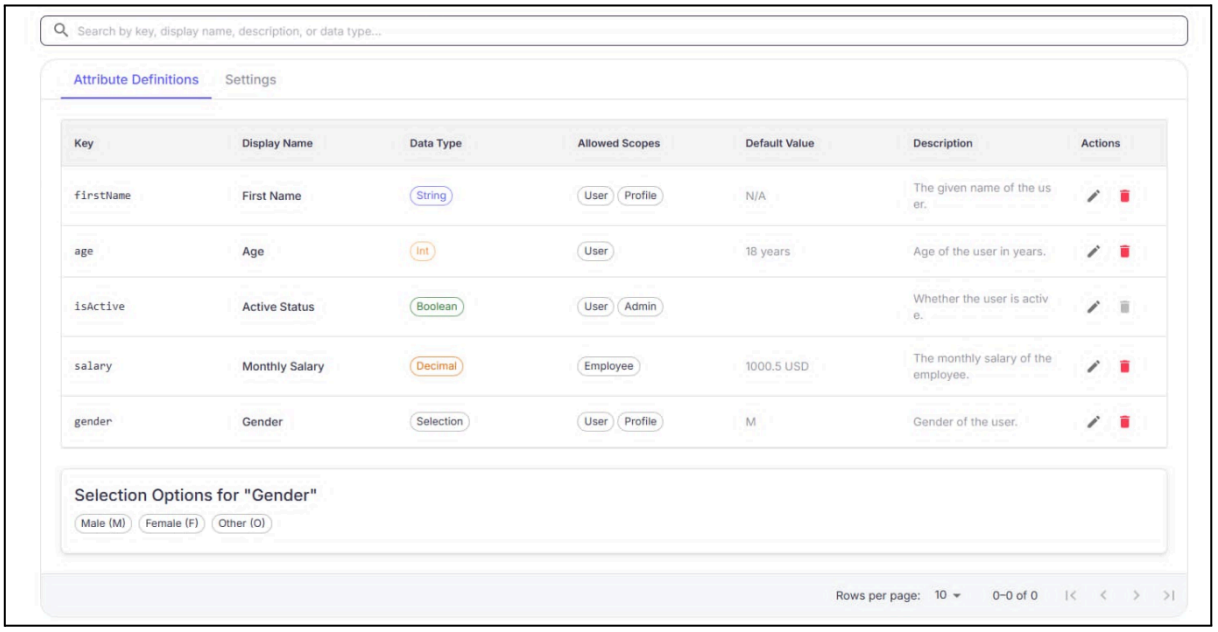


Figure 72: View Attribute Definitions

No	Field Name	Field Type	Description
1	**Page Title ("Configuration Management")**	Label	Header with helper text.
2	**Quick Actions**	Buttons	**Create Attribute Definition**, **Create Setting** (out of scope here).
3	**Search…**	Searchbox	Filter by key, display name, description, or data type.
4	**Tabs**	Tabs	**Attribute Definitions** (active), **Settings**.

No	Field Name	Field Type	Description
5	**Definitions Table**	Table	Columns below; shows all attributes.
6	**Pagination**	Control	Rows-per-page selector; Previous/Next.

Table 111 - View Attribute Definitions Layout

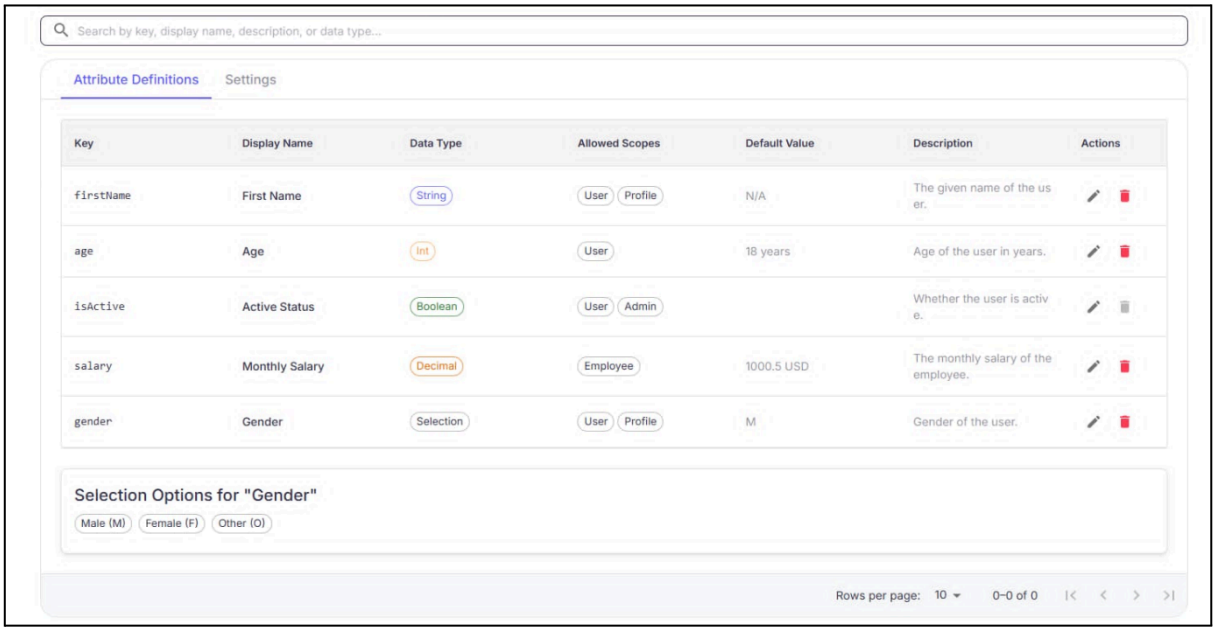


Figure 73: View Attribute Definitions Table

No	Column	Field Type	Description
1	**Key**	Text	Internal identifier.
2	**Display Name**	Text	Human-readable label.
3	**Data Type**	Badge	String, Int, Boolean, Decimal, Selection, …
4	**Allowed Scopes**	Badges	e.g., User, Profile, Employee, Admin.
5	**Default Value**	Text	Value used if none provided.
6	**Description**	Text	Purpose/meaning.
7	**Actions**	Icon buttons	**View (eye)**, Edit (pencil), Delete (trash). This UC uses **View**.

Table 112 - View Attribute Definitions Table

b. Use Case Specifications

ID and Name	UC68 - View Attribute Definitions		
Created By	DungND	**Date Created**	28/06/2025
Primary Actor	Admin	**Secondary Actors**	System

ID and Name	UC68 - View Attribute Definitions		
Trigger	User clicks **View (eye)** for an attribute on **Configuration Management** → **Attribute Definitions**.		
Description	As an authorized user, I want to view a specific attribute definition—its key, display name, data type, allowed scopes, default value, description, and (if applicable) selection options—so that I understand how it's configured and used.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to view configuration data. **PRE-3.** Target attribute definition exists.		
Postconditions	**POST- 1.** No configuration is modified by viewing. **POST- 2.** User can navigate back to the list.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	On **Configuration Management** → **Attribute Definitions**, click **View (eye)** for the desired attribute.
	2	System	Open **Attribute Definition Details** showing: **Key**, **Display Name**, **Data Type**, **Allowed Scopes**, **Default Value**, **Description**, and **Selection Options** if type = **Selection**.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Use search	Filter the list using the search box to locate the attribute quickly.
Exceptions	• **E1:** Attribute not found or deleted → show error and return to list. • **E2:** Permission error → "You don't have permission to view attribute definitions." • **E3:** Network/Server error → display error state with **Retry**.		
Frequency of Use	Medium		
Business Rules	BR-01, BR-02		
Other Information	• Read-only view; editing/deleting handled in separate UCs. • Dates/metadata (e.g., created/updated by) may be shown if available.		

ID and Name	UC68 - View Attribute Definitions
Assumptions	• Backend exposes endpoints to list and fetch attribute definition by key/ID. • "Allowed Scopes" are predefined enumerations enforced server-side. • Selection options are stored and returned in display order.

Table 113 - View Attribute Definition Specification

3.2.1.42 Create Attribute Definition

a. Screen Layouts

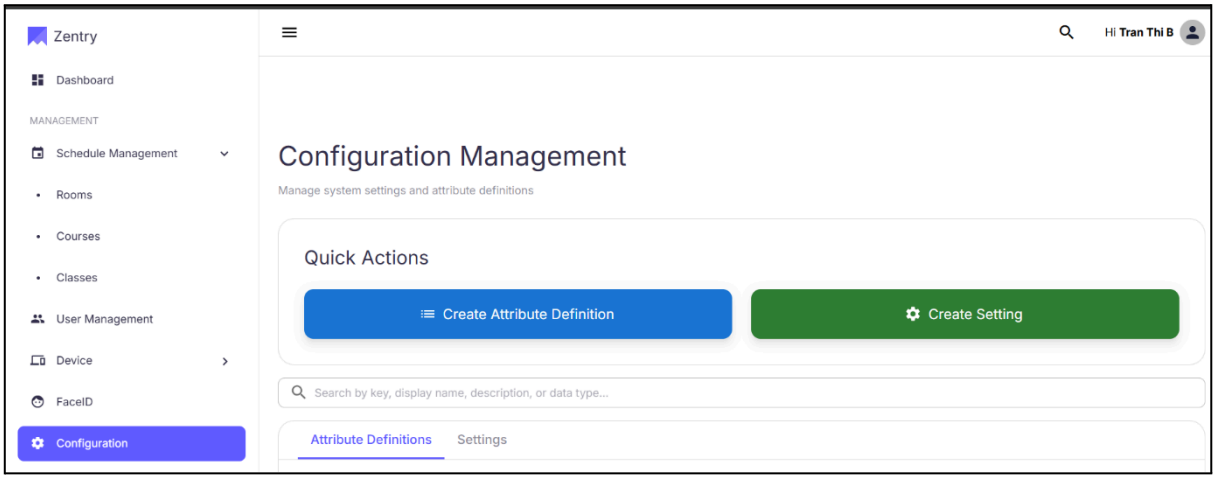


Figure 74: Create Attribute Definition Layout

No	Field Name	Field Type	Description
1	**Page Title ("Configuration Management")**	Label	Header and helper text.
2	**Quick Actions**	Buttons	**Create Attribute Definition**, **Create Setting**.
3	**Tabs**	Tabs	**Attribute Definitions** (active), **Settings**.
4	**Search…**	Searchbox	Filter list by key, display name, description, or data type.

Table 114 - Create Attribute Definition Layout

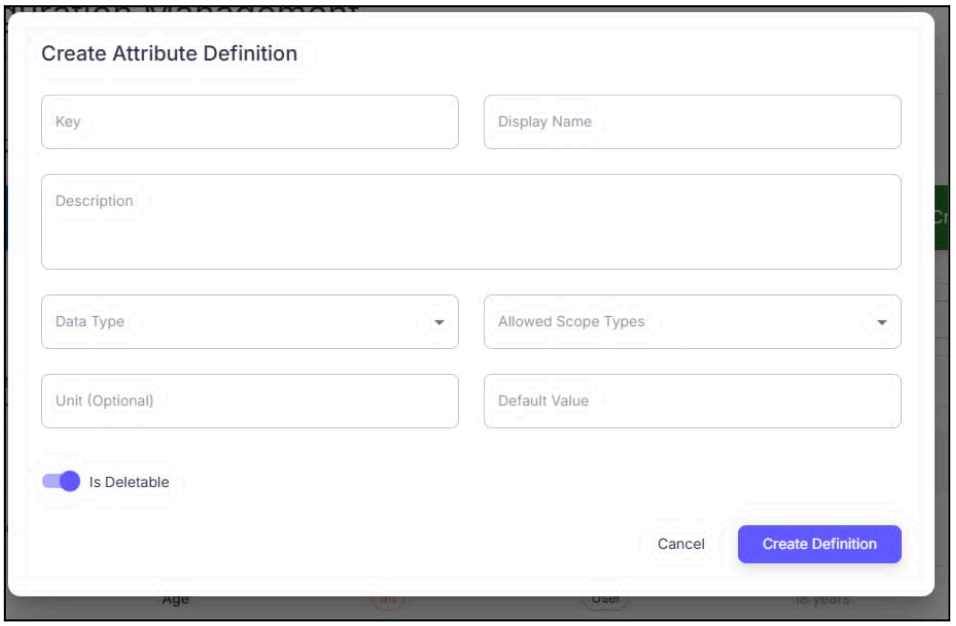


Figure 75: Create Attribute Definition

No	Field Name	Field Type	Description
1	**Title ("Create Attribute Definition")**	Label	Dialog header.
2	**Key**	Text (required)	Unique identifier; must be unique and follow key format.
3	**Display Name**	Text (required)	Human-readable name shown in UI.
4	**Description**	Textarea	Purpose/explanation of the attribute.
5	**Data Type**	Dropdown (required)	**String, Int, Boolean, Decimal, Date, Json, Selection**.
6	**Allowed Scope Types**	Multi-select / Checkboxes (required)	**Global, Course, Session**.
7	**Unit (Optional)**	Text	Measurement unit (e.g., %, hrs, USD).
8	**Default Value**	Dynamic input	Widget depends on **Data Type**.
9	**Is Deletable**	Toggle	Prevents standard deletion when off.
10	**Cancel**	Button (secondary)	Close without saving.
11	**Create Definition**	Button (primary)	Enabled when required fields are valid.
12	**Validation/Error Area**	Inline message	Field/type/uniqueness errors.

Table 115 - Create Attribute Definition

b. Use Case Specifications

ID and Name	UC69 Create Attribute Definition		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	On **Configuration Management**, the user clicks **Create Attribute Definition**.		
Description	As an authorized user, I want to define a new system attribute with key, display name, data type, scope, unit, and default value so that it can be reused across configured scopes.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to manage configuration **PRE-3.** Attribute **key** is unique.		
Postconditions	**POST- 1.** New attribute definition is saved and shown in **Attribute Definitions** list. **POST- 2.** Success notification is displayed; action is logged.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	On **Configuration Management**, click **Create Attribute Definition**.
	2	System	Open **Create Attribute Definition** dialog.
	3	User	Fill fields: **Key**, **Display Name**, **Description**, select **Data Type**, select **Allowed Scope Types** , set **Unit (Optional)**, **Default Value**, and **Is Deletable**.
	4	User	Click **Create Definition** to save, or **Cancel** to abort.
	5	System	Validate inputs (required/type/uniqueness). On success, persist and show green toast **“Attribute Definition created successfully.”** On failure, show inline error (e.g., **“Key already exists”**, **“Missing required field”**).
Alternative Flows	**Alt Step**	**Condition**	**Description**

ID and Name	UC69 Create Attribute Definition
Exceptions	• **E1:** Permission error → "You don't have permission to create attribute definitions." • **E2:** Network/Server error → show error; keep dialog data so the user can retry. • **E3:** Concurrency (key taken during save) → show **Key already exists** and request a new key.
Frequency of Use	Medium
Business Rules	BR-01, BR-02
Other Information	• Auto-suggest **Key** from **Display Name**; enforce allowed pattern (lowercase, hyphen/underscore, no spaces). • Numbers/dates use locale-aware inputs; client + server validation.
Assumptions	• Backend provides a create API enforcing uniqueness, type constraints, and scopes. • For **Selection** type, option values and default are created with the definition in the same request. • List view auto-refreshes to include the newly created definition.

Table 116 - Create Attribute Definition Specification

3.2.1.43 Update Attribute Definitions

a. Screen Layouts

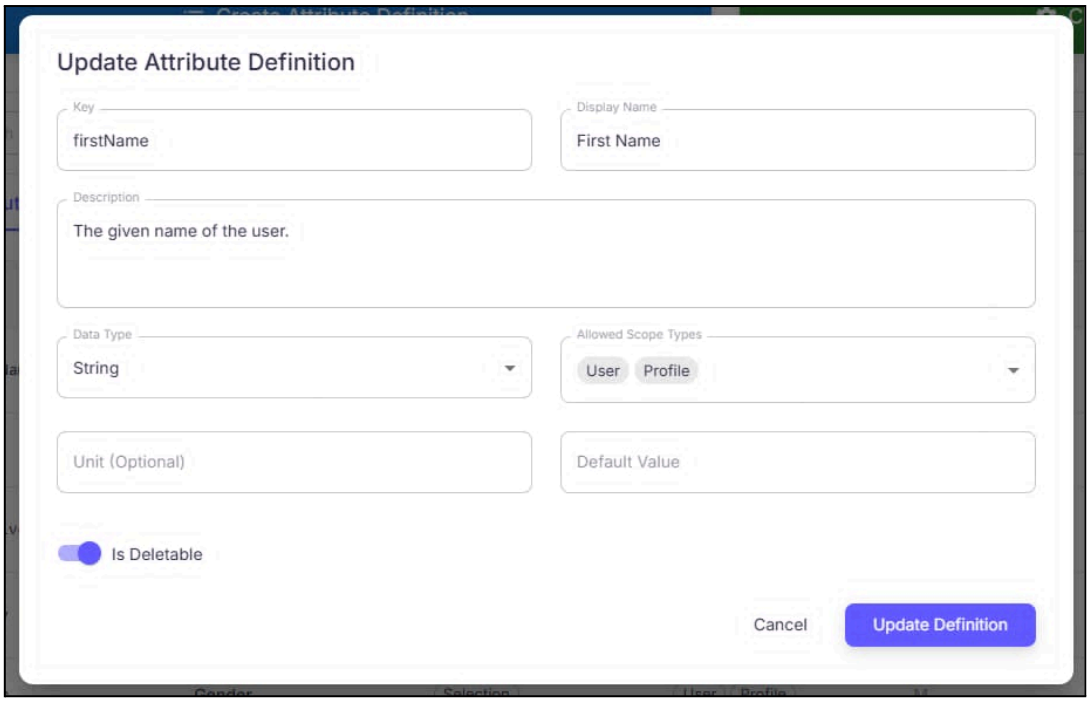


Figure 76: Update Attribute Definition

No	Field Name	Field Type	Description
1	**Title (“Update Attribute Definition”)**	Label	Dialog header.
2	**Key**	Text (read-only)	Unique identifier; non-editable.
3	**Display Name**	Text	Human-readable name.
4	**Description**	Textarea	Purpose/notes.
5	**Data Type**	Dropdown	**String, Int, Boolean, Decimal, Date, Json, Selection**. *(Figure xx – Select Data Type)*
6	**Allowed Scope Types**	Multi-select / Chips	**User, Profile, Global, Course, Session**. *(Figure xx – Allowed Scope Types)*
7	**Unit (Optional)**	Text	Measurement unit (%, hrs, USD…).
8	**Default Value**	Dynamic input	Widget depends on **Data Type**.
9	**Is Deletable**	Toggle	Allow/restrict deletion.
10	**Cancel**	Button	Close without saving.
11	**Update Definition**	Button (primary)	Validate and save.
12	**Validation/Error Area**	Inline message	Field/type/constraint errors.

Table 117 - Sign In - Sign in form

b. Use Case Specifications

ID and Name	UC70 Update Attribute Definitions		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	On **Configuration Management** → **Attribute Definitions**, the user clicks **Edit** for an attribute.		
Description	As an authorized user, I want to update an existing attribute definition (display name, description, data type, allowed scopes, unit, default value,		

<table>
<tr><td>ID and Name</td><td colspan="3">UC70 Update Attribute Definitions</td></tr>
<tr><td></td><td colspan="3">deletable option) so that its configuration stays accurate and usable across the system.</td></tr>
<tr><td>Preconditions</td><td colspan="3">PRE-1. User is authenticated.
PRE-2. User has permission to manage configuration.
PRE-3. Target attribute definition exists.
PRE-4. Key is immutable and uniquely identifies the definition.</td></tr>
<tr><td>Postconditions</td><td colspan="3">POST- 1. Changes are saved and reflected in the list.
POST- 2. Success notification is shown; action logged</td></tr>
<tr><td rowspan="6">Normal Flows</td><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>From the list, click Edit on the target attribute.</td></tr>
<tr><td>2</td><td>System</td><td>Open Update Attribute Definition dialog prefilled with current values.</td></tr>
<tr><td>3</td><td>User</td><td>Update fields: Display Name, Description, Data Type, Allowed Scope Types, Unit, Default Value, Is Deletable.</td></tr>
<tr><td>4</td><td>User</td><td>Click Update Definition to save (or Cancel to discard).</td></tr>
<tr><td>5</td><td>System</td><td>Validate inputs; on success persist changes, close dialog, refresh list, and show green toast “Attribute Definition updated successfully.” On failure, show inline errors.</td></tr>
<tr><td rowspan="6">Alternative Flows</td><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>Data Type = Selection</td><td>Show Options panel to manage choices and default option (require ≥1 choice).</td></tr>
<tr><td>A2</td><td>Change Data Type</td><td>If Default Value no longer matches the new type, system clears/blocks it and shows a type-mismatch warning.</td></tr>
<tr><td>A3</td><td>Scope change impact</td><td>If removing a scope that currently uses the attribute, system warns and may block or require confirmation.</td></tr>
<tr><td>A4</td><td>Unsaved changes</td><td>Closing the dialog with edits prompts a discard confirmation.</td></tr>
<tr><td>A5</td><td>No field changes</td><td>Update Definition disabled or results in no-op.</td></tr>
<tr><td>Exceptions</td><td colspan="3">• E1: Permission error → “You don’t have permission to edit attribute definitions.”</td></tr>
</table>

ID and Name	UC70 Update Attribute Definitions
	• **E2:** Network/Server error → show error toast; keep dialog open and allow retry. • **E3:** Invalid data (e.g., *Invalid data type*, *Missing required field*, type mismatch with default value) → highlight fields, block save. • **E4:** Concurrency conflict (definition changed/deleted by another user) → show notice and refresh list; reopen with latest data if applicable.
Frequency of Use	Medium
Business Rules	BR-01, BR-02
Other Information	• Key is immutable once created. • Inputs respect locale (numbers/dates). • Changes may affect downstream forms that consume the attribute; consider cache busting.
Assumptions	• Backend provides an **update** API that enforces type compatibility, scope constraints, and immutability of **Key**. • Selection options are editable when type = **Selection** and saved with the same request. • List auto-refreshes after a successful update.

Table 118 - Update Attribute Definitions Specification

3.2.1.44 Delete Attribute Definition

a. Screen Layout

Figure 77: Delete Attribute Definition

No	Field Name	Field Type	Description
1	**Title**	Label	"Delete Attribute Definition"
2	**Message**	Label	"Are you sure you want to delete the attribute definition **<AttributeName>**? This action cannot be undone."
3	**Cancel**	Button (secondary)	Close dialog; no changes.
4	**Delete**	Button (primary, red)	Confirms permanent deletion; disabled while processing or when policy blocks.
5	**Error Area (optional)**	Inline message	Shows reason if deletion is blocked.

Table 119 - Delete Attribute Definition

b. Use Case Specifications

ID and Name	UC71 Delete Attribute Definition		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	On **Configuration Management** → **Attribute Definitions**, the user clicks **Delete (🗑️)** for a specific attribute.		
Description	As an authorized admin, I want to permanently delete an attribute definition—when it is marked deletable—so that obsolete attributes are removed from the system.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has rights to manage configuration. **PRE-3.** Target attribute exists and is **Is Deletable = true**.		
Postconditions	**POST- 1.** If successful, the attribute is removed from the list and no longer available for use. **POST- 2.** A success toast is shown and the action is logged.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Locate the attribute and click **Delete (🗑️)**.
	2	System	Show **Delete Confirmation** dialog with the irreversible warning.
	3	User	Click **Delete** to confirm (or **Cancel** to abort).
	4	System	Validate deletability/constraints; on success, delete, refresh the list, and show green toast **"Attribute Definition deleted successfully."**
Alternative Flows	**Step**	**By**	**Step Details**
	A1	Admin	Click Cancel
Exceptions	• **E1:** Permission error → "You don't have permission to delete attribute definitions." • **E2:** Network/Server error → show error; keep dialog open and allow retry. • **E3:** Constraint violation (linked/critical use) → "This attribute cannot be deleted."		

ID and Name	UC71 Delete Attribute Definition
	• **E4:** Concurrency conflict (already deleted/changed) → inform user and refresh list.
Frequency of Use	Low–Medium
Business Rules	BR-01, BR-02
Other Information	• Consider exporting affected references in the error to help remediation. • Back navigation should not restore deleted rows from cache.
Assumptions	• Backend exposes a delete API that enforces **Is Deletable** and dependency checks, returning clear error messages. • List view auto-refreshes after deletion without a full page reload.

Table 120 - Delete Attribute Definition

3.2.1.45 View Settings

a. Screen Layouts

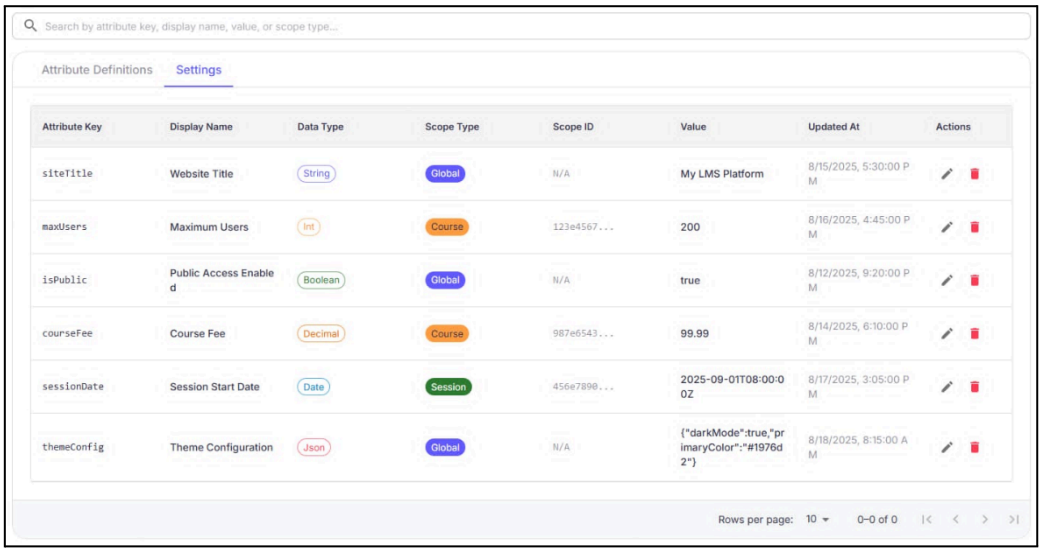


Figure 77: View Settings Layout

No	Field Name	Field Type	Description
1	**Page Title (“Configuration Management”)**	Label	Header with helper text.
2	**Tabs**	Tabs	**Attribute Definitions**, **Settings** (active).
3	**Search…**	Searchbox	Filter by key, display name, value, or scope type.
4	**Settings Table**	Table	Lists settings with columns below.

No	Field Name	Field Type	Description
5	**Pagination**	Control	Rows-per-page selector; Prev/Next arrows.

Table 121 - View Settings Layout

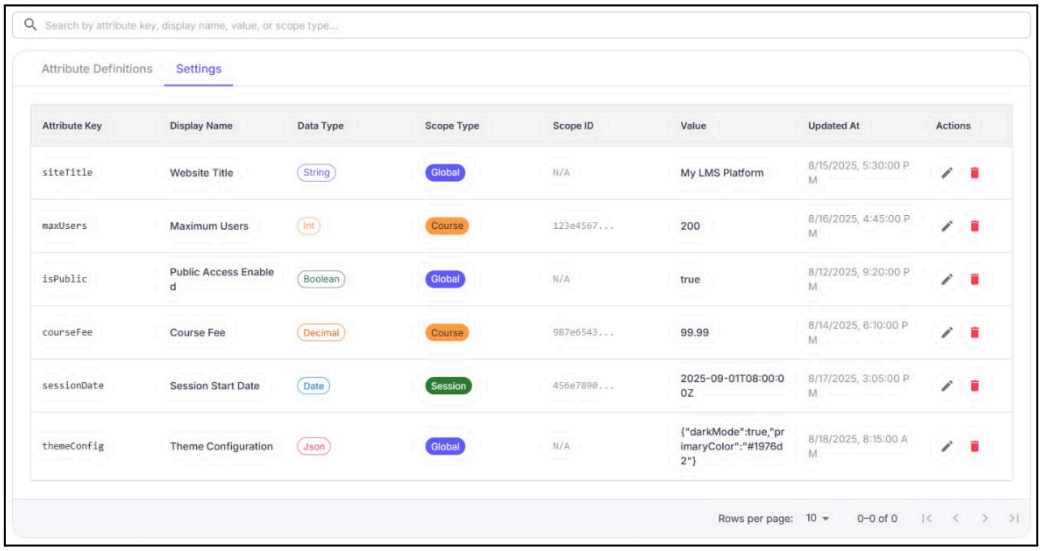


Figure 78: View Settings Table

No	Column	Field Type	Description
1	**Attribute Key**	Text	Unique ID (e.g., siteTitle, maxUsers).
2	**Display Name**	Text	Human-readable name.
3	**Data Type**	Badge	String, Int, Boolean, Decimal, Date, **JSON**.
4	**Scope Type**	Badge	Global, Course, Session, User, Profile…
5	**Scope ID**	Text	Identifier of scope (or **N/A** for Global).
6	**Value**	Text/Code	Current configured value (pretty-prints JSON).
7	**Updated At**	Date/Time	Last modification timestamp.
8	**Actions**	Icon buttons	**Edit**, **Delete**.

Table 122 - View Settings Table

b. Use Case Specifications

ID and Name	UC72 View Setting		
Created By	DungND	**Date Created**	28/06/2025
Primary Actor	Admin	**Secondary Actors**	System
Trigger			

<table>
<tr><th>ID and Name</th><td colspan="3">UC72 View Setting</td></tr>
<tr><td></td><td colspan="3">User navigates to Configuration Management → Settings.</td></tr>
<tr><th>Description</th><td colspan="3">As an admin, I want to view all system configuration settings in a structured table—showing key, display name, data type, scope, value, and last update—so I can understand and audit current configuration.</td></tr>
<tr><th>Preconditions</th><td colspan="3">PRE-1. User is authenticated.
PRE-2. User has permission to view configuration.</td></tr>
<tr><th>Postconditions</th><td colspan="3">POST- 1. No data is modified by viewing.
POST- 2. Filters and pagination reflect the current dataset.</td></tr>
<tr><th rowspan="6">Normal Flows</th><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>Open Configuration Management → Settings.</td></tr>
<tr><td>2</td><td>System</td><td>Display the Settings table with columns Attribute Key, Display Name, Data Type, Scope Type & Scope ID, Value, Updated At, Actions.</td></tr>
<tr><td>3</td><td>User</td><td>Use the Search bar to filter by key, display name, value, or scope type.</td></tr>
<tr><td>4</td><td>System</td><td>Dynamically update the table to reflect filters.</td></tr>
<tr><td>5</td><td>User</td><td>Use Pagination (rows per page, Prev/Next) to navigate large datasets.</td></tr>
<tr><th rowspan="4">Alternative Flows</th><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>Sort enabled</td><td>Sort by Updated At (desc by default) or other columns.</td></tr>
<tr><td>A2</td><td>Scope-focused review</td><td>Filter by Scope Type (e.g., Global only) using search or chips (if available).</td></tr>
<tr><td>A3</td><td>JSON values</td><td>Values of type JSON open a read-only pretty viewer on hover/click.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Permission error → “You don’t have permission to view settings.”
• E2: Network/Server error → Show inline error with Retry; keep current filters/pagination.
• E3: Data inconsistency (e.g., missing Scope ID/Value) → Render placeholders “—” and log for review.
• E4: Concurrency conflict (setting edited/deleted by another user) → Inform user and refresh the table to latest data.</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Medium</td></tr>
</table>

ID and Name	UC72 View Setting
Business Rules	BR-01, BR-02, BR-15, BR-16
Other Information	• Default sort **Updated At (desc)**; supports CSV export of the filtered view (admin only). • Clicking **Edit/Delete** transitions to their respective UCs.
Assumptions	• Backend provides a paginated Settings API returning display metadata, scope, value, and timestamps with server-side search/sort.

Table 123 - View Settings Specification

3.2.1.46 Create Setting

a. Screen Layouts

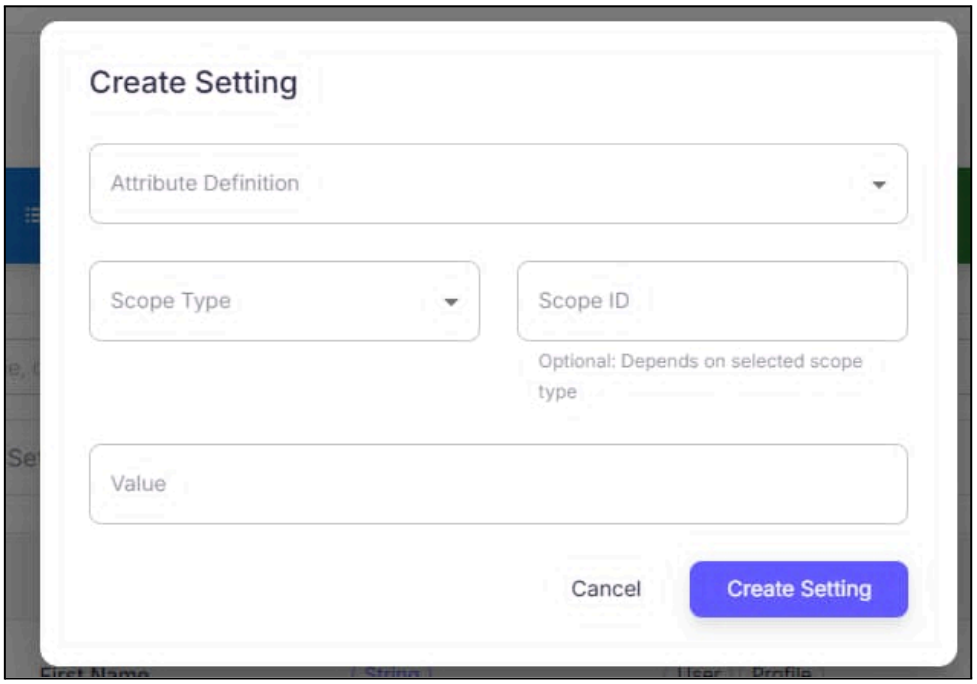


Figure 79: Create Settings Layout

No	Field Name	Field Type	Description
1	**+ Create Setting**	Button	Opens the creation dialog.
2	**Create Setting (dialog title)**	Label	Header of the modal dialog.
3	**Attribute Definition**	Dropdown (required)	Choose an existing attribute definition. The attribute's **data type** and **allowed scopes** drive the Value widget and scope options.
4	**Scope Type**	Dropdown (required)	**Global, Course, Session** (and others if enabled).

No	Field Name	Field Type	Description
5	**Scope ID**	Text (conditional)	Required when **Scope Type** ≠ Global. Shows helper text.
6	**Value**	Dynamic input (required)	Widget depends on the selected attribute's data type: String (text), Int/Decimal (number), Boolean (switch), Date (date picker), JSON (code editor), Selection (options list).
7	**Cancel**	Button	Close without saving.
8	**Create Setting**	Button (primary)	Saves when validation passes.
9	**Validation/Error Area**	Inline message	Shows format/type/required errors.

Table 124 - Create Settings Layout

b. Use Case Specifications

ID and Name	UC73 Create Setting		
Created By	DungND	**Date Created**	28/06/2025
Primary Actor	Admin	**Secondary Actors**	System
Trigger	On **Configuration Management** → **Settings**, the user clicks **+ Create Setting**.		
Description	As an authorized user, I want to create a new Setting by selecting an Attribute Definition, choosing a Scope Type (and Scope ID if needed), and providing a Value so that the configuration is applied to the chosen scope.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to manage settings. **PRE-3.** At least one **Attribute Definition** exists and supports the chosen scope.		
Postconditions	**POST- 1.** The new Setting is stored and appears in the **Settings** table. **POST- 2.** A success toast is shown; action is logged.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Click **+ Create Setting** on the Settings page.
	2	System	Open **Create Setting** dialog.
	3	User	Select **Attribute Definition** from dropdown.

ID and Name	UC73 Create Setting		
	4	User	Choose **Scope Type** (Global/Course/Session). If Course or Session, enter **Scope ID**.
	5	User	Enter **Value** using the widget that matches the attribute's data type.
	6	User	Click **Create Setting** to save (or **Cancel** to abort).
	7	System	Validate inputs and persist. **If successful:** add the row to the table and show "Setting created successfully." **If unsuccessful:** show inline error (e.g., "Invalid value format").
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Non-Global scope	**Scope ID** becomes required; system validates the referenced Course/Session exists.
	A2	Data-type driven value	Value control changes with type (Boolean switch, Date picker, JSON editor, Selection options).
	A3	Duplicate combination	If a Setting for *(Attribute Key, Scope Type, Scope ID)* already exists, system blocks creation and suggests editing the existing one.
	A4	Default value assist	If the definition has a **Default Value**, prefill the Value field (user may override).
Exceptions	• **E1:** Permission error → "You don't have permission to create settings." • **E2:** Invalid value format/type mismatch → highlight **Value** and show guidance. • **E3:** Invalid or missing **Scope ID** for non-Global scope → prompt to correct. • **E4:** Duplicate setting exists → "A setting for this attribute and scope already exists." • **E5:** Network/Server error → show error; keep dialog data and allow retry.		
Frequency of Use	Medium		
Business Rules	BR-01, BR-02		

ID and Name	UC73 Create Setting
Other Information	• Respect attribute's **allowed scope types**; disable unsupported scopes. • JSON values are validated with a parser; dates use locale/timezone aware pickers.
Assumptions	• Backend enforces uniqueness on *(attributeKey, scopeType, scopeId)* and validates value by data type. • Settings table refreshes immediately after successful creation.

Table 125 - Create Settings Specification

3.2.1.47 Edit Setting

a. Screen Layouts

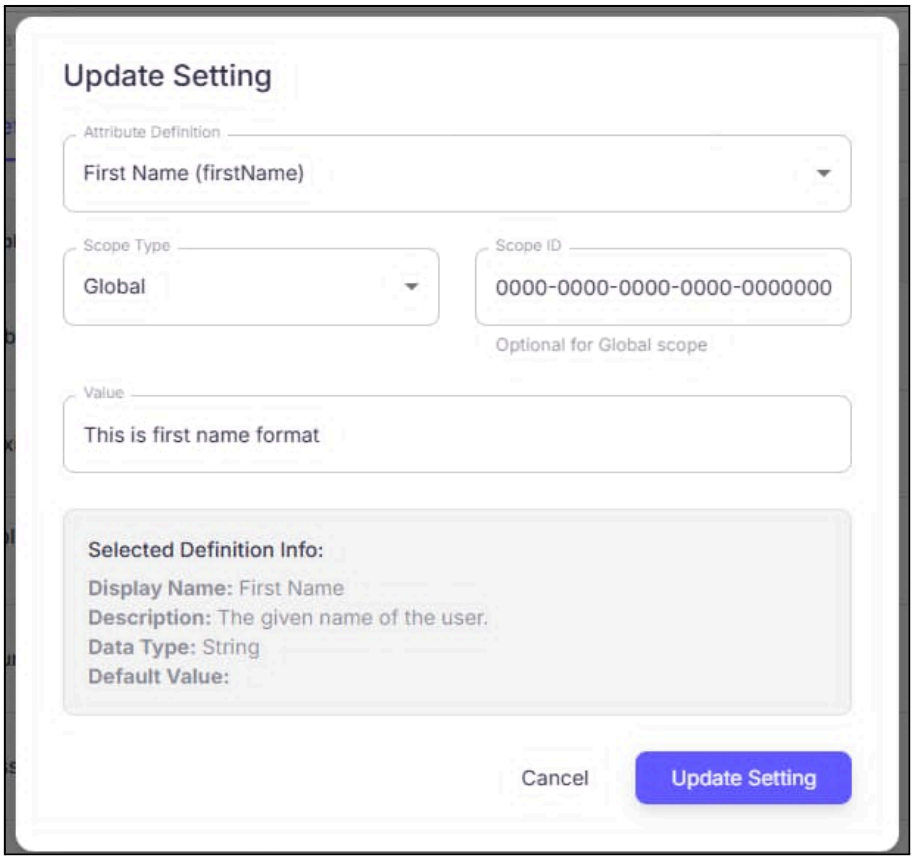


Figure 80: Update Settings Layout

No	Field Name	Field Type	Description
1	**Title ("Update Setting")**	Label	Dialog header.
2	**Attribute Definition**	Dropdown (read-only)	Fixed to the current attribute (not editable).
3	**Scope Type**	Dropdown	**Global, Course, Session** (and others if enabled).

No	Field Name	Field Type	Description
4	**Scope ID**	Text (conditional)	Required for non-Global scopes; optional for Global.
5	**Value**	Dynamic input	Widget follows attribute **Data Type** (text/number/switch/date/JSON/selection).
6	**Selected Definition Info**	Read-only panel	Shows **Display Name, Description, Data Type, Default Value** for reference.
7	**Cancel**	Button	Close without saving.
8	**Update Setting**	Button (primary)	Save when validation passes.
9	**Validation/Error Area**	Inline message	Type/format/scope errors.

Table 126 - Update Settings Layout

b. Use Case Specifications

ID and Name	UC74 - Update Setting		
Created By	DungND	**Date Created**	28/06/2025
Primary Actor	Admin	**Secondary Actors**	System
Trigger	On **Configuration Management** → **Settings**, the user clicks **Edit (✎)** on a row.		
Description	As an authorized user, I want to modify an existing Setting's scope (type/ID) and value—while keeping the attribute fixed—so that configuration remains correct for the intended scope.		
Preconditions	**PRE-1.** User is authenticated. **PRE-2.** User has permission to manage settings. **PRE-3.** Target Setting exists; **Attribute Definition** is immutable.		
Postconditions	**POST- 1.** On success, the Setting is updated in storage and the list reflects new values. **POST- 2.** Success toast is shown; action is logged		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Click **Edit (✎)** on the desired Setting.
	2	System	Open **Update Setting** dialog with pre-filled details: **Attribute Definition (fixed)**, **Scope**

ID and Name	UC74 - Update Setting		
			Type, **Scope ID**, **Value**, and **Selected Definition Info**.
	3	User	Modify **Scope Type/Scope ID** and/or **Value** as needed.
	4	User	Click **Update Setting** to save (or **Cancel** to discard).
	5	System	Validate and persist. **If successful:** update the table row. **If unsuccessful:** show inline error (e.g., "Invalid Scope ID").
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Non-Global scope	**Scope ID** becomes required; system validates that the referenced Course/Session exists.
	A2	Data-type driven value	**Value** control adapts to the attribute's **Data Type** (Boolean switch, Date picker, JSON editor, etc.).
	A3	Duplicate combination	Changing to a *(Attribute Key, Scope Type, Scope ID)* that already exists → block update and suggest editing that record instead.
	A4	Scope not allowed	If the attribute doesn't support the chosen scope, disable it or show a validation message.
	A5	Unsaved changes	Closing the dialog with edits prompts to confirm discarding.
	A6	No field changes	**Update Setting** disabled or results in no-op.
Exceptions	• **E1:** Permission error → "You don't have permission to edit settings." • **E2:** Invalid **Scope ID** (format or not found) → highlight field and show guidance. • **E3:** Value type/format mismatch with attribute's **Data Type** → block save and highlight **Value**. • **E4:** Network/Server error → keep dialog data; show retryable error. • **E5:** Concurrency conflict (setting changed/deleted elsewhere) → inform user and refresh list. • **E6:** Scope not allowed by definition → error: "This attribute cannot be applied to the selected scope."		
Frequency of Use	Medium		

ID and Name	UC74 - Update Setting
Business Rules	BR-01, BR-02, BR-15, BR-16
Other Information	• Show full timestamps in tooltips; localize numbers/dates. • Preserve filter/pagination state when returning to the list.
Assumptions	• Backend update API validates scope permissions, uniqueness on *(attributeKey, scopeType, scopeId)*, and value type. • Table auto-refreshes after success without reloading the page.

Table 127 - Sign In Specification

3.2.1.48 Delete Setting

a. Screen Layouts

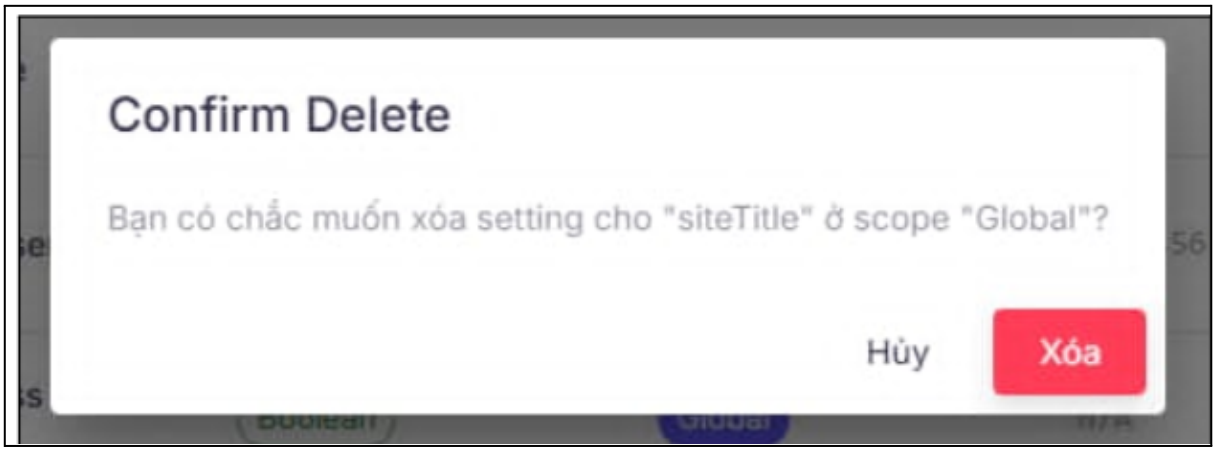


Figure 81 - Delete Setting

No	Field Name	Field Type	Description
1	**Settings Table**	Table	Columns: **Attribute Key, Display Name, Data Type, Scope Type, Scope ID, Value, Updated At, Actions**.
2	**Actions**	Icon buttons	**Delete (🗑️)** opens confirmation dialog.

Table 128 – Delete Setting Field Description Table

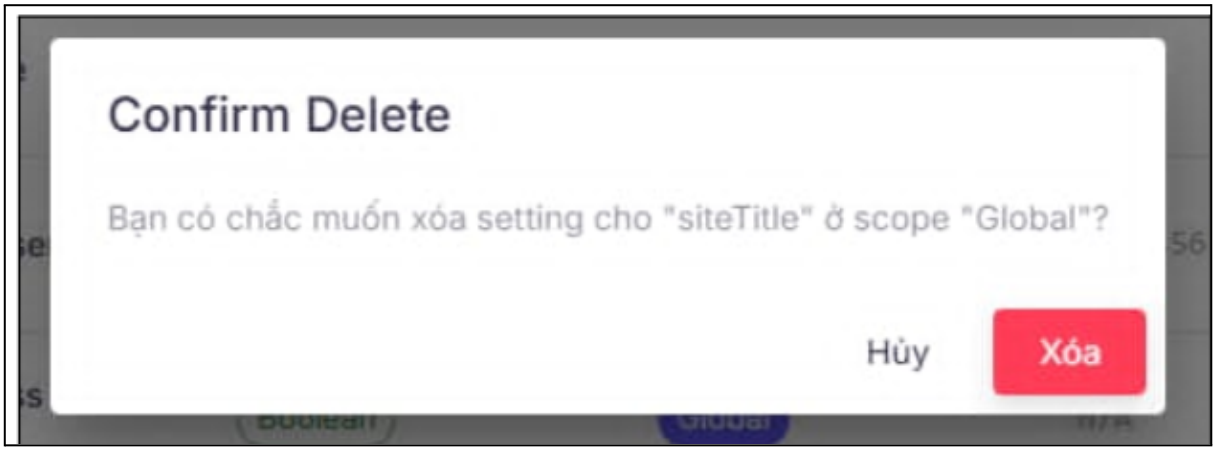


Figure 82: Confirm delete Layout

No	Field Name	Field Type	Description
1	**Title**	Label	"Confirm Delete".
2	**Message**	Label	Shows setting key and scope (e.g., siteTitle @ Global) and warns that the action cannot be undone.
3	**Cancel**	Button (secondary)	**Hủy / Cancel** – close without changes.
4	**Delete**	Button (primary, red)	**Xóa / Delete** – confirms permanent removal.
5	**Error Area**	Inline message	Displays reason if deletion is blocked.

Table 129 - Confirm delete Layout

b. Use Case Specifications

ID and Name	UC75 Delete Setting		
Created By	DungND	Date Created	28/06/2025
Primary Actor	Admin	Secondary Actors	System
Trigger	On **Configuration Management** → **Settings**, the user clicks **Delete (🗑️)** on a row.		
Description	As an authorized user, I want to permanently remove an existing Setting after confirming the action so that obsolete or incorrect configurations are cleared from the system.		
Preconditions	**PRE-1.** User is authenticated.		

<table>
<tr><th>ID and Name</th><td colspan="3">UC75 Delete Setting</td></tr>
<tr><td></td><td colspan="3">PRE-2. User has permission to manage settings.
PRE-3. Target setting exists.</td></tr>
<tr><th>Postconditions</th><td colspan="3">POST- 1. On success, the setting is deleted and removed from the list; success toast displayed.
POST- 2. Deletion is recorded for audit.</td></tr>
<tr><th rowspan="5">Normal Flows</th><td>Step</td><td>By</td><td>Step Details</td></tr>
<tr><td>1</td><td>User</td><td>In Settings list, click Delete (🗑️) for the desired setting.</td></tr>
<tr><td>2</td><td>System</td><td>Show Confirm Delete dialog with key and scope.</td></tr>
<tr><td>3</td><td>User</td><td>Click Xóa (Delete) to confirm, or Hủy (Cancel) to abort.</td></tr>
<tr><td>4</td><td>System</td><td>If confirmed and allowed, delete the setting, refresh the table, and show green toast “Setting deleted successfully.”</td></tr>
<tr><th rowspan="3">Alternative Flows</th><td>Alt Step</td><td>Condition</td><td>Description</td></tr>
<tr><td>A1</td><td>User cancels</td><td>Click Hủy / Cancel; dialog closes; no change.</td></tr>
<tr><td>A2</td><td>Soft-delete policy (optional)</td><td>System archives the setting instead of hard delete; hidden from default view.</td></tr>
<tr><th>Exceptions</th><td colspan="3">• E1: Permission error → “You don’t have permission to delete settings.”
• E2: Network/Server error → show error and allow retry; keep dialog open.
• E3: Constraint/lock (e.g., protected system setting) → “This setting cannot be deleted.”
• E4: Concurrency conflict (already deleted/changed) → inform user and refresh list.</td></tr>
<tr><th>Frequency of Use</th><td colspan="3">Low–Medium</td></tr>
<tr><th>Business Rules</th><td colspan="3">BR-01, BR-02, BR-15, BR-16</td></tr>
<tr><th>Other Information</th><td colspan="3">• Show full key and scope in dialog; include a short, irreversible warning.
• Preserve table filters/pagination after returning from deletion.</td></tr>
<tr><th>Assumptions</th><td colspan="3">• Backend delete API enforces protections and returns clear errors; list view updates without full page reload.</td></tr>
</table>

Table 130 - Sign In Specification

3.2.2. Mobile

3.2.2.1 Sign In

a. Screen Layouts

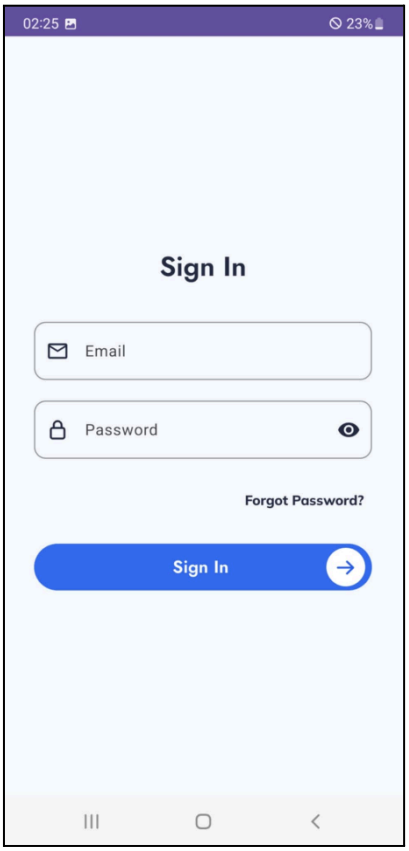


Figure 51: Sign in

No	Field Name	Field Type	Description
1	**Email**	Text Field	Input field for the user's registered email address. Includes an envelope icon.
2	**Password**	Password Field	Input field for the user's password, with an eye icon to show/hide password.
3	**Forgot Password?**	Text Link	Link that redirects the user to the password recovery process.
4	**Sign In Button**	Button	Action button to sign in, labeled "Sign In" with an arrow icon.

b. Use Case Specifications

ID and Name	UC – 01: Sign in		
Created By	Dungktn	Date Created	28/06/2025

ID and Name	UC – 01: Sign in		
Primary Actor	Student, Lecturer	Secondary Actors	System
Trigger	User opens the mobile app and chooses to log in.		
Description	The system allows users to sign in using their account credentials or via OAuth (e.g., Google). Upon successful authentication, users are redirected to their homepage.		
Preconditions	**PRE-1.** The user has already registered and has valid credentials.		
Postconditions	**POST-1.** The user is authenticated and redirected to the home screen.		
Normal Flow	**Step**	**By**	**Step Details**
	1	User	Launches the app and taps "Sign In."
	2	System	Displays login options: username/password or "Login with OAuth."
	3	User	Enters credentials or selects OAuth provider.
	4	System	Validates credentials or redirects to OAuth service.
	5	System	Upon successful authentication, stores session and redirects to home screen.
Alternative Flow	**Alt Step**	**Condition**	**Description**
	A1	User selects "Login with OAuth"	System redirects to OAuth provider. If successful, continue from Step 5.
Exceptions	**E1:** If login credentials are invalid, display: “Incorrect username or password.” **E2:** If authentication service is unavailable, show error: “Unable to connect. Please try again later.”		
Frequency of Use	High		
Business Rules	**BR-01, BR-02**		
Other Information	● Session tokens are stored securely in mobile storage. ● FaceID or biometric unlock may be added as an enhancement (not in this version).		
Assumptions	● The device is connected to the internet. ● Users have active accounts with correct roles assigned.		

Table 108: Sign in Specification

3.2.2.2 Sign Out

a. Screen Layouts

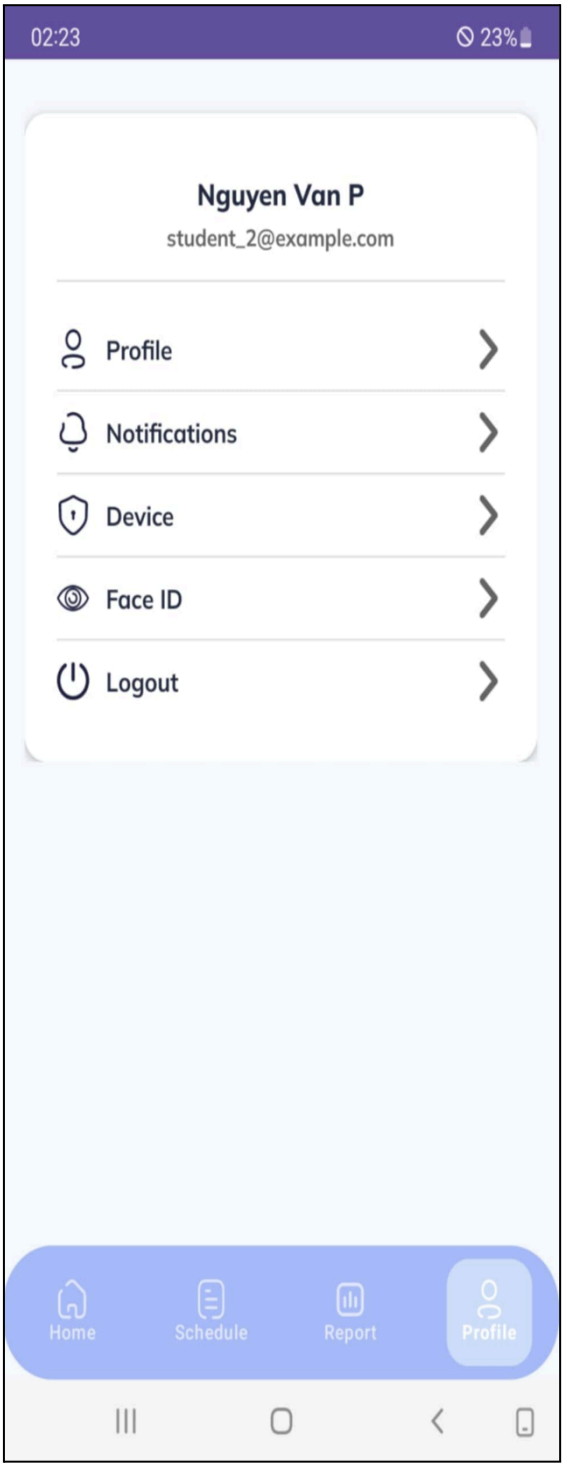


Figure 52: Sign out layout

No	Field Name	Field Type	Description
1	**Logout (menu item)**	Menu Item	Menu option in the Profile screen; tap to trigger immediate logout.

Table 109: Sign Out Field Table

b. Use Case Specifications

ID and Name	UC – 02: Sign Out		
Created By	Dungktn	Date Created	28/06/2025

ID and Name	UC – 02: Sign Out		
Primary Actor	Student, Lecturer	Secondary Actors	System
Trigger	User initiates a logout action from the settings menu.		
Description	The user can securely sign out from the application, ending the current session and clearing all locally stored authentication tokens.		
Preconditions	**PRE-1.** The user is currently signed in.		
Postconditions	**POST-1.** Session data is cleared. **POST-2.** User is returned to the login screen.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Opens settings and taps “Sign Out.”
	2	System	Displays confirmation dialog.
	3	User	Confirms action.
	4	System	Clears session data and stored tokens.
	5	System	Redirects user to login screen.
Alternative Flow	**Alt Step**	**Condition**	**Description**
	A1	User cancels confirmation	System closes dialog and keeps user signed in.
Exceptions	**E1:** If system fails to clear local storage, show: “Logout failed. Please try again.”		
Frequency of Use	Medium		
Business Rules	**BR-02**		
Other Information	• Tokens and session information must be securely removed from device storage. • Push notification tokens may be deregistered as part of logout.		
Assumptions	• User has access to settings and the app is functioning normally.		

Table 110: Sign out specification

3.2.2.3 View Student Setting

a. Screen Layouts

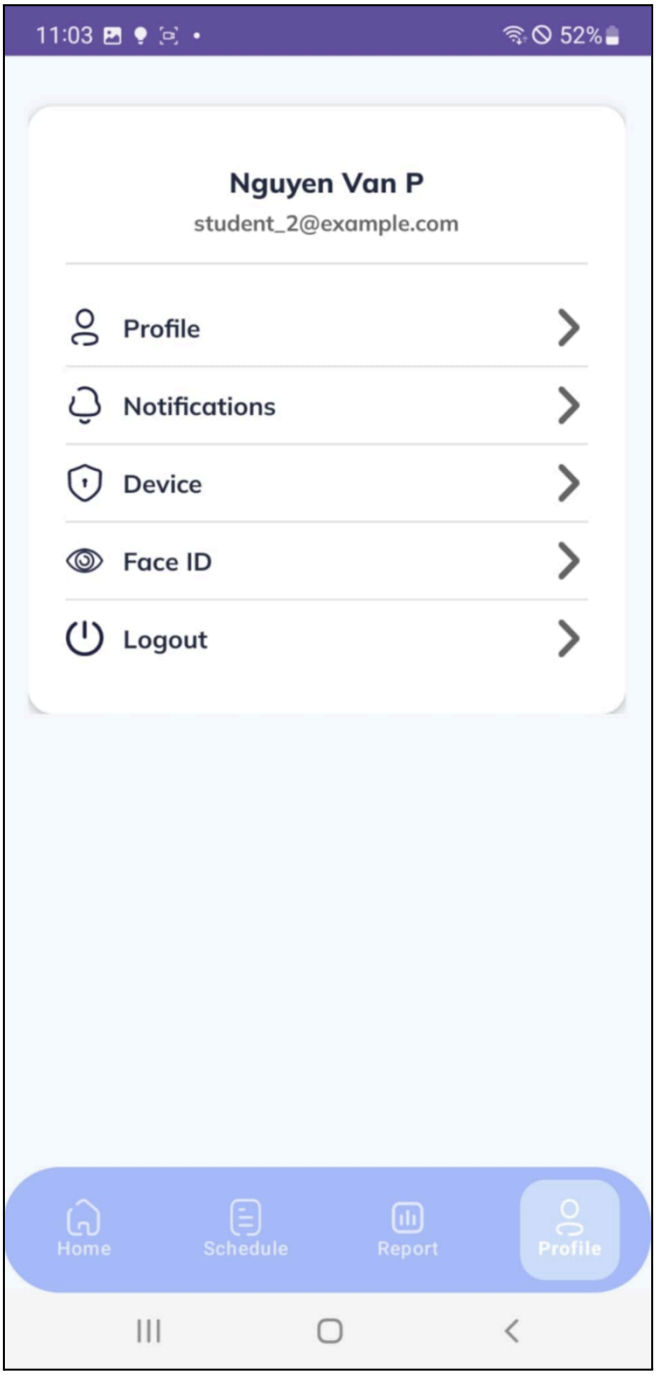


Figure 53: View Student Setting

No	Field Name	Field Type	Description
1	**User Name**	Text Label	Displays student's name (e.g., "Nguyen Van P").
2	**User Email**	Text Label	Displays student's email (e.g., student_2@example.com).
3	**Profile**	Menu Item	Option to view/edit personal profile.
4	**Notifications**	Menu Item	Option to access notification settings.
5	**Device**	Menu Item	Option to manage logged-in devices.

6	**Face ID**	Menu Item	Option to manage Face ID login settings.
7	**Logout**	Menu Item	Option to log out immediately.
8	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
9	**Bottom Tab – Schedule**	Navigation Tab	Shortcut to Schedule screen.
10	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
11	**Bottom Tab – Profile**	Navigation Tab	Highlights current Profile screen (active).

Table 111: View Student Setting Field Table

b. Use Case Specifications

ID and Name	**UC – 03: View Student Setting**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Student	**Secondary Actors**	System
Trigger	Students open the Settings tab from the navigation menu.		
Description	The student can access their settings, including personal profile, notification toggle, device and FaceID management, and logout option.		
Preconditions	**PRE-1.** Students are signed into the mobile app. **PRE-2.** The system displays all available settings for the student.		
Postconditions	**POST-1.** Accessibility permissions are granted, enabling automation scenarios.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Student	Taps the “Settings” icon in the bottom navigation bar.
	2	System	Displays the Settings screen with the following options: - View/Edit Profile - Manage Device - Manage FaceID - Notifications - Log out
	3	Student	Taps on any option to navigate to its detailed configuration page.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No internet connection	System shows cached data or displays offline message.
Exceptions	**E1.** If system cannot load settings, show error: “Unable to load settings. Please try again.”		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02, BR-03, BR-04, BR-12**		

ID and Name	UC – 03: View Student Setting
Other Information	● Navigation to sub-settings like FaceID and Device links directly to respective Use Cases (UC-08 to UC-14). ● Dark mode and app version info may also appear here (optional feature).
Assumptions	● All profile and system data is synced and accessible when settings are opened. ● Each student is linked to one active device and one FaceID (if registered).

Table 112: View Student Setting Specification

3.2.2.4 View Lecturer Setting

a. Screen Layouts

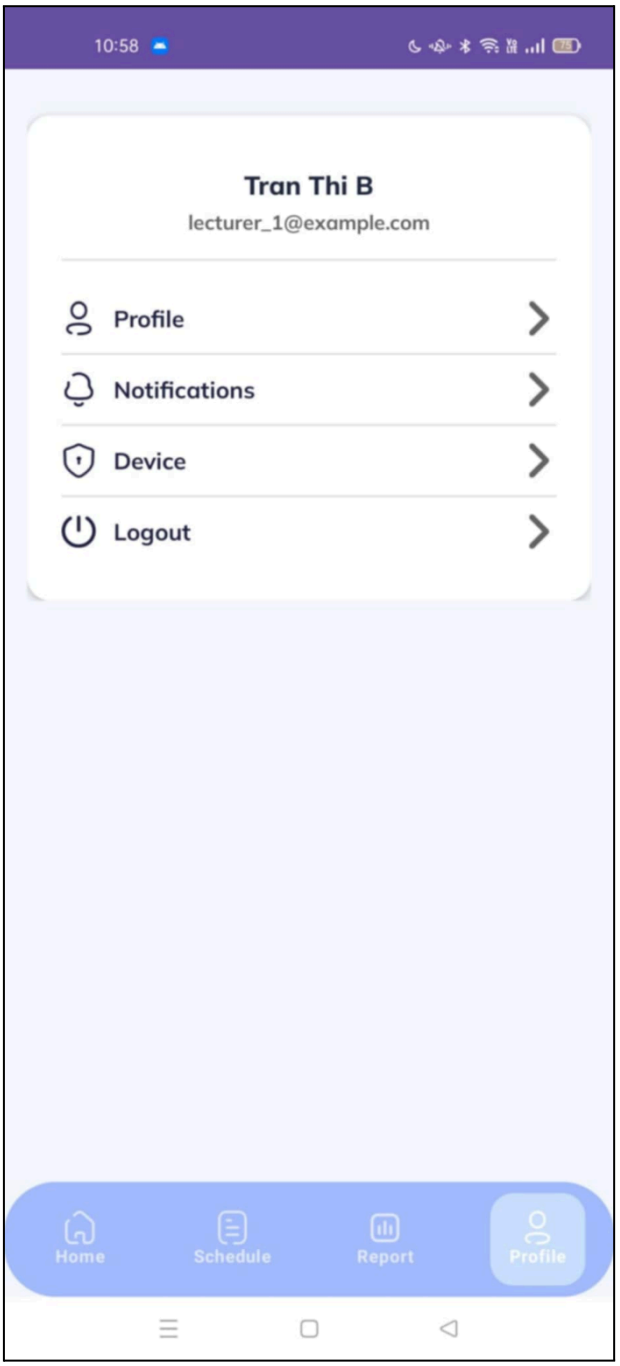


Figure 54: View Lecturer Setting

No	Field Name	Field Type	Description
1	**Lecturer Name**	Text Label	Displays lecturer's full name (e.g., "Tran Thi B").
2	**Lecturer Email**	Text Label	Displays lecturer's registered email (e.g., lecturer_1@example.com).
3	**Profile**	Menu Item	Option to view/edit lecturer's personal profile.
4	**Notifications**	Menu Item	Option to access notification settings.
5	**Device**	Menu Item	Option to view/manage logged-in devices.

6	**Logout**	Menu Item	Option to log out immediately.
7	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
8	**Bottom Tab – Schedule**	Navigation Tab	Shortcut to Schedule screen.
9	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
10	**Bottom Tab – Profile**	Navigation Tab	Highlights the active Profile tab (current screen).

Table 113: View Lecturer Setting Field Layout

b. Use Case Specifications

ID and Name	**UC – 04: View Lecturer Setting**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Lecturer	**Secondary Actors**	System
Trigger	Lecturer accesses the Settings page from the app.		
Description	Lecturer views and manages their account settings including profile, device, notifications, and logout options. Unlike students, lecturers do not manage FaceID here.		
Preconditions	**PRE-1.** Lecturer is signed into the mobile app.		
Postconditions	**POST-1.** The system loads and displays all relevant setting options.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	Taps on “Settings” from the bottom navigation bar.
	2	System	Displays a list of setting options: - View/Edit Profile - Manage Device - Notification Settings - Log Out
	3	Lecturer	Taps an item to open corresponding configuration screen.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No internet connection	System shows cached profile settings or a warning message.
Exceptions	• **E1:** If data fails to load, system displays: “Failed to load settings. Please try again.”		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02, BR-03, BR-12**		
Other Information	• The lecturer setting screen excludes FaceID sections. • Notification toggle is for receiving complaints and attendance-related alerts.		

ID and Name	UC – 04: View Lecturer Setting
Assumptions	• Lecturers use one device per session. • System settings are fetched from the server and cached locally.

Table 114: View Lecturer Setting Specification

3.2.2.5 View Profile

a. Screen Layouts

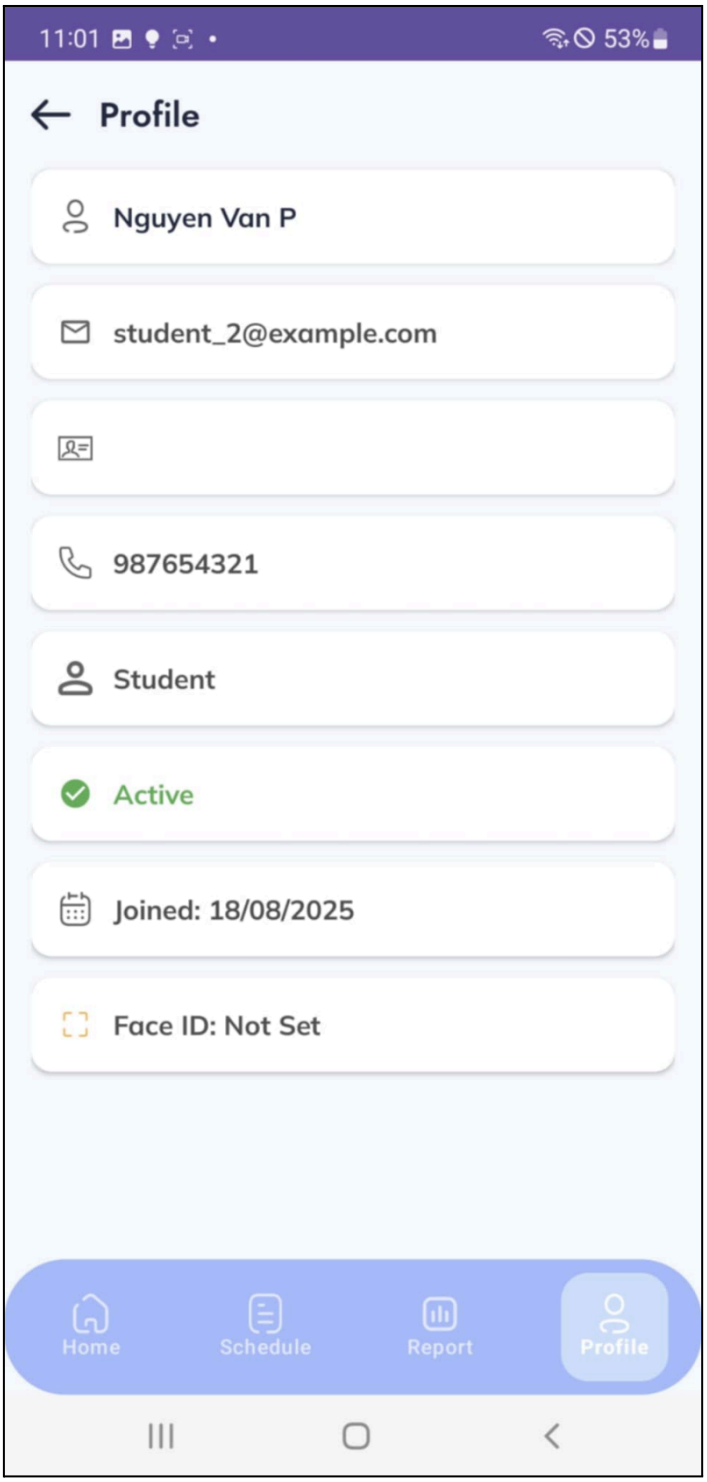


Figure 55: View Student Profile

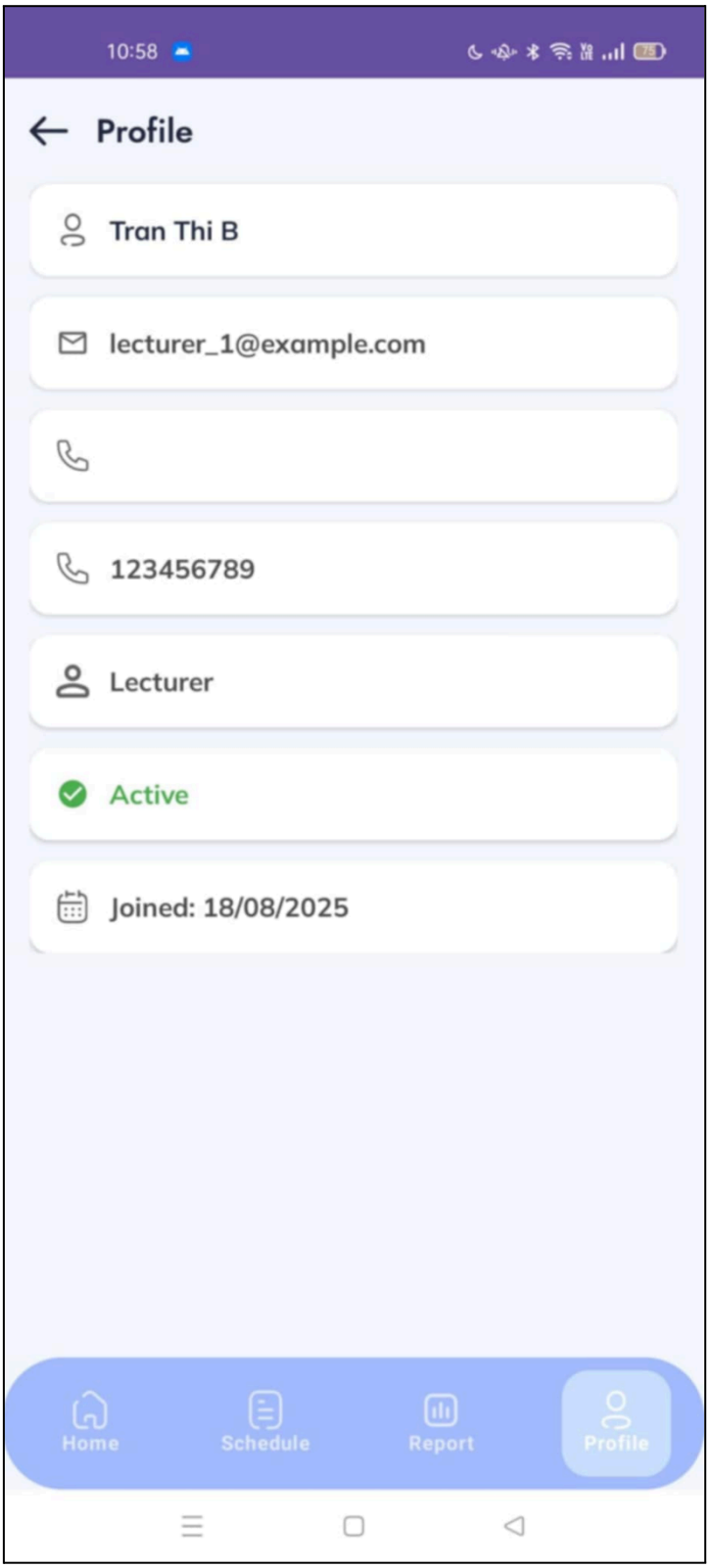


Figure 56: View Lecture Profile

No	Field Name	Field Type	Description
1	**User Name**	Text Label	Displays full name of the user (e.g., “Nguyen Van P”).

2	**User Email**	Text Label	Displays registered email of the user (e.g., student_2@example.com).
3	**Phone Number**	Text Label	Displays the user's phone number (e.g., 987654321).
4	**Role**	Text Label	Displays the role of the user (e.g., Student or Lecturer).
5	**Account Status**	Text Label	Shows whether the account is active/inactive (e.g., "Active").
6	**Joined Date**	Text Label	Displays the account creation/joined date (e.g., 18/08/2025).
7	**Face ID Status**	Text Label	Displays whether Face ID is set or not (e.g., "Not Set").
8	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
9	**Bottom Tab – Schedule**	Navigation Tab	Shortcut to Schedule screen.
10	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
11	**Bottom Tab – Profile**	Navigation Tab	Highlights the active Profile tab (current screen).

Table 115: View Profile Field Table

b. Use Case Specifications

ID and Name	**UC – 05: View Profile**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Lecturer, Student	**Secondary Actors**	System
Trigger	User selects the "View Profile" option from the settings menu.		
Description	Allows the user to view their personal profile information such as name, email, date of birth, phone number, gender, and role.		
Preconditions	**PRE-1.** User is signed into the mobile app.		
Postconditions	**POST-1.** The system displays the user's profile data.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Opens Settings and selects "View Profile."

ID and Name	UC – 05: View Profile		
	2	System	Retrieves and displays current profile info: full name, email, date of birth, phone number, gender, and role.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Profile not fully set up	System shows placeholders and suggests user to complete info.
Exceptions	• **E1:** If system fails to load data, show: "Unable to load profile. Please try again."		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02**		
Other Information	• Profile fields are read-only in this use case. Edit is handled in UC-07. • The screen may include a user avatar and "Edit Profile" button.		
Assumptions	• User information is synced and retrieved from the central database. • All fields are stored and updated by Admin or by user via edit flow.		

Table 116: View Profile Specification

3.2.2.6 View Device

a. Screen Layouts



Figure 57: View Device

No	Field Name	Field Type	Description
1	**Device Name**	Text Label	Shows the device name (e.g., TECNO BG6).
2	**Registered Date**	Text Label	Displays the date and time the device was registered (e.g., 18/08/2025 22:51).
3	**Model**	Text Label	Shows the device model (e.g., TECNO BG6).
4	**Manufacturer**	Text Label	Shows the device manufacturer (e.g., TECNO).
5	**Platform**	Text Label	Displays the operating system platform (e.g., Android).
6	**OS Version**	Text Label	Displays the version of the operating system (e.g., 13).
7	**App Version**	Text Label	Displays the installed application version (e.g., 1.0).
8	**Device Synced Button**	Button	Indicates sync status; pressed to re-sync or confirm sync with system.
9	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
10	**Bottom Tab – Schedule**	Navigation Tab	Shortcut to Schedule screen.
11	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.

Table 117: View Device Field Table

b. Use Case Specifications

ID and Name	UC – 06: View Device		
Created By	Dungktn	Date Created	28/06/2025
Primary Actor	Student, Lecturer	Secondary Actors	System
Trigger	User navigates to the “Device Management” section from settings.		
Description	Allows users to view information about their currently registered Bluetooth device used for attendance. Includes device name, ID, and registration date.		
Preconditions	**PRE-1.** User is signed in. **PRE-2.** User has previously registered at least one device.		
Postconditions	**POST-1.** System displays registered device information.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Navigates to Settings > Device Management.

ID and Name	UC – 06: View Device		
	2	System	Retrieves and displays registered device data: name, ID, registration date.
	3	User	Reviews the device status. If needed, can choose to request update (UC-10).
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No device is registered	System shows: “No device registered. Please register a device.”
Exceptions	• **E1:** If data retrieval fails, system shows: “Unable to load device information.”		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02, BR-03**		
Other Information	• Each user can have only one active device at a time. • Updating a device requires approval via UC-10.		
Assumptions	• Device info is synced from system database. • Device registration is linked to user authentication and attendance mechanism.		

Table 118: View Device Specification

3.2.2.7 Register Device

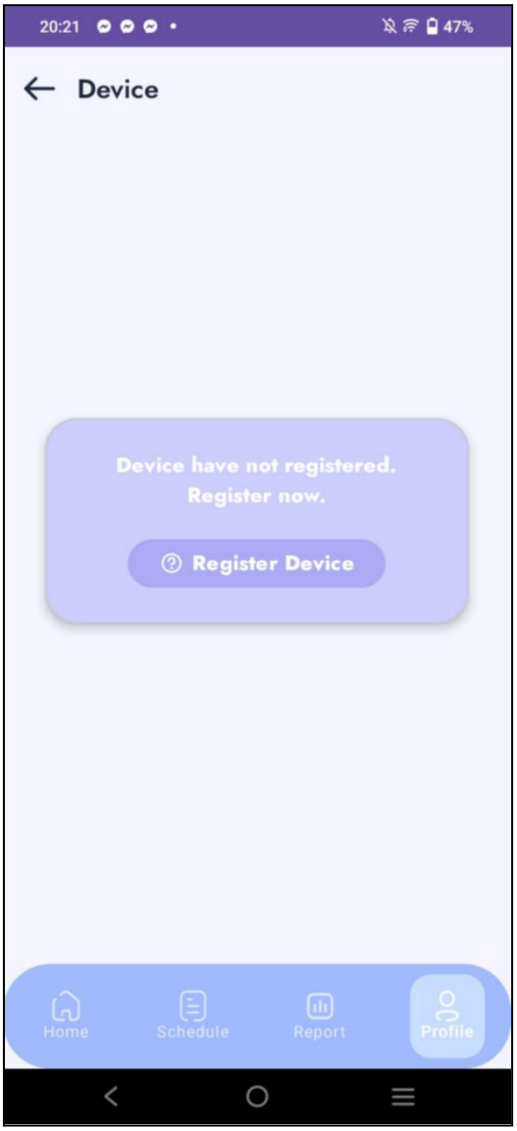


Figure 58: Register Device Layout

No	Field Name	Field Type	Description
1	**Device Not Registered Message**	Text Label	Displays message when no device is registered (e.g., “Device have not registered. Register now.”).
2	**Register Device Button**	Button	Action button to start device registration process.

Table 119: Register Device Field Table

b. Use Case Specifications

ID and Name	UC – 07: Register Device		
Created By	Dungktn	Date Created	28/06/2025
Primary Actor	Student, Lecturer	Secondary Actors	System
Trigger	Lecturer accesses the Settings page from the app.		

ID and Name	UC – 07: Register Device		
Description	Allow the user to register their Bluetooth-enabled mobile device for use in attendance sessions. The device will be linked to the user's identity and used for signal broadcasting or detection.		
Preconditions	**PRE-1.** User is signed in. **PRE-2.** Device is Bluetooth-capable.		
Postconditions	**POST-1.** Device is registered and stored in the system as the user's active device.		
Normal Flows	**Step**	**By**	**Step Details**
	1	User	Opens Settings > Device Management > Register Device.
	2	System	Detects available Bluetooth information (device ID, name).
	3	User	Confirms registration of detected device.
	4	System	Send registration data to server.
	5	System	Saves device as active and displays confirmation.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Device already registered	System shows warning: "You already have a registered device."
Exceptions	● **E1:** If Bluetooth is turned off, system prompts user: "Please enable Bluetooth to register device." ● **E2:** If server is unreachable, show: "Device registration failed. Please try again later."		
Frequency of Use	Low		
Business Rules	**BR-01, BR-02, BR-03, BR-10**		
Other Information	● Only one active device is allowed per user. ● Further changes to the device must go through UC-10 – Request Update Device.		
Assumptions	● Bluetooth is supported and available on the user's device. ● Admin approval may be required if the device is replaced.		

Table 120: Register Device Specification

3.2.2.8. View Face ID

a. Screen Layouts

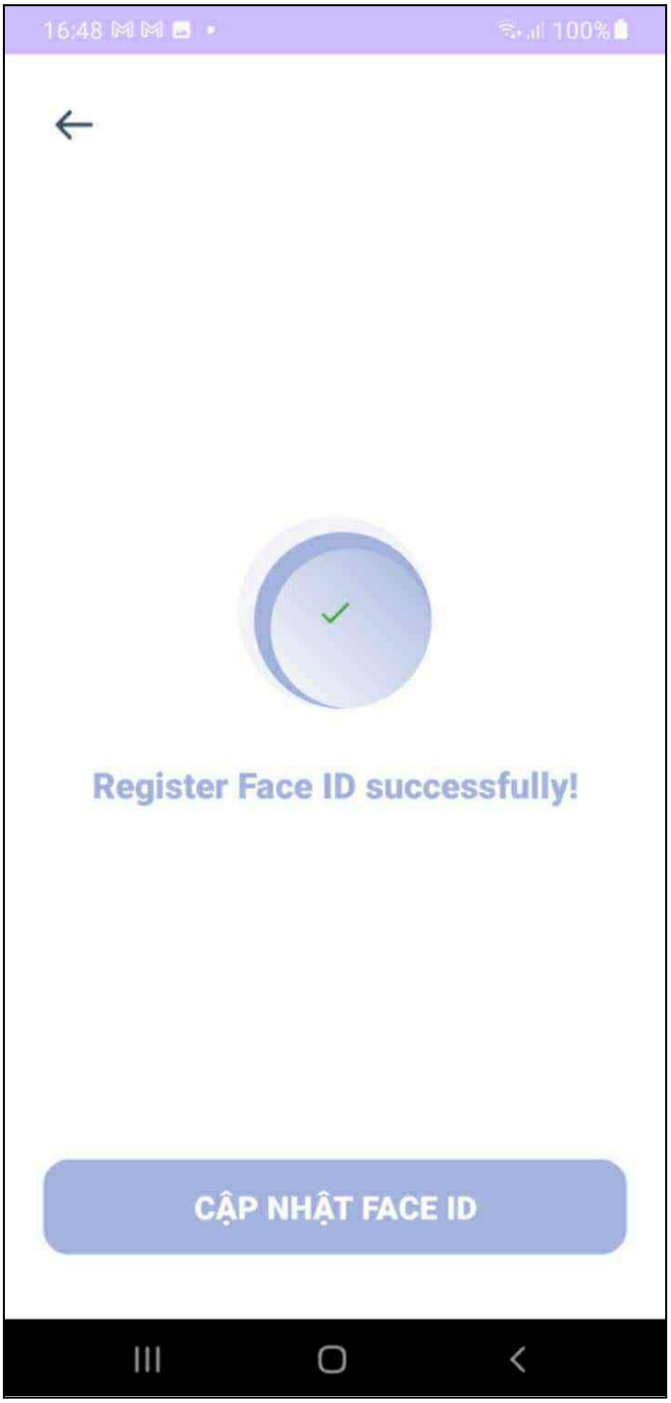


Figure 59: View Face ID Status Layout

No	Field Name	Field Type	Description
1	**Success Icon (Check mark)**	Graphic/Icon	Indicates that Face ID registration was successful.
2	**Success Message**	Text Label	Displays confirmation text (e.g., “Register Face ID successfully!”).
3	**Update Face ID Button**	Button	Allows the user to update/re-register Face ID if needed.

Table 121: View Face ID Field Table

b. Use Case Specifications

ID and Name	UC – 08: View FaceID		
Created By	Dungktn	Date Created	28/06/2025
Primary Actor	Student	Secondary Actors	System
Trigger	Student navigates to the FaceID section from the settings screen.		
Description	Student views their registered FaceID details, such as registration status and last update time. This FaceID is used for verifying identity in attendance sessions.		
Preconditions	**PRE-1.** User is signed in. **PRE-2.** A FaceID may or may not have been registered.		
Postconditions	**POST-1.** The system displays current FaceID status.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Student	Opens Settings > FaceID.
	2	System	Retrieves FaceID registration status and last update timestamp.
	3	Student	Reviews the displayed information.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	FaceID not registered	System displays: “FaceID not yet registered.” Prompts for UC-12.
Exceptions	• **E1:** If data retrieval fails, system shows: “Unable to load FaceID data. Please try again.”		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02, BR-04**		
Other Information	• Only one FaceID profile can be registered per student. • FaceID is used during in-class verification (e.g., UC-23). • Stored securely and protected under privacy policy.		
Assumptions	• The device camera and system support FaceID features. • FaceID is required for attendance verification flows.		

Table 122: View Face ID Specification

3.2.2.9. Register Face ID

a. Screen Layouts

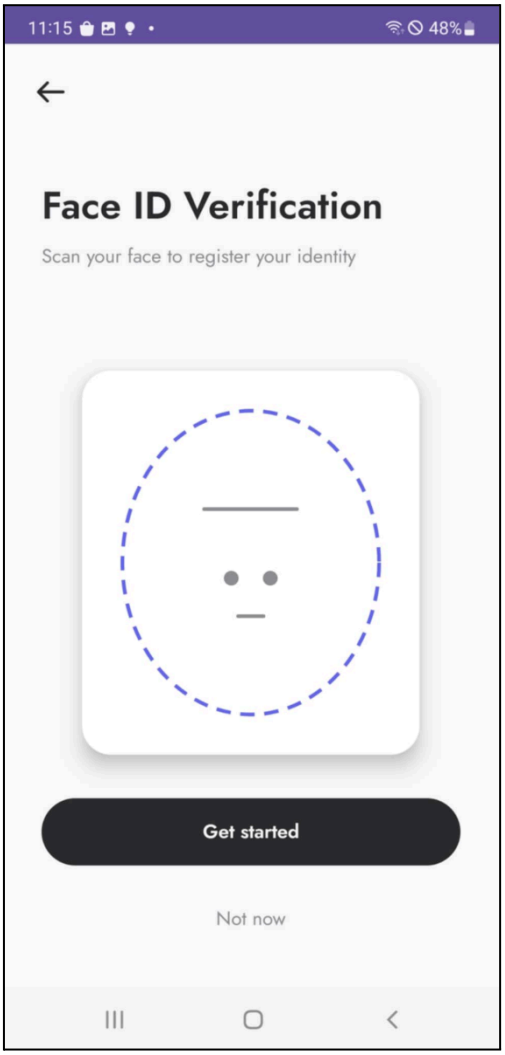


Figure 60: Register Face ID layout

No	Field Name	Field Type	Description
1	**Title – "Face ID Verification"**	Text Label	Screen title.
2	**Subtitle – instruction**	Text Label	Helper text: "Scan your face to register your identity."
3	**Face Frame/Illustration**	Graphic/Widget	Visual guide indicating where the face should align during capture.
4	**Get started**	Primary Button	Begins the Face ID registration flow (opens camera / capture steps).
5	**Not now**	Text Link	Skips Face ID registration and returns to the previous context.

Table 123: Register Face ID Field Table

b. Use Case Specifications

ID and Name	UC – 09: Register Face ID		
Created By	Dungktn	Date Created	28/06/2025
Primary Actor	Student	Secondary Actors	System
Trigger	Students choose to register FaceID from the FaceID setting screen.		
Description	Allow a student to register their facial data to be used for FaceID verification in attendance sessions.		
Preconditions	**PRE-1.** User is signed in. **PRE-2.** The device has a working front-facing camera. **PRE-3.** Student has not registered FaceID before.		
Postconditions	**POST-1.** FaceID is successfully stored and associated with student's profile.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Student	Navigates to Settings > FaceID > Register.
	2	System	Opens device camera with on-screen instruction for face capture.
	3	Student	Positions face and confirms capture.
	4	System	Processes and validates the image (e.g., quality, clarity).
	5	System	Saves FaceID and displays confirmation: "FaceID registered successfully."
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Face not properly detected	System prompts: "Face not clear. Please try again."
Exceptions	• **E1:** If camera permission is denied, show: "Please enable camera to register FaceID." • **E2:** If registration fails due to network/server error, show: "Failed to register. Try again later."		
Frequency of Use	Low		
Business Rules	**BR-01, BR-02, BR-04**		
Other Information	• FaceID data may be encrypted and stored securely. • Student can update FaceID later using UC-13 – Update FaceID.		
Assumptions	• Device supports biometric capture and has internet connection. • Students agree to privacy policy and biometric data usage terms.		

Table 124: Register Face ID Specification

3.2.2.10 Update Face ID

a. Screen Layouts

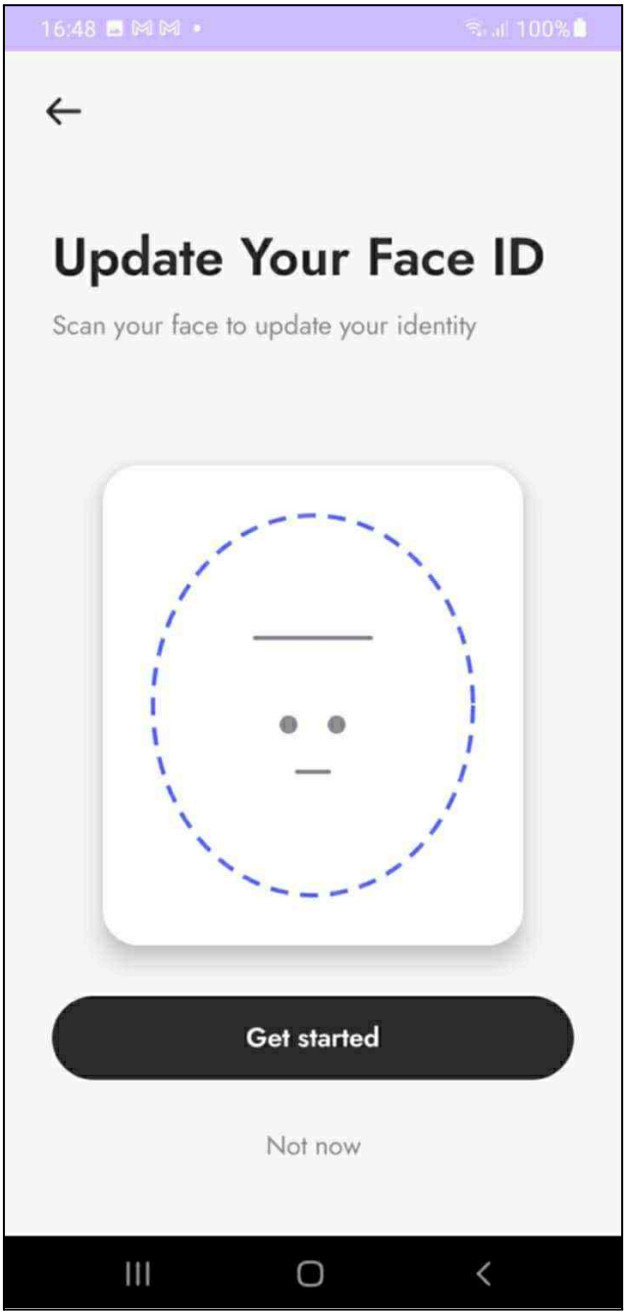


Figure 61: Update Face ID Layout

No	Field Name	Field Type	Description
1	**Title – "Update Your Face ID"**	Text Label	Screen title.
2	**Subtitle – instruction**	Text Label	Instruction text: "Scan your face to update your identity."
3	**Face Frame/Illustration**	Graphic/Widget	Visual guide showing where to position the face during scan.

4	**Get started**	Primary Button	Begins the Face ID update process (camera capture).

Table 125: Update Face ID Field Table

b. Use Case Specifications

ID and Name	**UC – 10: Update Face ID**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Student	**Secondary Actors**	System
Trigger	Students choose to re-register FaceID from the FaceID settings screen.		
Description	Allow a student to update their previously registered FaceID data. This may be required if their facial features have changed, or if the original registration failed.		
Preconditions	**PRE-1.** User is signed in. **PRE-2.** FaceID has already been registered. **PRE-3.** Device has a functioning front-facing camera.		
Postconditions	**POST-1.** The system displays current FaceID status.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Student	Opens Settings > FaceID > Update FaceID.
	2	System	Launches front-facing camera with on-screen guidance.
	3	Student	Positions face correctly and confirms image capture.
	4	System	Processes the image and checks quality (e.g., lighting, clarity).
	5	System	Saves updated FaceID and shows success message.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Image quality is insufficient	System prompts user to recapture: “Face not clear. Try again.”
Exceptions	● **E1:** If FaceID has not been registered, system shows: “No FaceID found. Please register first.” ● **E2:** If camera permission is denied, show: “Please allow camera access to update FaceID.” ● **E3:** If update fails due to network/server issue, system shows: “Update failed. Try again later.”		
Frequency of Use	Low		
Business Rules	**BR-01, BR-02, BR-04**		
Other Information	● FaceID update completely replaces the previous one. ● Timestamp of update may be stored for system audit.		
Assumptions	● The student’s device has a working camera. ● Students accept privacy policies related to facial data usage.		

Table 126: Update Face ID Specification

3.2.2.11 View Student Home Page

a. Screen Layouts

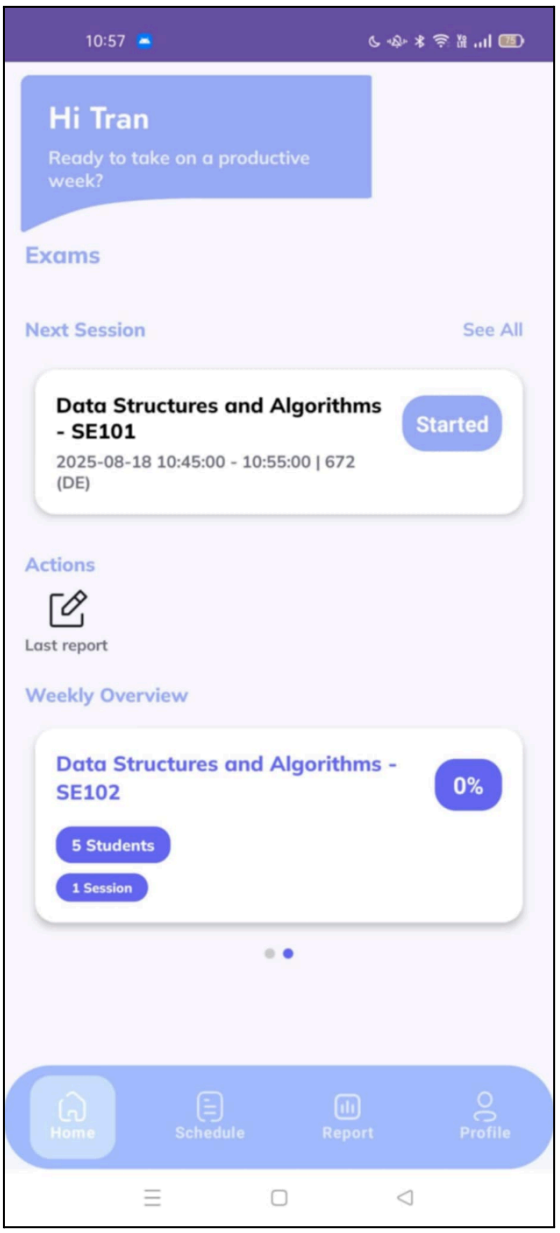


Figure 62: Homepage layout

No	Field Name	Field Type	Description
No	**Field Name**	Field Type	Description
1	**Greeting Section**	Text Label	Displays student's name and welcome message.
2	**Next Session**	Card Widget	Shows details of the upcoming class/session.
3	**See All**	Text Link	Navigates to full list of sessions/exams.
4	**Last Report**	Menu Item	Shortcut to view the latest report.
5	**Weekly Overview**	Card Widget	Displays weekly class information and progress.

6	**Bottom Tab – Home**	Navigation Tab	Active tab for Home screen.
7	**Bottom Tab – Schedule**	Navigation Tab	Shortcut to Schedule screen.
8	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
9	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.
No	**Field Name**	Field Type	Description

Table 127: View Student Home Page Field Table

b. Use Case Specifications

ID and Name	**UC – 11: View Student Home Page**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Student	**Secondary Actors**	System
Trigger	Students open the app or navigates to the home screen.		
Description	Displays an overview of the student's day, including today's class schedule, important announcements, and attendance warnings (if any).		
Preconditions	**PRE-1.** Students are signed in. **PRE-2.** Students have classes or announcements scheduled for the current date.		
Postconditions	**POST-1.** System shows personalized dashboard content.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Student	Launches app or taps "Home" from bottom navigation bar.
	2	System	Fetches today's schedule and latest announcements.
	3	System	Displays home screen with: – Today's class list – Announcements – Attendance warnings (if any)
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No classes or data for today	System shows message: "No scheduled classes or announcements."
Exceptions	• **E1:** If data fails to load due to network issues, show: "Unable to load homepage data. Please check your connection."		
Frequency of Use	High		
Business Rules	**BR-01, BR-02, BR-14, BR-17**		
Other Information	• This is the landing page after login. • Displays notification badge for pending complaints or session alerts.		

ID and Name	UC – 11: View Student Home Page
Assumptions	• Backend supports real-time sync of schedule and notification data. • Classes and announcements are time-sensitive and auto-refresh daily.

Table 128: View Student Home Page Specification

3.2.2.12 View Student Schedule

a. Screen Layouts

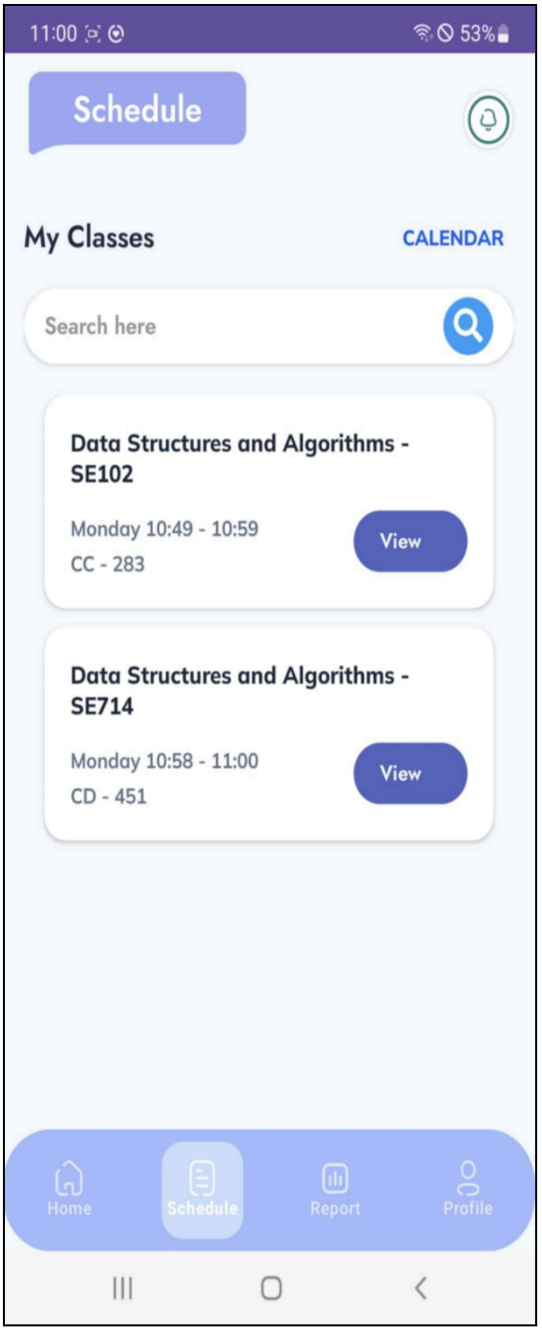


Figure 63: View Student Schedule Layout

No	Field Name	Field Type	Description
1	**Title – Schedule**	Text Label	Screen title.
2	**Calendar Switch**	Text Link/Button	Allows switching to calendar view.

3	**Search Bar**	Search Field	Search for a class by name/code.
4	**Class List**	Card Widget	Displays class information (course name, code, time, room). Each card includes a View button.
5	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
6	**Bottom Tab – Schedule**	Navigation Tab	Active tab, showing current Schedule screen.
7	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
8	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 129: View Student Schedule Field Table

b. Use Case Specifications

ID and Name	**UC – 12: View Student Schedule**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Student	**Secondary Actors**	System
Trigger	Student navigates to the “Schedule” section from the mobile app.		
Description	Displays an overview of the student’s day, including today’s class schedule, important announcements, and attendance warnings (if any).		
Preconditions	**PRE-1.** Students are signed in. **PRE-2.** Schedule has been uploaded and linked to the student.		
Postconditions	**POST-1**. Schedule information is retrieved and displayed.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Student	Taps “Schedule” from bottom navigation or Home screen.
	2	System	Loads current week’s timetable based on student’s class data.
	3	System	Displays each session with subject name, time, room, and status (e.g., upcoming, ongoing, completed).
	4	Student	Taps on a class item to view session details (linked to UC-22).
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No schedule available	System displays: “No classes scheduled for this period.”
Exceptions	• **E1:** If schedule fails to load, system displays: “Unable to retrieve schedule. Please try again.”		
Frequency of Use	High		
Business Rules	**BR-01, BR-02**		

ID and Name	UC – 12: View Student Schedule
Other Information	• May include filters (day/week view) and indicators for attendance status. • Supports redirection to attendance flow if session is active.
Assumptions	• System integrates schedule from central timetable or admin-imported files. • Student is assigned to at least one active class.

Table 130: View Student Schedule Specification

3.2.2.13 View Student Calendar

a. Screen Layouts



Figure 64: View Student Calendar Layout

No	Field Name	Field Type	Description

1	**Title – My Classes**	Text Label	Screen title.
2	**Calendar Widget**	Calendar UI	Displays monthly calendar with selectable dates.
3	**Date Indicator**	Highlight	Highlights the selected date (e.g., 18).
4	**Class List**	List Item	Shows sessions scheduled for the selected date (course name, code, time, room).
5	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
6	**Bottom Tab – Schedule**	Navigation Tab	Active tab, showing current Schedule/Calendar screen.
7	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
8	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 131: View Student Calendar Field Table

b. Use Case Specifications

ID and Name	**UC – 13: View Student Calendar**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Student	**Secondary Actors**	System
Trigger	Student access the calendar section from the app.		
Description	Allow students to view their class schedule in a calendar format, providing a monthly or weekly overview of all upcoming and past sessions.		
Preconditions	**PRE-1.** Students are signed in. **PRE-2.** Class schedule is available.		
Postconditions	**POST-1.** Calendar view is populated with class sessions.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Student	Taps “Calendar” icon/tab from the bottom menu or navigation drawer.
	2	System	Loads calendar view (default to current month).
	3	System	Displays all scheduled classes in date blocks with subject name and time.
	4	Student	Taps on a date to see daily class details and navigate to session info.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No events for selected date	System shows: “No classes scheduled for this date.”

ID and Name	UC – 13: View Student Calendar
Exceptions	• **E1:** If calendar data fails to load, system shows: "Unable to retrieve calendar. Please try again later."
Frequency of Use	High
Business Rules	**BR-01, BR-02**
Other Information	• May include color-coding for past, current, and future classes. • Calendar view supports navigation between months and weeks.
Assumptions	• Calendar data is synced from the same source as the weekly schedule (UC-16). • Time zone and date localization are handled by the app.

Table 132: View Student Calendar Specification

3.2.2.14 View Lecturer Home Page

a. Screen Layouts

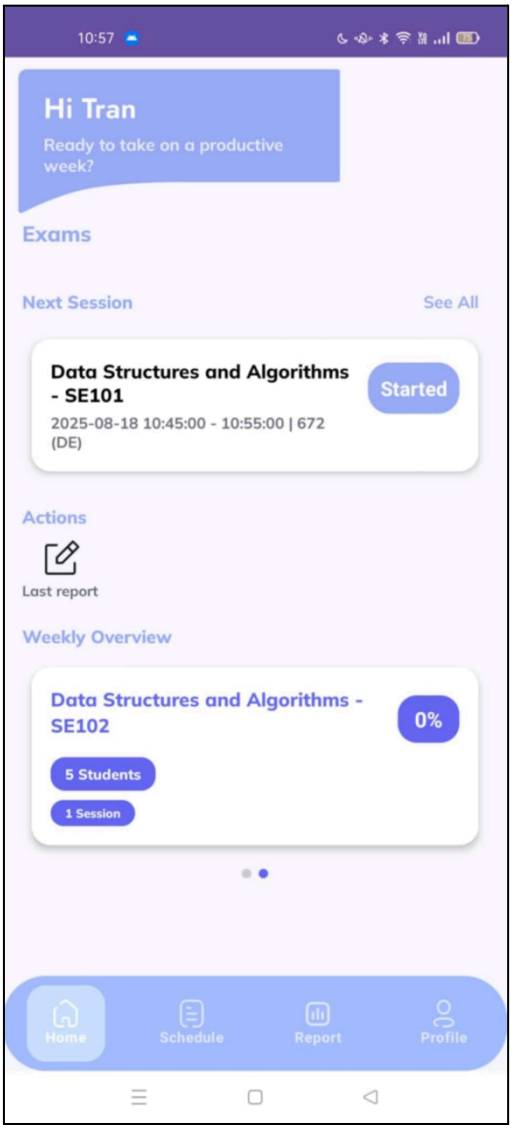


Figure 65: View Lecturer Home Page Layout

No	Field Name	Field Type	Description

1	**Greeting Section**	Text Label	Displays lecturer's name and welcome message.
2	**Next Session**	Card Widget	Shows details of the upcoming class/session.
3	**See All**	Text Link	Navigates to full list of sessions/exams.
4	**Last Report**	Menu Item	Shortcut to view the latest report.
5	**Weekly Overview**	Card Widget	Displays weekly class overview (subject, progress, student count, sessions).
6	**Bottom Tab – Home**	Navigation Tab	Active tab for Home screen.
7	**Bottom Tab – Schedule**	Navigation Tab	Shortcut to Schedule screen.
8	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
9	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 133: View Lecturer Home Page Field Table

b. Use Case Specifications

ID and Name	**UC – 19: View Lecturer Home Page**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Lecturer	**Secondary Actors**	System
Trigger	The lecturer opens the app or navigates to the home screen.		
Description	Displays an overview of today's teaching schedule, class status, student attendance summaries, and pending tasks such as complaints or sessions requiring finalization.		
Preconditions	**PRE-1.** Lectures are signed in. **PRE-2.** The lecturer has assigned classes for the day.		
Postconditions	**POST-1.** Home dashboard is shown with relevant teaching information.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	Opens the app or taps "Home" from navigation.
	2	System	Retrieves today's teaching schedule and attendance overview.
	3	System	Displays: – List of classes – Student attendance rates – Action items (e.g., complaints, sessions to finalize)
Alternative Flows	**Alt Step**	**Condition**	**Description**

ID and Name	UC – 19: View Lecturer Home Page		
	A1	No classes assigned for today	System shows message: “No classes scheduled for today.”
Exceptions	• **E1:** If data fails to load, system shows: “Unable to retrieve teaching data. Please try again.”		
Frequency of Use	High		
Business Rules	**BR-01, BR-02, BR-14**		
Other Information	• Entry point for accessing class sessions, launching attendance, and handling complaints. • May include summary KPIs like number of students absent today.		
Assumptions	• Lecturer’s classes are preloaded and synced via admin-imported schedule (UC-39). • Data refreshes automatically when app is opened.		

Table 134: View Lecturer Home Page Specification

3.2.2.15 View Lecturer Schedule

a. Screen Layouts

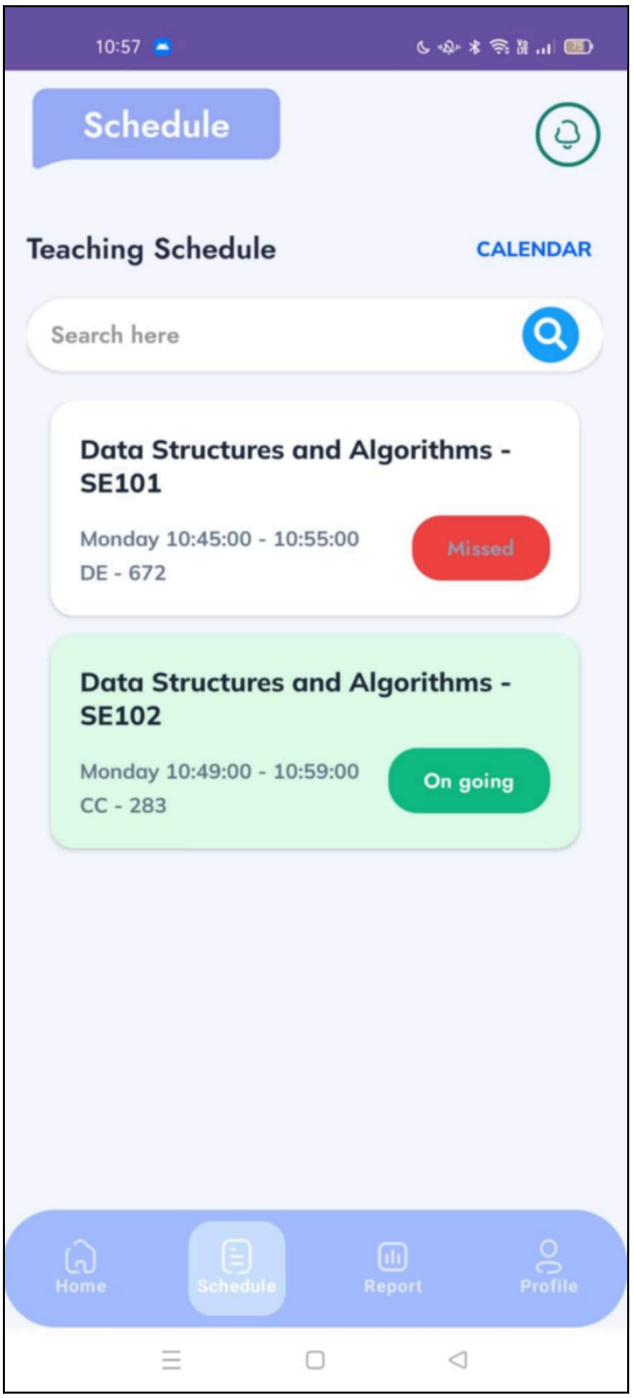


Figure 66: View Lecturer Schedule Layout

No	Field Name	Field Type	Description
1	**Title – Schedule**	Text Label	Screen title.
2	**Calendar Switch**	Text Link/Button	Allows switching to calendar view.
3	**Search Bar**	Search Field	Search for a teaching session/class.
4	**Teaching Schedule List**	Card Widget	Displays scheduled classes for the lecturer (course, time, room). Each card shows a status badge (e.g., Missed, On going).

5	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
6	**Bottom Tab – Schedule**	Navigation Tab	Active tab, showing current Schedule screen.
7	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
8	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.
1	**Title – Schedule**	Text Label	Screen title.

Table 135: View Lecturer Schedule Field Table

b. Use Case Specifications

ID and Name	**UC – 20: View Lecturer Schedule**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Lecturer	**Secondary Actors**	System
Trigger	Lecturer navigates to the “Schedule” section in the mobile app.		
Description	Allow lecturers to view their teaching schedule in a daily or weekly list, including class times, room information, and links to session detail or start session directly.		
Preconditions	**PRE-1.** Lectures are signed in. **PRE-2.** The schedule has been imported and linked to the lecturer’s account.		
Postconditions	**POST-1.** Teaching schedule is displayed.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	Opens the “Schedule” tab from the bottom navigation bar.
	2	System	Loads the current week’s teaching schedule from server.
	3	System	Displays subjects by day and time, including room info and session options.
	4	Lecturer	Taps a session to: – View session details (UC-27) – Start session (UC-35)
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No assigned classes this week	System shows: “No classes scheduled for this week.”
Exceptions	• **E1:** If schedule data fails to load, system shows: “Unable to retrieve schedule. Please try again later.”		
Frequency of Use	High		
Business Rules	**BR-01, BR-02**		
Other Information	• May include filtering by week/day and direct links to class actions.		

ID and Name	UC – 20: View Lecturer Schedule
	• Designed for fast access during class hours.
Assumptions	• Schedule data is synced from centralized timetable managed by Admin (via UC-39/40). • Internet connection required for real-time session launching.

Table 136: View Lecturer Schedule Specification

3.2.2.16 View Lecturer Calendar

a. Screen Layouts

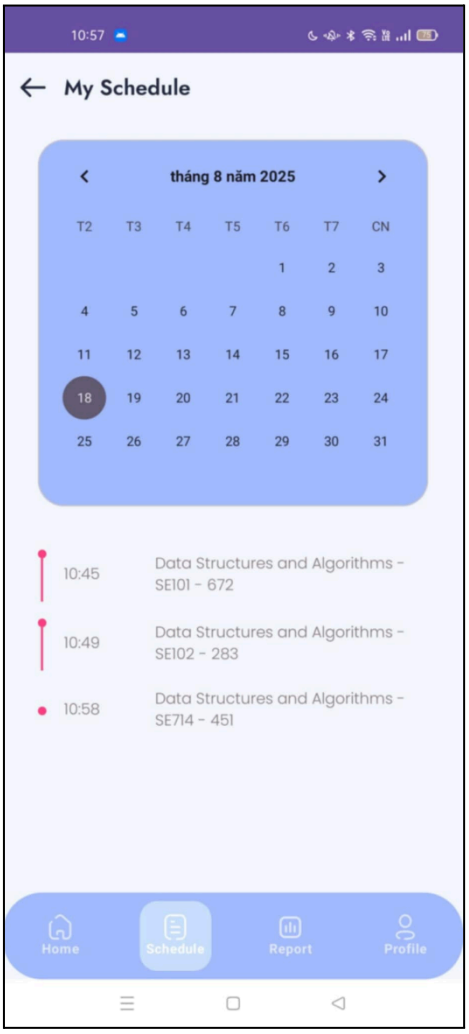


Figure 67: View Lecturer Calendar Layout

No	Field Name	Field Type	Description
1	**Title – My Schedule**	Text Label	Screen title.
2	**Calendar Widget**	Calendar UI	Monthly calendar allowing lecturers to pick a date.
3	**Date Indicator**	Highlight	Marks the selected date (e.g., 18).

4	**Class List**	List Item	Shows classes for the selected date (time, course code, room).
5	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
6	**Bottom Tab – Schedule**	Navigation Tab	Active tab, showing current Schedule/Calendar screen.
7	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
8	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 137: View Lecturer Calendar Field Table

b. Use Case Specifications

ID and Name	**UC – 21: View Lecturer Calendar**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Lecturer	**Secondary Actors**	System
Trigger	Lecturer opens the “Calendar” section from the app.		
Description	Provides lecturers with a full calendar view of their scheduled classes and events by day, week, or month, with the ability to navigate and view session details for any date.		
Preconditions	**PRE-1.** Lectures are signed in. **PRE-2.** Class schedule has been assigned and synchronized.		
Postconditions	**POST-1.** Teaching schedule is displayed.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	Taps “Calendar” icon/tab from main navigation.
	2	System	Loads calendar view with default to current month.
	3	System	Populates calendar with scheduled sessions, color-coded by status (upcoming, completed, canceled).
	4	Lecturer	Taps on a date to see: – Class list – Access session detail (UC-27) – Attendance launch (UC-35)
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No classes on selected date	System shows: “No classes on this date.”
Exceptions	• **E1:** If calendar data fails to load, show: “Unable to load calendar. Please check your connection.”		
Frequency of Use	Medium		

ID and Name	UC – 21: View Lecturer Calendar
Business Rules	**BR-01, BR-02**
Other Information	• View options: day, week, month. • May integrate reminders for pending attendance or unresolved complaints. • Supports visual indicators for attendance status (e.g., not started, in progress, completed).
Assumptions	• Calendar data is synced with master schedule from admin import (UC-39). • App handles local time zone and localization for international users.

Table 138: View Lecturer Calendar Specification

3.2.2.17 View Personal Attendance Report

a. Screen Layouts

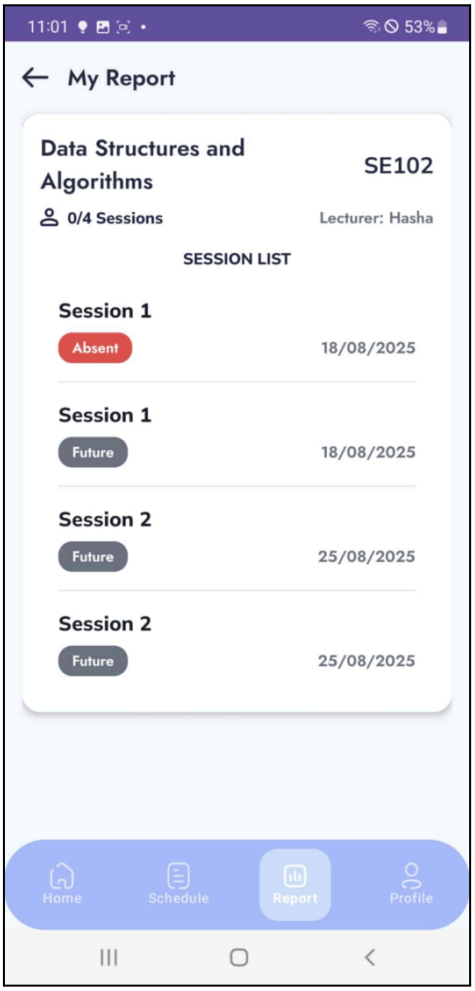


Figure 68: View Lecturer Calendar Layout

No	Field Name	Field Type	Description
1	**Title – My Report**	Text Label	Screen title.
2	**Course Info**	Text Label	Displays course name, code (e.g., SE102).

3	**Lecturer Info**	Text Label	Displays lecturer's name (e.g., Lecturer: Husha).
4	**Total Sessions**	Text Label	Shows total number of sessions (e.g., 0/4 Sessions).
5	**Session List**	List Item	Displays each session with status (Absent/Future/Present), session number, and date.
6	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
7	**Bottom Tab – Schedule**	Navigation Tab	Shortcut to Schedule screen.
8	**Bottom Tab – Report**	Navigation Tab	Active tab, showing current Report screen.
9	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 139: View Personal Attendance Report Field Table

b. Use Case Specifications

ID and Name	**UC – 14: View Personal Attendance Report**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Student	**Secondary Actors**	System
Trigger	Student selects the "Attendance" or "Report" section in the app.		
Description	Allow students to view their attendance summary, including attendance percentage by subject and semester, status breakdown, and warnings if thresholds are breached.		
Preconditions	**PRE-1.** Students are signed in. **PRE-2.** Attendance data has been recorded and synced.		
Postconditions	**POST-1.** The report is displayed with up-to-date attendance metrics.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Student	Opens "Attendance Report" section from navigation or profile.
	2	System	Retrieves attendance records across subjects for current semester.
	3	System	Populates calendar with scheduled sessions, color-coded by status (upcoming, completed, canceled). Displays: – % attendance per subject – Total attended / total sessions – Warning icon if attendance is below threshold

ID and Name	UC – 14: View Personal Attendance Report		
	4	Student	Taps on a subject to view session-level details (linked to UC-22).
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No attendance data available	System shows: “No attendance data available yet.”
Exceptions	• **E1:** If data retrieval fails, show: “Unable to load report. Please try again later.”		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02, BR-14, BR-15**		
Other Information	• Low attendance warnings may be triggered based on thresholds set by Admin (UC-36). • Session-level view includes history of attended, missed, and excused sessions.		
Assumptions	• Backend attendance data is continuously updated from sessions. • System stores attendance by student-subject-semester.		

Table 140: View Personal Attendance Report Specification

3.2.2.18 Join Attendance Session

a. Screen Layouts

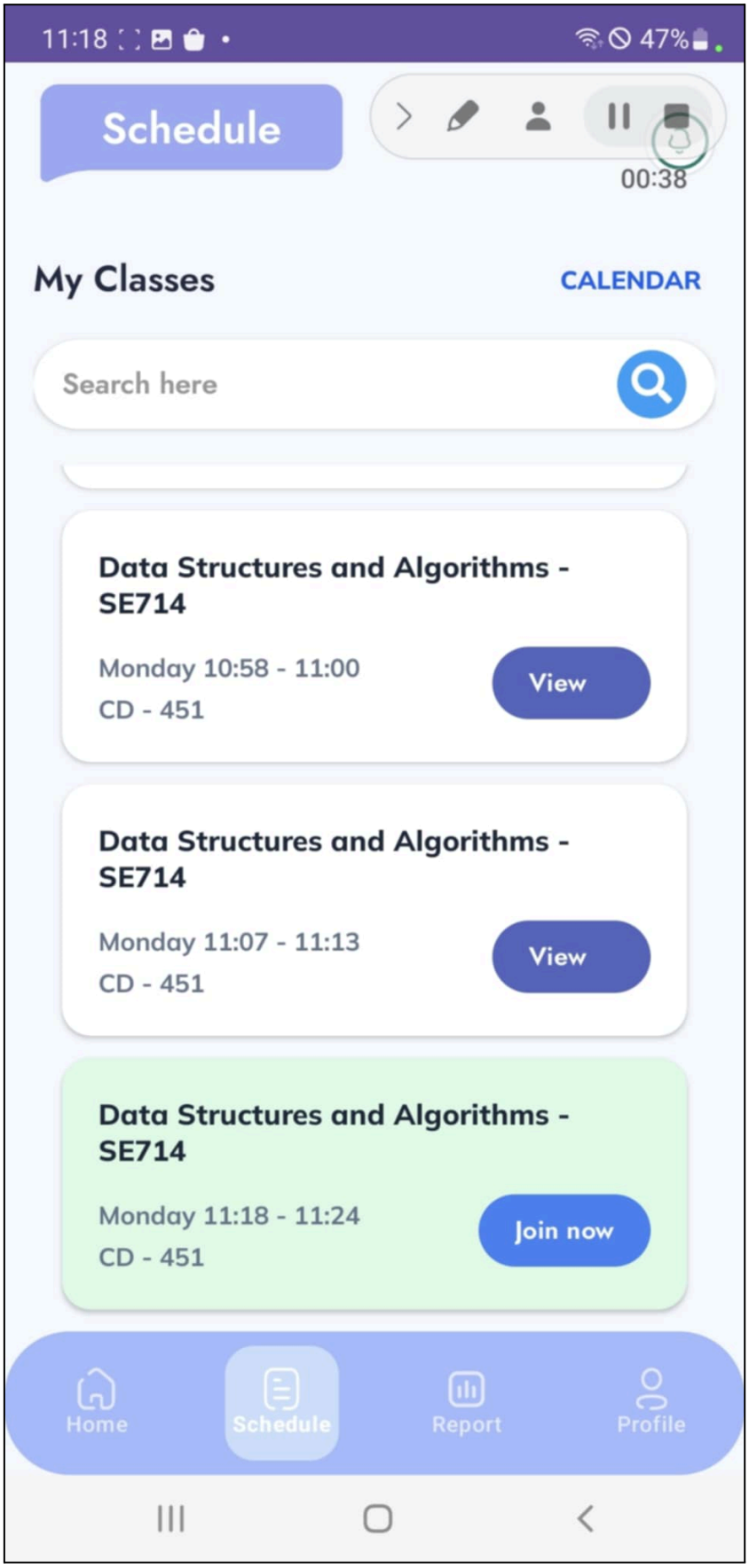


Figure 69: View Personal Attendance Report Layout

No	Field Name	Field Type	Description

1	**Title – Schedule**	Text Label	Screen title.
2	**Calendar Switch**	Text Link/Button	Allows switching to calendar view.
3	**Search Bar**	Search Field	Search for a class by name/code.
4	**Class List**	Card Widget	Displays session details (course name, code, time, room). Each card includes a button depending on the session status.
5	**View Button**	Button	Available for upcoming or past sessions, used to view details.
6	**Join Now Button**	Primary Button	Active for sessions currently ongoing, allowing the student to join attendance.
7	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
8	**Bottom Tab – Schedule**	Navigation Tab	Active tab, showing current Schedule screen.
9	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
10	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 141: Join Attendance Session Field Table

b. Use Case Specifications

ID and Name	**UC – 16: Join Attendance Session**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Student	**Secondary Actors**	System
Trigger	Student taps “Join Session” when class begins and signal is detected.		
Description	Allow students to join an ongoing attendance session via Bluetooth signal detection. System confirms the student’s presence and records the attempt.		
Preconditions	**PRE-1.** Students are signed in. **PRE-2.** A valid attendance session is active. **PRE-3.** Student’s device is registered and Bluetooth is enabled.		
Postconditions	**POST-1.** Attendance join request is submitted and pending FaceID if required.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Student	Taps “Join Attendance” on the active class session.
	2	System	Scans for lecturer’s signal (broadcasted via Bluetooth).

ID and Name	UC – 16: Join Attendance Session		
	3	System	Confirms signal received and device is valid.
	4	System	Logs the student's presence.
	5	System	Prompts for FaceID verification (UC-23) if configured.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No signal detected	System shows: "No session signal detected nearby."
	A2	Session not started yet	System shows: "Attendance session has not started."
Exceptions	• **E1:** If Bluetooth is turned off, system shows: "Please enable Bluetooth to join session." • **E2:** If device is not registered, show: "Your device is not registered for attendance."		
Frequency of Use	High		
Business Rules	**BR-01, BR-02, BR-03, BR-05, BR-04 (if FaceID is required)**		
Other Information	• This flow may be triggered manually or as part of automatic detection. • May include a countdown timer if joining is only available for limited time.		
Assumptions	• Bluetooth signal broadcast is stable and in-range. • Admin or lecturer has started the session using UC-35.		

Table 142: Join Attendance Session Specification

3.2.2.19 View Session Detail

a. Screen Layouts

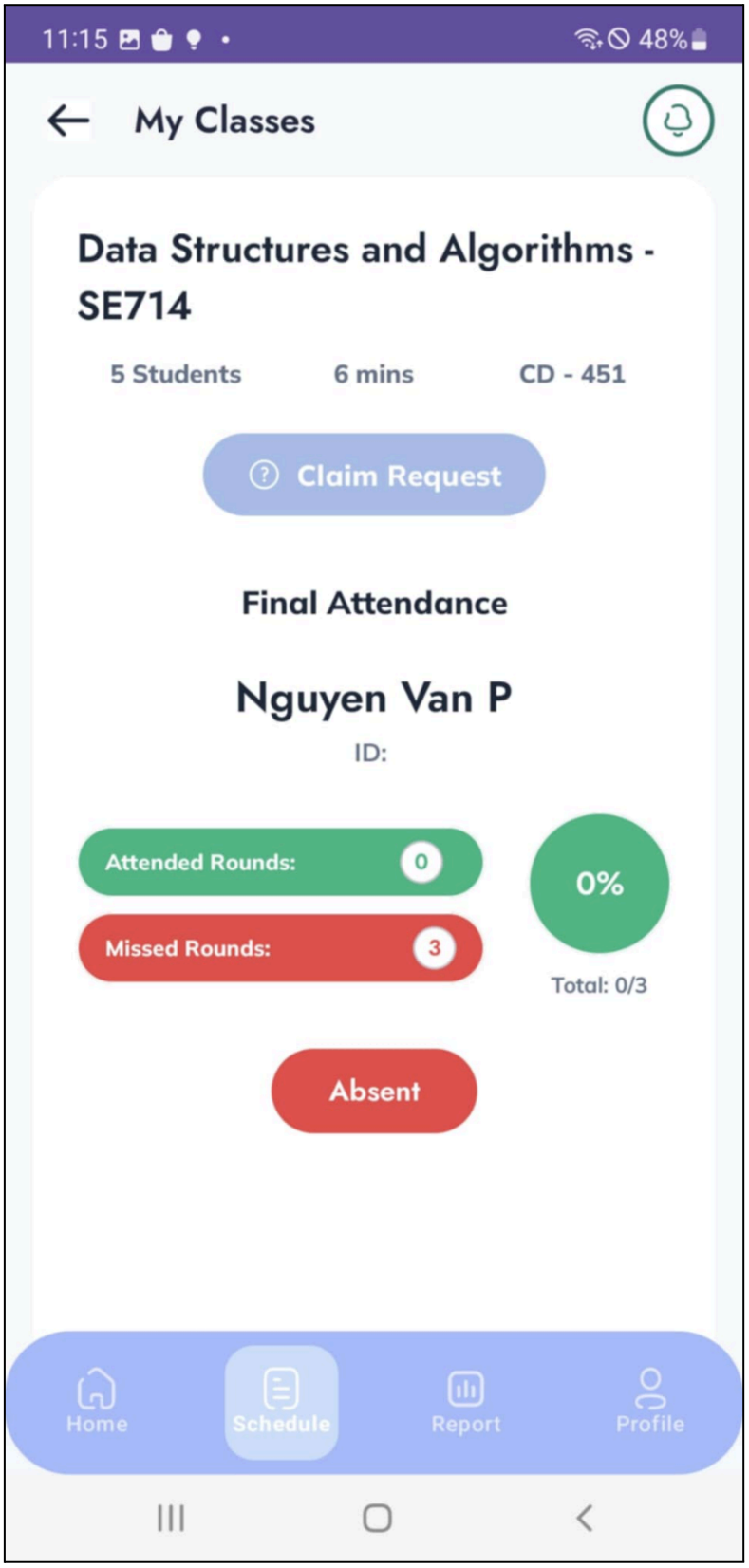


Figure 70: View Session Detail Layout

No	Field Name	Field Type	Description
S	**Title – My Classes**	Text Label	Screen title.

2	**Course Info**	Text Label	Shows subject name, code, number of students, duration, and room (e.g., Data Structures and Algorithms – SE714).
3	**Claim Request**	Button	Allows student to raise a request if attendance is incorrect.
4	**Student Info**	Text Label	Displays student's name and ID (e.g., Nguyen Van P).
5	**Attendance Stats**	Text Labels	Shows attended rounds, missed rounds, percentage, and total sessions.
6	**Attendance Status**	Status Label	Displays the attendance status (e.g., Absent).
7	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
8	**Bottom Tab – Schedule**	Navigation Tab	Active tab, showing current Schedule/Session Detail screen.
9	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
10	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 143: View Session Detail Field Table

b. Use Case Specifications

ID and Name	**UC – 17: View Session Detail**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Student	**Secondary Actors**	System
Trigger	Student taps on a specific session from the schedule or attendance history.		
Description	Allows students to view detailed information about a specific class session, including time, subject, location, attendance result, and session history.		
Preconditions	**PRE-1.** Students are signed in. **PRE-2.** The selected session has occurred or is active.		
Postconditions	**POST-1.** Session is retrieved and displayed.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Student	Selects a session from schedule or attendance report.
	2	System	Loads session information from server.
	3	System	Displays: – Subject, room, date/time

ID and Name	UC – 17: View Session Detail		
			– Attendance status (Present/Absent) – Join timestamp or verification log – Link to submit complaint (UC-19) if absent
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Session not linked to student	System shows: “Session not found or not assigned to you.”
Exceptions	• **E1:** If data retrieval fails, system shows: “Unable to load session details. Try again later.”		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02, BR-15**		
Other Information	• Useful for tracking attendance evidence. • Complaint access is only enabled if status is Absent and session is eligible (UC-19).		
Assumptions	• Attendance status is already processed and stored. • Session information is synced from admin-managed schedule (UC-39/40).		

Table 144: View Session Detail Specification

3.2.2.20 Verify Face ID Request

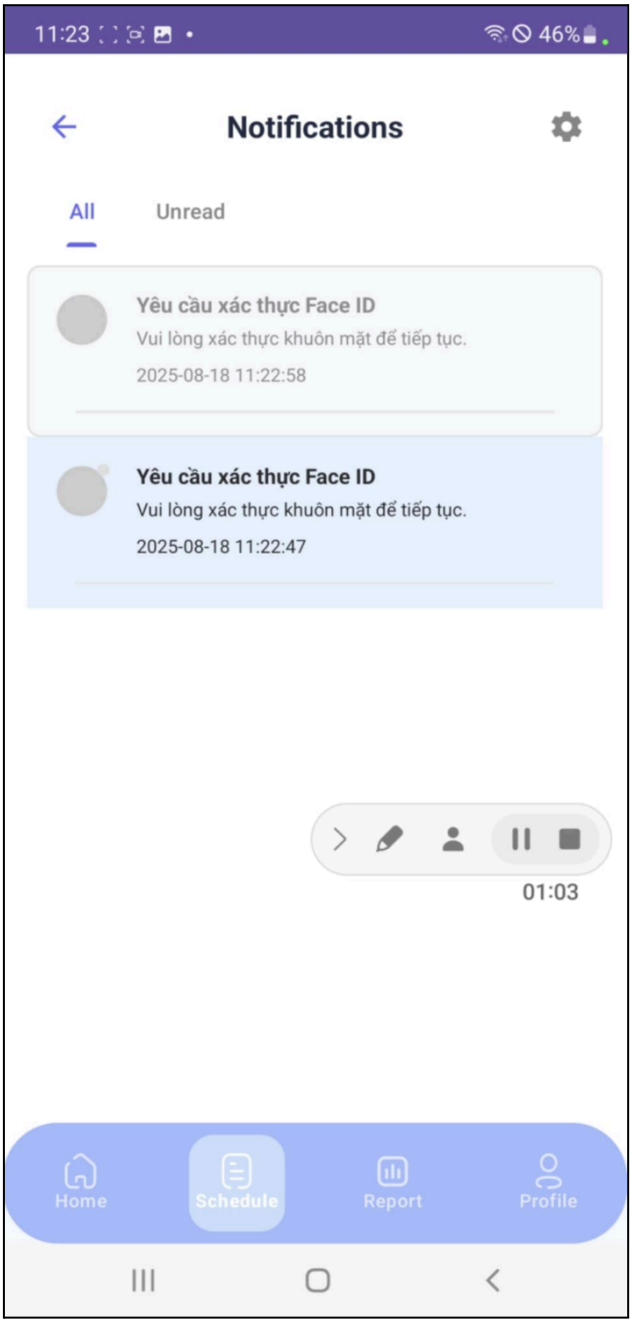


Figure 71: Notification Layout

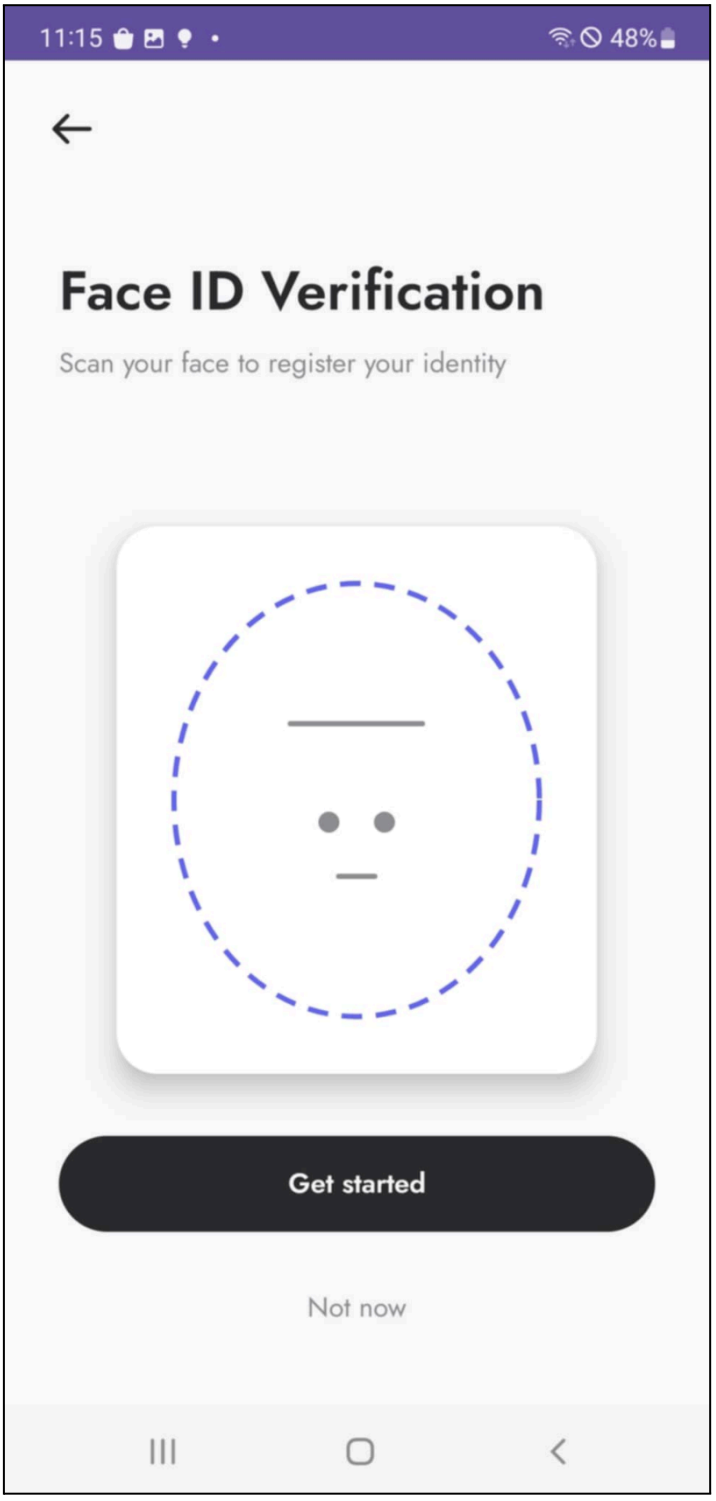


Figure 72: Verify Face ID Request Layout

a. Screen Layouts

No	Field Name	Field Type	Description
1	**Title – Notifications**	Text Label	Screen title showing notification list.
2	**Notification Tabs**	Tab Control	Tabs for All and Unread notifications.

3	**Notification Item**	List Item	Displays notification message (e.g., “Yêu cầu xác thực Face ID”) with timestamp.
4	**Settings Icon**	Icon Button	Opens notification settings.
5	**Face ID Verification Title**	Text Label	Title: “Face ID Verification”.
6	**Instruction Text**	Text Label	Helper text: “Scan your face to register your identity.”
7	**Face Frame/Illustration**	Graphic/Widget	Guide for face alignment during scanning.
8	**Get Started**	Primary Button	Starts the Face ID verification process.
9	**Not now**	Text Link	Skips the Face ID verification process.
10	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
11	**Bottom Tab – Schedule**	Navigation Tab	Shortcut to Schedule screen.
12	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
13	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 145: Verify Face ID Request Field Table

b. Use Case Specifications

ID and Name	**UC – 18: Verify Face ID Request**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Student	**Secondary Actors**	Lecturer, System
Trigger	The system sends FaceID verification requests during a class session.		
Description	Allow students to verify their identity via FaceID when requested by the lecturer, typically used during or after a session to confirm actual presence.		
Preconditions	**PRE-1.** Students are signed in. **PRE-2.** Attendance session is active. **PRE-3.** Student has registered FaceID.		
Postconditions	**POST-1.** FaceID verification is processed and result is stored.		
Normal Flows	**Step**	**By**	**Step Details**

ID and Name	UC – 18: Verify Face ID Request		
	1	System	Sends verification request to student's device.
	2	Student	Receives prompt and confirms FaceID scan.
	3	System	Launches front-facing camera and performs scan.
	4	System	Verifies scan result against registered data.
	5	System	Sends results (Success/Fail) to Lecturer.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Student declines request	System logs: "FaceID verification skipped by student."
Exceptions	• **E1:** If scan fails, system shows: "FaceID not recognized. Try again." • **E2:** If camera permission is not granted, system prompts for access.		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02, BR-04, BR-06**		
Other Information	• Optional but strongly recommended for identity integrity. • Verification may be used as audit evidence in disputes.		
Assumptions	• The device supports a camera and has network access. • Lecturers actively triggers the FaceID request from their session interface.		

Table 146: Verify Face ID Request Specification

3.2.2.21 View Lecturer Session Detail

a. Screen Layouts

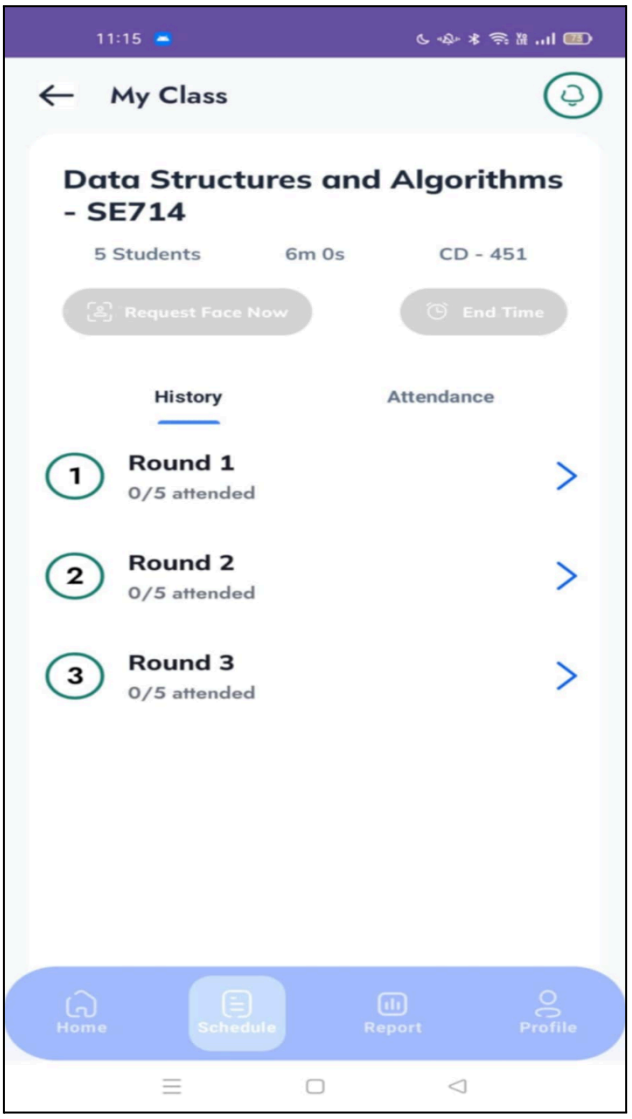


Figure 73: View Lecturer Session Detail Layout

No	Field Name	Field Type	Description
1	**Title – My Class**	Text Label	Screen title.
2	**Course Info**	Text Label	Displays subject name, code, number of students, duration, and room (e.g., Data Structures and Algorithms – SE714).
3	**Request Face Now**	Button	Allows lecturer to trigger a Face ID attendance request.
4	**End Time**	Button	Allows lecturer to end the attendance session.
5	**Tab – History**	Tab Option	Shows session history with list of rounds.
6	**Tab – Attendance**	Tab Option	Displays attendance details of students.

7	**Round List**	List Item	Each round shows round number and attendance count (e.g., 0/5 attended).
8	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
9	**Bottom Tab – Schedule**	Navigation Tab	Active tab, showing current Schedule/Session Detail screen.
10	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.
11	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 147: View Lecturer Session Detail Field Table

b. Use Case Specifications

ID and Name	**UC – 22: View Lecturer Session Detail**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Lecturer	**Secondary Actors**	System
Trigger	The lecturer selects a specific session from the schedule or calendar.		
Description	Allow lecturers to view detailed information for a class session, including subject, room, time, student list, attendance history, and final statistics.		
Preconditions	**PRE-1.** Lecturer is signed in. **PRE-2.** Session is assigned to the lecturer.		
Postconditions	**POST-1.** Session details are displayed and accessible for further actions.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	Opens session from schedule/calendar.
	2	System	Loads session data: class name, date, time, and location.
	3	System	Displays: - Student list - Attendance status (Present, Absent, Excused) - Duration - Final status (Completed/In Progress)
	4	Lecturer	May navigate additional actions (UC-35 Start Session, UC-33 Manual Edit).
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Session has not yet started	System shows upcoming session details only (no attendance yet).
Exceptions	• **E1:** If session data fails to load, system shows: “Unable to retrieve session details.”		
Frequency of Use	High		
Business Rules	**BR-01, BR-02, BR-07, BR-15**		

ID and Name	UC – 22: View Lecturer Session Detail
Other Information	• Supports access to edit, verify FaceID, or respond to complaints. • Session details may include performance graphs and summary stats.
Assumptions	• Attendance session is synced in real-time with system. • Lecturer has appropriate permissions to view and manage session data.

Table 148: View Lecturer Session Detail Specification

3.2.2.22 View Class Attendance Report

a. Screen Layouts

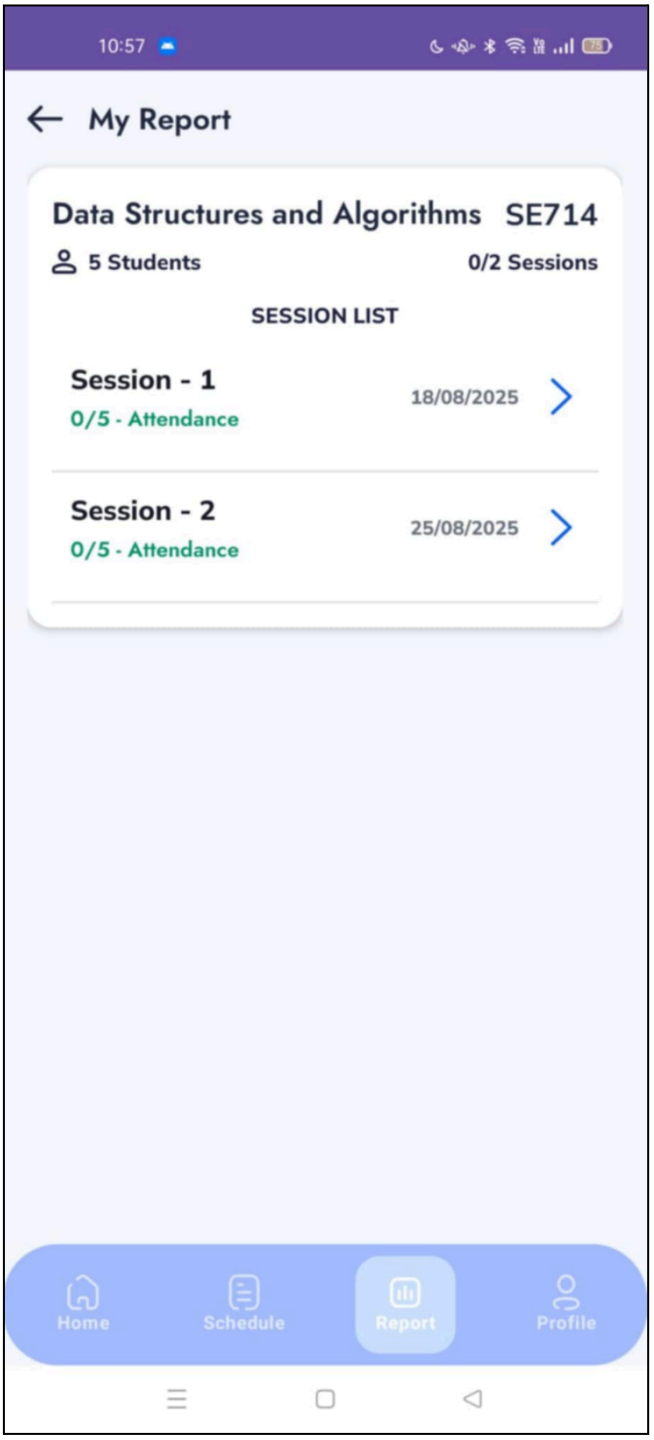


Figure 74: View Class Attendance Report Layout

No	Field Name	Field Type	Description
1	**Title – My Report**	Text Label	Screen title.
2	**Course Info**	Text Label	Shows subject name, code, and student count (e.g., Data Structures and Algorithms – SE714, 5 Students).
3	**Total Sessions**	Text Label	Displays the total number of sessions (e.g., 0/2 Sessions).
4	**Session List**	List Item	Displays each session with attendance count and date.
5	**Session Detail Link**	Button/Icon	Each session entry includes a link (arrow) to open detailed attendance for that session.
6	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
7	**Bottom Tab – Schedule**	Navigation Tab	Shortcut to Schedule screen.
8	**Bottom Tab – Report**	Navigation Tab	Active tab, showing current Report screen.
9	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 149: View Class Attendance Report Field Table

b. Use Case Specifications

ID and Name	UC – 23: View Class Attendance Report		
Created By	Dungktn	Date Created	28/06/2025
Primary Actor	Lecturer	Secondary Actors	System
Trigger	The lecturer opens an attendance report section and selects a class.		
Description	Enables lecturers to view aggregated attendance statistics for a specific class, including student participation rates, attendance breakdowns, and performance warnings.		
Preconditions	**PRE-1.** Lecturer is signed in. **PRE-2.** The lecturer is assigned to the selected class.		
Postconditions	**POST-1.** The system displays attendance statistics for the class.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	Navigates to “Reports” section in the app.
	2	System	Displays list of assigned classes.
	3	Lecturer	Selects a class to view attendance statistics.

ID and Name	UC – 23: View Class Attendance Report		
	4	System	Shows: – List of students – Attendance % per student – Number of sessions attended, missed, excused – Alerts for low attendance
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No attendance data available	System shows: “No attendance data available for this class.”
Exceptions	• **E1:** If report data fails to load, show: “Unable to load report. Please try again later.”		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02, BR-14, BR-15**		
Other Information	• May include class-wide attendance trends and semester summaries.		
Assumptions	• Attendance data has been consistently recorded and updated from sessions. • System supports real-time reporting and alert thresholds.		

Table 150: View Class Attendance Report Specification

3.2.2.23 View Session Attendance Report

a. Screen Layouts

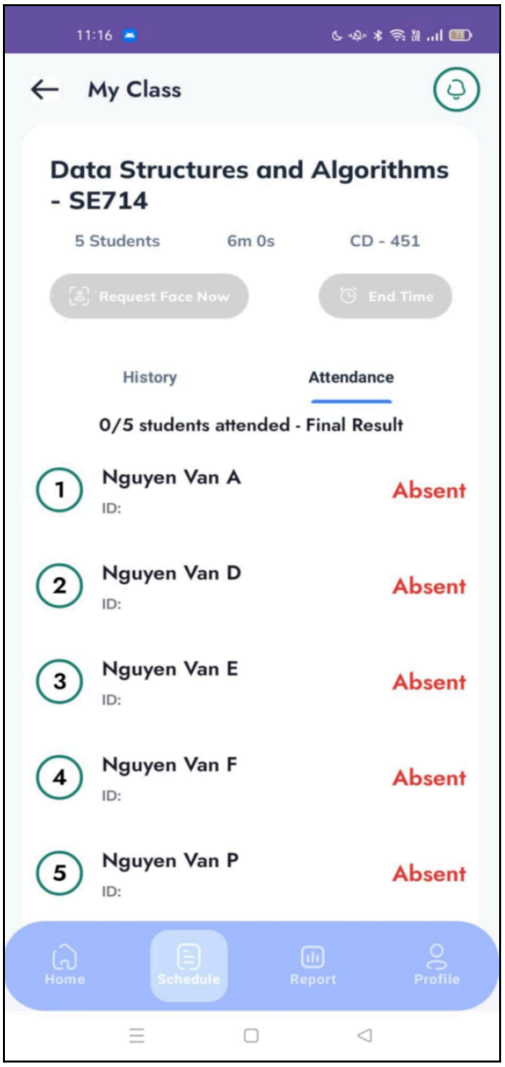


Figure 75: View Session Attendance Report Layout

No	Field Name	Field Type	Description
1	**Title – My Class**	Text Label	Screen title.
2	**Course Info**	Text Label	Shows subject name, number of students, duration, and room (e.g., Data Structures and Algorithms – SE714, 5 Students, 6 mins, CD-451).
3	**Request Face Now**	Button (disabled)	Option to trigger Face ID request (inactive when viewing past session).
4	**End Time**	Button (disabled)	Option to end attendance session (inactive when viewing past session).
5	**Tab – History**	Tab Option	Switch to view attendance history.
6	**Tab – Attendance**	Tab Option	Active tab, showing student attendance results.

7	**Attendance List**	List Item	Displays list of students with ID, attendance status (e.g., Absent).
8	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
9	**Bottom Tab – Schedule**	Navigation Tab	Shortcut to Schedule screen.

Table 151: View Session Attendance Report Field Tale

b. Use Case Specifications

ID and Name	**UC – 24: View Session Attendance Report**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Lecturer	**Secondary Actors**	System
Trigger	The lecturer opens a specific class session report to review attendance.		
Description	Enables lecturers to view attendance statistics for a specific class session, including individual student records, attendance statuses, and timestamps.		
Preconditions	**PRE-1.** Lectures are signed in. **PRE-2.** Lecturer is assigned to the session. **PRE-3.** The session has been completed, and attendance has been recorded.		
Postconditions	**POST-1.** The system displays full session attendance details.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	Navigates to a completed session via calendar or class detail.
	2	System	Retrieves attendance record for the selected session.
	3	System	Displays: – List of enrolled students – Status (Present, Absent, Excused) – Time joined – Method (Auto, Manual, FaceID)
	4	Lecturer	May optionally access complaint or FaceID verification history.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No attendance recorded	System shows: “Attendance has not been taken for this session.”
Exceptions	• **E1:** If session data retrieval fails, show: “Unable to load session report. Try again later.”		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02, BR-07, BR-15**		
Other Information	• May include breakdown of attendance method (Auto vs Manual). • Used to support complaint resolution and audit trails.		
Assumptions	• Session is finalized and synced. • Only the assigned lecturer has access.		

Table 152: View Session Attendance Report Specification

3.2.2.24 Broadcast and Scan Signal

ID and Name	UC – 25: Broadcast and Scan Signal		
Created By	Dungktn	Date Created	28/06/2025
Primary Actor	Lecturer, Student	Secondary Actors	System
Trigger	At the beginning of the session, the lecturer initiates a signal; students scan for it.		
Description	The lecturer broadcasts a Bluetooth signal to initiate the attendance session. Students scan for the signal to register their presence in real time.		
Preconditions	**PRE-1.** Both lecturers and students are signed in. **PRE-2.** Devices have Bluetooth enabled and are within range. **PRE-3.** Lecturer has started the session (UC-35).		
Postconditions	**POST-1.** Students who detect the signal are marked as present (pending verification if needed).		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	Opens the session and taps “Start Broadcast.”
	2	System	Broadcasts Bluetooth signal unique to session ID.
	3	Student	Opens session and taps “Scan Signal.”
	4	System	Detects broadcast and validates signal.
	5	System	Marks student as joined (subject to FaceID check in UC-23).
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Student device fails to detect signal	System shows: “No signal detected. Move closer or try again.”
Exceptions	• **E1:** If Bluetooth is turned off on either device, system shows appropriate prompt. • **E2:** If session hasn’t started, system alerts: “Attendance session not available yet.”		
Frequency of Use	High		
Business Rules	**BR-01, BR-02, BR-03, BR-05, BR-07**		
Other Information	• Signal is time-limited and rotates per session to prevent misuse. • Accuracy depends on Bluetooth hardware and proximity (typically <10m).		
Assumptions	• Mobile devices are paired with correct student accounts. • Lecturers start the session from their own device using UC-35.		

Table 153: Broadcast and Scan Signal Specification

3.2.2.25 Create Face ID Request

a. Screen Layouts

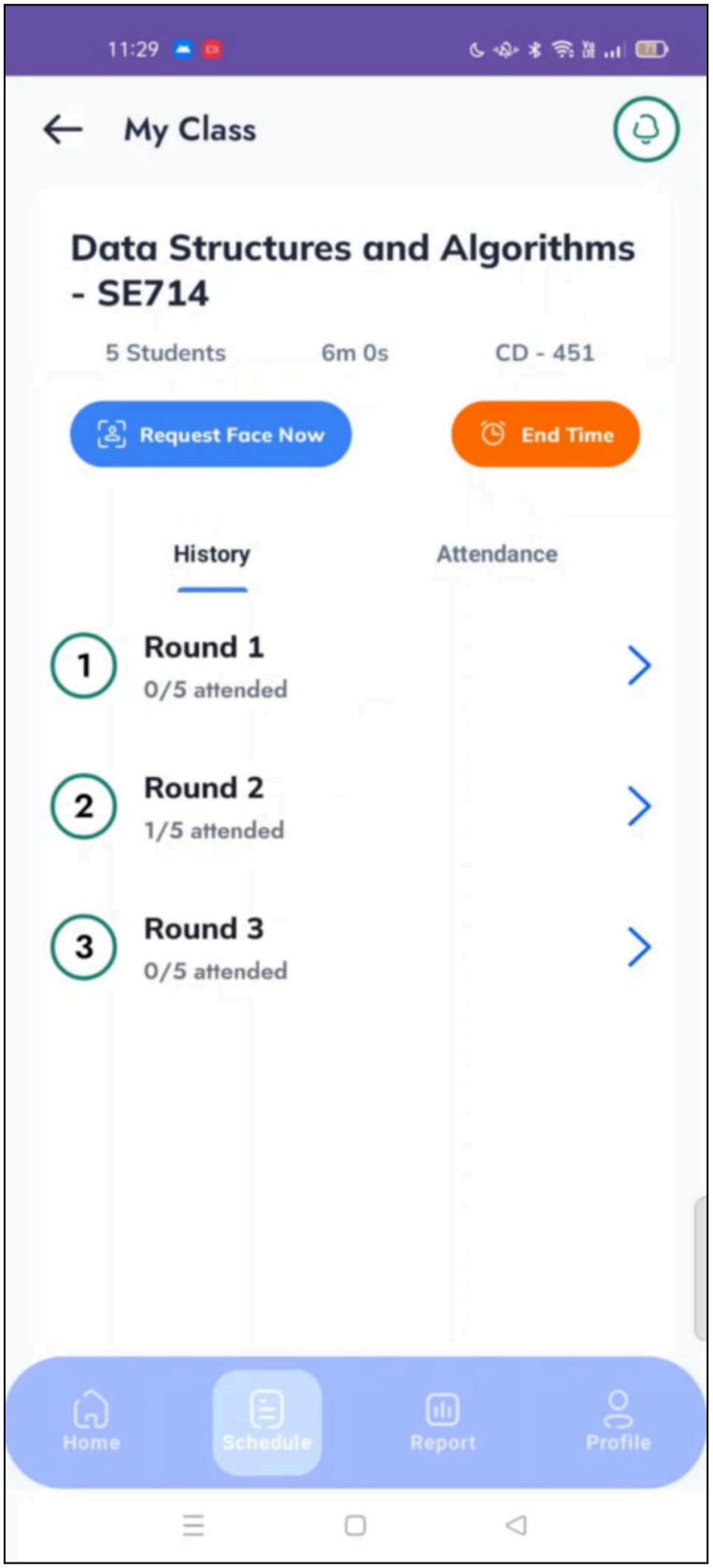


Figure 76: Schedule Layout

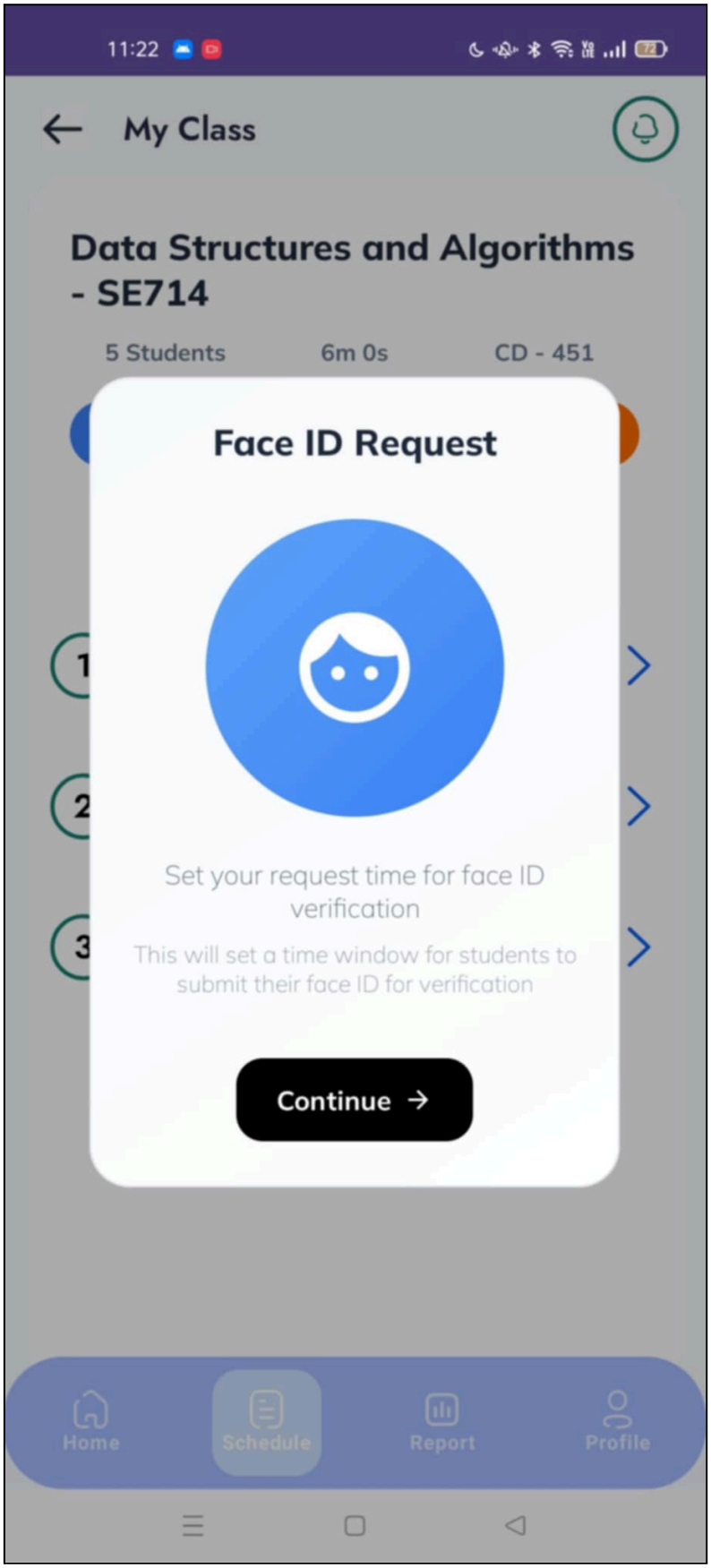


Figure 77: Create Face ID Request

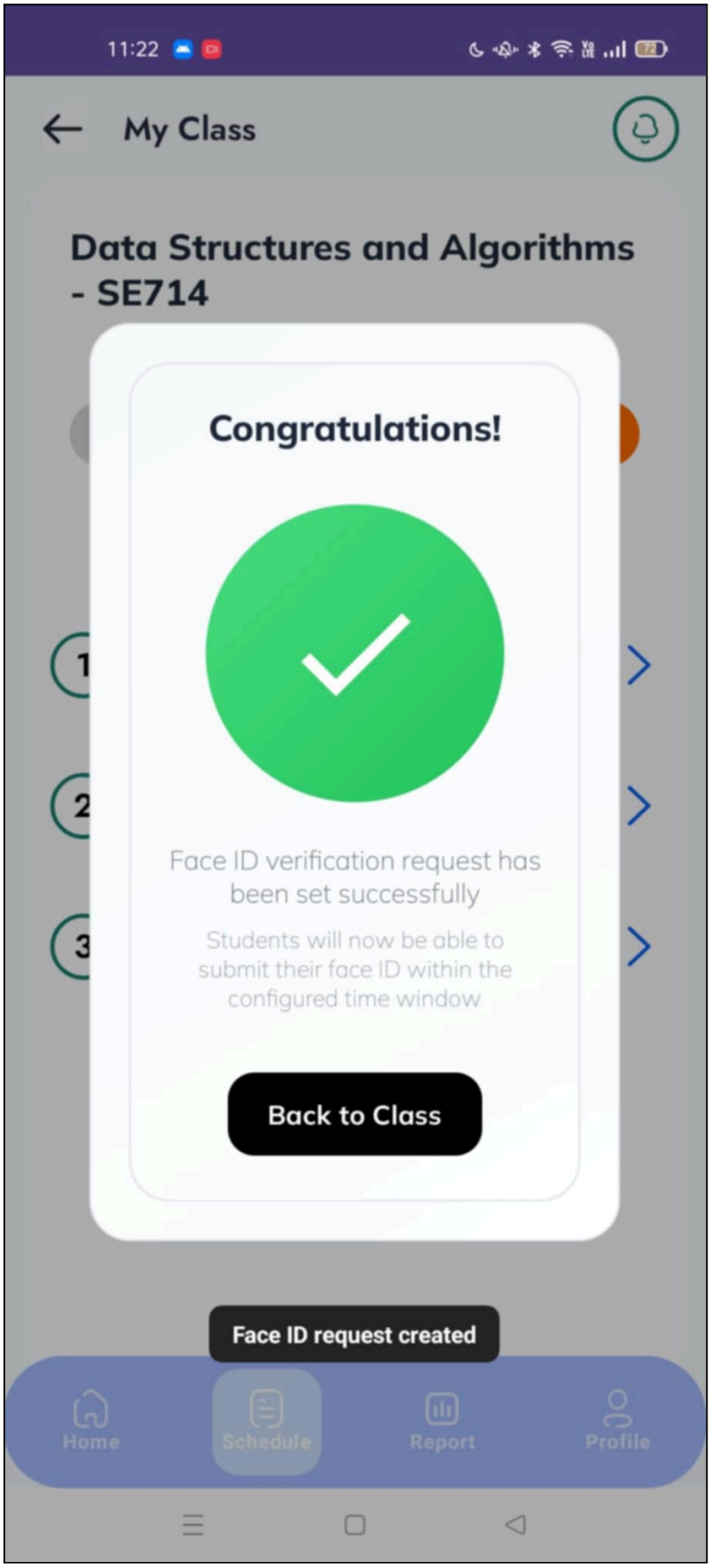


Figure 78: Create Face ID Request Successfully layout

No	Field Name	Field Type	Description

1	**Title – My Class**	Text Label	Screen title.
2	**Course Info**	Text Label	Displays course name, student count, session duration, and room (e.g., *Data Structures and Algorithms – SE714*).
3	**Request Face Now**	Primary Button	Allows lecturer to initiate a Face ID verification request.
4	**End Time**	Secondary Button	Ends the ongoing attendance session.
5	**Tab – History**	Tab Option	Shows list of rounds with attendance counts.
6	**Tab – Attendance**	Tab Option	Shows student attendance details.
7	**Round List**	List Item	Displays attendance per round (e.g., *0/5 attended*).
8	**Face ID Request Popup – Title**	Text Label	Title of popup (e.g., “Face ID Request”).
9	**Face ID Request Popup – Instruction**	Text Label	Explains the time window setup for Face ID submission.
10	**Face ID Request Popup – Continue**	Button	Confirms request initiation and proceeds.
11	**Success Popup – Message**	Text Label	Confirmation that Face ID request has been set successfully.
12	**Success Popup – Back to Class**	Button	Returns lecturer to the session detail screen.
13	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
14	**Bottom Tab – Schedule**	Navigation Tab	Active tab, showing current Schedule/Class screen.
15	**Bottom Tab – Report**	Navigation Tab	Shortcut to Report screen.

16	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 154: Create Face ID Request Field Table

b. Use Case Specifications

ID and Name	**UC – 26: Create Face ID Request**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Lecturer	**Secondary Actors**	Student, System
Trigger	Lecturer chooses to initiate a FaceID verification during an attendance session.		
Description	Allows the lecturer to send a FaceID verification request to one or more students during a session to verify actual presence and detect proxy attendance.		
Preconditions	**PRE-1.** Lecturer is signed in. **PRE-2.** Attendance session is active. **PRE-3.** Students have registered FaceID and are marked "joined."		
Postconditions	**POST-1.** Students receive verification requests and must respond immediately.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	From the session detailed screen, selects one or more students to verify.
	2	System	Sends FaceID requests to select student devices.
	3	Student	Receives request and performs FaceID scan (handled in UC-23).
	4	System	Processes scan and updates result in lecturer's session dashboard.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Students already verified in session	System blocks duplicate FaceID request for that student.
Exceptions	• **E1:** If FaceID request fails to send, system shows: "Unable to send request. Try again." • **E2:** If student device is offline, system flags: "Device unreachable."		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02, BR-04, BR-06**		
Other Information	• Used to prevent impersonation and enforce attendance integrity. • History of FaceID requests can be accessed in session detail for audits.		
Assumptions	• Students must respond in real time; late responses are logged. • Each request is traceable and timestamped.		

Table 155: Create Face ID Request Specification

3.2.2.26 Automotive Attendance

ID and Name	UC – 27: Automotive Attendance		
Created By	Dungktn	Date Created	28/06/2025
Primary Actor	Lecturer	Secondary Actors	Student, System
Trigger	Lecturer starts a session and chooses automatic attendance mode.		
Description	Enables lecturers to conduct attendance automatically using Bluetooth signal detection. Students in proximity with registered devices are marked present automatically without manual intervention.		
Preconditions	**PRE-1.** Lecturers are signed in. **PRE-2.** Attendance session has started (UC-35). **PRE-3.** Students' devices are registered and have Bluetooth enabled.		
Postconditions	**POST-1.** Attendance list is generated automatically.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	Starts class session and selects “Automatic Attendance.”
	2	System	Begins scanning for nearby student devices broadcasting signals.
	3	System	Matches signal to student registry and confirms device validity.
	4	System	Marks students as present automatically and logs time.
	5	Lecturer	Reviews auto-generated list and proceeds with optional FaceID verification (UC-31).
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No signals detected	System alerts: “No student devices detected nearby.”
Exceptions	• **E1:** If the system fails to match device, it is excluded and may require manual review. • **E2:** If student device is not broadcasting or out of range, student is not marked present.		
Frequency of Use	High		
Business Rules	**BR-01, BR-02, BR-03, BR-05, BR-07**		
Other Information	• The automatic method is time-bound (e.g., 10-minute window). • Attendance data is synced for reporting (UC-29, UC-28). • May be combined with FaceID requests for added validation.		
Assumptions	• Bluetooth signals are unique per student and registered. • All data is logged for transparency and audit trails.		

Table 156: Automotive Attendance Specification

3.2.2.27 Manual Attendance

a. Screen Layouts

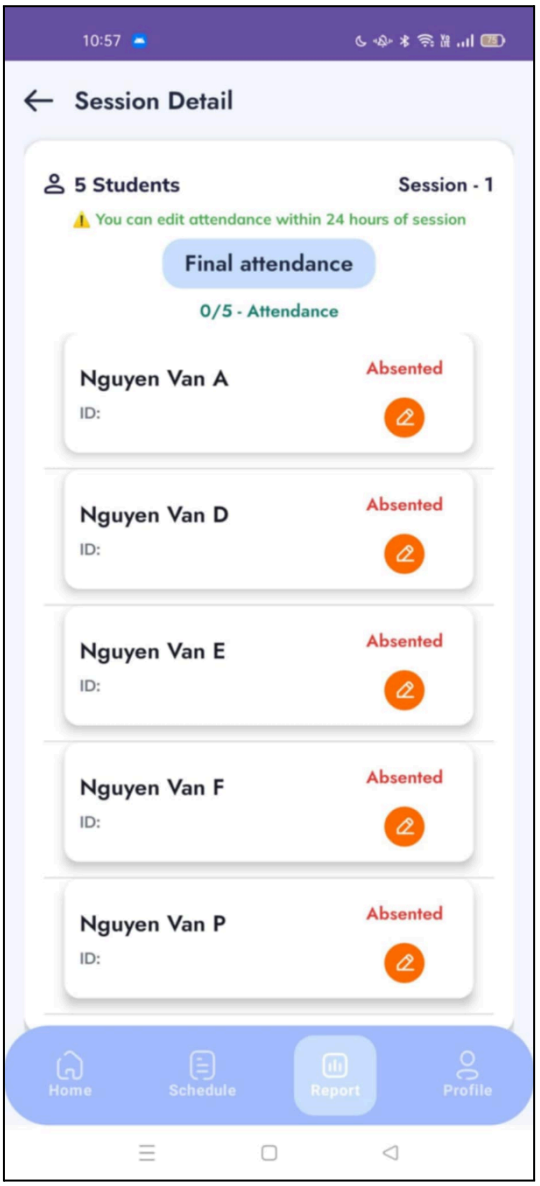


Figure 79: Manual Attendance layout

No	Field Name	Field Type	Description
1	**Title – Session Detail**	Text Label	Screen title.
2	**Session Info**	Text Label	Displays number of students and session number (e.g., 5 Students – Session 1).
3	**Edit Notice**	Text Label	Warning message that attendance can be edited within 24 hours of the session.
4	**Final Attendance**	Button/Label	Section header for final attendance results (e.g., 0/5 Attendance).
5	**Student List**	List Item	Displays student name, ID, and attendance status.

6	**Attendance Status**	Status Label	Indicates attendance status (e.g., Absented).
7	**Manual Edit Icon**	Action Icon	Allows lecturer to manually edit or override attendance status.
8	**Bottom Tab – Home**	Navigation Tab	Shortcut to Home screen.
9	**Bottom Tab – Schedule**	Navigation Tab	Shortcut to Schedule screen.
10	**Bottom Tab – Report**	Navigation Tab	Active tab, showing current Report/Session screen.
11	**Bottom Tab – Profile**	Navigation Tab	Shortcut to Profile screen.

Table 157: Manual Attendance Field Table

b. Use Case Specifications

ID and Name	**UC – 28: Manual Attendance**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Lecturer	**Secondary Actors**	System
Trigger	The lecturer chooses to manually adjust attendance during or after a session.		
Description	Enables lecturers to manually mark, adjust, or correct attendance records for students in special situations, such as technical issues, late arrivals, or verification failures.		
Preconditions	**PRE-1.** Lecturer is signed in. **PRE-2.** Session is active or has ended within the allowed adjustment window. **PRE-3.** Lecturer has proper authorization to edit attendance.		
Postconditions	**POST-1.** Adjusted attendance is saved and reflected in session records.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	Opens session detail and selects “Manual Attendance.”
	2	System	Displays full student list with current attendance status.
	3	Lecturer	Edits statuses (e.g., Absent → Present, adds Excused) and optionally adds notes.
	4	System	Validates and saves updated attendance records.
	5	System	Logs the changes with timestamp and editor identity for audit.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Session outside editable window	System shows: “Manual editing not allowed after 24 hours.”

ID and Name	UC – 28: Manual Attendance
Exceptions	• **E1:** If save fails, system shows: “Update failed. Please try again or contact admin.” • **E2:** If lecturer lacks permission, system blocks edit and displays: “Access denied.”
Frequency of Use	Medium
Business Rules	**BR-01, BR-02, BR-08, BR-11, BR-15**
Other Information	• Notes on edits help in resolving attendance complaints (UC-34). • Manual edits are visible in session history with reason tags.
Assumptions	• Session is editable within 24 hours of end time (BR-08). • Lecturer decisions override automatic detection when justified.

Table 158: Manual Attendance Specification

3.2.2.28 Start Attendance Session

a. Screen Layouts

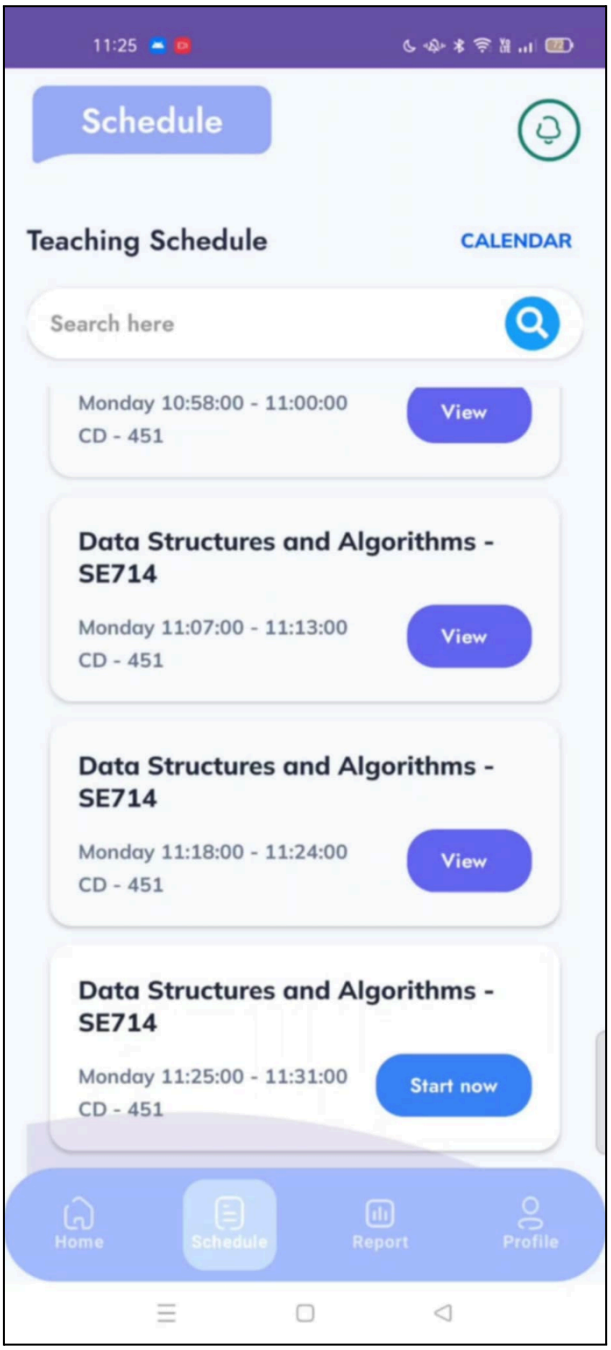


Figure 80: Start Attendance Session layout

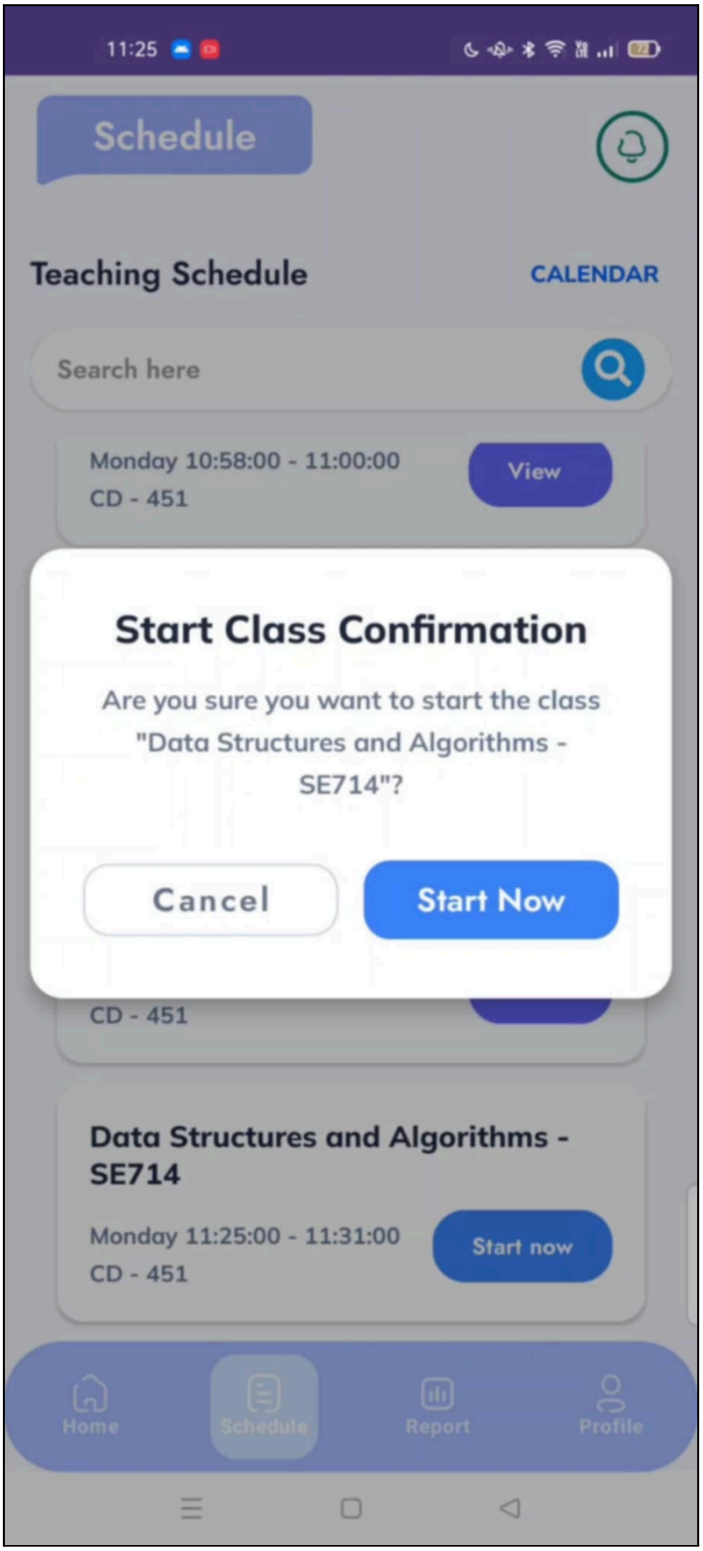


Figure 81: Confirm start layout

No	Field Name	Field Type	Description
1	**Header Title**	Label/Text	Displays the title “Schedule”.
2	**Calendar Button**	Icon/Button	Shortcut to switch to calendar view.

3	**Search Bar**	Textbox + Icon	Input field to search class sessions with a search icon.
4	**Class Card**	Container	Each scheduled class displayed as a card.
5	**Subject Name**	Label/Text	Shows subject name (e.g., Data Structures and Algorithms – SE714).
6	**Class Time**	Label/Text	Displays session start and end time.
7	**Classroom Code**	Label/Text	Displays classroom code (e.g., CD – 451).
8	**Action Button**	Button	"View" for normal sessions, "Start Now" for starting attendance.
9	**Bottom Navigation**	Navigation Bar	Contains 4 buttons: Home, Schedule, Report, Profile.

Table 159: Start Attendance Session Field Table

b. Use Case Specifications

ID and Name	**UC – 29: Start Attendance Session**		
Created By	Dungktn	**Date Created**	28/06/2025
Primary Actor	Lecturer	**Secondary Actors**	System
Trigger	The lecturer selects a scheduled class and initiates the attendance session.		
Description	Allow lecturers to start a new attendance session manually or automatically, enabling students to join and register attendance within a valid time window.		
Preconditions	**PRE-1.** Lecturer is signed in. **PRE-2.** The selected session is scheduled and assigned to the lecturer. **PRE-3.** The current time is within allowed timeframe.		
Postconditions	**POST-1.** Attendance session is active and accepting student joins.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Lecturer	Opens upcoming session from schedule/calendar.
	2	Lecturer	Selects "Start Session" and chooses mode: Automatic or Manual.
	3	System	Activates session and begins tracking attendance inputs.
	4	System	Broadcasts signal (if applicable) and opens join window.
	5	System	Displays real-time student join status to lecturer.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Session has already started	System shows: "Session already active or completed."
Exceptions	• **E1:** If network fails, system shows: "Unable to start session. Check connection."		

ID and Name	UC – 29: Start Attendance Session
	• **E2:** If schedule mismatch occurs, system alerts: “Session not found or not assigned.”
Frequency of Use	High
Business Rules	**BR-01, BR-02, BR-05, BR-07, BR-17**
Other Information	• Session duration and end rules are defined in admin configuration (UC-42). • Signal may be auto disabled after a timeout.
Assumptions	• Lecturer starts session only once. • Students will use UC-21 to join after session is active.

Table 160: Start Attendance Session Specification

3.2.2.29 Backfill Data

ID and Name	UC – 15: Backfill Data		
Created By	Dungktn	Date Created	28/06/2025
Primary Actor	System	Secondary Actors	Student, Lecturer
Trigger	Device reconnects to the internet after being offline.		
Description	Automatically detects and synchronizes attendance data that was captured and stored locally on a mobile device while it was offline. Ensures that no attendance records are lost due to temporary connectivity issues.		
Preconditions	**PRE-1.** The device has stored offline data locally. **PRE-2.** Internet connection is re-established.		
Postconditions	**POST-1.** The report is displayed with up-to-date attendance metrics.		
Normal Flows	**Step**	**By**	**Step Details**
	1	System	Detects that device has reconnected to the internet.
	2	System	Scans local storage for pending attendance data.
	3	System	Verifies data integrity and timestamps.
	4	System	Uploads valid records to central server.
	5	System	Marks records as synced and clears local cache.
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	No data found locally	System skips sync process.
Exceptions	• **E1:** If upload fails due to server error, data is queued for retry.		
Frequency of Use	Automatic		
Business Rules	**BR-01, BR-02, BR-13, BR-15, BR-19**		
Other Information	• Retry queue supports exponential backoff strategy. • All operations are transparent to the user; no manual action is required.		
Assumptions	• Device storage is secure and accessible. • Local records include all necessary metadata for validation.		

Table 161: Backfill Data Specification

3.2.2.30 Process Attendance Data

ID and Name	UC – 76: Process Attendance Data		
Created By	Dungktn	Date Created	28/06/2025
Primary Actor	System	Secondary Actors	Student, Lecturer
Trigger	Students join sessions or attendance records are submitted from mobile devices.		
Description	The system receives raw attendance data (e.g., Bluetooth signal, join timestamps, FaceID results), validates it against session configuration and business rules, and stores final attendance records in the database.		
Preconditions	**PRE-1.** Attendance session is active or recently ended. **PRE-2.** Devices are authorized and synced.		
Postconditions	**POST-1.** Validated attendance is stored in the central database.		
Normal Flows	**Step**	**By**	**Step Details**
	1	System	Receives raw attendance signals or records from devices.
	2	System	Validates device identity, signal time, and session matching.
	3	System	Applies FaceID verification (if required).
	4	System	Checks business rules (e.g., time window, session overlap).
	5	System	Stores finalized attendance record with status (Present, Absent, Excused).
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Data duplicates or already processed	System ignores and logs for review.
Exceptions	• **E1:** If data is incomplete or invalid, system logs it and excludes from attendance. • **E2:** If session not found, system discards record and alerts admin.		
Frequency of Use	High		
Business Rules	**BR-01, BR-02, BR-03, BR-04, BR-05, BR-06, BR-07, BR-15, BR-19**		
Other Information	• Every record includes timestamp, device ID, and verification status. • All processing is automated, no human intervention required.		
Assumptions	• Network and synchronization with mobile app are stable. • All students use registered devices and follow session rules.		

Table 162: Process Attendance Data Specification

3.2.2.31 Store Local Attendance Data

ID and Name	UC – 77: Store Local Attendance Data		
Created By	Dungktn	Date Created	28/06/2025
Primary Actor	System	Secondary Actors	Student, Lecturer
Trigger	Students join an attendance session while the mobile device is offline.		
Description	When a student joins an attendance session without internet connectivity, the mobile app stores the attendance record locally on the device. This data will be automatically synchronized when a connection is restored (UC-20).		

ID and Name	UC – 77: Store Local Attendance Data		
Preconditions	**PRE-1.** Students join sessions offline. **PRE-2.** Session and student data is caught locally.		
Postconditions	**POST-1.** Attendance data is stored in encrypted local storage.		
Normal Flows	**Step**	**By**	**Step Details**
	1	Student	Opens the app and joins attendance session (offline).
	2	System	Records session ID, student ID, timestamp, and verification result (if any).
	3	System	Store the record securely in local storage.
	4	System	Flags the record for sync once connectivity is restored (see UC-20).
Alternative Flows	**Alt Step**	**Condition**	**Description**
	A1	Student declines request	System logs: “FaceID verification skipped by student.”
Exceptions	• **E1:** If local write fails, system logs error and alerts user to retry later.		
Frequency of Use	Medium		
Business Rules	**BR-01, BR-02, BR-03, BR-13, BR-15, BR-19**		
Other Information	• Data includes checksum to prevent tampering. • Sync process is handled by UC-20 (Backfill Data). • Offline records are not shown to lecturers until synced.		
Assumptions	• App has local storage permission and enough space. • Student device maintains accurate time for timestamping.		

Table 163: Store Local Attendance Data Specification

4. Non-Functional Requirements

No	Category	Requirement
AAS Web System		
01	**Performance**	The web system must respond to user actions within 1 second for 95% of requests.
02	**Scalability**	The system must support at least 10,000 concurrent users during peak load.
03	**Availability**	The system must maintain 99.9% uptime monthly, excluding scheduled maintenance.
04	**Reliability**	Attendance records and configuration settings must not be lost even in the event of server failure.
05	**Security**	All access must be protected via HTTPS, and only authenticated users can access the system.
06	**Maintainability**	The codebase must be modular and follow clean architecture to enable updates without downtime.
07	**Usability**	Admin dashboard must be accessible with clear labels, icons, and support for basic accessibility.
08	**Data Integrity**	Any change to schedules, devices, or user data must be logged with timestamps and actors.

No	Category	Requirement
09	**Compliance**	The system must comply with institutional data retention policies (e.g., keeping logs for 6 months).
10	**Backup & Recovery**	Daily backups must be stored and recoverable within 4 hours of data loss incident.
AAS Mobile System		
01	**Performance**	App must respond within 1.5 seconds for 95% of user interactions (e.g., tap, navigation).
02	**Offline Capability**	Apps must support offline attendance and syncing when connection is restored (see UC-20, UC-46).
03	**Battery Efficiency**	Bluetooth broadcasting and scanning must minimize battery drain during sessions.
04	**Security**	Apps must enforce biometric or passcode login where applicable (e.g., FaceID, fingerprint).
05	**Platform Support**	The app must run smoothly on Android 10+ and iOS 13+ with backward compatibility.
06	**Data Integrity**	Local data must be encrypted and tamper-proof while awaiting sync.
07	**Usability**	Interface must be designed for quick one-hand use and be intuitive for non-technical users.
08	**Notification System**	Students and lecturers must receive timely push notifications for attendance and complaints.
09	**Accessibility**	Apps must be used by individuals with color blindness and provide screen reader support.
10	**Recovery**	In case of app crash, data must persist and automatically resume the previous state on restart.

Table 164: Non-Functional Requirements Table